MONTHLY ROLLUP • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

October 2026 | Entry window 1–31 October 2026, 31 daily issues | 523 records pooled

523 RECORDS POOLED (31 DAILY ISSUES)

130 EXPOSURE ASSESSMENT (25%), LARGEST BAND

113 EFFECT ESTIMATES (113 WITH A CI)

75 TIER-A RECORDS (14%)

112 METADATA-ONLY RECORDS (21%)

Signal of the month

October pulled the exposure variable apart at both ends: the epidemiology moved from $PM_{2.5}$ mass to its components, its number concentration and its source, while the measurement literature showed that the references every sensor is calibrated against carry biases of their own.

The health side. In five weeks the effect was carried by something other than mass in every large cohort that could separate it. Never-smoker lung cancer in the China Kadoorie Biobank: nitrate HR **1.35** (1.23–1.48) per IQR against 1.13 for the mixture (3 Oct). Dementia mortality in 2.1 million adults: HR **1.22** (1.17–1.27) per 10,000 cm^{-3} of ultrafine particle number (5 Oct). Incident ADRD in 39.7 million Medicare beneficiaries: HR **1.029** (1.024–1.035) per 1 $\mu g\,m^{-3}$ of smoke $PM_{2.5}$ but **1.090** (1.077–1.103) per 1 $\mu g\,m^{-3}$ of smoke EC+OC (W42). eGFR decline in 307,639 adults at HR **1.27–1.30** for sulfate, nitrate and ammonium; ADHD following elemental carbon at HR **1.24** independent of $PM_{2.5}$ (W43). Prenatal wildfire $PM_{2.5}$ and autism in 893,212 Medicaid pairs: HR **1.30** (1.07–1.59) per 0.75 $\mu g\,m^{-3}$, ADHD null (*JAMA Pediatr*, 26 Oct). **Every one of these component, number and smoke exposures is modelled** — CTM, satellite–ML, smoke-fraction regression or source apportionment — not measured.

The measurement side. A three-level meta-analysis of 96 paired comparisons put professional photometers at **36 % high** against gravimetry: lnR 0.31 (0.25–0.37), I^2 **78.4 %**, prediction interval 0.11–0.63; site calibration was the strongest moderator (factory 43 % high, site 13 %) and no eligible study tested a sub-$500 sensor (*Ann Work Expo Health*, 23 Oct). Gravimetry itself retains strongly bound water through standard conditioning, a **+3 to +18 %** mass bias largest for PM_1 (17 Oct). The PurpleAir housing warms the PMS5003 detection zone and dries the sample before it is counted, so RH corrections fitted on bare sensors assume the wrong humidity (27 Oct). **The month's verdict: a network that reports corrected mass is converging on a quantity the epidemiology is leaving behind, and is converging on it against a reference that is itself biased by composition and water.**

What changed in October

- **Volume fell 16 % and the mix moved toward modelling and the lung.** 523 records against 622 in September, on Sunday–Saturday week totals of 59 (1–3 Oct, partial), 87, 123, 119 and 135. Daily volume ran from **1** (Sunday 11 Oct) to **30** (1 and 27 Oct). *Exposure assessment & modelling* grew in every week (16, 23, 26, 31, **34**) and is the largest band at 130 (25 %); *Respiratory & allergic* rose from **1 to 17** per week. *Sensing* fell to 77 (15 %, from 19 %), *Cardiovascular & metabolic* 46 → 28 and *Occupational & indoor* 72 → 50. The measurement share (sensing plus exposure) read 42, 44, 39, 39 and **37 %** by week: the measurement side is now mostly models, not instruments.
- **Calibration papers started reporting the hold-out that matters — September's prediction (2) met.** Mercado-Velazquez et al. (*Sensors*, 27 Oct) co-located low-, mid-cost and reference instruments for ten months at two London sites: $PM_{2.5}$ calibrations trained April–October held through winter and transferred between sites at $R^2 \geq 0.56$, while NO_2 and CO did not ($R^2 \leq 0.28$); linear T/RH regression matched ML. Sylvester et al. (*ACS ES&T Air*, 14 Oct): the national Barkjohn correction beat local fits in Philadelphia (RMSE 1.64 $\mu g\,m^{-3}$, R^2 0.683), and R^2 fell to 0.567 after a year in the field. Alturki et al. (22 Oct) reached spatially blocked R^2 0.82–0.84 for a GEOS-CF emulator — and a random forest on latitude and longitude alone scored **0.91**, which is the leakage argument in one number. A conformal $PM_{2.5}$ map lost nominal coverage (0.794) under temporal hold-out and its preregistered hypothesis failed (8 Oct). Fang et al.'s 66-study review (10 Oct) gives the field's baseline: median calibrated R^2 0.92 in the laboratory against **0.72** in the field, median RMSE 2.6 against **12.3** $\mu g\,m^{-3}$.
- **The reference became the object of study.** Beyond the photometer meta-analysis and the filter-water bias: a fluorescence–Raman–Mie lidar that gates its extinction-to-mass regression on aerosol type cut surface RMSE by 38 % and bias by ∼83 % under date-separated validation (*AMT*, 16 Oct); four prototype electrostatic mass monitors at one power-plant site agreed only within pairwise ratios of 0.33–2.83 for $PM_{2.5}$ (single-unit ICC 0.42 for PM_{10}), with no reference to say which was right (20 Oct); and a chassis-dynamometer tire-wear campaign saw its reference CPC's response double over the campaign (15 Oct).

- **Number concentration acquired endpoints.** Beside the dementia cohort: kidney-disease ED visits in New York State rose 0.74 % (0.28–1.20) per IQR of CTM-modelled UFP at lag 0–3 across 269,945 visits (29 Oct); source-resolved lung deposition ranged from 76 % (new particle formation) to 27 % (solid fuel) at equal ambient number (8 Oct); 3-D printing emits almost entirely below the optical cut of consumer sensors (25 Oct). Ultrafine and size-resolved number records rose to 23 (4.4 %, from 3 %).
- **Wildfire smoke separated from ambient $PM_{2.5}$ as its own exposure** — with its own composition (the EC+OC ADRD contrast), its own lags (a deconvolved US mortality response peaking three years after a smoke pulse, 1.7 million deaths 2011–2023) and its own endpoints (head-and-neck cancer RR 1.41, 1.24–1.60; autism). The month also produced informative nulls: incident depression in 179,218 All of Us participants (preregistered, 30 Oct) and ADHD in the Medicaid cohort. On the measurement side, 32 Idaho AQS sites over 20 years show fire-season $PM_{2.5}$ rising 2.54 $\mu g\,m^{-3}$ per decade while the annual mean stayed flat (31 Oct).
- **Scale and inventories.** Yu et al. (*Lancet Planet Health*, 12 Oct): 346.5 million admissions in 949 communities, RR **1.080** (1.074–1.086) per 10 $\mu g\,m^{-3}$, with malignant neoplasms (1.108) and mental disorders (1.100) above respiratory (1.067); heterogeneity described as substantial but not reported with the pooled estimate. China's BC burden was put at >540,000 deaths in 2015, twice prior estimates (*Nat Commun*, 8 Oct), and road tire-and-road-wear $PM_{2.5}$ at 425 kt in 2023, projected to reach 67 % of traffic $PM_{2.5}$ emissions by 2050 (*Nat Commun*, 9 Oct).
- **One network measured the fraction the epidemiology points to.** Manchanda et al. (*Sci Adv*, 28 Oct) assimilated dense mobile and fixed black-carbon measurements (6.7 sites km^{-2}) into hourly 150 m emission maps for West Oakland: the posterior raised the BC inventory ∼33 % and quadrupled the exposure share of neighbourhood and port sources (5 → 20 %); ∼10 fixed monitors plus mobile maps recovered the result. It is not a low-cost network with a season-long field calibration, so September's prediction (1) is **not met** — but it is the closest record yet.

Corpus analytics — October 2026

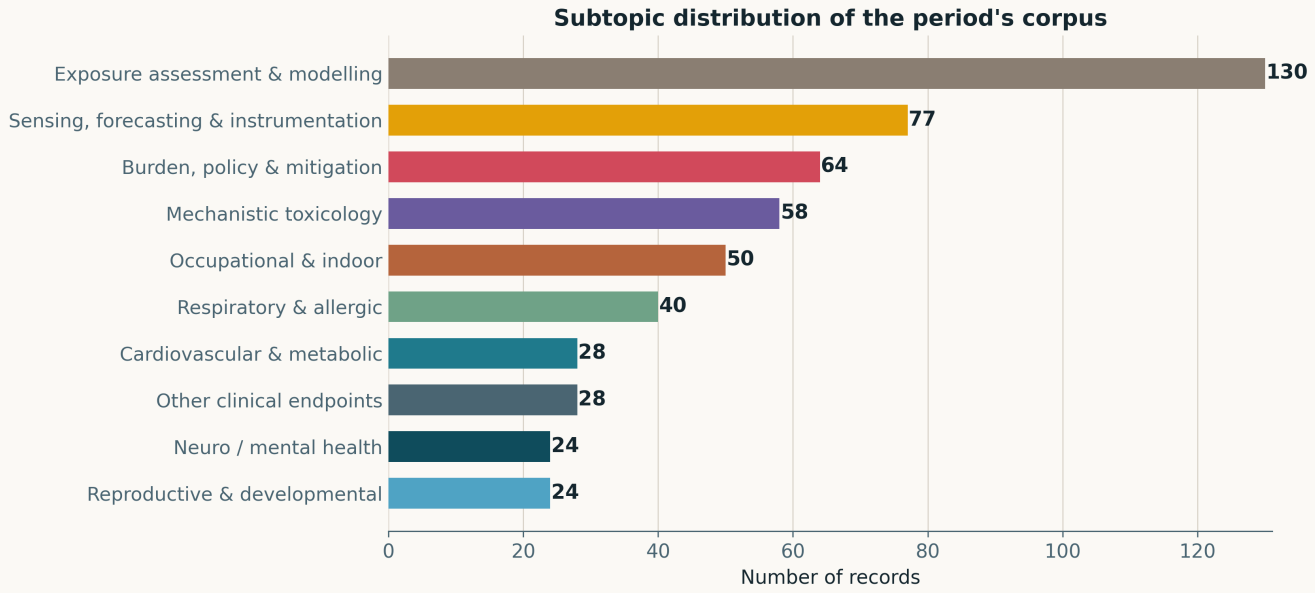


Fig. Subtopic distribution across all 523 pooled records; all ten subtopics populated. *Exposure assessment & modelling* 130 (25 %) and *Sensing, forecasting & instrumentation* 77 (15 %) together hold 40 % (September 43 %, August 48 %). *Burden, policy & mitigation* 64, *Mechanistic toxicology* 58, *Occupational & indoor* 50, *Respiratory & allergic* 40, *Cardiovascular & metabolic* and *Other clinical* 28 each, *Neuro / mental health* and *Reproductive & developmental* 24 each. Single-label assignment by dominant contribution.

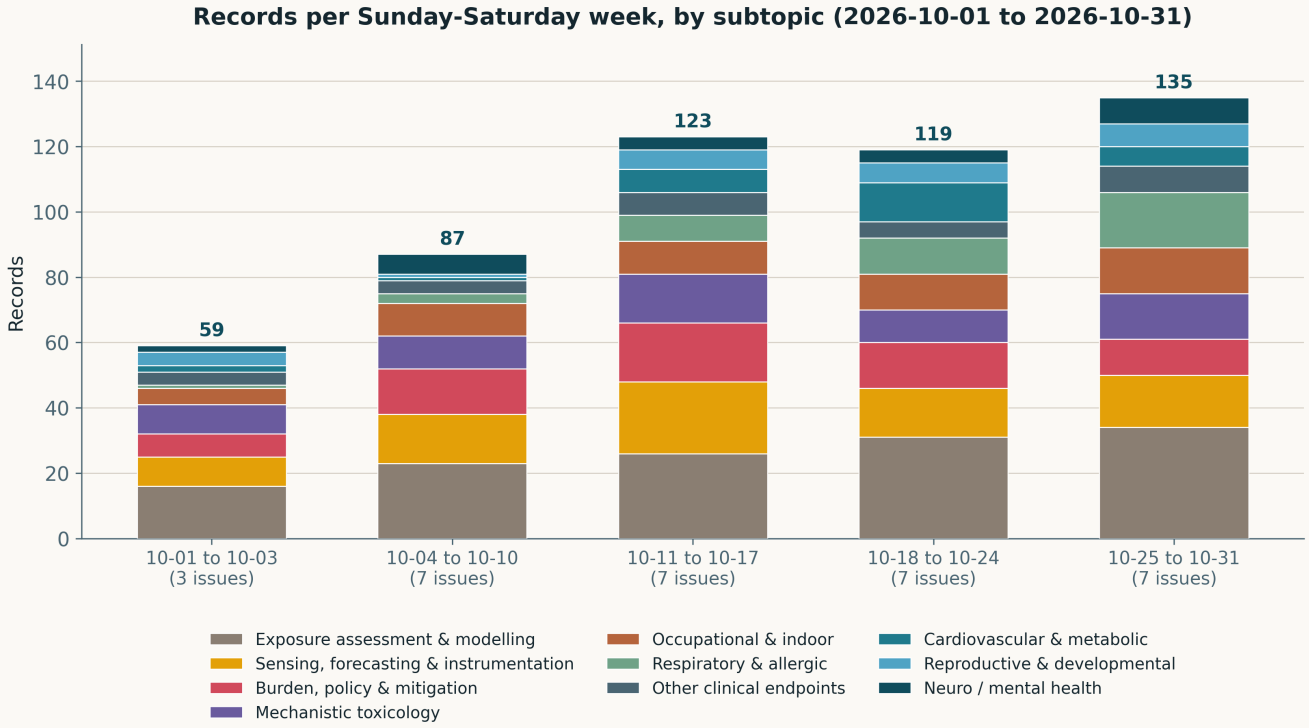


Fig. Records per Sunday–Saturday week by subtopic across the 31 October issues (the first week is partial): 59 for 1–3 Oct, then 87, 123, 119 and 135. *Exposure* rose in every week (16 to 34) and *Respiratory* from 1 to 17; *Sensing* peaked at 22 in 11–17 Oct. Daily volume ranged from 1 (Sunday 11 Oct) to 30 and tracks publisher deposit timing. Issues for 3–8 Oct and 23–29 Oct were built late on recovery runs; each kept its own entry-date window.

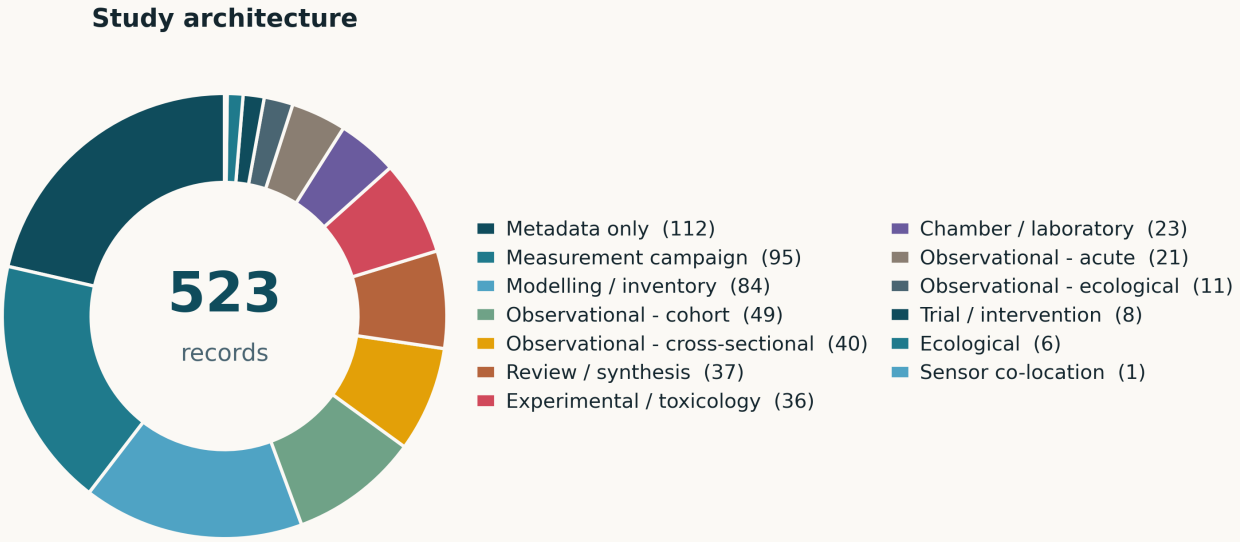


Fig. Study architecture. Metadata-only 112, measurement campaigns 95, modelling/inventory 84, cohorts 49, cross-sectional 40, reviews 37, experimental/toxicology 36, chamber/laboratory 23, acute designs 21, ecological 17 (drawn as two bars, "Observational – ecological" 11 and "Ecological" 6: a label-map defect, not two designs), trials or interventions 8, sensor co-location 1. **Metadata-only rose again, 12 % → 16 % → 21 %** across August–October. Trials: 8 in 523, unchanged in absolute terms from September.

Health endpoint distribution — not drawn

The endpoint figure is omitted for the third month running: the canonical endpoint map is still not capped, so at month scale the chart is several dozen rows at an unreadable size. The subtopic weekly series (above) carries the shift that matters — respiratory endpoints rising through the month and cardiometabolic endpoints falling. **Fix still owed**: cap the map at ∼20 categories.

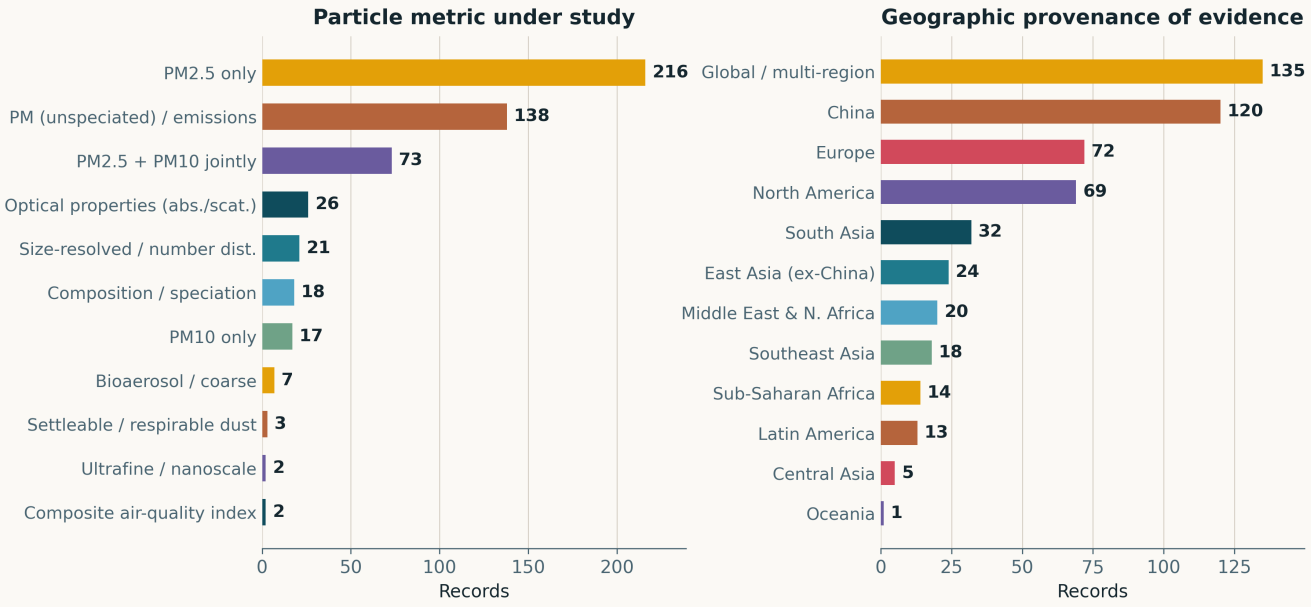


Fig. Left — particle metric. $PM_{2.5}$ alone 216, unspeciated PM or emissions 138, $PM_{2.5}$ with PM_{10} 73, optical properties 26, **size-resolved number 21 and ultrafine 2** (4.4 % together, against 3 % in September), composition/speciation 18, PM_{10} alone 17, bioaerosol/coarse 7, respirable dust 3, composite indices 2. Composition is under-counted here: component exposures inside cohort records are filed under $PM_{2.5}$. Right — geography. Global / multi-region 135 (inflated by metadata-only records), **China 120 (23 %, from 27 %)**, Europe 72, North America 69, South Asia 32, East Asia ex-China 24, Middle East & N. Africa 20, Southeast Asia 18, **Sub-Saharan Africa 14** (22 in September), Latin America 13, Central Asia 5, Oceania 1.

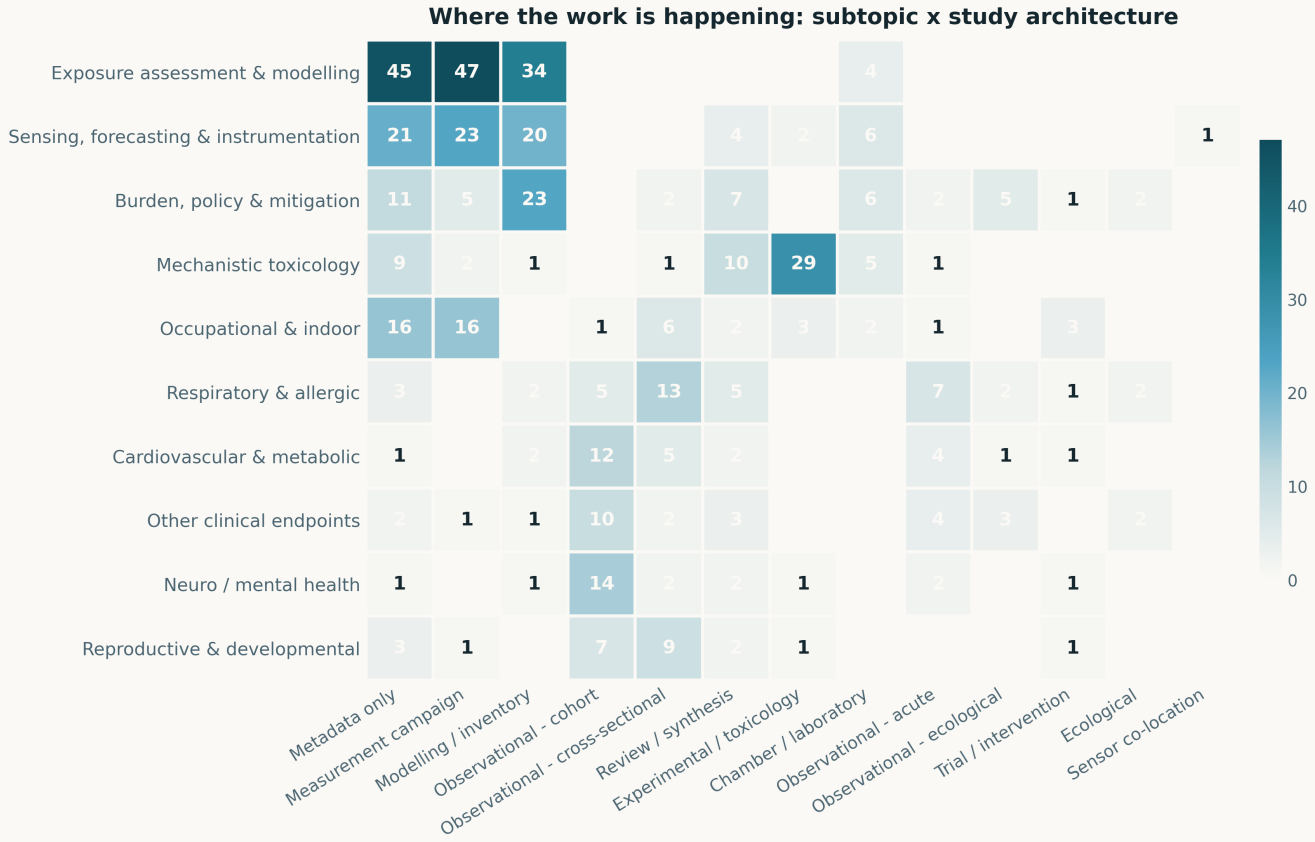


Fig. Subtopic × study-architecture cross-tabulation. Dense cells: exposure × modelling, sensing × measurement campaign, and metadata-only across every row. The trial column remains nearly empty in every health subtopic, and the sensing and exposure rows still contribute almost nothing to the effect-estimate pool.

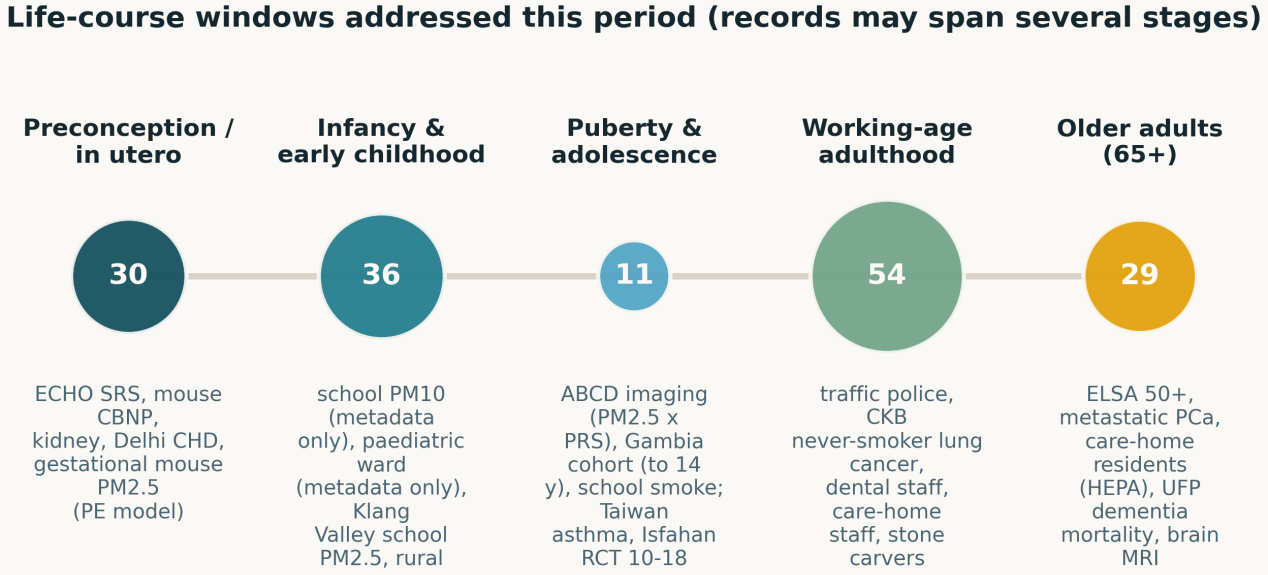


Fig. Life-course windows across the month (non-exclusive; only records with an identifiable human window): in utero 30, infancy/childhood 36, adolescence 11, working age 54, older adults 29 (September 41, 35, 10, 77, 40). **Adolescence remains the thinnest window** for the fourth month running.

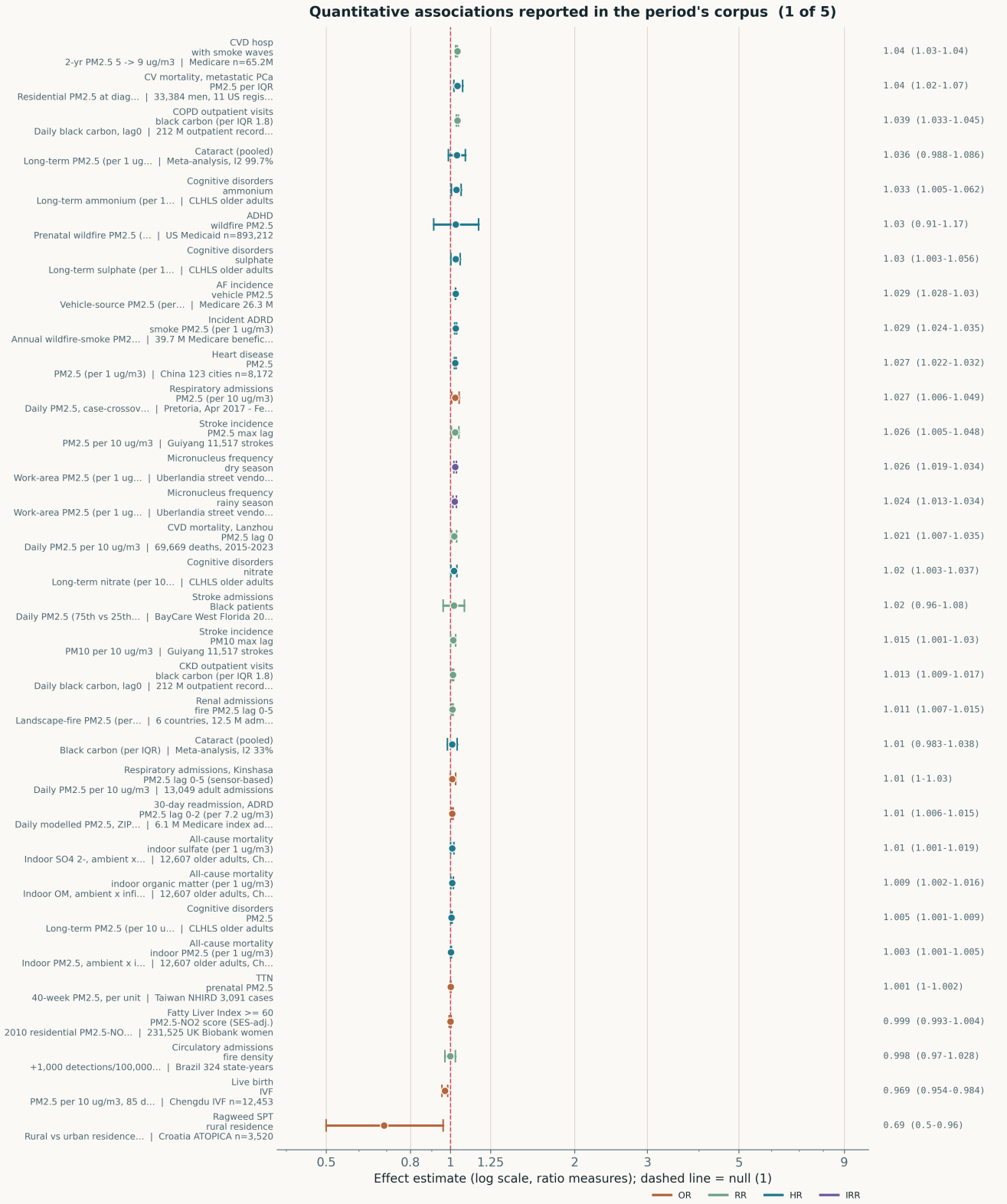


Fig. Quantitative effect estimates extracted across October, panel 1 of 5. **This is not a meta-analysis and no pooled summary is drawn.** Contrasts are not commensurable (per 1, per 10, per IQR, per 10,000 cm^{-3}, threshold, quartile, fuel-type, smoke-fraction contrasts) and the outcomes are unrelated, so **between-study heterogeneity is not estimable and none is reported**. The one pooled estimate in the set carries its own: the photometer–gravimetric lnR 0.31 at I^2 **78.4 %**; the six-country admissions RR (Yu et al.) was published without its heterogeneity statistic. All 113 estimates carry an interval. Of 97 on ratio scales (HR 45, OR 30, RR 19, IRR 3), **82 exclude the null in the harmful direction, 2 fall below it and 13 cross it**.

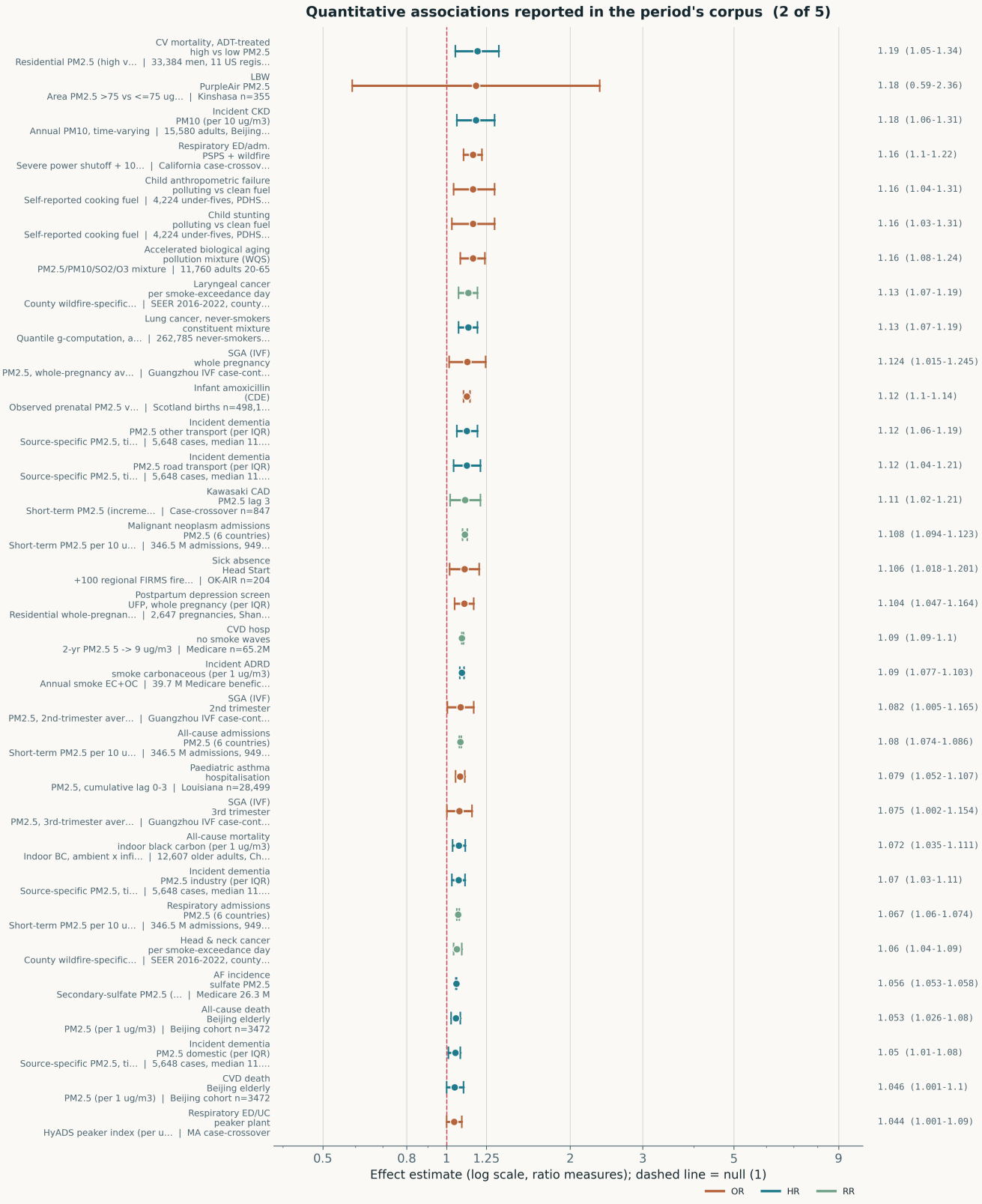


Fig. Effect estimates, panel 2 of 5. Same caveats as panel 1.

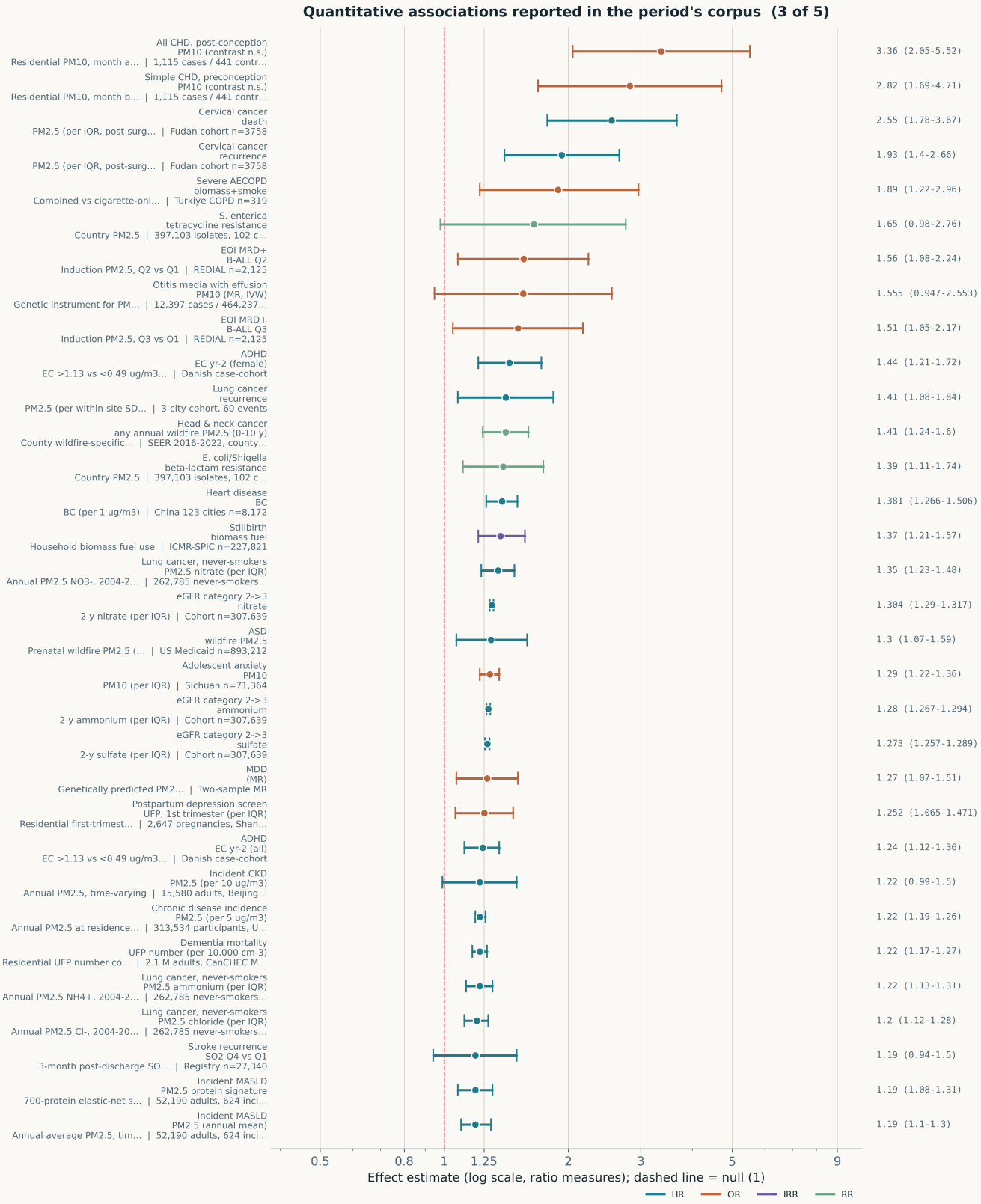


Fig. Effect estimates, panel 3 of 5. Same caveats as panel 1.

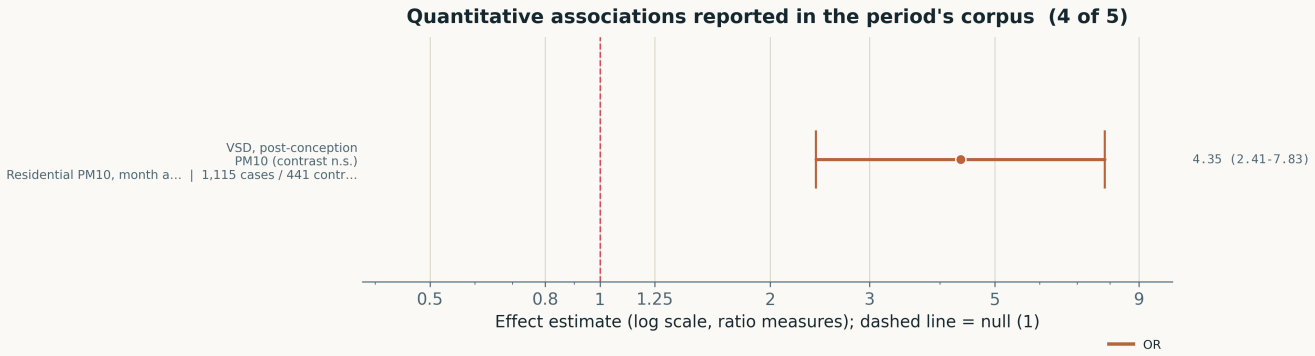


Fig. Effect estimates, panel 4 of 5. Same caveats as panel 1.

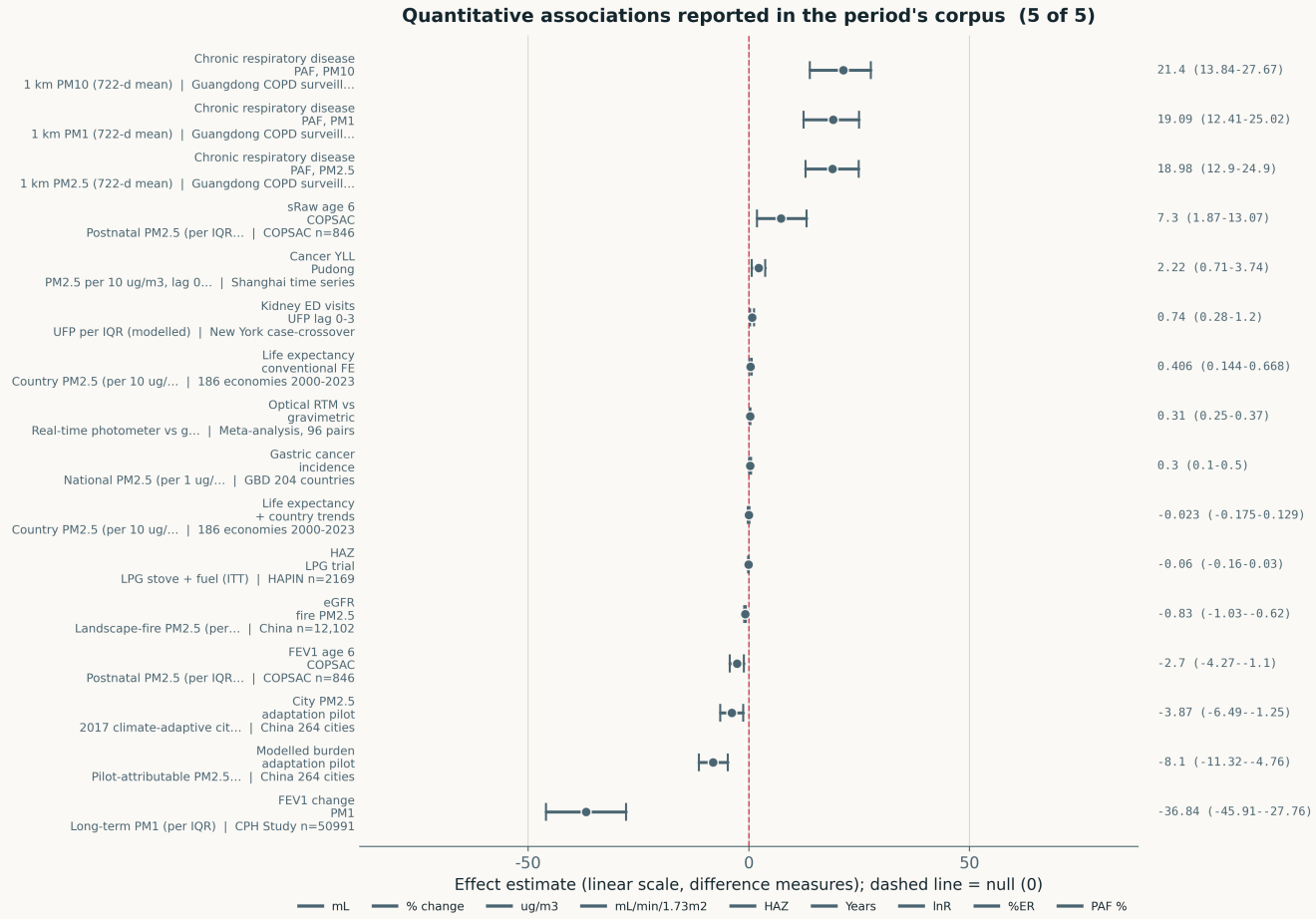


Fig. Effect estimates, panel 5 of 5. Neither ratio estimate below the null is a protective PM effect: IVF live birth OR 0.969 (0.954–0.984) is harm on a decreasing scale, and ragweed sensitisation OR 0.69 (0.50–0.96) is a protective co-exposure (rural residence). Of the 16 non-ratio estimates (FEV_1, eGFR, years, PAF, %-change, HAZ), 14 exclude zero. The largest ratios — OR 4.35 (2.41–7.83) for VSD after post-conception PM_{10} in a Delhi case-control set, and HR 2.55 (1.78–3.67) for cervical-cancer death in a 3,758-patient Fudan cohort — come from single-centre designs with wide intervals and, in the Delhi case, a non-significant window contrast.

Emerging, fading, and recurring

Emerging. **(i) The exposure as a vector, not a scalar** — secondary inorganics, EC, smoke carbon and UFP number each carried a named effect in a large cohort; the leading constituent differs by outcome (BC for stroke and heart failure in September, nitrate for lung cancer, sulfate/nitrate/ammonium for kidney, EC for ADHD), which is either outcome specificity or model-to-model instability in the component surfaces — no record yet distinguishes the two. **(ii) Reference-side metrology** — photometer bias pooled with a prediction interval, filter-bound water, type-gated lidar mass, reference CPC drift, housing-induced sample drying. **(iii) ML emulators and forecasters of the CTM** (ProNet across 338 cities; the Foehn generative km-scale forecaster; a CMAQ nitrate U-Net) — the reason *Exposure* grew every week. **(iv) Respiratory endpoints** ($1 \rightarrow 17$ per week), led by early-life lung function (COPSAC FEV_1 $-2.7\,\%$, -4.27 to -1.1, at age 6), COPD and interstitial lung disease. **(v) Non-exhaust traffic PM**: TRWP inventory, dynamometer tire-wear particles.

Fading. **Sensing share** $25 \rightarrow 19 \rightarrow$ **15 %** across August–October. Part of this is the Crossref AMT/ACP channel and part is real: the instrumentation records that did appear were reviews and validation audits rather than new devices. **Cardiovascular & metabolic** halved ($46 \rightarrow 28$) and **tier A** fell from 23 % to **14 %** — the latter partly a metadata-only effect, since a record without an abstract cannot be tier A. Mass-only single-pollutant time series continue to lose significance under co-pollutant adjustment; that observation has now held for three months and is no longer news.

Recurring groups and venues. **No author-cluster analysis is possible**: the corpus still has no affiliation field, and surname repeats are not investigator clusters. What does recur is a design family — national US claims cohorts (Medicare, Medicaid) with modelled smoke-fraction or component exposure, which produced the ADRD, atrial-fibrillation and autism results. Venue concentration: *Environ Pollut* 27, *Atmosphere* 24, *Environ Sci Technol* 23, *Atmos Environ* 23, *Atmos Pollut Res* 20, *Atmos Chem Phys* 20, *Front Public Health* 15, *Build Environ* 13, *Environ Res* 12, *Environ Monit Assess* 12, *Sensors* 8. **Preprints rose to 33** (Research Square 22, medRxiv 11), so a larger share of the month is unreviewed. **Most-cited work is not reported**: citation counts on records days to weeks old measure indexing lag. The month's highest-venue records: Yu et al. (*Lancet Planet Health*), Papatheodorou et al. (*JAMA Pediatr*), Manchanda et al. (*Sci Adv*), and the two *Nat Commun* inventories (China BC burden; TRWP).

Trial registry watch

Registry sweeps (`LastUpdatePostDate` windows) ran through October with gaps covered by multi-day windows; `state/trials.json` now holds **20 trials**, **four first seen in October**. The pattern carried from August and September holds: **interventions framed around PM whose primary endpoint is not a measured PM exposure.**

NCT07536178 ("Lungs on Fire", University of British Columbia, controlled exposure, n=24, crossover; first seen 29 Oct): lodgepole-pine woodsmoke against filtered air in people with asthma, COPD or lung cancer. `analyze_endpoints`: co-primary **FeNO and CANTAB cognition**; secondaries include sputum cells, resting-state fMRI default-mode connectivity, ^{129}Xe MRI, a NULISAseq CNS panel, NETs, epigenetic age and glial extracellular vesicles — the first registered protocol in this watch putting lung and brain endpoints under one smoke exposure, which is exactly the endpoint pairing the October smoke epidemiology implies. **NCT02763917** (AIRWEIGHS, Johns Hopkins, n=198, placebo- controlled home purifier in obese vs non-obese asthmatic children): record updated 21–22 Oct, primary completion September 2026 — **results posting is the event to watch**. **NCT07500948** (EPIC-AIR, Leicester, feasibility RCT, n=80): real-time air-quality guidance during exercise rehabilitation; the co-primary is **modelled** ambient PM at GPS location during activity, not a personal measured dose, and there is no health co-primary. **NCT07886346** (Taif COPD prevalence survey, observational): logged, low relevance.

Published trial-class records: **8 in 523**, the most relevant a randomised crossover of short-term indoor $PM_{2.5}$ on cognition and HRV (*iScience*, 26 Oct). The final sweep (30 Oct–1 Nov) returned no hits. **Registry and literature now agree on one thing: the intervention evidence base for PM is being built on filtration and smoke chambers, and the registered exposure endpoint, where one exists, is increasingly modelled.**

Open gaps and read on direction

Gaps in the literature

(i) Component, number and smoke exposures are all modelled. Not one October effect estimate on a constituent, UFP number or smoke carbon rests on measured speciation, a CPC network or an absorption channel. Model-to-model disagreement in component surfaces is unquantified, and it could produce the outcome-specific "leading constituent" pattern on its own. **(ii) Reference uncertainty is not propagated.** Calibrations report sensor error against a reference treated as truth; October quantified photometer (+36 %), filter-water (+3 to +18 %) and housing-RH biases, and no calibration paper carries them into its error budget. **(iii) Number instruments still lack long records** — no low-cost PN or CPC network with a field calibration of one season or more appeared (September's gap (iii), unchanged). **(iv) RH correction is fitted to the wrong humidity** for housed sensors; corrections developed on bare PMS5003 units should not transfer to PurpleAir without re-fitting. **(v) Trials register modelled or no exposure** (EPIC-AIR; three September registrations).

Gaps and defects this pipeline creates

(i) Duplicates. 523 records, **523 unique DOIs**. The 30 Oct backfill screened 23–29 Oct in parallel and double-admitted ∼40 cross-day re-surfacings; the DOI gate and an earliest-issue rule removed all of them before shipping, and backfills are now screened sequentially. **(ii) 112 records (21 %) are metadata-only**, the third monthly rise (12, 16, 21 %). Scholar Gateway returned access errors when called and was skipped on most runs — it needs the user to reconnect it; Elsevier day-0 deposits remain abstract-free. **(iii) Run continuity.** Runs stalled for 6–9 Oct and 23–29 Oct; both gaps were backfilled day by day with their own entry windows, so coverage is contiguous but several issues were built 1–7 days late. **(iv) Source legs.** OpenAlex returned 429 on every run; Europe PMC showed day-0 index lag on several days (0–3 records) and re-surfaces earlier PubMed records on later days; arXiv returned 150 kB with no in-window entry on most days; Consensus returned **0 new** records on every sweep. **(v) Label maps.** Ecological designs split across two labels (f2a); the endpoint map is uncapped (f2b omitted); no affiliation field, so no author clusters.

Scoring September's predictions. (1) *A low-cost network with an absorption/BC or particle-number channel and a field calibration of at least one full season* — **not met**. The nearest record is Manchanda et al.'s West Oakland BC network (*Sci Adv*, 28 Oct): network-scale BC with Bayesian inversion, but not low-cost nodes and no season-long field calibration reported. (2) *A calibration paper reporting a temporally blocked or prospective hold-out as its primary metric* — **met**: Mercado-Velazquez et al. report seasonal hold-out (April–October training, winter test) and cross-site transfer as headline results; the type-gated lidar used date-separated validation, and a Nagoya PAH model used time-blocked CV.

Read on direction. August read the two halves of the field as optimising different objectives; September narrowed the gap. October widened it again, from the health side: the epidemiology now names components, number and source as the exposure, while the instruments that could measure those at network scale are absent from the month's record and the instruments that measure mass were shown to be calibrated against references with composition-dependent biases. The practical consequence for a distributed PM network is that **density is no longer the binding constraint — the measured quantity and the reference are**. **Falsifiable predictions for November:** (1) a constituent-resolved or number-resolved health estimate whose exposure is *measured* (speciation network, CPC network, or network absorption/BC channel) rather than modelled; and (2) a sensor-calibration paper that propagates reference-instrument uncertainty (gravimetric water, FEM bias or photometer bias) into its reported error. If (2) fails, reported LCS errors should be read as lower bounds.

Paper-level digest — the month by subtopic

Every pooled record appears below. Records that carry a full assessed summary have it lifted verbatim from the daily issue that first published it; a title-only or extractive entry is **not an assessment** and the absence of criticism is not approval. Metadata-only records are a pointer to a DOI and nothing more.

Sensing, forecasting & instrumentation — 77 records

Carried from the daily issues — full summaries (77 of 77 records in this subtopic)

Aakko-Saksa et al. 2026 Environ Sci: Atmos

Aggregated non-methane organic and inorganic gases as indicators of aged particulate matter induced by traffic

METADATA ONLY. Metadata-only record: RSC deposited title, authors and DOI on 1 Oct with no abstract; PubMed efetch, Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The deposit carries a one-line teaser: aggregate NMVOC emissions were found to indicate the organic fraction of the subsequently aged PM. If it holds, a gas-phase total (cheap to sense) could flag secondary organic PM that optical counters see only after ageing; the data and the lag structure are unknown. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1039/d6ea00024j

Ai optical trapping Raman single carbonaceous aerosol smoke 2026 Sensors (Basel)

In-Situ Single-Particle Sensing of Carbonaceous Aerosols in Freshly Generated Smoke Using Optical Trapping-Raman Spectroscopy

Optical trapping-Raman spectroscopy captured smoke particles from nylon rope and plastic fibre ignited by the trapping laser itself, and analysed them airborne within seconds. Particles of several micrometres (~10 pg) gave full Raman fingerprints; band analysis separated two particle types, and spatially resolved spectra showed organic carbon and soot segregated within single particles. Limitation: freshly generated, supermicron, laser-ignited particles from two polymer fuels in an open chamber; ambient carbonaceous PM is mostly accumulation-mode, aged and far lighter, and the trapping laser may heat and alter what it measures. No throughput or detection-limit figures. Relevance: low-moderate - a reference technique for particle chemistry, not a field sensor.

doi: 10.3390/s26206527

Bachour WRF-Chem AOD post-correction hot desert 2026 Atmos Pollut Res

Statistical post-correction of WRF-Chem aerosol optical depth under hot-desert conditions

METADATA ONLY. No abstract at harvest. Title indicates statistical bias correction of WRF-Chem AOD in a hot-desert setting (author group based in Qatar). Recorded for coverage; no estimate inferred. Relevance: moderate - dust-dominated AOD is where both models and optical LCS fail, and corrected CTM fields are common fusion inputs.

doi: 10.1016/j.apr.2026.103233

Bei UPLC-MS/MS 74 antibiotics in $PM_{2.5}$ Beijing 2026 J Chromatogr A

Optimization and application of a sensitive UPLC-MS/MS method for the quantitative determination of antibiotics in $PM_{2.5}$

Method for 74 antibiotics in $PM_{2.5}$: T3 column, formic-acid mobile phase, Oasis HLB extraction with Na2EDTA. Recoveries 69.7-131.7%, RSD 1.0-18.1% at three spike levels, quantification limits down to 0.37 pg/m3. On 29 Beijing 24-h samples, 22 compounds exceeded detection limits (max 121.38 pg/m3); roxithromycin was in every sample and oxytetracycline had the highest mean (14.48 pg/m3). Limitation: 29 samples from one city and season give occurrence, not exposure; recoveries near 70% for some classes bias low, and the link to resistance spread is asserted, not tested. Relevance: low for sensing - a filter-bound chemical fraction no optical sensor sees, but co-located filter sampling at sensor nodes could support it.

doi: 10.1016/j.chroma.2026.467564

Birinci Bosphorus bridge traffic ML proxy $PM_{2.5}$ NO_2 2026 Atmosphere

Machine-Learning Proxy of Air-Quality Variability Associated with Intercontinental Bridge Traffic over the Bosphorus in Istanbul

17,520-h panel (2022-2023) pairing six bridge traffic sensors with the nearest monitors. A Prophet traffic-only baseline gave negative R2 for every bridge x pollutant. LightGBM, XGBoost and a BiLSTM-attention model with meteorology, calendar, precipitation lags and short autoregressive terms reached mean hold-out R2 ~0.77 (0.58-0.86), RMSE down 47-63%; the three families were practically equivalent. SHAP ranked autocorrelation, season, wind direction and heavy vehicles above total traffic. Limitation: autoregressive pollutant terms make this nowcasting persistence, not a traffic proxy - the headline R2 largely reflects lagged concentrations, and 'closest monitor' distance is unstated. Relevance: moderate - shows traffic counts alone do not substitute for measurement in complex coastal flow.

doi: 10.3390/atmos17101024

CONIOPOL CL61 aerosol typing 2026 Atmos Meas Tech

New operational perspective to identify aerosols and hydrometeors in real-time with a pioneering algorithm (CONIOPOL) based on single-wavelength polarization lidar (CL61)

Automated classification from a polarised automatic lidar-ceilometer combining depolarisation, attenuated backscatter and cloud base to separate aerosol, cloud phase and precipitation and to type aerosol as dust, smoke/marine, continental or mixed. One year at Uccle showed strong temporal and vertical correspondence with CAMS for transported dust and smoke. Limitation: single wavelength cannot separate smoke from marine aerosol, and validation against CAMS is model-conditioned agreement rather than in-situ truth. Relevance: high - ceilometers are already networked across Europe; vertical typing tells a surface sensor network when its column assumption is broken.

doi: 10.5194/amt-19-6415-2026

Cai self-supervised stacking particulate nitrate sparse observations 2026 Atmos Environ

A Self-supervised Stacking Framework for Estimating Particulate Nitrate in Two Chinese Urban Agglomerations under Sparse Ground Observations

METADATA ONLY. Title and authors only (Cai, He, Li, Liu et al.). No abstract was retrievable from Crossref, Europe PMC or Semantic Scholar at harvest. The title describes a self-supervised stacking ensemble that estimates particulate nitrate in two Chinese urban agglomerations from sparse ground observations. Predictors, validation design and skill cannot be stated. Limitation: unverifiable beyond the title; with sparse speciation sites, random cross-validation would overstate skill (see the 22 Oct blocked-CV record). Relevance: medium - component estimation under sparse reference sites is the same problem as filling between LCS anchors.

doi: 10.1016/j.atmosenv.2026.122433

Calabar ZigBee WSN 2026 Research Square (preprint)

Design and Performance Evaluation of a ZigBee-Based Wireless Sensor Network for CO and $PM_{2.5}$ Air Quality Monitoring in Calabar Metropolis

A ZigBee mesh of low-cost CO and $PM_{2.5}$ nodes, with analytical models for sensing, communication, energy use and dispersion exercised in MATLAB. Reported: reliable links, coverage growing with node count, a lifetime of about 4,065 transmissions, mean CO 30 ppm, mean $PM_{2.5}$ 40 $\mu g\,m^{-3}$, AQI 112.5, and a composite 'performance index' of 76.5%. Limitation: the central weakness is that no calibration, co-location or reference comparison is reported at all - a 40 $\mu g\,m^{-3}$ mean from uncalibrated optical nodes in a humid tropical city is not a measurement, and a self-defined performance index is not a validation; the dispersion component is simulated, not observed. Relevance: moderate as a record of the deployment gap in West Africa, where ambient data is genuinely absent; low as evidence. This is the failure mode this watch exists to flag.

doi: 10.21203/rs.3.rs-10637757/v1

Chang et al. 2026 Environ Pollut

Wintertime vertical stratification of air pollutants and meteorological associations within an urban forest canopy in Shenyang, China

A three-level monitoring mast inside an urban forest in a cold-climate northern Chinese city through the winter heating season. $PM_{2.5}$ and PM_{10} were generally higher at 10 and 20 m than at pedestrian level, O_3 and CO higher at 1.5 m, and NO_2 and SO_2 peaked mid-canopy; relative humidity and wind speed dominated the statistical and machine-learning associations, but height-dependently. Limitation: one site, one winter, no replication mast and no flux measurement, so the inversion of the usual near-ground maximum is described rather than explained - and at wintertime humidity, uncorrected optical PM readings are themselves height-correlated through the humidity profile. Relevance: high - siting guidance for low-cost networks assumes a well-mixed layer below canopy height; this is direct evidence that the 1.5 m reading is not the 20 m reading.

doi: 10.1016/j.envpol.2026.129270

Chemyakin et al. 2026 Opt Lett

Single-scattering look-up table for remote sensing and in situ black carbon studies

Tabulates single-scattering properties of fractal-like soot aggregates (refractive index 1.95-0.96i; fractal dimension 1.4-2.8; 1-4,096 monomers of 15-25 nm radius) across 0.35-2.3 um for PACE retrievals, giving a speed-up of at least 10^6 over direct computation. As a test it reproduces the low single-scattering albedo of in situ ACTIVATE aircraft aerosol. Limitation: bare, identical, touching monomers only - no coating, no polydispersity, no necking - so aged, internally mixed BC (the ambient case) and its absorption enhancement fall outside the table; agreement with one campaign is a plausibility check, not validation. Relevance: moderate - fast BC optics is what multi-wavelength absorption channels on low-cost nodes need to convert attenuation to mass with a defensible morphology assumption.

doi: 10.1364/ol.609835

Chen et al. 2026b J Aerosol Sci

Linking Aerosol Chemical Composition to Volatility: A Predictive Parameterization and Implications for Future Clean Air

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 3 Oct without an abstract. By title, a parameterisation predicting aerosol volatility from chemical composition. Volatility is the quantity that decides how much of an ambient aerosol survives the heated inlet of a reference monitor or the warm optical cavity of a low-cost sensor, so a composition-based predictor is directly relevant to the semi-volatile bias that separates sensor and FEM mass. Relevance: potentially high; on the retry list.

doi: 10.1016/j.jaerosci.2026.106902

Chiappini WRF-Chem nudging 2026 Atmosphere

Regime-Dependent Effectiveness of Spectral Nudging in WRF-Chem: Meteorological and Saharan Dust Skill over the Central Mediterranean

A two-tier design separates the meteorological response to spectral nudging (a 104-day chemistry-free pair) from the dust-transport response (paired event simulations). Nudging improves sea-level pressure most (daily RMSE 2.08 to 1.53 hPa) and 10 m wind less (2.50 to 2.20 m/s), while 2 m temperature does not improve (2.05 to 2.15 C); the benefit recurs episodically rather than growing with simulation age. Dust skill is conditional: in the weakly forced June case all seven AERONET sites improved (station-mean RMSE 0.457 to 0.235, correlation 0.778 to 0.881), while in March RMSE fell at all eight sites but mean correlation was unchanged. Limitation: two outbreaks is a thin basis for a regime claim, and AOD is a column integral that constrains surface dust PM only weakly. Relevance: moderate - dust is the event class where satellite-and-model PM products diverge most from surface nodes.

doi: 10.3390/atmos17100992

Chyzhykov gravimetric filter water Karl Fischer bias 2026 Research Square (preprint)

Impact of Mass Loading and Water Speciation on Precision and Artifacts in Gravimetric Measurements of Atmospheric Particulate Matter

Thirty filters (n=5 per fraction per site) plus five field blanks, weighed under PN-EN 12341:2024 conditioning, then stepwise thermal water release at 50, 120, 180 and 250 C. Between-filter variability was U-shaped in mass loading, minimal near 1330-1360 ug; PM_1 was the most variable fraction (CV up to 53%). Strongly bound water (180-250 C) made up 51-60% of filter water, survives standard conditioning, and adds a systematic +3% to +18% mass bias, largest for PM_1 at the steelworks. Limitation: 35 filters at two industrial sites; the bias is not yet resolved by composition or RH history, so transfer to urban background aerosol is untested. Relevance: high - every LCS calibration inherits the reference's error; a positive, composition-dependent water bias in gravimetric PM_1 and $PM_{2.5}$ propagates directly into co-location slopes.

doi: 10.21203/rs.3.rs-11157411/v1

Cover solvatochromic fluorophores in situ aerosol fluorescence 2026 Aerosol Sci Technol

Evaluation of solvatochromic fluorophores for in situ fluorescence characterization of aerosols

METADATA ONLY. Title and authors only (Cover, Gibbons, Boadu, Ohno et al.). No abstract was retrievable from Crossref or Semantic Scholar at harvest. The title describes evaluating solvatochromic dyes, whose emission shifts with local polarity, as in-situ probes of aerosol particle properties. Analytes, particle types and performance metrics cannot be stated. Limitation: unverifiable beyond the title. Solvatochromic readouts also confound polarity with viscosity and water content, so any phase-state inference needs independent calibration. Relevance: low-medium - a laboratory composition probe, not a field-deployable sensor channel.

doi: 10.1080/02786826.2026.2746515

Dai et al. 2026 Atmos Chem Phys

Cross-phase partitioning of sulfur-nitrogen ratios and aerosol mixing-state evolution based on single-particle observations

SPA-MS through winter and summer emission-control periods and matched normal periods, with three sulfur-to-nitrogen metrics (gas gSNR, particle pSNR, number-based nSNR) analysed by causal inference plus interpretable machine learning. Control periods cut NOx while SO_2 stayed roughly flat, raising gSNR, lifting pSNR by about 57% across particle classes and expanding mean vacuum aerodynamic diameter by 27%; enrichment was greatest in particles containing both black and organic carbon and least in BC-free inorganic particles. Limitation: emission-control periods are confounded with season and meteorology, and 'causal inference' on an observational time series with co-varying controls is a strong label for what remains an association. Relevance: high - a 27% diameter shift under a control policy is a direct change in the optical cross-section a nephelometric sensor measures, i.e. a calibration drift caused by policy rather than by hardware.

doi: 10.5194/acp-26-14015-2026

Dayare Navi Mumbai 2026 Preprints.org (preprint)

Weather, Co-Pollutants and Daily $PM_{2.5}$ in a Coastal Indian City: Multi-Station Evidence for Meteorology-Triggered Air-Quality Warnings in Navi Mumbai

VAR models with exogenous weather on screened, unimputed station data. Lagged co-pollutants add little: equivalence tests bound six of seven effects within +/-5% per SD; lagged ozone raises $PM_{2.5}$ by 4%. Rain and boundary-layer height lower same-day $PM_{2.5}$ by 10-24% per SD; date effects shared across stations explain 78% of variance, three-quarters of it weather. A stagnation index plus persistence warns as well as XGBoost, which does not beat persistence. Limitation: a preprint; ERA5 boundary-layer height over a coastal city is coarse, and five stations limit the spatial claim. Relevance: high - a clean demonstration that 'driver' inference from co-movement is weather confounding, and that persistence is the baseline any network forecast must beat.

doi: 10.20944/preprints202610.0209.v1

Drone boundary-layer profiles Germany 2026 Atmos Chem Phys

Influence of irradiance and mixing layer height on the vertical trace matter distribution in the lower planetary boundary layer - drone-based investigation

Two 2.5-week summer campaigns: night profiles in the lowest 120 m showed multi-layer fine-scale stratification; O_3 and potential temperature were the most reliable mixing-height markers, and trace species outperformed several meteorological variables. Local sources and sinks affected in-layer distribution more than irradiance, but aerosol concentrations homogenised under strong radiation even at low mixing heights. Limitation: rural summer only; drone-borne particle sensors are themselves low-cost optical units with flow and RH artefacts that the abstract does not address. Relevance: high - directly tests when a ground node represents the mixed layer.

doi: 10.5194/acp-26-14417-2026

Dust lidar asymptotics 2026 Atmos Chem Phys

Asymptotic behavior of lidar scattering properties of absorbing dust aerosols across Rayleigh and geometrical-optics regimes: theory and implications

TAMUdust2020 and triaxial-ellipsoid calculations give shape-insensitive limits: Rayleigh S ~ x^-3, SSA ~ x^3, small depolarisation; geometric-optics depolarisation falls, SSA -> 1/2 and lidar ratio rises with strong absorption. Large, strongly absorbing dust can therefore mimic weakly depolarising classes and bias classifiers toward under-counting coarse and super-coarse dust. Limitation: theory only, no field test. Relevance: moderate - the same physics explains why optical PM sensors under-read coarse absorbing dust; the bias direction carries over.

doi: 10.5194/acp-26-14487-2026

Epple DLR tire-wear particle chassis dynamometer 2026 Atmosphere

Temperature and Inflation Pressure Effects on Airborne Tire Wear Particle Measurements on a Chassis Dynamometer

Two-level factorial (inflation 1.8/3.8 bar x chamber 11/35 C) on the DLR TRWP system, in two blocks of toe setting. Higher temperature raised cumulative MCPC response 3.6-fold (95% CI 1.8-7.2); higher pressure cut OPS number 18% (12-24%); OPS mass proxy fell ~17% with temperature. Toe >48 arcmin produced rising signals and irregular wear. Reference-condition MCPC response doubled over the campaign, so drift monitoring and randomised blocked run order are prerequisites. Limitation: one tire, one rig; the OPS mass proxy assumes density and refractive index; the ultrafine mode may be volatile, not wear. Relevance: moderate - non-exhaust PM metrology with explicit instrument drift; the same discipline applies to LCS.

doi: 10.3390/atmos17101011

Fang LCS calibration review 2026 Curr Environ Health Rep

Calibration of Low-Cost Particulate Matter Sensors for Exposure Assessment: A Critical Review

Sixty-six studies, 32 laboratory evaluations and 34 field studies. Median calibrated R2 was 0.92 (n = 31) in the laboratory against 0.72 (n = 29) in the field, and median RMSE 2.6 $\mu g\,m^{-3}$ (n = 7) against 12.3 $\mu g\,m^{-3}$ (n = 9) - a factor of nearly five in error on crossing from bench to deployment. The authors are explicit that these are summaries of a heterogeneous literature, not a pooled effect: aerosol mixture, comparison instrument, concentration range, averaging interval and validation design all differ. RH and temperature were handled in most studies; ML and hybrid models appear more often in field work and sometimes lower error, but independent validation across time, place and sensor unit is uneven. Limitation: only 7 and 9 studies reported RMSE at all, so the headline error figures rest on a small and probably favourable subset. Relevance: high - this is the reference statement that calibration does not transfer, which is the standing assumption behind every distributed network claim in this corpus.

doi: 10.1007/s40572-026-00566-2

Fluorescence-Raman-Mie lidar $PM_{2.5}$ profiles Beijing 2026 Atmos Meas Tech

Classification-constrained retrieval of $PM_{2.5}$ vertical profiles from fluorescence-Raman-Mie polarization lidar

Beijing lidar separates anthropogenic pollution aerosol from desert and mineral dust using particle depolarisation ratio plus fluorescence capacity, then applies an extinction-to-$PM_{2.5}$ regression only to the anthropogenic fraction. Date-separated near-surface validation (411 hourly samples, 69 days): RMSE 17.32, MAE 13.54, bias 3.30 $\mu g\,m^{-3}$; versus the unconstrained model RMSE -38.2%, MAE -45.0%, bias about -83%. Limitation: validation is at the surface only, so the vertical profile itself is unverified, and residual scatter of the empirical extinction-mass relation, not the classification, dominates uncertainty. Relevance: high - the same coarse-dust mass bias wrecks optical LCS; type-gated calibration is directly transferable to network QC during dust events.

doi: 10.5194/amt-19-6517-2026

Ghaffarpasand Pope vehicle remote sensing real-world emissions high emitters 2026 Atmos Pollut Res

Emerging Evidence on Real-World Exhaust Emissions and High Emitting Vehicles from Remote Sensing Observations

METADATA ONLY. Title and authors only (Ghaffarpasand, Pope); no abstract at harvest. Appears to synthesise vehicle emission remote-sensing (VERSS) evidence on real-world exhaust and high emitters. Whether PM/smoke opacity or only NOx/CO/HC is covered cannot be stated. Limitation: unverifiable beyond the title; VERSS PM is inferred from UV opacity with large per-pass uncertainty, and single-pass snapshots overweight hot-running conditions. Relevance: moderate - high-emitter fractions explain the kerbside spikes roadside LCS record but cannot attribute.

doi: 10.1016/j.apr.2026.103244

Ghajari fuzzy DL Tehran 2026 Research Square (preprint)

A Novel Fuzzy Hybrid Deep Learning Architecture for Spatiotemporal $PM_{2.5}$ Prediction

A fuzzy transformer-CNN-GRU with cross-branch attention for one-day-ahead $PM_{2.5}$ in Tehran, trained on 2014-2015 and tested on 2016, outperforming seven comparator models with a reported RMSE of 0.73 $\mu g\,m^{-3}$. Limitation: an RMSE below 1 $\mu g\,m^{-3}$ for next-day $PM_{2.5}$ in Tehran is implausible for a genuine forecast and suggests leakage (e.g. same-day inputs or normalised units); no persistence baseline is reported, and the data end in 2016. Relevance: low - the same-day Navi Mumbai record shows why persistence must be the reference; this one does not test against it.

doi: 10.21203/rs.3.rs-11058151/v1

HMAC message authentication vs LoRa telemetry poisoning 2026 Sensors (Basel)

Securing AI-Driven Systems for Environmental Sustainability in IoT Applications: A Lightweight Message-Authentication Scheme Against Telemetry Poisoning

Proposes an encryption-free HMAC-SHA-256 tag over device ID, timestamp, monotonic counter and payload for raw LoRa links, with a durable server-side counter high-water mark to reject forgeries and replays. Host-side reference tests: zero false accepts/rejects over 25,000 adversarial messages. In a ten-seed poisoning study on public environmental datasets, a value-based outlier defence raised $PM_{2.5}$ forecast RMSE by up to ~104% by discarding real extremes, whereas authentication held the clean baseline. Cost: payload 34 to 99 B, airtime ~2.2x. Limitation: never run on the ESP32-S3 hardware; energy unquantified. Relevance: moderate - smoke-event extremes are precisely what outlier filters in LCS pipelines wrongly remove.

doi: 10.3390/s26196276

Hazelwood et al. 2026 J Aerosol Sci

Size-Resolved Particle Dry Deposition Velocities in a Controlled Spherical Chamber: Effects of Turbulence, Surface Complexity, and Particle Physicochemical Properties

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 5 Oct with no abstract. By title, a chamber determination of size-resolved dry deposition velocity against turbulence intensity, surface complexity and particle physicochemical properties. Deposition velocity is the term that sets indoor particle loss rates and therefore the inferred air-change rate in every filtration study that does not measure one, and it is the largest uncertainty in indoor-outdoor infiltration models. Relevance: potentially high; on the retry list.

doi: 10.1016/j.jaerosci.2026.106903

He dust-storm PM_{10} deep learning 2026 Atmos Environ

Transport-aware spatiotemporal deep learning for accurate PM_{10} forecasts during dust storms

METADATA ONLY. No abstract at harvest. Title indicates a spatiotemporal deep-learning PM_{10} forecaster with explicit transport awareness, targeted at dust storms where station-only models lag the plume. Recorded for coverage; no metric is inferred. Relevance: moderate - upwind sensor placement is the obvious data complement to transport-aware models.

doi: 10.1016/j.atmosenv.2026.122409

Hu et al. 2026 J Environ Sci (China)

Columnar optical properties of locally emitted light-absorbing carbonaceous aerosols

Paired surface and mountain-station aethalometer (AE33) measurements in Beijing under pollution controls; an ensemble HYSPLIT + concentration-weighted-trajectory screen isolates locally emitted events, and a wavelength-dependent AAE method derives BrC absorption at 370 and 470 nm. BrC falls off more steeply with height than BC, so the BC share of absorption rises aloft. During local events, noon AOD and AAOD rise 1.91- and 1.93-fold and SSA shows a noon minimum. Limitation: columnar optics are reconstructed from two point sites plus Mie theory for scattering, not measured by a sun photometer or lidar, and trajectory screening of 'local' is model-dependent. Relevance: moderate - surface BrC/BC ratios from ground nodes do not transfer to the column that satellite fusion uses.

doi: 10.1016/j.jes.2026.02.005

Huang foliar particulate UV-Vis-NIR calibration transfer 2026 Environ Monit Assess

Nondestructive UV-Vis-NIR assessment of foliar particulate retention in urban vegetation: cross-species transfer and calibration correction

Reflectance models built on 180 Ligustrum leaves and transferred to 45 Photinia leaves from park, street and industrial zones. Zone classification reached 91.1% after excluding endpoint and water bands. Direct mass prediction on the target species failed (R2 0.265, positive bias); a linear output correction using 15 target reference leaves (five per zone), tested on the other 30 over 50 splits, gave R2 0.797 +/- 0.042. Limitation: target-based scheme selection and the correction set share zones with the test set, and deposit properties are unmeasured, so this is reference-assisted local calibration, not transfer. Relevance: moderate - the same slope-and-offset recalibration logic, and the same failure of naive transfer, governs moving LCS calibrations between sites.

doi: 10.1007/s10661-026-16062-x

Intra four co-deployed electrostatic PM monitors power plant agreement 2026 Research Square (preprint)

Multi-point monitoring of PM_{10}, $PM_{2.5}$, and PM_1.0 at a combined-cycle power plant: field agreement between four co-deployed electrostatic monitors

Four prototype three-channel electrostatic mass monitors at a combined-cycle plant, Feb-Jun 2020, 92-100% daily recovery; station means PM_{10} 44.9-54.1, $PM_{2.5}$ 24.8-30.8, PM_1.0 7.5-11.9 $\mu g\,m^{-3}$. Fixed station differences explained 0.7% of log 24-h PM_{10} variance but the day-by-station interaction 57.1%; pairwise 95% limits of agreement spanned ratios 0.35-3.26 (PM_{10}) and 0.33-2.83 ($PM_{2.5}$); single-unit ICC 0.42 for PM_{10}, 0.75 for the four-unit mean. Disagreement did not shrink with averaging. Limitation: preprint; no reference instrument, so agreement says nothing about accuracy; units were not co-located side by side first, so unit drift and real micro-gradients are separated only by inference. Relevance: high - the same argument applies to any LCS network: station contrasts need routine co-location checks.

doi: 10.21203/rs.3.rs-10901106/v1

Kalisa East Africa passive sampler network PAHs metals 2026 Environ Pollut

First Results from a Passive Air Monitoring Network across Five East African Countries for Polycyclic Aromatic Hydrocarbons (PAHs) and Metals

Ten sites, 30-day dry-season deployments everywhere and paired wet-season samples at four. PAH ring profiles and metal loadings were spatially heterogeneous; water-soluble metals were a small share of total. HYSPLIT put >90% of air masses from the S-SE, and adjacent cross-border sites saw near-identical air, so concentration differences are local-source driven; PCA components explained 46.5% and 19.9% of variance. Wet-season levels were lower. Screening risk was metal-dominated at some sites and PAH-dominated at Bukavu. Limitation: PUF-PAS uptake rates for particle-bound species are uncertain and site-specific, so concentrations are semi-quantitative; no PM mass. Relevance: high - electricity-free composition sampling is the natural complement to optical LCS mass networks in data-sparse Africa.

doi: 10.1016/j.envpol.2026.129361

Kang et al. 2026 Build Environ

Moisture-coupled aerosol filtration in moss-based porous media: enhanced particle capture and aerodynamic resistance trade-off

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 3 Oct with no abstract. By title, a filtration study of moss-based porous media in which moisture improves particle capture at a cost in aerodynamic resistance. The efficiency-vs-pressure-drop trade-off is the same constraint that governs inlet and filter design in gravimetric samplers and in filtered-air intervention trials. Relevance: low-moderate; on the retry list.

doi: 10.1016/j.buildenv.2026.115363

Kara Pythagorean fuzzy $PM_{2.5}$ monitoring station placement Trabzon 2026 J Air Waste Manag Assoc

A Pythagorean Fuzzy decision making framework for $PM_{2.5}$ monitoring station placement in urban areas: A case study of Trabzon, Turkiye

Modified Delphi to set eight criteria, PF-SWARA to weight them and PF-CODAS to rank five candidate sites in Trabzon. $PM_{2.5}$ level weighted highest, then population density and distance to existing stations; a shopping-mall site ranked first. Limitation: the criterion weights are expert judgment; there is no objective function (coverage, representativeness, variance reduction) and no test that the chosen site improves exposure estimation. Weighting '$PM_{2.5}$ concentration' highest is circular when the network is being sited to measure it, and five alternatives make the ranking near-trivial. Relevance: medium - a contrast case for data-driven siting (kriging variance, information gain) that LCS deployments use.

doi: 10.1080/10962247.2026.2754941

Kheder Foehn generative km-scale global $PM_{2.5}$ forecast 2026 arXiv (preprint)

Global probabilistic forecasting of air pollution

Foehn, a generative model trained on global kilometre-scale $PM_{2.5}$ estimates that also computes air-mass origin. In one year of retrospective forecasts it beat the operational forecast at three of four stations; its ensembles are reported as calibrated, and the ensemble mean detects local pollution extremes four days ahead about as well as the operational system does one day ahead. Limitation: preprint; 'kilometre-scale estimates' are themselves satellite/model fusion products, so skill is partly measured against the training distribution, and the authors concede real-time km-scale inputs do not yet exist - operational use is hypothetical. Skill metrics and station counts are not given in the abstract. Relevance: high - calibrated LCS networks are the plausible real-time km-scale input this model says it lacks.

doi: 10.48550/arXiv.2610.25941

Li runway visual range model brown carbon aerosol chromatic effects 2026 Opt Lett

Refined runway visual range measurement model incorporating brown carbon aerosol effects

Optics Letter: conventional runway visual range (RVR) algorithms treat black carbon and scattering aerosol only; brown carbon's wavelength-selective absorption introduces two systematic errors - extinction spectral mismatch between the transmissometer/forward-scatter wavelength and the photopic band, and a chromatic shift in the visual threshold. A refined RVR model including BrC chromatic effects is proposed and 'experimentally validated'. Limitation: no error magnitudes, BrC loadings or validation data in the abstract. Relevance: moderate - the same single-wavelength assumption underlies optical PM sensors, whose mass conversion is biased by absorbing, spectrally selective aerosol.

doi: 10.1364/ol.611004

Loh et al. 2026 Mar Pollut Bull

Seasonal characteristics of submicron aerosols in the coastal atmosphere of southeastern Korea: Influence of urban-port outflow

Seasonal HR-ToF-AMS measurement at a coastal receptor site southwest of Busan, winter 2021 to autumn 2022. PM_1 ranged 4.23-10.0 $\mu g\,m^{-3}$; PMF resolved five organic factors including hydrocarbon-like, oxidised primary (OPOA), two oxygenated and biomass-burning. Ultrafine growth events were frequent, most in autumn, when OPOA reached 36% alongside elevated black carbon under north-easterly transport from the port. Limitation: one receptor site with no upwind reference, so 'port influence' rests on wind sector rather than on a source-resolved tracer, and AMS does not measure refractory sea salt or dust - the submicron mass is non-refractory by construction. Relevance: high - growth events at 4-10 $\mu g\,m^{-3}$ of mass are invisible to a mass-reporting network, which is the recurring theme of this week's issues.

doi: 10.1016/j.marpolbul.2026.120437

MISR Research Product Algorithm 1.1 km AOD 2026 Atmos Meas Tech

The MISR Research Product Algorithm - producing global, self-consistent, pixel-level aerosol retrievals for the Multi-angle Imaging SpectroRadiometer

Automates the MISR research aerosol retrieval into a production algorithm (MRPA) at 1.1 km pixels versus 4.4 km for the standard product. Against AERONET over land, MRPA mid-visible AOD has roughly half the RMSE and a higher correlation than the standard algorithm, at the cost of somewhat higher bias; Angstrom exponent improves, especially at high AOD. Smoke regions retrieve small-medium spherical absorbing particles, dust regions large non-spherical weakly absorbing ones. Limitation: automation reduces particle-type sensitivity, SSA and shape are validated only qualitatively, and MISR's 9-day global revisit limits daily $PM_{2.5}$ use. Relevance: moderate - type-resolved AOD is a plausible covariate for compositional correction of LCS-satellite fusion.

doi: 10.5194/amt-19-6503-2026

Malik MPSS calibration 2026 Aerosol Sci Technol

Traceable calibration of MPSS in fast-scan mode: Recommendations for PNSD measurement at urban traffic hot spots

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 9 Oct with no abstract. By title, a metrology-grade calibration route for MPSS/SMPS operated in fast-scan mode, with recommendations for particle number size distribution at traffic hot spots. Fast scans trade sizing accuracy for time resolution - the scan-direction smearing and DMA transfer-function errors that dominate when concentrations change faster than a scan are the reason roadside PNSD is rarely comparable across sites. Relevance: high - a traceable reference for sub-100 nm number is the missing anchor for any UFP sensor co-location; on the abstract-retry list.

doi: 10.1080/02786826.2026.2726892

Manchanda hyperlocal BC sensing Bayesian inverse modeling West Oakland 2026 Sci Adv

Hyperlocal sensing and inverse modeling reveal community impacts of urban air pollutant emissions

Dense mobile and fixed-site BC measurements were assimilated in a Bayesian inversion to produce hourly 150 x 150 m emission maps for West Oakland, a freight-corridor community. The posterior raised total BC emissions by ~33% over the bottom-up inventory and quadrupled the share of population-weighted exposure attributed to neighbourhood and port-related sources (5% to 20%), concentrated in low-income blocks. Sensitivity runs with ~10 fixed monitors (~1 site/km2) plus time-averaged mobile maps recovered similar results. Limitation: the posterior is only as good as the transport kernel and prior error covariance; unpermitted sources are inferred, not verified on the ground, and one community limits transfer. Relevance: high - a direct recipe for turning a sparse low-cost BC/PM network into source-resolved emission corrections.

doi: 10.1126/sciadv.aeb7647

Mei et al. 2026 Atmos Chem Phys

Synthesis of the tethered balloon system and other TRACER campaign measurements elucidates aerosol property profiles

149 tethered-balloon flights over greater Houston in summer during the DOE ARM TRACER campaign, with back-trajectory k-means clustering into marine, mixed and urban/long-range air masses. Profiles vary strongly with cluster, modulated by boundary-layer depth and coastal circulation; the marine cluster had CCN at 0.8% supersaturation below 1,000 cm^{-3}, urban and mixed clusters higher. Limitation: one coastal city, one summer, and CCN is inferred from size distribution under an assumed kappa rather than measured, so composition heterogeneity propagates straight into the activated fraction; tethered-balloon sampling is intermittent and fair-weather-biased. Relevance: high - this is the measurement that exposes the central fiction of a surface network, namely that a 2 m reading represents the column a satellite retrieval sees.

doi: 10.5194/acp-26-13885-2026

Mercado-Velazquez low-cost sensors three-tier London calibration transferability 2026 Sensors (Basel)

How Accurate Are Low-Cost Air Quality Sensors? A 10-Month Study of Three-Tier Comparison with Three Calibration Methods and Cross-Site Transferability

Ten-month co-location of low-cost (MOS, optical), mid-cost electrochemical and reference instruments at two London sites. Calibration by manufacturer curve, T/RH linear regression and ML. Field-calibrated $PM_{2.5}$ explained over half of reference variance at both sites, and linear calibration matched ML. $PM_{2.5}$ calibrations trained April-October 2025 held through winter, whereas NO_2 and CO did not; transferred between sites, $PM_{2.5}$ reached R2 >=0.56 and gases R2 <=0.28. The mid-cost NO_2 and low-cost O_3 sensors failed under all methods. Limitation: two sites 1.6 km apart in one city share aerosol type and humidity regime, so 'transferability' is tested at its easiest. Relevance: high - supports simple T/RH regression for $PM_{2.5}$ nodes and mandatory co-location for gas channels.

doi: 10.3390/s26206441

Metris and Metris 2026 PeerJ

Fungus above us: exploring the atmospheric mycobiome by aircraft

Aircraft bioaerosol sampling across seasons, sequenced and joined to vegetation-productivity and meteorological data. Diversity fell with altitude and with increasing vegetation productivity; decomposers and pathogens dominated, with over 40% of taxa putative plant or animal pathogens (Neoascochyta, Cladosporium, Alternaria, Penicillium among the abundant genera); guild composition shifted with productivity and surface wind. Limitation: a small number of flight tracks stand in for a continent, relative abundance from amplicon data is not concentration, and viability is untested - 'pathogen' here is taxonomic assignment, not infectivity. Relevance: moderate - aircraft campaigns set the vertical context that ground bioaerosol sensors, which cannot resolve taxa at all, are implicitly assuming.

doi: 10.7717/peerj.21746

Nobin Bangladesh GNN $PM_{2.5}$ 2026 PLoS One

Explainable spatiotemporal Graph Neural Networks: Multi-City $PM_{2.5}$ forecasting in Bangladesh with geographic graph topology validation

Graph-diffusion GRU on four years of daily multi-source data at 12 cities. Up to 3-day lead: MAE 8.85 $\mu g\,m^{-3}$, R2 0.666, beating non-graph baselines (FDR-corrected Diebold-Mariano p<0.001) and DCRNN when pooled across horizons. The useful result is negative for complexity: a fixed distance-adjacency graph beat adaptive, wind-informed and hybrid graphs, and occlusion-based influence tracked geographic adjacency (r=0.928). Limitation: 12 nodes is too sparse to discriminate graph topologies - transport structure that matters at 10-50 km scale cannot be resolved by stations hundreds of km apart. Relevance: high - a direct test of what graph priors a denser LCS network would need before learned adjacency pays off.

doi: 10.1371/journal.pone.0351928

Ouimette PMS5003 PurpleAir internal heating hygroscopic mass reduction 2026 Preprints.org (preprint)

Air Flow, Heat Transfer, and Hygroscopic Aerosol Mass Reduction in the Plantower PMS5003 Sensor and PurpleAir Monitor

Lab and field work on temperature in the particle detection zone of the PMS5003, standalone and inside a PurpleAir. The PurpleAir housing warms the detection zone more than a bare PMS5003, lowering sample RH; a wraparound USB heater pushed sample RH down to regulatory-monitor conditions. In Santa Rosa and Winston-Salem (against a T640x), external heating lowered reported $PM_{2.5}$ at high ambient RH, and the unheated PurpleAir already read below a standalone PMS. CFD characterised airflow, residence time and temperature. Limitation: two sites, no quantified bias reduction or growth-factor fit in the abstract; preprint. Relevance: high - EPA-style RH corrections assume ambient RH, but the sensor sees a warmed, drier sample; corrections built on bare-PMS data may not transfer to housed units.

doi: 10.20944/preprints202610.2325.v1

Pavlidis cold start 2026 Environ Sci: Atmos

Measurements of Real-World Cold Start Emissions in a Parking Garage

Only the deposited lead sentences are available: motor-vehicle cold starts contribute disproportionately to total vehicle emissions, and this work quantifies them under real-world conditions by measuring in a parking garage rather than on a dynamometer. A semi-enclosed garage gives the dilution control a roadside site lacks while keeping real engines, real soak times and real driver behaviour. Limitation: the abstract is truncated at deposit, so emission factors and fleet composition cannot be read; garage ventilation sets the dilution and is rarely characterised well enough to convert concentrations to per-vehicle factors. Relevance: high - cold starts are short, intense, local events, the exact signal a dense node network resolves and a regulatory monitor averages away. The 8 October aged-parking-garage-aerosol record sampled the same kind of microenvironment for oxidative potential.

doi: 10.1039/d6ea00044d

Pires quantum NN 2026 arXiv (preprint)

An Annealing-based Quantum Neural Network with Discrete Wavelet Transforms for Enhanced Air Pollution Nowcasting

Hybrid variational quantum circuits with quantum annealing, scaled to 172 qubits on a neuromorphic platform, nowcast $PM_{2.5}$ from UK urban data and are reported to outperform MLP and LSTM on raw high-entropy inputs, more so with wavelet decomposition. Limitation: the abstract gives no error metric, horizon or persistence baseline, and beating an MLP/LSTM on unpreprocessed data is a weak comparator; 'quantum utility' claims need a classical model tuned on the same wavelet features. Relevance: low - nowcasting at node level is a solved problem for persistence-plus-meteorology.

doi: 10.48550/arXiv.2610.13882

Pyae Tin Saw portable $PM_{2.5}$ cross-site prediction fixed station 2026 Environ Pollut

Short-term $PM_{2.5}$ data continuity: Cross-site prediction of portable measurements using a nearby fixed monitoring station

METADATA ONLY. Title and authors only (Pyae Tin Saw, Kallawicha); day-0 Elsevier deposit with no abstract anywhere at harvest. The title describes predicting short-term portable $PM_{2.5}$ records from a nearby fixed monitoring station to maintain data continuity. Sensor model, distance, n, and skill metrics cannot be stated. Limitation: unverifiable beyond the title; cross-site transfer functions are typically valid only within the regional-background regime and fail for local sources, which is exactly when portable data matter. Relevance: high - gap-filling an LCS or portable record against a regulatory anchor is a core network-operations problem.

doi: 10.1016/j.envpol.2026.129384

Qian MAG-MISD probabilistic multi-station $PM_{2.5}$ forecasting 2026 Preprints.org (preprint)

MAG-MISD: A Mode-Specific Adaptive Graph Framework for Multiscale Interaction and Station Dependence in Probabilistic $PM_{2.5}$ Forecasting

Variational mode decomposition into five components, cross-mode Transformer per station, and a separate directed graph per mode (geographic, persistent adaptive and sample-conditioned dynamic edges) before fusion into conditional quantiles. Against the same-input VMD-iTransformer, RMSE fell 4.76%, 2.78%, 0.13% and 0.72% at 1, 6, 12 and 24 h, with better pinball loss and interval score and ~70% fewer parameters; the persistent adaptive relation drove most of the gain in ablation. Limitation: the improvement vanishes beyond 6 h, and decomposition of the full series before splitting risks leakage unless done causally - the abstract does not say. Relevance: moderate - station-graph forecasting transfers to LCS networks, where per-mode dependence could also flag drifting nodes.

doi: 10.20944/preprints202610.1533.v1

Rawat et al. 2026 Atmos Environ

Integrated Assessment of $PM_{2.5}$ in the Central Himalaya using Satellite, Reanalysis, and a new In-situ CUPI Sensor network

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 3 Oct with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still refusing the connection. By title this is exactly the fusion problem this watch tracks - a new in-situ sensor network (CUPI) evaluated against satellite and reanalysis $PM_{2.5}$ in complex mountain terrain, where both gridded products are weakest. Relevance: potentially high; on the retry list, and worth a full-text read when the abstract appears.

doi: 10.1016/j.atmosenv.2026.122395

Ren physics-constrained UQ urban $PM_{2.5}$ prediction driving factors 2026 Build Environ

Integrating Implicit Physics Constraints and Uncertainty Quantification for Urban $PM_{2.5}$ Prediction and Driving Factor Analysis

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. The title describes an urban $PM_{2.5}$ prediction model with implicit physics constraints and uncertainty quantification, plus driving-factor attribution. Architecture, city, input features, horizon, skill metrics and whether the uncertainty is calibrated cannot be stated. Limitation: 'implicit' physics constraints in ML usually mean loss-function regularisation, which does not guarantee physical consistency out of distribution; nothing beyond the title is verifiable. Relevance: moderate - calibrated predictive uncertainty is what allows LCS-fed forecasts to propagate sensor error honestly.

doi: 10.1016/j.buildenv.2026.115449

Risti Jambi PMS5003 PMS7003 vs BAM-1020 2026 Lingkar: J Environ Eng

Evaluation of Low-Cost $PM_{2.5}$ Sensor Accuracy in Comparison with Government Reference Instrument

Consensus sweep. IoT nodes with two Plantower sensors co-located with the BMKG BAM-1020 at the Sultan Thaha Jambi climatology station, December 2025-January 2026, followed by linear-regression correction. Both tracked the reference moderately (R2 0.60 PMS5003, 0.59 PMS7003) and overestimated: PMS7003 MBE 10.46, MAE 12.53, RMSE 18.02 $\mu g\,m^{-3}$; PMS5003 MBE 11.84, MAE 13.60, RMSE 19.60 $\mu g\,m^{-3}$. The authors attribute the positive bias to high humidity. Limitation: two wet-season months at one site, no RH-stratified or kappa-Kohler correction and no held-out validation of the regression, so the correction will not transfer to the dry or haze seasons. Relevance: high - an equatorial reference co-location adds to the regional-calibration evidence; the 10 $\mu g\,m^{-3}$ bias matches humid-tropics PurpleAir behaviour.

doi: 10.22373/ljee.v7i1.10530

SEARO air-quality early-warning review 2026 Front Public Health

Prediction accuracy of available early warning systems for air pollution across WHO South-East Asia region: a systematic review

PROSPERO-registered review (to Feb 2026): 14 modelling studies from 839 records, PROBAST+AI risk-of-bias. Hybrid deep-learning forecasters reported the lowest RMSE (0.84-4.61 $\mu g\,m^{-3}$) and R2 up to 0.998 for $PM_{2.5}$; ML reached 98.3% categorical accuracy; CTMs cut bias ~70% only with satellite data assimilation. Most studies were at high risk of bias from absent external validation and overfitting; heterogeneity precluded meta-analysis. Limitation: an R2 of 0.998 for daily $PM_{2.5}$ is a leakage signature, and the review reports such figures without re-analysis. Relevance: moderate - operational forecast skill in the region is unproven; LCS networks could provide the out-of-sample validation these systems lack.

doi: 10.3389/fpubh.2026.1945362

Saharan fresh dust optics 2026 Atmos Chem Phys

Optical properties and shape-dependent retrievals of the imaginary part of the refractive index of freshly emitted Saharan dust

In-situ optics during active emission events (Sept 2019): SSA for $PM_{2.5}$ (PM_{10}) 0.95 (0.94) at 370 nm and 0.97 (0.96) at 660 nm; AAE to 2.5 (2.0) from iron oxides; MAE fell from 0.30 to 0.15 m2/g at 370 nm with size. The imaginary index was retrieved under prescribed shape and density. Limitation: the retrieval depends on the shape assumption it is titled after, and a single campaign. Relevance: moderate - mass scattering efficiency shifting with size is the physics behind optical sensors' coarse-dust under-response; these values are usable priors for a dust correction.

doi: 10.5194/acp-26-14531-2026

Schafer AERONET lunar AOD assessment 2026 Atmos Meas Tech

An assessment of lunar photometry in AERONET

CIMEL model-T lunar AOD at 492 sites since 2014 with an updated empirical correction to the ROLO lunar irradiance model. At all long-record sites, day-night AOD differences are typically within +/-0.01 at 440 nm and as good or better at longer wavelengths; in 3 h day/night transition windows under low, stable AOD, AOD, Angstrom exponent, water vapour and SDA fine/coarse AOD show statistically negligible differences. Limitation: agreement is shown only under stable low AOD - smoke and dust nights, where night AOD matters most, are untested, and lunar-phase coverage leaves gaps. Relevance: high - validated night-time AOD removes the daytime-only constraint on AOD-$PM_{2.5}$ fusion and gives LCS networks a night-time column reference.

doi: 10.5194/amt-19-6487-2026

Schmidt AOD gap filling iterative singular spectrum analysis AERONET Cuiaba 2026 Environ Monit Assess

Gap filling in long-term AOD time series using Iterative Singular Spectrum Analysis at a tropical AERONET site

Applies iterative singular spectrum analysis (ISSA) to fill a 2005-2024 daily AOD record from the Cuiaba-Miranda AERONET station, where ~47% of days are missing, mostly in the cloudy wet season. Climatological (day-of-year mean) initialisation beat global-mean initialisation, giving lower error and better temporal consistency, and the cross-validation design materially changed the optimal reconstruction parameters. No error magnitudes are given in the abstract. Limitation: gaps are not missing at random - they coincide with cloud and wet-season aerosol regimes - so a climatology-anchored fill will bias toward dry-season behaviour exactly where data are absent. Relevance: low-medium - the same informative-missingness problem affects sensor networks with humidity-triggered dropouts and satellite-surface fusion.

doi: 10.1007/s10661-026-16067-6

Shai Nanozen RT vs gravimetric 2026 Ann Work Expo Health

Comparison of respirable dust concentrations between gravimetric weighing and real-time dust monitoring equipment in non-mining industries

20 paired task-based stationary samples at 11 industrial sites. No significant difference between gravimetric and raw real-time (Wilcoxon p=0.067) or density-adjusted (p=0.277); proportional bias grew with concentration but was absent below 1 mg/m3 (R2 0.06 and 0.02). Limitation: n=20, and the gravimetric reference is the instrument's own cassette downstream of the same inlet, so inlet losses and size-selection error are shared and invisible to the comparison; Wilcoxon non-significance at n=20 is not agreement. Relevance: moderate - the 1 mg/m3 cut is a usable operating limit, and the shared-inlet design is a pattern to avoid in LCS co-location.

doi: 10.1093/annweh/wxag086

Sharma Singapore aerosol type classification ground remote sensing reanalysis 2026 Aerosol Air Qual Res

Aerosol Types Classification in the Tropical Climate of Singapore by Integrating Ground-Based Remote Sensing and Reanalysis Datasets

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. Aerosol-type classification in tropical Singapore combining ground-based remote sensing (AERONET sun photometer presumably) with reanalysis (MERRA-2/CAMS). Classification scheme, period and the share of transboundary smoke cannot be stated. Limitation: unverifiable beyond the title; columnar typing does not map one-to-one onto surface PM. Relevance: moderate - aerosol type drives optical-LCS response (hygroscopicity, refractive index) and so the calibration regime a tropical network needs.

doi: 10.1007/s44408-026-00164-9

Sim real-time dust monitor bias gravimetric meta-analysis 2026 Ann Work Expo Health

Systematic bias in real-time dust monitoring: a meta-analysis of measurement discrepancy and its implications for occupational exposure assessment.

PRISMA meta-analysis of 112 occupational studies; 96 paired RTM-gravimetric comparisons pooled in a three-level random-effects model. Professional photometers read 36% high on average: pooled lnR 0.31 (0.25-0.37), I2 78.4%, prediction interval 0.11-0.63. Factory-calibrated units read 43% high, site-calibrated units 13% high; site-specific calibration was the strongest moderator (beta -0.24, P<0.001), followed by log GSD of the size distribution (beta 0.15) and RH (beta 0.002 per %). No eligible study compared sub-$500 sensors against gravimetry. Limitation: geometric-mean ratios from heterogeneous aerosols and campaign lengths; the prediction interval never crosses zero but its width means a single uncalibrated deployment can be off by 12-88%. Relevance: high - the size-distribution and RH moderators are the same terms LCS calibration models need, now with a pooled prior.

doi: 10.1093/annweh/wxag089

Spinosa volcanic Saharan dust biomass burning aerosol advections Naples 2026 Atmos Res

Multiple advections of volcanic, Saharan dust and biomass burning aerosols over Naples

METADATA ONLY. Title and authors only (Spinosa, Manzo, Amoruso, Sannino et al.). No abstract was retrievable at harvest. The title describes overlapping advections of volcanic, Saharan dust and biomass-burning aerosol over Naples; the author group operates lidar there, so a profiling case study is likely but is not confirmed. Instruments, dates and surface PM impacts are not available; nothing is inferred. Relevance: medium - mixed dust/smoke/volcanic layers are the cases where optical low-cost sensors misreport mass most, because refractive index and size change simultaneously.

doi: 10.1016/j.atmosres.2026.109425

Srivastav ion particle size distribution NPF Indo-Gangetic Plain 2026 Atmos Chem Phys

Ion and particle size distribution measurements in the Indo-Gangetic Plain: seasonal insights and challenges in new particle formation event characterization

Ion and particle spectrometer measurements at a suburban IGP site, May-December 2023. NPF occurred on 30/35 valid days (86%) in May-June but 20/73 (28%) in winter/post-monsoon. Morning median condensation sink was lower on event days (0.0095 vs 0.0292 s-1). Growth rates rose with size, 14.7 nm/h (3-7 nm) to 19.0 nm/h (7-20 nm). Pollution plumes intermittently masked or distorted events, so both visual and automated classification and growth-rate fitting were unreliable. Limitation: no precursor gas (H2SO4, amines, HOMs) measurements, so mechanism is unconstrained; eight months at one site. Relevance: medium - in high-CS settings the UFP number seen by condensation or diffusion-charger sensors is plume- not nucleation-driven.

doi: 10.5194/acp-26-14917-2026

Stutz Fairbanks inversion layers 2026 Atmos Chem Phys

Vertical distribution of pollution trapped by wintertime surface inversions in Fairbanks, Alaska, during the ALPACA experiment

Long-path DOAS profiles during ALPACA 2022 resolve polluted surface layers 20-40 m deep persisting for days, with SO_2 to 35 ppb, NO_2 60 ppb, HONO 2.5 ppb, HCHO 7 ppb and depleted ozone. PACT-1D, with mixing parameterised from surface temperature gradient and wind, reproduces the profiles and gives a 1-4 h residence time. Limitation: gas-phase only - particulate profiles, the health-relevant quantity in Fairbanks, are not retrieved - and one winter at one city. Relevance: high for siting - a 20-40 m layer means a node on a rooftop and one at breathing height sample different air masses under inversions.

doi: 10.5194/acp-26-14345-2026

Sylvester PurpleAir Philadelphia 2026 ACS EST Air

Validation of PurpleAir $PM_{2.5}$ Sensors in Temperate Urban Environments

Co-location in Philadelphia across summer and spring plus a one-year re-evaluation. Best performance came from the nationwide Barkjohn (2021) correction applied to the mean of the two channels' atm outputs: RMSE 1.64 $\mu g\,m^{-3}$, MBE 0.10 $\mu g\,m^{-3}$, R2 0.683 at 1 h and 24 h, ahead of locally derived corrections. RH source (on-board vs weather station) changed R2 by only 0.04. After a year in the field R2 fell to 0.567. Limitation: R2 0.68 at low temperate concentrations means RMSE is the honest metric and the R2 drop partly reflects a narrower concentration range in the later period, not drift alone - the abstract does not separate them. Applying Barkjohn, derived on cf_1, to atm outputs is off-label. Relevance: high - direct evidence that a national correction can beat local fitting, and that year-one degradation needs scheduled re-co-location.

doi: 10.1021/acsestair.5c00443

Taiwan transboundary PM_{10} forecast 2026 Environ Pollut

Forecasting transboundary PM_{10} pollution in northern Taiwan using upwind observations and transport timing

Upwind-transition-receptor framework with temporally separated training and evaluation for 1-4-day PM_{10} forecasts. Nested models attribute most added skill to distributed upwind observations; trajectory timing adds a smaller, receptor-dependent refinement. Best model (CatBoost with upwind data): RMSE 11.04 $\mu g\,m^{-3}$, 13.0% below persistence. Limitation: 13% over persistence at 1-4 days is modest, and upper-tail precision is limited by event scarcity - the episodes that matter are the ones the model sees least. Relevance: high - quantitative evidence that upwind observation density, not trajectory sophistication, buys forecast skill; a siting argument for upwind nodes.

doi: 10.1016/j.envpol.2026.129340

Trivanovic periodic technical inspection particle number devices gasoline exhaust 2026 J Aerosol Sci

Assessment of existing PN-PTI devices to measure gasoline exhaust

METADATA ONLY. Title and authors only (Trivanovic, Melas, Auderset, Hammer et al.). No abstract was retrievable at harvest. The title indicates an evaluation of existing particle-number periodic-technical-inspection (PN-PTI) instruments - developed and type-approved for diesel vehicles with particle filters - when applied to gasoline exhaust, whose size distribution sits lower and nearer the counters' 23 nm or 10 nm cut. Instruments, test vehicles and error magnitudes are not available; nothing is inferred. Relevance: high - PN-PTI counters are the first mass-produced, low-cost, legally metrological particle counters, and their size-cut behaviour is the same question for community UFP sensing.

doi: 10.1016/j.jaerosci.2026.106901

UAV rotor-wash bias in airborne pollutant sensing 2026 Sensors (Basel)

UAV-Induced Measurement Bias in Air Pollution Sensing: A Physics-Informed Framework for Characterization, Sensor Placement, and Correction

Actuator-disk momentum theory and literature constraints are combined into a four-zone rotor-wash contamination map, a sensor-placement rule and a two-stage bias correction for nine pollutants. Hover and 5 m/s co-location flights on a 21-inch quadrotor against a fixed reference showed bias of 3-10% in hover and 6-15% in forward flight across NO_2, CO, O_3, PM_{10} and $PM_{2.5}$, directionally matching predictions, but correlation fell to r = 0.44 in forward flight. Limitation: a single airframe, few flights and no wind tunnel - the authors concede qualitative rather than quantitative support. Relevance: high for vertical-profile LCS work - mounting position is a first-order error term and r = 0.44 is below any usable validation threshold.

doi: 10.3390/s26196178

Vannini biochar passive biomonitor potentially toxic elements lichens 2026 Atmos Environ

Wood-derived biochar as a potential sustainable alternative to lichens for monitoring airborne potentially toxic elements

METADATA ONLY. Title and authors only (Vannini, Sidoli, Pagano, Magnani). No abstract was retrievable at harvest. The title proposes wood-derived biochar as a standardisable passive accumulator of airborne potentially toxic elements in place of transplanted lichens. Exposure design, elements, comparison data and agreement are not available; nothing is inferred. Relevance: medium - a cheap passive element sampler could add composition to dense optical-mass networks, provided uptake can be calibrated to air concentration.

doi: 10.1016/j.atmosenv.2026.122457

Waghmare machine learning explainable AI $PM_{2.5}$ prediction Nagpur 2026 MethodsX

Machine learning models with explainable AI for $PM_{2.5}$ prediction for Urban air quality assessment

Six regressors (decision tree, extra trees, random forest, KNN, XGBoost, SVR) trained on ~4,000 observations of $PM_{2.5}$, NO_2, SO_2, NH_3, O_3 and meteorology in Nagpur, 70/30 split with 10-fold CV. A single decision tree scored best: test R2 0.90, RMSE 7.5-8.9 $\mu g\,m^{-3}$, MAE 5.4-6.6 $\mu g\,m^{-3}$; SHAP ranked NO_2, O_3, RH and NH_3 top. Limitation: random splits of autocorrelated time series leak information, so skill is inflated; co-pollutants measured at the same station make this a nowcast, not a forecast, and a lone tree beating ensembles is a warning sign. Relevance: low-medium - illustrates the validation discipline (blocked or spatial CV) that sensor-network ML calibration needs.

doi: 10.1016/j.mex.2026.104217

Wang $PM_{2.5}$-bound PAH association rules machine learning forecasting Nagoya 2026 Environ Monit Assess

Characterizing statistical association patterns and local forecasting of $PM_{2.5}$-bound polycyclic aromatic hydrocarbon concentrations in Nagoya, Japan

Hourly $PM_{2.5}$-bound PAHs in Nagoya with association-rule mining and ML under time-blocked cross-validation. Low wind speed was the most consistent condition for PAH accumulation; BC and NOx peaked at rush hours. XGBoost and RF predicted high-molecular-weight PAHs best (R2 0.81-0.93, RMSE 0.02-0.11); SVR and MLP were poor. BaP dominated toxicity; lifetime cancer risks were below the USEPA negligible threshold. Limitation: a 10-day observation period, so the models learn one synoptic episode, and the authors themselves restrict the rules to site-specific description. Relevance: moderate - a BC-plus-meteorology proxy for PAHs is the kind of virtual sensor a network could run, if trained on far longer records.

doi: 10.1007/s10661-026-16083-6

Washeem machine learning condensation bioaerosol sampler collection volume 2026 J Aerosol Sci

Machine-Learning-Guided Prediction and Temperature Optimization for Collection Volume Control in a Condensation-Based Bioaerosol Sampler

METADATA ONLY. Title and authors only (Washeem, Li, Jansen, Shirkhani et al.). No abstract was retrievable at harvest. The title describes using machine learning to predict and control the liquid collection volume of a condensation-growth bioaerosol sampler by optimising its temperature set points. Instrument, training data and performance are not available; nothing is inferred. Relevance: low - sampler control engineering for bioaerosol, adjacent to but outside mass-concentration sensing.

doi: 10.1016/j.jaerosci.2026.106907

Wildfire smoke onset detection fixed cameras 2026 Sensors (Basel)

Detecting Wildfire Smoke Onset from Fixed View Camera Networks

Reframes smoke detection as onset detection: a scene model learned from unlabelled fire-free video yields per-tile prediction residuals, read through four cues (residual magnitude, opacity accretion, motion coherence, temporal persistence), with detection delay and false alarms in the loss. On FIgLib the first alarm comes 1.52 min after ignition at 0.21 false alarms per camera-day, transferring to PyroNear-2025 and SKLFS-WildFire without labels. Limitation: ignition-to-alarm is benchmarked on curated sequences; night, fog and low-cloud performance is not reported. Relevance: low-moderate - an upstream trigger, not a PM measurement, but camera onset times could gate event-mode sampling and plume attribution in LCS networks.

doi: 10.3390/s26196020

Wu ProNet physics-AI coupled CTM $PM_{2.5}$ O_3 forecasting 338 cities 2026 Natl Sci Rev

Fusing physics and AI enhances accuracy and diagnostic capabilities in air quality forecasting

ProNet embeds observation-constrained neural correctors inside the governing equations of a chemical transport model rather than post-processing its output. On 2022-23 data across 338 Chinese cities it beat the physical CTM and AI-only baselines, generalised better to extreme events outside training, and surpassed forecaster-adjusted operational forecasts by 6.1% ($PM_{2.5}$) and 3.0% (O_3). The correctors yield process-level error bookkeeping that exposes systematic CTM parameterisation biases. Limitation: '6.1% accuracy' is an unspecified metric; one two-year period invites temporal leakage, and process attribution from learned correctors is not unique - compensating errors can be assigned to the wrong process. Relevance: high - in-equation assimilation is the natural slot for dense calibrated LCS observations.

doi: 10.1093/nsr/nwag570

Wu source-dependent organic aerosol hygroscopicity eastern China biomass burning 2026 Atmos Chem Phys

Source-dependent organic aerosol hygroscopicity in eastern China: observational evidence for conditional enhancement under biomass-burning influence

Winter field campaign at a rural eastern-China site retrieving organic-aerosol hygroscopicity from humidified nephelometry and ZSR closure. kappa_OA ranged from ~0 to 0.49 (mean 0.11 +/- 0.11) and correlated only weakly with O:C; instead it rose with biomass-burning influence, and ageing enhanced kappa_OA only when biomass burning dominated. Highly oxidised OA under weak biomass influence stayed weakly hygroscopic. Limitation: kappa_OA is a residual after subtracting inorganic contributions, so errors in the inorganic closure load onto the organic term; one site, one season. Relevance: high - humidity correction of optical PM sensors usually assumes a fixed kappa; this shows kappa shifts with source, so smoke-affected periods need a different growth correction.

doi: 10.5194/acp-26-15333-2026

Yan et al. 2026 Atmosphere

Baseline Determination for Threshold-Based Interpretation and Groups of Single-Particle Aerosol Fluorescence Measurements

Tests deionised water, NaCl and SiO2 as low-fluorescence baselines for the SwisensPoleno Jupiter and compares mu+3sigma, P99 and MAD thresholds. NaCl gives the broadest baseline (mu+3sigma upper thresholds ~52.0, 39.3, 38.2 x10^-3 a.u. across excitation sources, vs 6.6/10.1/3.1 for H2O and 12.1/9.1/2.6 for SiO2), and the effect persists within a shared 10-30 um size range; baseline choice changes the fraction of fluorescence-positive particles in both test materials and summer/winter ambient data. P99 tracks mu+3sigma; MAD diverges. Limitation: one instrument model, and no ground truth for which threshold best recovers biological particles. Relevance: moderate - fluorescence-positive counts from different networks are not comparable unless baseline material and threshold rule are reported, which is a metadata standard LCS bioaerosol channels lack.

doi: 10.3390/atmos17100971

Yang et al. 2026 Sensors (Basel)

An NB-IoT-Based Architecture with Spatial-Statistical Analytics for Cross-Domain Air and Water Quality Monitoring in Aquaculture

Edge nodes with low-cost air (PM_1/2.5/10, O_3, TVOC, eCO_2, T/RH, UV) and water sensors stream over NB-IoT and MQTT to a cloud database with IDW mapping and alerting. A deployment at one aquaculture facility logged 14,400 one-minute records with 99.24% raw packet delivery; analyses correct for autocorrelation (N_eff ~892) and FDR, and the headline findings are air-temperature/dissolved-oxygen and humidity/turbidity correlations. Limitation for this watch: no PM sensor is co-located with a reference, calibrated or humidity-corrected, so the PM channel is uncharacterised; the PM data enter none of the reported results. Relevance: low - the telemetry (gateway-free NB-IoT, packet delivery) is the transferable part; it says nothing about PM data quality.

doi: 10.3390/s26185964

Yang et al. 2026 Environ Sci Technol Lett

Development and Diagnosis of a Temporal-Context U-Net Deep Learning Surrogate Model for CMAQ Particulate Nitrate over China

An attention U-Net takes the target hour, the previous 2 h and a 96 h summary to emulate CMAQ surface nitrate over China; it beats baseline surrogates, especially at high concentrations. Emission-reduction tests reproduce the direction of July national-mean nitrate decreases but underestimate their magnitude, and agreement is poor for January NOx responses outside training months. Residuals correlate with CMAQ NH_3 and particle H+, and adding them helps. Limitation: it emulates a model, not the atmosphere - CMAQ's own nitrate biases are inherited - and the weak emission-response skill is exactly what a policy surrogate needs. Relevance: moderate - nitrate is the PM component optical sensors misread most under humidity, and fast surrogates are a route to composition-aware calibration.

doi: 10.1021/acs.estlett.6c00755

Zhang et al. 2026e Appl Opt

Full field of view temporal calibration for the directional polarimetric camera onboard the GaoFen-5 (02) satellite based on a data reconstruction method

METADATA ONLY. Metadata-only record: PubMed indexed title, authors and DOI on 6 Oct; Applied Optics deposited no abstract and the retry leg recovered none. By title, a full-field-of-view temporal calibration for the DPC instrument, correcting on-orbit response drift by data reconstruction. Polarimetric retrievals are the satellite side of aerosol loading, and a drifting on-orbit calibration propagates straight into the AOD that surface $PM_{2.5}$ models are trained against - the same class of problem as field drift in a ground sensor, at a different scale. Relevance: moderate-high; on the retry list.

doi: 10.1364/ao.597069

Zhao spin-trapping EPR radicals atmospheric fine particulate matter cavity quality factor 2026 Aerosol Sci Technol

Optimized spin-trapping method for radicals detection induced by atmospheric fine particulate matter: Critical roles of temporal evolution and cavity quality factor

METADATA ONLY. Title and authors only (Zhao, Chen, Ding, Zhang et al.); no abstract retrievable from Crossref, Europe PMC or Semantic Scholar at harvest. The title describes optimising EPR spin-trapping for radicals generated by $PM_{2.5}$, identifying the time course of adduct formation and the cavity quality factor as critical. Sample type, n and performance metrics cannot be stated. Limitation: unverifiable beyond the title; spin-trap adducts decay on minute timescales, so inter-lab oxidative-potential comparisons hinge on exactly these timing choices. Relevance: low - offline oxidative-potential metrology.

doi: 10.1080/02786826.2026.2750406

zum Berge multicopter vibration optical particle counter accuracy 2026 Aerosol Sci Technol

Influence of multicopter-induced vibrations on the measurement accuracy of optical particle counter

METADATA ONLY. Title and authors only (zum Berge, Platis, Bange). No abstract was retrievable from Crossref or Semantic Scholar at harvest. The title describes quantifying how airframe vibration from a multicopter degrades the accuracy of an optical particle counter carried aloft - relevant to the growing use of drone-borne OPCs (often the same Alphasense/Plantower-class sensors used in ground networks) for vertical PM profiling. Design, sensor model, vibration spectrum and error magnitudes are not available; nothing is inferred. Relevance: high - vibration is an uncharacterised error term for mobile and UAV-mounted low-cost OPCs; retry the abstract first.

doi: 10.1080/02786826.2026.2747765

Exposure assessment & modelling 130 records

Carried from the daily issues — full summaries (130 of 130 records in this subtopic)

Accra $PM_{2.5}$ elemental composition PMF source shift e-waste burning 2026 Environ Health Perspect

Shifting Dominant Air Pollution Sources in Sub-Saharan Africa's Changing Urban Environment: Analysis of Elemental Composition and Source Contributions of Ambient Fine Particulate Matter ($PM_{2.5}$) Pollution in Accra, Ghana

Citywide campaign April 2019-June 2020: >600 weekly gravimetric $PM_{2.5}$ filters from 146 sites, ED-XRF and PMF, compared with four legacy sites from 2007-2008. Harmattan crustal dust was 52-68% of elemental mass. Outside Harmattan, 4-6 factors: biomass burning 11-32%, traffic/road dust 9-36%, solid and e-waste combustion 0-35%. Over a decade the biomass contribution fell by ~16% (5.2-25.8) while waste burning rose by 23 points (15.9-30.0). Limitation: ED-XRF elements miss the carbonaceous fraction, so the shares are of elemental mass, not $PM_{2.5}$ mass, and the decadal comparison rests on four sites. Relevance: high - a 146-site gravimetric network is the speciation reference a West African LCS network needs; e-waste burning is a source optical sensors cannot distinguish.

doi: 10.1021/ehp.6c00323

Aerosol Fe isotopes 2026 Atmos Chem Phys

Isotopic composition of aerosol iron from anthropogenic sources: implications for source apportionment of aerosol iron

Iron isotope endmembers measured for desert dust and several anthropogenic sources: dust averaged +0.14 +/- 0.10 per mil (n=7, consistent with prior work), power-plant coal fly ash +0.26 +/- 0.18 (n=28) and steelwork fly ash -0.07 +/- 0.41, with a wide anthropogenic range overall. Limitation: the endmembers overlap heavily - a steelwork standard deviation of 0.41 per mil against a dust mean of +0.14 means a mixed urban sample cannot be apportioned by delta-56Fe alone, which the spread itself makes plain; small n for dust. Relevance: moderate - soluble aerosol iron matters for oxidative potential and for ocean biogeochemistry, and isotopic apportionment is the kind of offline chemistry that fixed networks will never do but that calibrates what they see.

doi: 10.5194/acp-26-14073-2026

Akhtar et al. 2026 Environ Monit Assess

Spatiotemporal characteristics of ozone and $PM_{2.5}$ pollution in Jilin City: insights from observations in 2020 and 2023

WRF-CMAQ at 3 km plus surface observations compare May-July 2020 (COVID low-emission baseline) with 2023 in a petrochemical city in Northeast China. Meteorology explains ~35% of O_3 variability (temperature-dominated) but <1% of $PM_{2.5}$; $PM_{2.5}$ rose in May and fell sharply in July, staying high in Huadian, Chuanying and Jiaohe. Limitation, stated by the authors: emissions and meteorology both changed between the two years, so attribution is associational, and O_3 is systematically underestimated; <1% explained $PM_{2.5}$ variance means the chosen predictors simply do not describe $PM_{2.5}$ here. Relevance: low - mid-size industrial cities are under-monitored, but this adds two seasons, not a method.

doi: 10.1007/s10661-026-15929-3

Aktypis nucleation-mode particles marine atmosphere downwind urban 2026 Atmos Environ

Frequent Appearance of Nucleation-Mode Particles in the Marine Atmosphere Downwind of an Urban Area

METADATA ONLY. Title and authors only (Aktypis, Vasilakopoulou, Matrali, Kaltsonoudis et al.). No abstract was retrievable at harvest. The title reports frequent nucleation-mode particles in marine air downwind of an urban area, implying urban plumes seed new-particle formation or carry fresh emissions over water. Site, instrumentation and event statistics are not available; nothing is inferred. Relevance: low - sub-25 nm particles are invisible to optical sensors and carry negligible mass.

doi: 10.1016/j.atmosenv.2026.122458

Aloraini negative control PSCF CWT trajectory dust attribution Riyadh 2026 Atmosphere

Transport Pathways Versus Dust Sources: A Negative-Control Test of Trajectory-Statistical Attribution in Riyadh, Saudi Arabia

Ten years of Riyadh METARs separated locally lifted dust (BLDU/BLSA, 179 days) from suspended dust (DU/SA, 345). ERA5 models predicted local lifting better (held-out AUC 0.914 vs 0.765). PSCF and CWT fields from 14,612 HYSPLIT trajectories were then run on the local class as a negative control: it was more enriched over a priori source regions than the suspension class (permutation p <=0.007), and wind matching erased most structure in both. Dust days also fell 85.9% in 2023-2024 without model explanation. Limitation: METAR codes are visibility-based, not PM. Relevance: high - trajectory 'source maps' from single receptors, common in LCS papers, mostly reflect transport climatology.

doi: 10.3390/atmos17111055

Alturki ensemble ML emulation GEOS-CF $PM_{2.5}$ Arabian Peninsula spatial transfer 2026 Atmosphere

Interpretable Ensemble Machine Learning for Statistical Emulation of Summer $PM_{2.5}$ Change in GEOS-CF over the Arabian Peninsula: How Far Does Spatially Blocked Skill Transfer?

Emulated the 2020-2025 change in summer $PM_{2.5}$ over 3,933 GEOS-CF cells from 13 satellite and reanalysis predictors. Tree ensembles reached spatially blocked CV R2 0.82-0.84 (linear 0.55), but a Random Forest on latitude and longitude alone scored higher (R2 0.91). The predictor model lost all skill once cells within 300 km were excluded, failed on an unseen eastern sector (r 0.11) and did not transfer to other summers. Plain interpolation filled gaps better than any emulator. Split-conformal 90% intervals covered 82-98% of test cells, but only 60% in the east. Limitation: the target is model output, not observations. Relevance: high - a clean demonstration that blocked-CV R2 in PM mapping can be pure interpolation; the same diagnostics belong in sensor-fusion papers.

doi: 10.3390/atmos17101035

Alvi PMF trace elements PM_{10} Faisalabad health risk 2026 J Hazard Mater

PMF Resolved Source Identification & Health Risk Assessment of Trace-elements in Airborne Particulate Matter (PM_{10}) from South-Asian Megacity

Filter PM_{10} in Faisalabad analysed by ICP-OES: mean 517 +/- 206 $\mu g\,m^{-3}$, ~10x WHO, higher in summer than autumn or winter. Element order Fe > Al > Si > Zn > Cu > Pb > Cd. PMF resolved seven sources: metal works, vehicle exhaust and non-exhaust, industrial fossil fuel, municipal waste incineration and industrial mixed dust. Cumulative carcinogenic risk 1.22e-3 (adults) and 5.05e-4 (children). Limitation: ICP-OES detection limits for As, Cr and Cd are marginal at these loadings, PMF with ~20 elements and no carbon or ions cannot separate combustion sources robustly, and n of samples is not given. Relevance: low-moderate - coarse, crustal-dominated PM_{10} at these levels saturates optical LCS assumptions.

doi: 10.1016/j.jhazmat.2026.144032

Anjum Idaho $PM_{2.5}$ 2005-2024 wildfire vs wood-smoke regimes 2026 Research Square (preprint)

Long-Term Spatiotemporal Variability of Fine Particulate Matter ($PM_{2.5}$) Across Idaho, USA, Using Statistical and Geospatial Tools

EPA AQS daily $PM_{2.5}$ at 32 Idaho sites, 2005-2024, aggregated to 4,260 site-months; Mann-Kendall/Theil-Sen trends, PCA and IDW. Mean 7.53 $\mu g\,m^{-3}$ (site range 3.16-12.08). Bimodal climatology: an August-September wildfire peak and a November-February wood-smoke/inversion peak; July-September carries 36.6% of annual mass. Cold-season and background series declined while fire-season $PM_{2.5}$ rose 2.54 $\mu g\,m^{-3}$ per decade, holding the annual mean flat; 32.7% of site-years exceed the 9.0 $\mu g\,m^{-3}$ standard. Limitation: unreviewed preprint; IDW over 32 sites in mountainous terrain is not an exposure surface, and the 'regimes' are PCA loadings, not source apportionment. Relevance: high - two diverging seasonal regimes in inversion valleys are precisely where sparse FRM coverage fails and a calibrated PurpleAir layer (smoke-specific correction) adds most.

doi: 10.21203/rs.3.rs-11373778/v1

Anwar Pretoria FTIR 2026 Aerosol Air Qual Res

$PM_{2.5}$ Composition and Organic Matter Sources in Pretoria, South Africa Employing Fourier Transform-Infrared Spectroscopy

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 10 Oct with no abstract. By title, FT-IR functional-group analysis of $PM_{2.5}$ organic matter in Pretoria with source attribution. FT-IR on routine filters is one of the few speciation routes cheap enough to run network-wide, and sub-Saharan African composition data are scarce enough that any new site matters. Relevance: high for the composition gap this corpus repeatedly hits; on the abstract-retry list.

doi: 10.1007/s44408-026-00172-9

Arora rural PAH 2026 Atmos Environ

Seasonal PAH Exposure and Health Risks at Central European Rural Background Sites Driven by Combustion Sources and Transport Pathways

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 9 Oct with no abstract. By title, seasonal particle-bound PAH exposure and incremental lifetime cancer risk at Central European rural background sites, attributed to combustion sources and transport. Rural background PAH is dominated by residential solid-fuel burning, the same source the same-day Belgrade apportionment puts at 21% of PM_{10}. Limitation unverifiable from metadata. Relevance: moderate - PAH is a composition endpoint no optical node reaches; on the abstract-retry list.

doi: 10.1016/j.atmosenv.2026.122419

Balabat et al. 2026 World J Microbiol Biotechnol

Urban Road Dust as a Microbial Hotspot: Spatial Variation of Bacterial Communities and Metal Associations in the Philippines

Road dust from nine sites across three metropolitan areas profiled by 16S rRNA metabarcoding with PICRUSt2 functional prediction and trace-metal pollution indices. Proteobacteria and Actinobacteria dominated everywhere with a persistent core of Paracoccus, Acinetobacter, Kocuria and Deinococcus; alpha diversity did not differ between cities (Kruskal-Wallis $p>0.05$) but composition did (PERMANOVA $R2=0.466$, $p=0.004$). Paracoccus rose toward the most metal-laden Cagayan de Oro site and correlated with cobalt and nickel alongside raised predicted denitrification. Limitation: three sites per city, no replication within site, PICRUSt2 infers function rather than measuring it, and road dust is a reservoir - nothing here shows what becomes airborne or is inhaled. Relevance: low-moderate - the resuspended coarse fraction that low-cost optical nodes do see is biologically non-inert.

doi: 10.1007/s11274-026-05307-z

Baliaka Eaton urban firestorm airborne toxic metals lead Los Angeles 2026 Sci Adv

Fire at the boundary: Airborne toxic metals from the 2025 Los Angeles urban firestorm

Hourly airborne trace-element measurements during the January 2025 Eaton fire tracked the transition from structure to wildland burning via the K/Pb ratio. Relative to wildfire smoke, the urban plume was enriched >50-fold in lead, ~7-fold in halogens and ~3-fold in arsenic, attributed to vehicles, electronics and building materials. An estimated 7 million residents (range 4.3-8.0 million) experienced 24-h lead above 50 ng/m3, >80 times background. Limitation: population exposure is extrapolated from a small number of monitoring points through a dispersion estimate, and the 24-h lead threshold has no health-based standard counterpart (NAAQS is a 3-month mean). Relevance: high - optical PM sensors would register the smoke but not its metal toxicity; speciation is needed.

doi: 10.1126/sciadv.aei4455

Barnett-Itzhaki hourly compound heat air pollution extremes 2026 Atmos Pollut Res

Hourly Resolution Reveals Hidden Temporal Windows of Compound Heat-Air Pollution Extremes Masked by Daily Averaging

METADATA ONLY. Title and author only; no abstract retrievable at harvest. The title claims that hourly data expose compound heat-air-pollution extreme windows that daily means hide. Pollutants, location, thresholds, record length and how many events are missed by daily averaging cannot be stated. Limitation: unverifiable beyond the title; compound-extreme definitions are threshold-dependent, and hourly co-occurrence of heat and O_3 is expected by photochemistry, so the PM-specific contribution is unknown. Relevance: moderate - sub-daily exposure windows are precisely what continuous low-cost monitors resolve and daily regulatory filters do not.

doi: 10.1016/j.apr.2026.103243

Barthaux & Zuend 2026 Environ Sci: Atmos

Estimating Speciated Input Concentrations for Biomass Burning Organic Aerosol Modeling

METADATA ONLY. Metadata-only record: RSC deposited title, authors and DOI on 1 Oct with no abstract; PubMed efetch, Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The deposit carries only an opening sentence on biomass burning VOC and primary organic aerosol inputs for increasingly complex chemical models. Relevant to smoke-PM modelling, which the 30 Sep issue showed carries the mortality excess. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1039/d6ea00135a

Bei air moisture particulate matter winter Guanzhong Basin 2026 Atmos Pollut Res

Contributions of air moisture to particulate matter pollution during wintertime in the Guanzhong Basin, China

METADATA ONLY. Title and authors only (Bei, Chen, Jiang, Tie et al.). No abstract was retrievable at harvest. The title reports an analysis of how air moisture contributes to wintertime particulate pollution in the Guanzhong Basin. Design, mechanism, metrics and magnitude are not available; nothing is inferred. Relevance: potentially high - humidity-driven aerosol water uptake is the dominant error term in optical low-cost PM sensing, so a basin-scale moisture budget bears directly on correction. Retry abstract first.

doi: 10.1016/j.apr.2026.103256

Bloise water-soluble carbon nitrogen PM_{10} Chennai street vertical 2026 Atmosphere

Seasonal, Spatial, and Near-Surface Vertical Variability of Water-Soluble Carbon and Nitrogen in PM_{10} in Chennai, India

Three Chennai street environments across three campaigns, sampled at nominal heights of 3, 16 and 37 m: 76 PM_{10} samples, 75 WSOC/WSTN and 74 WSIN determinations. Means: PM_{10} 65.3 $\mu g\,m^{-3}$, WSOC 2.53 ug C/m3, WSTN 1.28 and WSIN 1.26 ug N/m3. WSOC fell from 4.01 to 0.62 ug C/m3 between campaigns while WSIN rose from 1.12 to 1.86. Site contrasts were strong for PM_{10}, and vertical differences were not significant. Limitation: about 25 samples per campaign split across sites and heights gives little power for the vertical null. Campaigns confound season with meteorology. Relevance: medium - no vertical gradient between 3 and 37 m supports rooftop siting of street-canyon LCS for PM_{10}, though the null is underpowered.

doi: 10.3390/atmos17101033

Budapest deposition 2026 Environ Sci Technol

Aerosol Deposition in the Human Respiratory System from Distinct Urban Sources Using Multiple Exposure Metrics

Source apportionment of particle number at an urban background site feeds a lung deposition model under reference breathing conditions. Most particles reach the deep lung with maximum deposition at airway generations 17-21 depending on activity; deposition fraction varies by source from 76% (new particle formation) to 27% (solid fuel combustion). Exercise raises total deposition and shifts it from extra-thoracic to alveolar; deposition rates are more sensitive to activity than deposition fractions, and number and surface-area curves are similar in shape but rank sources differently. Limitation: a deposition model, not a measurement - ICRP-type models are calibrated on healthy adults at rest and carry large uncertainty below 20 nm, exactly where the new-particle-formation fraction sits. Relevance: high - a factor of nearly three in deposited fraction between sources at the same ambient number concentration is the clearest statement in this backfill that mass, and even number, is not dose.

doi: 10.1021/acs.est.6c10850

Carnec adjoint nodal IAQ 2026 Indoor Environments

Adjoint-based corrective approach for nodal indoor air quality predictions

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 9 Oct with no abstract. By title, an adjoint-based correction for nodal indoor air quality models, which treat each room as one well-mixed node and are fast but systematically wrong where a room is not well mixed. An adjoint formulation gives sensitivity of predicted concentrations to each input at the cost of one extra solve, which is what makes correction from sparse sensor data tractable. Relevance: high - this is the data-assimilation route by which a handful of indoor nodes can correct a whole-building model, the indoor analogue of the conformal-coverage question raised on 8 October. On the abstract-retry list.

doi: 10.1016/j.indenv.2026.100199

Chakraborty fog intensity droplet size aqueous SOA Kanpur 2026 Environ Sci Technol

Fog Intensity and Droplet Size-Resolved Aqueous Secondary Organic Aerosol (aqSOA) Formation in Polluted Winter Fog: New Insights from a Combined Observational and Modeling Study

Two winters of size-resolved fogwater at Kanpur analysed by offline HR-AMS and PMF; three highly oxidised factors (O/C 0.95-1.19) were summed as an aqSOA proxy and scaled by liquid water content and WSOC. Across seven events aqSOA was 3.1-8.4 $\mu g\,m^{-3}$, a median ~7% of ambient SOA and 5% of OA, up to 13% of SOA in the densest fogs. Fine droplets carried 43-84% of aqSOA, linked to dissolved Fe and Cu and longer residence. Limitation: the aqSOA proxy is a PMF construct, and offline AMS of fogwater may not represent residual particle mass after evaporation. Relevance: medium - Indo-Gangetic fog episodes are when optical PM sensors fail outright from droplet interference, while secondary mass is being added.

doi: 10.1021/acs.est.6c09700

Chaudhary India meteorology 2026 Environ Monit Assess

Meteorological modulation of ground-level air quality in India based on in situ measurements

Eight years of in-situ measurements across Indian monitoring sites with Sen's slope trends. $PM_{2.5}$ fell at several Indo-Gangetic sites (-0.893 to -0.097 $\mu g\,m^{-3}$/month) and PM_{10} likewise (-1.320 to -0.169), while Punjab rose on local sources; ozone increased (0.041 to 0.529 $\mu g\,m^{-3}$/month), read as aerosol inhibition weakening plus more precursors. Declining boundary-layer height and wind speed over north and east India modulate all of it. Limitation: Sen's slope on eight years of CPCB data cannot separate emission change from meteorological trend, which is the paper's own point turned against its trend estimates; station siting and instrument changes over that period are undocumented. Relevance: moderate - the regional version of the same-day argument that a concentration trend is not an emissions trend.

doi: 10.1007/s10661-026-15915-9

Chavan and Lataye 2026 Environ Sci Pollut Res Int

Application of Openair package in R programming and PCA: unveiling air quality dynamics and meteorological influences

A full year of regulatory monitoring data for a tier-2 Indian city analysed with openair and PCA. Winter AQI reached 492 (severe) in February with monsoon improvement; varimax PCA gave two components explaining 73.1% of variance, read as anthropogenic combustion (PM, NO_2) and photochemistry (O_3, temperature); $PM_{2.5}$ and PM_{10} exceeded CPCB standards on 27-29 days per winter month. Limitation: descriptive reanalysis of one station-year with no source apportionment, no receptor model and no validation - PCA components here are correlation structure, not sources, and the policy recommendations do not follow from the analysis. Relevance: low as method, moderate as a reminder that tier-2 cities have one regulatory station and no spatial information at all.

doi: 10.1007/s11356-026-38275-w

Chen Chengdu WSON partitioning 2026 Atmos Chem Phys

Simultaneous measurements of gas- and aerosol-phase water-soluble organic nitrogen in the winter urban atmosphere of Chengdu, China

Winter urban campaign: particulate WSON 2.3+/-1.4 ugN/m3, twice gaseous, rising with $PM_{2.5}$; particle fraction 0.70+/-0.14. WSON partitions into organic phase under warm-dry and into aerosol liquid water under cold-humid conditions; amines are mainly gaseous but shift to particles as aminium sulfate/nitrate during episodes. ALWC dominated partitioning over T, RH and pH; $NH4NO_3$ was the main ALWC contributor. Limitation: one site, one season, and thermodynamic S-curve confirmation is model-on-model. Relevance: moderate - ALWC is precisely what inflates optical PM readings at high RH, so the nitrate-ALWC link is a calibration covariate as well as a chemistry result.

doi: 10.5194/acp-26-14329-2026

Chen ambient $PM_{2.5}$ representativeness children microenvironments 2026 Build Environ

Evaluating the representativeness of ambient $PM_{2.5}$ for children's exposure and source characteristics across microenvironments during school semesters

METADATA ONLY. No abstract at harvest. The title indicates a comparison of ambient $PM_{2.5}$ with children's exposure and source profiles across school-semester microenvironments. Recorded for coverage; no estimate inferred. Relevance: high in principle - quantifying the ambient-to-personal gap for children is precisely what portable low-cost monitors are deployed for, and the full text should be checked.

doi: 10.1016/j.buildenv.2026.115452

Chen primary brown carbon China 2005-2020 trends radiative effect 2026 Atmos Chem Phys

Long-term evolution and effects of primary brown carbon aerosol in China

CTM simulation of primary BrC across China, 2005-2020, with an internal-mixing optical model. Mean surface PBrC 0.81 ugC/m3; residential, industrial and agricultural combustion contributed 57%, 22% and 18%, together 91% of column PBrC in 2010. PBrC fell 20.8% over the period vs 8.1% for $PM_{2.5}$; direct radiative effect fell from +0.032 to +0.023 W/m2 (-25.7%), 84.2% anthropogenic. Limitation: BrC emission factors and absorption Angstrom exponents are poorly constrained, and evaluation against measured absorption is not described in the abstract. Relevance: medium - multi-wavelength absorption (BrC) channels on sensor nodes could test the residential-combustion dominance.

doi: 10.5194/acp-26-14867-2026

Cheng BTH multi-source satellite $PM_{2.5}$ stacking 2026 Atmosphere

Machine Learning-Based Retrieval of $PM_{2.5}$ Concentrations Using Multi-Source Satellite Data in the Beijing-Tianjin-Hebei Region of China

BTH 2019-2021: RF, SVR, XGBoost and linear stacking on four predictor sets with an 80:10:10 split. Test R2 rose from 0.51-0.54 with AOD alone to 0.83-0.84 with all eight predictors; stacking best at R2 0.84, RMSE 12.4, MAE 9.7 $\mu g\,m^{-3}$. Southern/southwestern Hebei highest; winter > summer. Missing AOD limited spatial completeness. Limitation: a random split leaks spatial and temporal autocorrelation, so the test R2 overstates out-of-site skill; no spatial-block CV, and gap filling - the real problem - is left to future work. Relevance: moderate - AOD alone explains half the variance; ground nodes and meteorology carry the rest.

doi: 10.3390/atmos17101010

Cheng LETKF WRF-Chem CALIOP component-resolved aerosol assimilation East Asia 2026 Atmos Chem Phys

Constraining the vertical profiles of aerosol component and heating rate over east Asia through assimilating CALIOP observations

Component-resolved LETKF assimilation in WRF-Chem of CALIOP-MODIS synergistic species-specific extinction profiles (dust, sea salt, BC, water-soluble), with spatial and observational constraints. The analysis reconstructs multi-layer vertical structure and corrects severe underestimation of elevated BC plumes, including persistent free-tropospheric BC over South Asia; AERONET and AD-Net lidar validation shows gains in AOT, extinction and single-scattering albedo. Limitation: CALIOP's narrow curtain and 16-day repeat leave most of the domain unobserved, the species classification is retrieval-model dependent, and the abstract gives no quantitative skill figures; surface $PM_{2.5}$ is not evaluated. Relevance: moderate - column and vertical constraints are what satellite-surface fusion with LCS networks currently lacks.

doi: 10.5194/acp-26-14671-2026

China 20 cities building green $PM_{2.5}$ 2026 Environ Pollut

City-size-dependent interactive effects of building and green-space on $PM_{2.5}$ pollution: Evidence from 20 representative Chinese cities

METADATA ONLY. No abstract at harvest. Title indicates interactions of building form and green space with $PM_{2.5}$ that depend on city size. Recorded for coverage. Relevance: low-moderate.

doi: 10.1016/j.envpol.2026.129344

Conformal $PM_{2.5}$ mapping 2026 Research Square (preprint)

Preregistered External Validation of Spatially Valid Conformal Prediction for Air-Pollution Exposure Mapping in China

A preregistered external validation - rare in this literature - of spatially valid conformal prediction against gradient boosting, with an acceptance band of [0.85, 0.95] on pooled coverage declared in advance. Under spatial hold-out the spatially weighted method reached 0.873 coverage at nominal 90% (gradient boosting: 0.486), 5 of 7 macro-regions met the block criterion (worst, North China, 0.719), and interval widths matched equal-weight conformal prediction, so coverage was not bought with wider intervals. Under temporal hold-out coverage fell to 0.794, outside the band, and the preregistered joint hypothesis failed - traced to the sharp post-2017 $PM_{2.5}$ decline. Limitation: province-level annual means are a very coarse spatial unit; coverage there says little about a 1 km surface, and the temporal failure is the one that matters for forecasting. Relevance: high - every exposure map in this corpus ships point estimates with no interval, and this is the first record in the series to preregister the test and report the failure.

doi: 10.21203/rs.3.rs-11182286/v1

DC 1 km WRF-SMOKE-CMAQ 2026 ACS EST Air

Neighborhood-Scale O_3, NO_2, and $PM_{2.5}$ in Washington, D.C. from a 1 x 1 km2 WRF-SMOKE-CMAQ Framework

1 km CMAQ for 2017: hourly O_3 r up to 0.7; NO_2 biased -14 to -29% from unresolved near-road gradients. NO_2's coefficient of variation exceeded O_3's and was more than three times $PM_{2.5}$'s; urban-core O_3 was 4-11% lower from titration. Moving from 4 to 1 km changed population-weighted exposure by only -1.5 to -3% for $PM_{2.5}$. Limitation: evaluation against a handful of AQS sites in one city cannot test intra-urban gradients - the very thing 1 km is meant to add. Relevance: high - quantifies that $PM_{2.5}$ is the spatially smoothest pollutant, so dense $PM_{2.5}$ networks buy less spatial information than NO_2 networks would.

doi: 10.1021/acsestair.6c00155

Dayanandana CMIP6 dust optical depth MENA evaluation 2026 Atmos Environ

Evaluation of Dust Optical Depth in CMIP6 Simulations over the Middle East-North Africa Region

METADATA ONLY. Title and authors only (Dayanandana, Parottil, Abbas, Al-Harrasi et al.); no abstract at harvest. Evaluates CMIP6 dust optical depth over MENA, presumably against MODIS/MERRA-2 DOD. Models, metrics and biases cannot be stated. Limitation: unverifiable beyond the title; column optical depth maps to surface PM_{10} only through assumed vertical profiles and size distributions, so climate-model DOD skill says little about surface exposure. Relevance: low-moderate - MENA dust is where optical LCS mass error from coarse particles is largest.

doi: 10.1016/j.atmosenv.2026.122431

Delhi-NCR COVID transitions 2026 Sci Prog

Time series evidence on air quality transitions under urban shock during COVID-19 shutdown in Delhi-NCR, India

Ten regulatory stations across pre-, during- and post-COVID periods analysed with ARIMA, PCA, change-point detection and recurrence quantification: significant reductions and altered temporal dynamics during lockdown, with meteorology and secondary formation relatively more important. Limitation: no deweathering is described, so lockdown effects are confounded with seasonal meteorology; the lockdown literature on Delhi is saturated. Relevance: low.

doi: 10.1177/00368504261498073

Dhital on-road black carbon PM exposure Nepal route travel mode 2026 J Air Waste Manag Assoc

On-road exposure to black carbon and particulate matter in Nepal: Effects of route attributes and travel modes

Mobile monitoring across Nepali road types, gradients and travel modes. Mean on-road eBC 45.7 +/- 32.5 $\mu g\,m^{-3}$, ~11x background; eBC made up 51% of on-road $PM_{2.5}$ and 75% of PM_1. Concentrations rose from residential streets to intercity highways. Upgraded orbital-highway segments had lower $PM_{2.5}$ and PM_{10} but unchanged eBC and PM_1, so paving cut resuspended dust rather than exhaust. High-gradient (4.5%) segments: median eBC 86.7 vs 41.5 $\mu g\,m^{-3}$ on 0.6% segments. Motorcycles gave the highest exposure, then non-AC and AC buses. Limitation: no n of runs or season control is reported, and the old-vs-upgraded comparison is not a before-after design. Relevance: high - shows optical $PM_{2.5}$ nodes miss the exhaust-dominated BC contrast that drives on-road exposure.

doi: 10.1080/10962247.2026.2754932

Fang end of uniformity spatial climatic toxicological redefinitions air quality 2026 Environ Sci Technol

The End of Uniformity: Spatial, Climatic, and Toxicological Redefinitions of Air Quality

METADATA ONLY. Title and authors only (Fang, Song, Weon). No abstract was retrievable at harvest; the format is likely a viewpoint or perspective. The title argues against uniform air-quality definitions along three axes - spatial heterogeneity, climate-driven change and toxicity beyond mass - which maps directly onto hyperlocal sensing, wildfire-era exposure and component- or oxidative-potential-based metrics. Arguments and any proposals are not available; nothing is inferred. Relevance: medium - if it advocates spatially resolved or toxicity-weighted standards, it is a policy argument for distributed networks; retry the text on a later run.

doi: 10.1021/acs.est.6c14170

Fazelvalipour LA microplastics 2026 Preprints.org (preprint)

Size-Resolved Characteristics and Source Influences of Airborne Microplastics in Central Los Angeles

Two seasonal campaigns in central Los Angeles: microplastics were in every sample at 2.2-4.0 MP/m3 (cool season) and 2.2-3.4 MP/m3 (warm), dominated by fibres and by the 5-50 and 50-100 um size fractions. Total MP correlated strongly (rho >= 0.79) with crustal (Fe, Si, Mn, Ca, Al, Mg), traffic (Ba, Cu, Ni) and carbonaceous (OC, EC) tracers, while short-term meteorology showed no consistent relationship, implying MPs travel as part of the resuspended urban particle mixture rather than as a separate contaminant. Limitation: a preprint; microscopy-based MP counts carry large operator and blank uncertainty, correlation with crustal tracers is also what resuspension artefact looks like, and particles under 5 um - the inhalable fraction - are not resolved. Relevance: moderate-high - supports the same-day tyre-wear argument that the growing non-exhaust, coarse, resuspended fraction is what networks increasingly measure badly.

doi: 10.20944/preprints202610.0577.v1

Five megacities $PM_{2.5}$ O_3 trends meteorology 2018-2023 Atmosphere

Urban Air Quality Trends and Meteorological Influences Across Five Global Megacities, 2018-2023

Descriptive 2018-2023 comparison. Shanghai $PM_{2.5}$ fell from 43.5 to 27.0 $\mu g\,m^{-3}$ (2018-2022) then rose to 30.3 in 2023; Delhi stayed above 87 $\mu g\,m^{-3}$ throughout; Paris declined; Sao Paulo rose by 2023. $PM_{2.5}$ exceedance days: Delhi 2,025, Shanghai 1,881, Los Angeles 366. Temperature correlated negatively with $PM_{2.5}$ in Delhi (r = -0.66), sea-level pressure positively (r = 0.65); wind speed was negative everywhere. Limitation: pairwise correlations without meteorological normalisation, so trends conflate emissions and weather; exceedance counts summed over pollutants and stations are hard to compare. Relevance: low - context only.

doi: 10.3390/atmos17101021

GOES-R $PM_{2.5}$ CONUS 2026 GeoHealth

Machine Learning Predictions of $PM_{2.5}$ and Their Applications to Exposure and Health Assessment in the CONUS

A two-phase framework: a U-Net-like partial convolutional network inpaints missing geostationary AOD (agreement with AERONET R=0.75, paired observations raised from 430,603 to 728,115 by iterative gap-filling), then a random forest predicts hourly surface $PM_{2.5}$ at ~1 km for 2019 onward. Limitation: the inpainting is validated against AERONET, which is sited where retrievals already work - gap-filled cells are by definition cloudy, humid or smoke-covered, and their skill is not demonstrated; a random forest on gap-filled inputs inherits that uncertainty without propagating it. Relevance: high - hourly rather than daily is the step that makes a satellite product comparable to a sensor network's native time resolution, and the same day's conformal-prediction record is the missing half of this one.

doi: 10.1029/2026gh001884

Ghazaryan explainable ML annual $PM_{2.5}$ Europe Central Asia 2026 Preprints.org (preprint)

Explainable Machine Learning for $PM_{2.5}$ Prediction and Comparative Country-Level Risk Assessment in Europe and Central Asia

Fifty-nine country-year observations (ten cities, 2020-2025) with eleven predictors; six models under five-fold CV. Linear regression had the lowest RMSE and highest R2, Extra Trees the lowest MAE; SHAP attributed added information to development indicators beyond meteorology. The authors call the result exploratory. Limitation: 59 observations and 11 predictors with random-fold CV on autocorrelated city panels cannot support model ranking or attribution; numeric skill is not reported in the abstract. Relevance: low - illustrates why annual aggregate predictors are no substitute for measured spatial coverage.

doi: 10.20944/preprints202610.1559.v1

Ghosh meter-scale heterogeneity new particle formation 3D model 2026 Environ Sci Technol

How Microscale Flow and Vapor Transport Shape New Particle Formation and Growth.

A 10 m resolution 3D multiphysics aerosol model (3D-ICAM) compared with an equivalent box model across emission geometries. Under weak winds (<1 m/s), it reproduced nucleation hotspots, intermittent multi-episode nucleation and persistent bimodal size distributions that the homogeneous case lacks. Resolving spatial structure raised predicted sub-3 nm growth rates up to ~20-fold, bringing them into the observed range, because transport-generated microenvironments relax the growth-scavenging competition. Limitation: idealised geometries; agreement with observations is at the level of growth-rate ranges, not site-specific validation. Relevance: medium - implies point UFP measurements in streets sample sub-grid hotspots; dense number-sensor networks are the natural test.

doi: 10.1021/acs.est.6c10206

Gray-green infrastructure $PM_{2.5}$ MSPA 2026 Build Environ

Spatially Explicit Morphological Regulation of Gray-Green Infrastructure for $PM_{2.5}$ Mitigation in High-Density Cities: An Interpretable MSPA-Machine Learning Framework

METADATA ONLY. No abstract at harvest. Title indicates morphological spatial pattern analysis plus interpretable ML relating gray-green infrastructure to $PM_{2.5}$ in dense cities. Recorded for coverage. Relevance: low.

doi: 10.1016/j.buildenv.2026.115406

Hanoi $PM_{2.5}$-bound PCDD/Fs 2026 Atmos Pollut Res

Levels and contributing sources to $PM_{2.5}$-bound PCDD/Fs and dl-PCBs in ambient air of Hanoi

METADATA ONLY. No abstract at harvest. Title indicates measurement and source attribution of particle-bound dioxins, furans and dl-PCBs in Hanoi $PM_{2.5}$. Recorded for coverage. Relevance: low - trace organics require filter sampling.

doi: 10.1016/j.apr.2026.103235

Harbin ammonium $PM_{2.5}$ and precipitation sources 2026 Environ Pollut

From atmosphere to river: Tracing the origins and deposition of ammonium in both $PM_{2.5}$ and precipitation in Harbin

METADATA ONLY. No abstract at harvest. Title indicates source tracing of ammonium in $PM_{2.5}$ and precipitation in Harbin and its deposition to river systems. Recorded for coverage; no estimate inferred. Relevance: low - secondary inorganic aerosol source apportionment.

doi: 10.1016/j.envpol.2026.129335

Hayli Gubbi eruption 2026 Atmos Environ

Short-term Atmospheric Impacts of the Hayli Gubbi Volcanic Eruption 2025

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 7 Oct with no abstract. By title, the short-term atmospheric impact of the 2025 Hayli Gubbi eruption. Volcanic SO_2 and ash plumes are the natural experiment that tests whether a sparse monitoring network can detect and attribute a known, dated, large perturbation - and East African ground monitoring is close to absent, so the answer is informative either way. Relevance: moderate; on the retry list.

doi: 10.1016/j.atmosenv.2026.122406

He NYC-METS size-resolved organic aerosol offline AMS 2026 Aerosol Sci Technol

Characteristics of size-resolved organic aerosols using offline aerosol mass spectrometry: NYC-METS

METADATA ONLY. Title and authors only; no abstract in Crossref, Europe PMC or Semantic Scholar at harvest. The record reports size-resolved organic aerosol characterised by offline aerosol mass spectrometry - re-aerosolised filter or impactor extracts - within the NYC-METS campaign in New York City. Sampling period, size cuts, source apportionment and sample counts cannot be stated. Limitation: offline AMS loses water-insoluble and semi-volatile fractions in extraction, so size-resolved OA mass is a lower bound; no inference beyond the title is possible. Relevance: moderate - size-resolved organic composition governs refractive index and hygroscopicity, the two terms optical LCS corrections approximate.

doi: 10.1080/02786826.2026.2746516

Hopke et al. 2026 Atmos Pollut Res

Characterization of PM Pollution Episodes in Kuwait

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 5 Oct with no abstract. By title, a characterisation of PM pollution episodes in Kuwait. Dust-dominated Gulf environments are where optical low-cost sensors fail hardest - coarse mineral dust sits outside the calibrated size and refractive-index range of nephelometric nodes - and the episode statistics are the input any regional network design needs. Relevance: potentially high; on the retry list.

doi: 10.1016/j.apr.2026.103226

Imran Chinese cities urban air quality determinants interpretable ML 2014-2023 J Environ Manage

Spatio-temporal variation and determinants of urban air quality in Chinese cities during 2014-2023: an interpretable machine learning analysis

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. The title describes a decade (2014-2023) of spatio-temporal variation in Chinese urban air quality with interpretable ML (SHAP-type attribution) of its determinants. Pollutants, number of cities, predictors and model skill cannot be stated. Limitation: unverifiable beyond the title; SHAP attributions on correlated socio-economic and meteorological predictors are not causal, and the 2014-2023 decline is dominated by the 2013 and 2018 action plans. Relevance: low - monitor-network secondary analysis.

doi: 10.1016/j.jenvman.2026.131184

Jabbari-Hichri on-road emissions light-duty vehicles particulate gaseous pollutants 2026 Atmos Pollut Res

On-road emissions of light-duty vehicles: detailed analysis of particulate and gaseous pollutants

METADATA ONLY. Title and authors only (Jabbari-Hichri, Azizi, Guiot, Boreave et al.); no abstract retrievable at harvest. The title describes detailed on-road measurement of particulate and gaseous emissions from light-duty vehicles. Vehicles, instruments and emission factors cannot be stated. Limitation: unverifiable beyond the title; on-road (PEMS) particle number measurements usually exclude volatile and sub-23 nm particles, which matters for comparing with roadside data. Relevance: low.

doi: 10.1016/j.apr.2026.103252

Julaha vertical aerosol chemical composition 230 m 4 m ACSM rural background 2026 Environ Pollut

Meteorological Effects on the Vertical Variability of Aerosol Chemical Composition at a Rural Background Site

Concurrent dual-height speciation, December 2019-May 2020: compact ToF-AMS at 230 m, ToF-ACSM at 4 m, Sunset OCEC. Every component was higher at 4 m; median 230/4 m ratios 0.55-0.82. Carbonaceous species had the steepest gradients and winter peaks; secondary inorganics were more homogeneous. Under inversions with mixing height <230 m, surface concentrations were 2-3x those aloft. Limitation: two different instruments at the two heights (AMS vs ACSM) - inter-instrument bias contributes directly to the ratio, and the abstract gives no co-location intercomparison. One site, one half-year. Relevance: high - a 2-3x surface/aloft contrast under shallow mixing is the bias any satellite-AOD or tower-based calibration of ground sensors must model.

doi: 10.1016/j.envpol.2026.129431

Kalkavouras ammonia sulfuric acid BVOC urban new particle formation Athens 2026 Environ Pollut

Assessing the combined roles of ammonia, sulfuric acid, and biogenic volatile organic compounds in urban new particle formation

Seven months (June 2023-January 2024) of particle number size distributions, NH_3 and BVOCs at the Thissio urban background site in Athens, with a proxy for H2SO4. Mean NH_3 2.3 +/- 0.7 $\mu g\,m^{-3}$. NPF occurred on 19% of days, most often in autumn (29-33%) when ventilation and condensation sink were low. Precursor levels barely differed between event and non-event days; the highest J14 came at intermediate NH_3 and H2SO4 with low BVOCs, while growth rates rose with NH_3 and isoprene/monoterpenes. Limitation: H2SO4 is a proxy, not measured, and the record lacks a full annual cycle. Relevance: medium - NPF bursts dominate urban number concentration but are invisible to mass-based optical sensors, which see nothing below ~300 nm.

doi: 10.1016/j.envpol.2026.129453

Kaskaoutis in situ aerosol optical properties eastern Mediterranean 2026 Atmos Environ

In situ aerosol optical properties in the background eastern Mediterranean: seasonal variability, regional transport, and source attribution

METADATA ONLY. Title and authors only (Kaskaoutis, Baalbaki, Bimenyimana et al.). No abstract was retrievable at harvest. The title describes in situ scattering/absorption measurements at a background eastern Mediterranean site, with seasonal variability, regional transport and source attribution. Site, instruments and values cannot be stated. Limitation: unverifiable beyond the title. Relevance: low-medium - background optical properties set the mass-scattering efficiency that optical PM sensors assume.

doi: 10.1016/j.atmosenv.2026.122436

Kim SpaMixed spatial mixed models microbiome $PM_{2.5}$ 2026 Genome Med

Spatial mixed models for assessing environmental exposure effects on the microbiome

SpaMixed is a spatial mixed model for microbiome count data that handles both spatial dependence across regions (conditional autoregressive priors) and ecological correlation among taxa. In simulations it achieved high feature-selection power with low false-positive rates and lower estimation error than existing methods. Applied to two $PM_{2.5}$ studies, it found 'biologically plausible' $PM_{2.5}$-associated taxa. Limitation: the abstract gives no sample sizes, regions or effect sizes for the applications. CAR priors absorb the spatial variance that carries the $PM_{2.5}$ contrast, so the method trades confounding for attenuation. Relevance: low - statistical method, no sensing content.

doi: 10.1186/s13073-026-01770-3

Kirsanov COSMO-ART Saint Petersburg UHI pollution PM_{10} feedback 2026 Atmosphere

Assessing Urban Pollution and Heat Island Feedback in a Northern Megacity Using the Fully Coupled COSMO-ART Model

COSMO-ART with TERRA_URB for winter and summer episodes in Saint Petersburg. UHI-driven mixing and deeper PBL cut near-surface primary pollutants 15-54% (winter) and 5-33% (summer), PM_{10} peaks by >44%, while O_3 rose. Aerosol radiative feedback on the UHI was weak (hundredths of a degree, peaks <0.3 C), cooling in summer and warming over winter snow, under a fifth of canopy forcing. Limitation: two episodes, model-only; no comparison with observed PM_{10} is reported in the abstract, and PM_{10} in a snowy city is dominated by road-dust resuspension that emission inventories handle poorly. Relevance: moderate - urban/rural LCS gradients conflate emission density with UHI-enhanced dilution.

doi: 10.3390/atmos17101030

Kwon seasonal $PM_{2.5}$ secondary inorganic aerosol livestock rural Ulsan 2026 Atmosphere

Seasonal Composition of $PM_{2.5}$ and Secondary Inorganic Aerosols at a Rural Site near Livestock Facilities in Ulsan, South Korea

Weekly 24 h filters, June 2023-November 2025, at a rural site near livestock facilities. $PM_{2.5}$ peaked in spring; nitrate peaked in winter and was lowest in summer (seasonal means 3.00 and 0.26 $\mu g\,m^{-3}$). Elevated ammonium-to-sulfate ratios were frequent in spring and winter, but nitrate exceeded sulfate by mass far more often in winter, so the $PM_{2.5}$ maximum and the nitrate maximum do not coincide. Limitation: one sample a week, no gas-phase NH_3/HNO_3 matched in time and no thermodynamic modelling, so ammonia-limited vs nitrate-limited regimes cannot be assigned, as the authors state. Relevance: low-medium - semi-volatile nitrate is a known source of LCS-vs-FEM divergence.

doi: 10.3390/atmos17111042

Langford cooking dominant primary PM_1 eddy covariance London Beijing Delhi 2026 npj Clim Atmos Sci

Traffic and combustion emission controls leave cooking as a dominant local source of primary PM_1 in megacities

Eddy-covariance fluxes with aerosol mass spectrometry and a new source-attribution method in three megacity neighbourhoods. In Delhi, traffic and combustion dominate primary PM_1 emission. In London and Beijing, cooking is the largest local primary source at mealtimes: up to 83% of midday and 87% of early-evening emission in London, 61% and 76% in Beijing. Cooking is absent from national, European and global inventories, though the London Atmospheric Emissions Inventory's estimate matched. Primary emission sources differ from those controlling ambient PM_1 mass, which is secondary-dominated. Limitation: one flux footprint per city and non-refractory PM_1 only (no BC or dust). Relevance: high - source-resolved flux benchmarks for inventory checks; cooking plumes are a known optical-sensor artefact at street level.

doi: 10.1038/s41612-026-01583-6

Li Guangzhou 3D vegetation 2026 Atmosphere

Scale-Dependent Effects of Three-Dimensional Urban Vegetation on Land Surface Temperature and Air Pollutants: Synergy and Trade-Off Zones in Guangzhou, China

Field LAI in 570 quadrats showed MODIS LAI is uninformative (r=0.15), so 3-D greenness was proxied by canopy height x tree cover. MGWR at 1-5 km: one SD of vegetation volume associated with 1.3 C lower LST, 0.30 $\mu g\,m^{-3}$ lower $PM_{2.5}$ and 1.3 $\mu g\,m^{-3}$ lower NO_2; O_3 null citywide with offsetting local responses. Limitation: $PM_{2.5}$ is a 1 km product whose own predictors include land cover, so the vegetation coefficient is partly the model reading its inputs back. Relevance: low-moderate - 0.3 $\mu g\,m^{-3}$ is below the error floor of any LCS network, which is itself the point.

doi: 10.3390/atmos17101001

Li atmospheric cadmium concentration deposition China modelling 2026 Atmos Environ

Modeling the spatial distributions and seasonal variations of atmospheric cadmium concentration and deposition in China

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. National modelling of atmospheric cadmium - carried almost entirely on fine particles - with its spatial and seasonal distribution and deposition. Model, inventory, resolution and evaluation against measured Cd cannot be stated. Limitation: unverifiable beyond the title; metal inventories for smelting and coal are the dominant uncertainty. Relevance: low - mass-only LCS cannot see toxic trace-metal fractions, which is why such composition models remain needed.

doi: 10.1016/j.atmosenv.2026.122427

Lian firework barium aerosols single-particle SPAMS Guangzhou 2026 Atmos Chem Phys

Evolution of firework-related barium aerosols: insights from single-particle analysis and mass concentration monitoring

Online Ba mass monitoring and SPAMS at four Guangzhou sites through firework events. Ba rose 1-3 orders of magnitude above background. The urban-core site under a complete firework ban peaked lower than a suburban site with time- and place-restricted use, despite higher population density - consistent with, but not proof of, a ban effect. Chloride-rich and nitrate-dominated Ba-containing particle classes were interpreted as fresh and aged, tracking chloride displacement and nitrate uptake. Limitation: four sites and a few events; the ban contrast is confounded by local emissions and transport, as the authors acknowledge, and SPAMS is semi-quantitative. Relevance: medium - firework episodes are a known optical-sensor artefact (hygroscopic, metal-rich particles), and this characterises their chemistry.

doi: 10.5194/acp-26-15101-2026

Liang deep learning low-visibility events fog-haze eastern China 2026 Atmos Chem Phys

Classification and quantification of low-visibility events using deep learning over eastern China

A unified CNN fuses geostationary imagery, meteorology and surface $PM_{2.5}$ to map and classify low-visibility events, separating polluted fog-haze from clean fog over land and sea. Scenario analysis: a 20% cut in $PM_{2.5}$ shrinks winter fog-haze area by 43% but population exposure to fog-haze only by 15%, because the most populated and most polluted regions coincide; >40% $PM_{2.5}$ reduction is needed there. Limitation: the scenario perturbs $PM_{2.5}$ as a model input, so it inherits the learned correlation rather than fog microphysics (hygroscopic growth, aerosol-fog feedback); classification skill metrics are not in the abstract. Relevance: medium - humidity-driven haze is the regime where optical LCS fail.

doi: 10.5194/acp-26-15003-2026

Liu Beijing autumn-winter $PM_{2.5}$ oxidative potential source predictors 2026 Atmos Environ

Autumn-winter variability and source-related predictors of $PM_{2.5}$ oxidative potential in urban Beijing

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. An autumn-winter campaign relating $PM_{2.5}$ oxidative potential in urban Beijing to source-related predictors (heating, traffic, secondary formation presumably). Assay (DTT, AA, GSH), sample count and predictive model cannot be stated. Limitation: unverifiable beyond the title; a single season and site. Relevance: high - follows the 19 Oct Guangdong OP finding; if OP is predictable from source tracers, a subset of networked sensors with composition proxies could map it.

doi: 10.1016/j.atmosenv.2026.122428

Liu $PM_{2.5}$ sources mixing state Chengdu Kangding altitude 2026 Environ Monit Assess

Altitudinal and seasonal differences in PM(2.5) sources, secondary formation, and particle mixing states in the basin-plateau transition zone of Southwest China.

Summer and winter $PM_{2.5}$ sampling at Chengdu (500 m) and Kangding (2,500 m). Winter $PM_{2.5}$ 79.4 vs 15.2 $\mu g\,m^{-3}$, summer 29.1 vs 8.1. Inorganic ions dominated in Chengdu, carbonaceous species in Kangding; sulfate in summer gave way to nitrate in winter. PMF: Chengdu combustion 26.0%, traffic 32.4%, secondary 35.3%; Kangding combustion 47.9%. Single-particle data showed more SIA particles in Chengdu (29.0% vs 17.6%) and more organic-SIA internal mixing in Kangding. Limitation: two seasons, two sites; the altitude contrast is confounded with city size and heating fuel. Relevance: low - descriptive regional chemistry.

doi: 10.1007/s10661-026-16102-6

Liu et al. 2026a J Environ Sci (China)

Persistent cold air pools and associated sustained haze pollution in Taiyuan Basin

Builds a refined detector for persistent cold-air pools from multi-station and sounding data in the Taiyuan Basin (2018-2022), validates WRF for PCAP meteorology, and identifies 56 events, mostly September-March, winter ones commonly lasting >48 h. Events carry stronger surface inversions, larger valley heat deficit and suppressed ventilation; $PM_{2.5}$ peaks lag heat-deficit intensification by 12 h ($R = 0.55$). Warm advection aloft maintains PCAPs and lee-side subsidence warming reinforces them; strong cold advection ends them. Limitation: $R = 0.55$ on a single basin, with emissions held implicit, so the 12-h lag is descriptive. Relevance: moderate - a 12-h lead from a sounding-derived metric is a usable forecasting feature, and basin networks must site nodes at more than one elevation to see the pool.

doi: 10.1016/j.jes.2026.03.027

Liu et al. 2026b J Environ Sci (China)

Metropolitan atmospheric microplastics: Characteristics and its inputs to fluvial system

Deposition sampling across urban, suburban and exurban Chengdu: 246, 192 and 177 items/m2/day respectively, dominated by fibres and fragments. PCA finds three source patterns (73.9% of variance), the largest a mixed daily-plastic-products source (55.1%). Abundance tracks precipitation intensity, weakly wind direction, not temperature or RH, and shows no relation to other air pollutants. Rainy-season input to the river system is estimated at 689-1,088 kg. Limitation: deposition flux, not airborne concentration or inhalable size fraction, so there is no exposure metric; identification method and lower size cut are not given in the abstract. Relevance: low - the absent correlation with regulated pollutants means PM networks are no proxy for airborne plastics.

doi: 10.1016/j.jes.2026.03.077

Lu heterogeneous ammonia uptake SOA WRF-Chem North China Plain 2026 Atmos Chem Phys

Impacts of the heterogeneous ammonia uptake on air quality in the North China Plain

A first-order NH_3 uptake on SOA forming NOCs was added to WRF-Chem for November 2017, with uptake coefficient gamma 1e-5 to 1e-3. Laboratory decline in reaction efficiency and measured NOC burdens support gamma=1e-5 as a conservative upper limit. At that value, NH_3 fell 0.62%, inorganic aerosol and $PM_{2.5}$ fell <0.06%, and the pathway produced 1.2 $\mu g\,m^{-3}$ of NOCs. Limitation: one month, one region, and the constraining gamma is chosen partly from the NOC burden it is meant to explain. Relevance: low - a negligible $PM_{2.5}$ mass term; useful only as a null for the chemistry.

doi: 10.5194/acp-26-15257-2026

Lu street-view greenery street canyon PM concentrations 2026 Environ Sci: Atmos

Association between urban greenery and particulate matter concentrations in street canyons: a local-scale study based on street view imagery

Street-view imagery quantified street greenery quantity and configuration, which were related to PM concentrations in street canyons; the authors attribute the associations to low-level vegetation barriers restricting ventilation. The record's abstract is two sentences: no city, sample size, PM metric, measurement method or effect direction and magnitude are given. Limitation: on what is available, the inference is correlational and the proposed ventilation mechanism is untested; treat as near-metadata until the full text is read. Relevance: moderate - siting of street-level sensors relative to hedges and low canopy is a known representativeness issue for LCS networks, and image-derived greenery is a cheap covariate for spatial calibration.

doi: 10.1039/d6ea00086j

Lucknow Holika Dahan $PM_{2.5}$ 2026 Environ Monit Assess

Impact of Holika Dahan on ambient $PM_{2.5}$ and associated health risks in an urban environment of Lucknow, India

Pre-, during- and post-Holika Dahan nights at a residential and a commercial site: $PM_{2.5}$ rose 2.0x and 1.4x; burning tracers 1.2-2.7x; PCA/HCA identified festival biomass burning as the dominant source; AERMOD placed hotspots out to 1.2-1.5 km. Hazard indices stayed below USEPA limits. Limitation: two sites over three nights, and a chronic-dose hazard index applied to a single-night event is the wrong risk metric. Relevance: moderate - a short, intense, spatially bounded event of exactly the kind a dense network resolves and two filter sites do not.

doi: 10.1007/s10661-026-16041-2

Luo et al. 2026 Environ Pollut

Primary nitrated phenol emissions from heavy-duty diesel vehicles based on real-world activity data in Hunan, China

Combines tandem filter/PUF exhaust emission factors with real-world heavy-duty diesel activity data to inventory primary nitrated phenols in Hunan for 2025: 1,196.8 kg/yr, 48.3% from container trucks, 22.9% freight trucks, 17.7% buses; China IV and V vehicles 84.1%. 4-Nitrophenol dominates (334.9 kg/yr). Monthly totals range 76.8 kg (Feb) to 114.4 kg (Dec), and five cities emit 53.9%. Limitation: emission factors from a limited test fleet are scaled to a province, the particle/gas split is not reported in the abstract, and no uncertainty is given. Relevance: low - nitrated phenols are a light-absorbing BrC component near freight corridors, relevant to multi-wavelength absorption channels, but the mass involved is trivial against $PM_{2.5}$.

doi: 10.1016/j.envpol.2026.129268

Magalhaes extreme rainfall landslides PM_{10} $PM_{2.5}$ Sao Sebastiao 2026 Environ Monit Assess

Can extreme rainfall associated with landslides affect air quality? A case study in a coastal city (Sao Sebastiao, Sao Paulo State, Brazil)

Hourly PM and meteorology from December 2021 to February 2025, with k-means meteorological clusters and openair polar and time-variation analysis around the February 2023 extreme rainfall and landslides. Rain first scavenged PM. The post-landslide period then showed PM_{10} up 124% and $PM_{2.5}$ up 93.3%. Two comparison rain events showed similar scavenging but different recovery, so the authors do not attribute the rise to landslides alone. Limitation: a single station and one event; clean-up traffic, resuspension of dried debris and meteorology are not separated, and there is no composition data. Relevance: medium - post-disaster resuspension is a case for rapidly deployable sensor nodes.

doi: 10.1007/s10661-026-16071-w

Mai population-weighted $PM_{2.5}$ O_3 China 2015-2024 preprint 2026 Research Square (preprint)

Population Weighted Ambient $PM_{2.5}$ and O_3 Concentrations in China from 2015 to 2024

Ecological geospatial analysis of annual ~1 km CHAP $PM_{2.5}/O_3$ rasters matched to WorldPop counts, 2015-2024. Population-weighted $PM_{2.5}$ fell from 53.06 to 31.25 $\mu g\,m^{-3}$ (-41.1%) while population-weighted O_3 rose from 77.06 to 96.75 $\mu g\,m^{-3}$ (+25.6%); the population-weighted minus geographic gap narrowed for $PM_{2.5}$ (5.77 to 2.43) and widened for O_3 (1.01 to 2.72). 97.4% of 0.2-degree blocks show falling $PM_{2.5}$ with rising O_3 at a 0.5 $\mu g\,m^{-3}$/y threshold. Limitation: purely descriptive and inherits CHAP's ML smoothing; no uncertainty on the weighted means; preprint, not peer reviewed. Relevance: moderate - the narrowing urban-rural $PM_{2.5}$ gap shifts LCS value toward peri-urban and rural siting.

doi: 10.21203/rs.3.rs-10964316/v1

Makacha PRECISE personal ambient $PM_{2.5}$ monitoring framework pregnancy SSA 2026 Front Glob Womens Health

An integrated climate-structured personal and ambient air pollution monitoring framework embedded within a pregnancy and postpartum cohort in Sub-Saharan Africa

PRECISE cohort: 343 pregnant or postpartum women (Gambia 160, Kenya 105, Mozambique 78), two 5-day wearable deployments in opposing seasons recording $PM_{2.5}$, T, RH and GPS/accelerometry; six fixed health-facility monitors. Calibration by repeated pre-deployment co-location with reference instruments and in-country Deming-regression harmonisation. Yield 3,190 person-days personal and 328 ambient days; inter-sensor R2 >0.72 pre-deployment and >0.75 in country; ambient capture >85% in Gambia and Kenya, lower in Mozambique (connectivity). Limitation: inter-sensor R2 is precision, not accuracy - the reference co-location bias and slope, and the RH correction applied, are not in the abstract; 5-day windows in a purposive subsample limit pregnancy-average inference. Relevance: high - a reusable LCS QA protocol for low-resource cohorts.

doi: 10.3389/fgwh.2026.1849628

Mandujano change-point detection trend estimation long-term PM records 2026 Environ Sci Atmos

Assessing the Effect of Change-Point Detection Techniques on Trend Estimation in Long-Term Particulate Matter Records

METADATA ONLY. Mandujano, Gonzalez, Hernandez, Guillen et al. Only the opening sentence of the abstract was retrievable from Crossref or Semantic Scholar: long-term PM trend analyses assume a single trend direction across the record, which can mask structural changes. The title states that change-point detection techniques are compared for their effect on trend estimates. Location, data set, methods and results are not available at harvest; nothing is inferred. Relevance: medium - instrument changes and station relocations are exactly the breakpoints that contaminate trends from mixed reference/LCS archives; retry the abstract.

doi: 10.1039/d6ea00110f

Mao et al. 2026 Atmosphere

Seasonal and Pollution-Level Variations in $PM_{2.5}$-Bound PFASs and Associated Health Risks in Jinan, a Typical Northern Chinese Industrial City

163 $PM_{2.5}$ filters over a year (Dec 2023-Nov 2024) in Jinan, 17 PFASs by LC-MS/MS. Total PFAS ranks winter 133.8 > autumn 123.6 ~ summer 123.5 > spring 97.1 pg/m3; in summer and autumn concentrations were numerically higher on clean than on slightly polluted days. No monotonic relation of total PFAS with single meteorological variables; compound responses depend on chain length. PFOA and short-chain PFCAs make up 73.1% of estimated intake and PFOA 95.6% of risk. Limitation: 'numerically higher' differences are not tested, one site, and risk uses reference doses for ingestion-derived toxicity. Relevance: moderate - a $PM_{2.5}$ hazard that is decoupled from $PM_{2.5}$ mass is invisible to any mass sensor, whatever its calibration.

doi: 10.3390/atmos17100967

Mielnik et al. 2026 Research Square (preprint)

Comparing geospatial interpolation techniques for estimating ambient $PM_{2.5}$ mass concentrations from sugarcane burning in South Florida

Low-cost sensors at 46 locations (8 Jun 2024 - 1 Dec 2025) spanning burn and non-burn seasons in the Everglades Agricultural Area; IDW and kriging variants compared by leave-one-out cross-validation for 24 h $PM_{2.5}$. Daily R2 ranged 0-0.99; IDW won on most days, and a per-day best-model ensemble was most robust. Smoke presence raised estimated $PM_{2.5}$ by 1.65 $\mu g\,m^{-3}$ (SE 0.13), concentrated inland near burn areas; the coast was cleanest. Limitation: sensor type, calibration and humidity correction are not stated in the abstract - in smoke, optical sensor bias is composition-dependent, so interpolation error and sensor error are confounded; LOOCV on a network this sparse rewards smoothness; preprint. Relevance: high - a direct test of how far a rural LCS network can be stretched into a health-study exposure surface for an episodic agricultural source.

doi: 10.21203/rs.3.rs-10743669/v1

Monsoon aerosol Pb isotopes rainwater offsets 2026 Atmos Environ

Monsoon-related variability of water-soluble aerosol Pb isotopes and systematic aerosol-rainwater offsets at Peng Chia Yu

METADATA ONLY. No abstract at harvest. Title indicates monsoon-resolved variation in water-soluble aerosol Pb isotope ratios and systematic offsets between aerosol and rainwater signatures. Recorded for coverage; no estimate inferred. Relevance: low - source apportionment input, not a monitoring method.

doi: 10.1016/j.atmosenv.2026.122425

Mukhamediyarov et al. 2026 Atmosphere

Elemental Composition, Sources, and Health Risks of Ambient $PM_{2.5}$ in Ust-Kamenogorsk (Oskemen) and Pavlodar, Kazakhstan

PCA separates a non-ferrous metallurgical profile in Ust-Kamenogorsk from coal fly ash, ferrous metallurgy and petrochemicals in Pavlodar. The risk arithmetic then reports cumulative hazard indices exceeding 1 by more than 500-fold, driven by Zn (HQinh 304.7) and Al (129.5), and carcinogenic risks for Cr and As up to 9x the 1e-4 threshold. Limitation: a hazard quotient of 300 for zinc is a sign that the reference concentration is being misapplied, not that residents are inhaling a toxic zinc dose - EPA inhalation RfCs do not exist for several of these elements and the substitutes used are oral values scaled by default breathing rates; sampling duration and n are not stated. Relevance: moderate for the Central Asian composition data, which is genuinely scarce; the risk numbers should not be propagated.

doi: 10.3390/atmos17100977

Mulena dust intrusion regimes north-central Argentina vertical structure optical properties 2026 Atmos Res

Dust intrusion regimes over north-central Argentina: Vertical structure, optical properties, and transport pathways

METADATA ONLY. Metadata only: no abstract at harvest. The title describes a classification of dust-intrusion regimes over north-central Argentina by vertical structure, optical properties and transport pathway, likely from lidar or sun-photometer records with trajectory analysis. No data can be summarised. Limitation: not appraisable; column and vertical optical properties do not by themselves give surface PM. Relevance: low-medium - dust-regime classification is the prior a satellite-surface fusion model needs for coarse-mode episodes in an under-monitored region.

doi: 10.1016/j.atmosres.2026.109404

Multi-scale $PM_{2.5}$ nonlinear time-series two metros 2026 Atmos Pollut Res

Deciphering multi-scale $PM_{2.5}$ temporal dynamics and cross-scale interactions in two climatically distinct metropolitan areas: insights from advanced nonlinear time-series analysis

METADATA ONLY. No abstract at harvest. Title indicates nonlinear multi-scale decomposition of $PM_{2.5}$ series and cross-scale interactions in two climatically distinct metropolitan areas (author affiliations suggest Vietnam/Taiwan). Recorded for coverage; no estimate inferred. Relevance: low-moderate - scale separation is how network redundancy and siting density are usually argued.

doi: 10.1016/j.apr.2026.103238

Munteanu et al. 2026 Sensors (Basel)

Evaluating Graph-Centrality Context for Station-Level Correction of CAMS Air Quality Reanalysis in Lombardy from 2018 to 2021

Corrects CAMS European reanalysis to ARPA Lombardia station observations (2018-2021) by treating stations as nodes of a multiplex graph and feeding six centrality measures (including an OpenStreetMap-context adapted PageRank) into GTWR, XGBoost and random forest with spatio-temporal kriging, under station, temporal and combined holdout. Station-holdout XGBoost reaches R2 up to 0.66 for NO_2; OSM-context centrality helps NOx and $PM_{2.5}$, is mixed for NO_2, and helps PM_{10} only temporally with RF-STK. Limitation: the abstract gives no $PM_{2.5}$ R2 or RMSE, gains are described qualitatively, and centrality is computed on the regulatory station layout itself, so transfer to a denser or differently sited network is untested. Relevance: moderate - graph features of node placement are directly computable for low-cost networks.

doi: 10.3390/s26185844

N-containing OA 2026 Atmos Environ

Unveiling the Precursor-Specific Aqueous Formation Mechanisms of Nitrogen-Containing Organic Aerosols

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 7 Oct with no abstract. By title, a mechanistic account of how particular precursors form nitrogen-containing organic aerosol in the aqueous phase. Aqueous-phase secondary formation is humidity-dependent, which makes it the physical process underneath the humidity correction every optical low-cost sensor applies - the correction treats growth as hygroscopic swelling when some of it is mass that was not there before. Relevance: moderate-high; on the retry list.

doi: 10.1016/j.atmosenv.2026.122408

Nassir et al. 2026 Environ Pollut

A multi-sensor satellite assessment of air quality during the 2026 Middle East Conflict across nine regional countries

A dual-baseline design (14-month annual and seasonally matched prior year) applied to the 39-day 2026 conflict period across nine countries (~217 million people). Domain-mean anomalies: NO_2 -15.1%, SO_2 -45.6%, CO -8.6%, AOD -19.6%, CH4 +0.37% (noise), ordered by atmospheric lifetime - the signature of an emission change; cold spots sit over Iranian, Kuwaiti and Qatari hydrocarbon fields. The NO_2 burden estimate (-7,849; 95% CI -35,641 to +13,551) crosses zero, and the $PM_{2.5}$ burden is called indeterminate because dust and anthropogenic aerosol cannot be separated at CAMS resolution. Limitation: 39 days, no surface PM validation; a wind reversal is documented. Relevance: moderate - an honest statement that column AOD cannot apportion PM in dust-dominated regions without ground speciation.

doi: 10.1016/j.envpol.2026.129258

Netto metal distribution $PM_{2.5}$ smelting Rio de Janeiro solubility risk 2026 Atmos Pollut Res

Metal distribution in atmospheric particulate matter ($PM_{2.5}$) from urban areas with smelting activities in the Rio de Janeiro state, Brazil: Solubility, correlation and risk assessment

METADATA ONLY. Title and authors only (Netto, da Silva, Diniz et al.); no abstract retrievable at harvest. The title describes $PM_{2.5}$ metal concentrations, their soluble fractions and risk assessment in urban areas with smelting in Rio de Janeiro state. Concentrations, n and risk estimates cannot be stated. Limitation: unverifiable beyond the title; risk assessments on total metal mass overstate dose unless the bioaccessible (soluble) fraction is used, which the title suggests is addressed. Relevance: low - filter chemistry.

doi: 10.1016/j.apr.2026.103250

Nolles IGP vs NCP $PM_{2.5}$ pollution forming efficiency CAMS 2026 Environ Sci Pollut Res Int

Indo-Gangetic Plain experiences higher aerosol pollution than North China Plain despite lower emissions

CAMS EAC4 reanalysis, CAMS-GLOB-ANT emissions and ERA5 for 2019. The NCP emitted 65% more primary PM and precursors (28,860 vs 17,464 kt) yet the IGP had higher annual $PM_{2.5}$ (123.6 vs 113.2 $\mu g\,m^{-3}$); NCP logged more bad-air days (251 vs 238) but the IGP longer episodes, up to 170 consecutive days around Delhi. A 'pollution forming efficiency' was nearly double in the IGP in winter, attributed to winds <6 km/h and PBL <300 m. Limitation: both concentration and emissions are model products for one year; EAC4 assimilates AOD and is biased in the IGP, and emission inventories under-count agricultural and residential burning there, which inflates the efficiency ratio. Relevance: moderate - quantifies what reanalysis cannot replace ground networks for.

doi: 10.1007/s11356-026-38305-7

North Macedonia biomass PMF 2026 Environ Pollut

Biomass burning drives urban $PM_{2.5}$ across the Western Balkans: evidence from a six-city source apportionment study in North Macedonia

1,195 filters at six urban-background sites over two year-long campaigns. Campaign means 29.1-62.2 $\mu g\,m^{-3}$ (1.2-2.5x the EU annual limit); residential wood combustion and open/waste burning resolved as distinct factors everywhere, together 21-35% of $PM_{2.5}$, with heating-season RWC up to 33x non-heating. Biomass exceeded 50% of $PM_{2.5}$ on 83% of Skopje episode days; an undocumented winter signal in Strumica fits greenhouse boilers. Limitation: PMF factor naming relies on tracer interpretation, and RWC/OBB separation without levoglucosan isotopes or 14C is assumption-laden. Relevance: high - a wood-smoke regime with bimodal evening peaks to 93 $\mu g\,m^{-3}$, where optical sensor response to fresh smoke is the calibration problem.

doi: 10.1016/j.envpol.2026.129339

OA molecular composition predicts ROS yield MCR-VK 2026 Atmos Chem Phys

Linking molecular composition of organic aerosols to reactive oxygen species formation using an MCR-VK framework

Ambient fine PM from four Chinese megacities, a German semi-urban site and a boreal forest, plus chamber SOA, analysed by ultrahigh-resolution MS and H2O2/radical assays. Volume-normalised ROS rose with PM mass and dominates exposure in polluted cities, but mass-normalised ROS varied independently of loading and tracked the ratio of oxidised to unsaturated organic fractions (RF_OOC/RF_UOC); biogenic aerosol carried the highest per-mass yield. Limitation: correlational across few sites; cell-free ROS is a surrogate, and transition metals - a major ROS driver - are outside the framework. Relevance: moderate - mass-based LCS cannot see per-mass toxicity variation of this size; a case for pairing networks with periodic speciation.

doi: 10.5194/acp-26-14655-2026

Oliveira et al. 2026 Atmos Pollut Res

Nanomineralogical and geochemical signatures of coal-derived particles in playground dust

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 1 Oct with no abstract; PubMed efetch, Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes nanomineralogy of coal-derived particles in playground dust. Relevant to child resuspension exposure, which ambient nodes do not measure. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.apr.2026.103209

$PM_{2.5}$ and GPP 2026 Environ Pollut

Climate-dependent nonlinear relationships between $PM_{2.5}$ and ecosystem productivity across China's urban agglomerations

Satellite observations and interpretable machine learning relate $PM_{2.5}$ to gross primary productivity across climate and pollution gradients. The relationship is robust everywhere but changes sign by region: moderate aerosol loading goes with enhanced productivity in arid environments, while elevated $PM_{2.5}$ generally suppresses it elsewhere, jointly shaped by climate and pollution regime. Limitation: the authors flag the central problem themselves - $PM_{2.5}$, radiation, humidity and temperature co-vary, so a diffuse-radiation fertilisation effect cannot be separated from co-varying meteorology by feature attribution on observational data; both $PM_{2.5}$ and GPP are satellite products with shared retrieval dependencies. Relevance: low for health, moderate as a reminder that aerosol has non-health consequences that enter policy arithmetic.

doi: 10.1016/j.envpol.2026.129305

Pani Southeast Asian biomass burning smoke aerosol light extinction apportionment high-altitude western North Pacific 2026 Atmos Res

Impact of peninsular Southeast Asian biomass-burning smoke on chemical apportionment of aerosol light extinction at a high-altitude remote environment in the western North Pacific

METADATA ONLY. Title and authors only (Pani, Lee, Lin); no abstract retrievable at harvest. The title describes apportioning aerosol light extinction by chemical component at a high-altitude remote site in the western North Pacific under transported Southeast Asian biomass-burning smoke. Concentrations and apportionment fractions cannot be stated. Limitation: unverifiable beyond the title; IMPROVE-type mass extinction efficiencies are fixed assumptions that aged smoke violates. Relevance: medium - mass-extinction relations are what optical PM sensors implicitly assume.

doi: 10.1016/j.atmosres.2026.109421

Park $PM_{2.5}$ composition sources COVID-19 social distancing Seoul 2026 Toxics

Restructuring of $PM_{2.5}$ Composition and Source Contributions Across COVID-19 Social Distancing Periods: A Case Study at an Urban Monitoring Site in Seoul, South Korea

Feb 2018-Mar 2024 at one Seoul site, split into pre-, during- and post-distancing periods; PMF with GAM adjustment for meteorology and season. $PM_{2.5}$ stayed lower during and after distancing. Secondary inorganic aerosol remained dominant with no period effect after adjustment; metal industry fell then partly rebounded; construction dust stayed elevated and paved-road dust rose post-distancing. Limitation: one site, and 'post-distancing' is also the period of Korea's seasonal-management PM policy and Chinese emission reductions, so the periods confound policy and transboundary change with social activity. No effect sizes in the abstract. Relevance: medium - non-exhaust road dust is a coarse-mode source optical LCS undercount.

doi: 10.3390/toxics14100886

Park Paris UFP number emissions mass-number conserving 2026 J Hazard Mater

Reconciling mass and number emissions through particle size: Implications for urban ultrafine particle modelling

Summer 2021 simulations from regional background to street scale, with a mass- and number-conserving algorithm converting sectoral PM mass emissions to sectional PN. Scenarios: COPERT vs on-board PN factors for exhaust, non-exhaust traffic, and non-combustion industrial processes including construction. The reference run, using mass emissions plus representative sector diameters, reproduced total PN at background and traffic stations; non-exhaust contributed about a third of traffic PN; the size distribution near 10 nm remained poorly reproduced. Limitation: one summer, and agreement on total PN can hide compensating errors across modes; sub-10 nm emission factors are missing. Relevance: moderate - PN, not $PM_{2.5}$ mass, is the UFP exposure metric, and low-cost condensation counters are the only route to network density.

doi: 10.1016/j.jhazmat.2026.143879

Perez-Vizcaino PM_{10} sources Riotinto mining multi-time-resolution PMF 2026 J Hazard Mater

Characterization of PM(10) sources in the Riotinto copper mining district (SW Spain) using multi-time-resolution receptor modelling.

Year-long (July 2024-July 2025) campaign combining an hourly XRF metals monitor and an Aethalometer with daily high-volume quartz-filter chemistry, merged in multi-time-resolution PMF. Seven sources: waste rock 4.9, regional + traffic 3.6, open pit 1.6, combustion 1.2, aged marine 1.2, dust 1.1 and copper concentrate 0.7 $\mu g\,m^{-3}$. Daily species identified traffic tracers and SOA that the hourly subset alone missed. Arsenic was the main geochemical anomaly. Limitation: one receptor site; multi-resolution PMF weights the daily data heavily and factor uncertainty (bootstrap/DISP) is not reported in the abstract. Relevance: medium - hourly XRF + BC is the reference tier that low-cost networks around mines would need to be anchored to for metal-rich dust.

doi: 10.1016/j.jhazmat.2026.143869

Petrovic Belgrade PMF 2026 Atmos Chem Phys

Chemical characterization and source apportionment of PM_{10} in Belgrade, Serbia: influence of local and regional anthropogenic and natural sources

USEPA PMF 5.0 on PM_{10} at an urban-background site using 34 inputs including organic tracers - broader than earlier Belgrade apportionments that used elements or ions alone. Biomass burning leads at 21% of PM_{10}, concentrated in the heating season, then ammonium sulphate 18%, mineral dust 17% and mixed traffic/industry 15%, with primary biological and biogenic secondary factors resolved separately. Limitation: one site, and a mixed traffic-industry factor means PMF could not separate the two sources most amenable to local control; PM_{10} mass weighting under-represents the fine combustion fraction that drives health effects. Relevance: moderate - a 21% residential biomass share in a Balkan capital is the case where low-cost optical networks track the winter smoke signal well but need composition-aware RH correction.

doi: 10.5194/acp-26-14165-2026

Pourkiaei Santiago CMAQ EDGAR vs INEMA inventories 2026 Research Square (preprint)

High-Resolution Air Quality Modeling over Santiago, Chile: A Comparative Assessment of EDGAR and INEMA Anthropogenic Emission Inventories

Global EDGAR v8.1 (2022) vs regional INEMA v2 (2020) in CMAQ with three one-way nests to 1 km. Aggregate $PM_{2.5}$ emissions were similar but sectoral splits differed sharply; INEMA reproduced PM better, EDGAR daytime O_3. After scaling emissions to mean observations, source apportionment shifted between industrial combustion (EDGAR) and transport (INEMA), with building energy use important in both. DDM showed VOC-limited O_3 and sector-specific non-linear second-order $PM_{2.5}$ sensitivity to NOx. Limitation: calibrating emissions to the observed mean forces agreement and then attributes sources within that constrained state, so apportionment differences reflect inventory structure, not validated truth; performance statistics are not in the abstract. Relevance: moderate - 1 km CTM fields are the prior that dense LCS networks are fused with in Andean basins.

doi: 10.21203/rs.3.rs-10259950/v1

Pu Lima hybrid transfer-learning $PM_{2.5}$ 2026 ACS EST Air

A Hybrid Transfer Learning Approach for Ambient $PM_{2.5}$ Exposure Modeling in LMIC Contexts: A Case Study from Lima, Peru

Lima 2010-2023 daily $PM_{2.5}$ surface from an instance-weighted random forest that fuses reference and low-cost monitor data, satellite AOD and WRF-Chem. Sampling bias toward monitored districts is handled by geo-environmental similarity transfer learning that borrows weighted training instances from the US. Cross-validated R2 0.76; predicted mean $PM_{2.5}$ in mountainous districts fell from 34.19 to 29.23 $\mu g\,m^{-3}$ versus the unweighted model. Limitation: CV R2 is computed on the same biased monitor locations the method claims to correct, so the mountain-region gain is unverified by held-out sites there; how low-cost data were calibrated is not stated. Relevance: high - a template for using sparse LCS networks plus transfer learning in data-poor cities.

doi: 10.1021/acsestair.6c00125

Qu $PM_{2.5}$-bound heavy metals source apportionment Laibin winter 2026 Toxics

Source Apportionment and Health Risk Assessment of $PM_{2.5}$-Bound Heavy Metals in Wintertime Laibin, Guangxi, China

42-day winter 2025 campaign: mean $PM_{2.5}$ 47.6 $\mu g\,m^{-3}$, higher at night with lower wind and shallower boundary layer; nine metals summed 363.0 ng/m3, dominated by Zn, Cr and Ni, with severe Cd and significant Ni enrichment. PMF: non-ferrous smelting plus coal combustion 43%, industrial combustion 34%, traffic 23%; traffic share rose at night. Hazard index <1 for all groups; carcinogenic risk slightly above 1e-4 under winter scenarios, Cr dominant. Limitation: PMF on nine species and ~84 day/night samples is under-determined (the authors say 'preliminarily'), and Cr risk depends on an assumed Cr(VI) fraction. Relevance: low-medium - nocturnal boundary-layer peaks are visible to any LCS node.

doi: 10.3390/toxics14100923

Rocha Volta Redonda 2026 Environ Monit Assess

Long-term trends and seasonal variability of particulate matter (PM_{10} and $PM_{2.5}$) in Volta Redonda (RJ), Brazil: a 15-year temporal analysis

METADATA ONLY. Metadata-only record: Springer deposited title, authors and DOI on 9 Oct with no abstract, and the retry chain (PubMed, Europe PMC, Semantic Scholar) recovered none. By title, a 15-year PM_{10} and $PM_{2.5}$ trend and seasonality analysis for a steel-industry city in Rio de Janeiro state. Long regulatory records from Latin American industrial cities are scarce, and instrument changes over 15 years are the usual confounder of any trend claim. Relevance: moderate; on the abstract-retry list.

doi: 10.1007/s10661-026-15988-6

Rosu multi-instrument Saharan dust event South-Eastern Europe 2026 Atmos Res

A multi-instrument analysis of a Saharan dust event in South-Eastern Europe

Metadata only: no abstract at Crossref, Europe PMC or Semantic Scholar on the issue date (Elsevier day-0 deposit). The title reports a multi-instrument analysis of a Saharan dust intrusion over South-Eastern Europe; instruments, sites, PM metrics and results are not available here and are not inferred. Limitation (structural): single-event case studies cannot establish event frequency or generalise dust contributions to PM_{10} exceedances. Relevance: medium - dust events are where optical LCS under-read coarse mass most strongly, so multi-instrument events are useful calibration benchmarks. Retry the abstract before the weekly; if $PM_{10}/PM_{2.5}$ ratios are reported, they matter for the coarse-mode LCS correction.

doi: 10.1016/j.atmosres.2026.109409

Salem Riyadh $PM_{2.5}$ vegetation 2026 Environ Monit Assess

Spatiotemporal variation of $PM_{2.5}$ and its relationship with vegetation cover and meteorological factors during spring (2016-2022) in Riyadh, Saudi Arabia

16 stations, spring 2016-2022: seasonal mean $PM_{2.5}$ ranged from 94.6 (2018) to 55.0 $\mu g\,m^{-3}$ (2020), highest in southern and central Riyadh. Meteorological correlations were non-significant; the vegetation-$PM_{2.5}$ inverse association reached significance in one year only (rho=-0.539, p=0.031, n=16). Limitation: n=16 Spearman correlations per year cannot separate vegetation from land use, and dust events dominate spring variance; the one significant year out of seven is consistent with chance. Relevance: low-moderate - documents a dust-dominated regime where optical sensors' coarse-mode bias would govern any network.

doi: 10.1007/s10661-026-16057-8

Sao Paulo 2024 fire extremes 2026 J Environ Manage

Compound fire-meteorology extremes drove megacity air pollution during the 2024 South American fire season

Three multi-day 2024 episodes with daily $PM_{2.5}$ of 80-100 $\mu g\,m^{-3}$ in metropolitan Sao Paulo. Fire-off simulations missed the $PM_{2.5}$ and CO peaks; fire-attributable increments reached 17-21 $\mu g\,m^{-3}$ state-wide and a cumulative metropolitan burden of ~547 $\mu g\,m^{-3}$-day. Episodes differed in transport configuration, and the worst coincided with heat, dryness and weak ventilation. Limitation: all fire inventories underestimated the most extreme episode, so the attribution is a lower bound with unquantified error; a global model is coarse for a megacity receptor. Relevance: moderate - the inventory shortfall at the extreme is exactly where a dense surface network constrains attribution.

doi: 10.1016/j.jenvman.2026.131131

Savolainen road traffic emission factors impact factors Helsinki street canyon 2026 Atmos Chem Phys

From direct emission factors to inverse and indirect impact factors of road traffic: influence of emerging and unregulated pollutants

Wintertime supersite measurements in a central Helsinki street canyon. Fleet emission factors were derived by normalising pollutant increments to CO_2 as a fuel-consumption tracer, across particle number and size, BC, PM_{10}, $PM_{2.5}$, organic and inorganic PM components, NOx and VOCs including PAHs. PM and NOx emission factors were well above the post-2014 Euro limits, so older and heavy-duty vehicles dominate. Traffic showed negative 'impact factors' for O_3 and sub-2 nm small ions and contributed terpenoid VOCs. Limitation: one canyon in one season; the CO_2-ratio method assumes co-emission and common dilution, which fails for secondary and non-exhaust species. Relevance: high - a CO_2-normalised increment is a method LCS nodes with NDIR CO_2 can apply directly.

doi: 10.5194/acp-26-14823-2026

Sheidaei et al. 2026 Environ Sci Technol

A Multi-Stage Deep Learning Framework for Daily $PM_{2.5}$ Estimation at 100 m Resolution Across the Contiguous United States, 2000-2024

Three-stage pipeline: a reanalysis-informed U-Net fills missing satellite AOD, a bidirectional LSTM refines its temporal structure and terrain downscales it to 100 m; a multistream network then predicts daily $PM_{2.5}$ over >766 million cells for 2000-2024. Site-held-out validation: R2 = 0.82, RMSE = 2.85 $\mu g\,m^{-3}$, MAE = 1.84 $\mu g\,m^{-3}$; spatial R2 0.94, temporal R2 0.78. Limitation: the 100 m detail comes from downscaled AOD and land-use covariates, not from 100 m observations - validation sites are regulatory monitors, which cannot test sub-kilometre gradients near roads or smoke plumes; the temporal R2 (0.78) is the binding figure for short-term epidemiology, and performance during wildfire days is not reported. Relevance: high - a 25-year 100 m baseline is the natural prior that dense low-cost networks should be fused against, and the place their residuals are informative.

doi: 10.1021/acs.est.6c05375

Shi et al. 2026 Environ Pollut

Compositional transition and coupled formation pathways of $PM_{2.5}$ pollution in a typical industrial city: An empirical study based on Handan, China

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 5 Oct with no abstract. By title, an empirical account of how $PM_{2.5}$ composition in a heavily industrial Hebei city has shifted and how its formation pathways couple. A compositional transition from primary to secondary is the slow change that silently alters the relationship between an optical sensor's scattering signal and gravimetric mass over years of a network's life. Relevance: moderate; on the retry list.

doi: 10.1016/j.envpol.2026.129296

Shoukat et al. 2026 Atmos Pollut Res

Suspended Atmospheric Microplastics in the Outskirts of Seoul City: Seasonal Variability, Correlation with Meteorology

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 1 Oct with no abstract; PubMed efetch, Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes seasonal suspended (airborne) microplastic measurements near Seoul and their meteorological correlates. Complements the Chengdu deposition record in this issue with an airborne concentration metric. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.apr.2026.103222

Somsunun $PM_{2.5}$-bound metals health risk haze season Chiang Mai 2026 Toxics

Health Risk Assessment of Ambient $PM_{2.5}$-Bound Metals During Haze Seasons in Urban and Rural Chiang Mai, Thailand

101 24 h samples, 15 Jan-22 Apr 2020 haze season, at urban (CMU), agricultural rural (Mae Chaem) and roadside rural (Chom Thong) sites. Median $PM_{2.5}$ 50.3, 42.1 and 64.9 $\mu g\,m^{-3}$. Despite lower mass, Mae Chaem had HI 2.10 and TCR 2.37e-4 vs CMU 1.34 and 1.18e-4; HI >1 and TCR >1e-4 at all sites, highest at Chom Thong (2.69; 3.63e-4), driven by Cr. Limitation: risk is dominated by total Cr with an assumed Cr(VI) share, and spring 2020 overlaps COVID-19 restrictions. Relevance: medium - a direct case that equal $PM_{2.5}$ mass from a sensor does not mean equal hazard across land uses.

doi: 10.3390/toxics14100869

Szalata waste fire toxic emissions Gaussian plume environmental risk 2026 Toxics

Environmental Risk Assessment of Toxic Emissions into the Air Resulting from Waste Fires

Receptor-oriented screening: Gaussian plume (Pasquill classes) for a 14,000 kg municipal-waste fire burning 3.5 h, compared against Polish, US EPA, OSHA and NDS limits. Under the worst case analysed (class F, 1 m/s southerly) the residential receptor 700 m away stayed below reference values. Limitation: emission factors for mixed-waste fires are highly uncertain, and a steady-state Gaussian plume ignores fire-driven plume rise, lofting and near-field downwash, and comparing to occupational limits is not a public-health criterion. No measurement validates the field. Relevance: medium - rapid-deployment LCS around waste fires are the only practical check on such screening models.

doi: 10.3390/toxics14100892

Szczepanik Warsaw wintertime air quality improvement 2016-2025 2026 Aerosol Air Qual Res

Improvement in wintertime air quality in urban and suburban areas of Warsaw within 2016-2025

METADATA ONLY. Title and authors only (Szczepanik, Deptula, Klapinski, Rakowska). No abstract was retrievable at harvest. The title reports improved wintertime air quality across urban and suburban Warsaw over 2016-2025, the period of Poland's anti-smog resolutions and boiler-replacement programmes. Pollutants, data sources and magnitudes are not available; nothing is inferred. Relevance: medium - suburban solid-fuel heating is where Polish LCS networks (Airly and others) are densest; the paper may use them.

doi: 10.1007/s44408-026-00173-8

Temirzhanova Pavlodar snow aerosol 2026 Atmosphere

Geochemical Characteristics of the Pollution of Snow Cover with Aerosols in an Industrial Center (Pavlodar, North-Eastern Kazakhstan)

55 snow samples across industrial, buffer, non-industrial and background zones. Pb exceeded background 2-100-fold, with Pb, Sr and V peaking in industrial and buffer zones; PC1 loaded on rare-earth elements. Limitation: snowpack integrates wet and dry deposition over an unknown accumulation period, so it is a deposition map, not an air concentration map, and cannot be converted to inhalation exposure. Relevance: low - useful as a cheap spatial prior for siting composition samplers where no network exists.

doi: 10.3390/atmos17100995

Torii Kalimantan peat fire super-linear scaling 2026 Research Square (preprint)

Peat Burning Scales Super-linearly with Fire-Season Magnitude in the West Kalimantan-Sarawak Border Region, 2012-2025

Fourteen fire seasons of VIIRS detections against a peat map: peat-fire counts scale with total detections as a power law, exponent 1.96 (95% CI 1.57-2.36; R2 0.89), so doubling fire activity quadruples peat burning; robust to alternative predictors (2.36) and a Suomi-NPP-only subset (1.98). Peat share rises from <4% in La Nina years to 30-51% in El Nino years (ONI r 0.78), though ONI is collinear with season size. 24.3% of cells burning >=7 years gave 57.7% of cumulative FRP; scaling survives back-trajectory transport (1.59). Limitation: single-author preprint, detections not emissions, and no PM observations close the chain. Relevance: moderate - receptor-site sensors in Sarawak could test whether haze PM really tracks the squared term.

doi: 10.21203/rs.3.rs-11311408/v1

Tripathi satellite-reanalysis sulfate aerosol India ground measurements 2026 Environ Pollut

Improving Satellite-Reanalysis Sulfate Aerosol Estimates over India Using Ground Measurements

METADATA ONLY. Title and authors only (Tripathi, Sahu, Sharma, Gupta); no abstract indexed at harvest. The title indicates ground measurements used to correct satellite-reanalysis (likely MERRA-2/CAMS-type) sulfate aerosol over India. Data sources, method and skill cannot be stated. Limitation: unverifiable beyond the title; Indian speciation networks are sparse and urban-biased, so any correction trained on them risks transferring urban bias to rural cells. Relevance: moderate - component-resolved surfaces are the complement LCS mass networks lack; pairs with the sulfate-dominant AF signal in today's Medicare cohort.

doi: 10.1016/j.envpol.2026.129391

Tsai mineral dust biomass burning aerosol convergence East Asia western North Pacific 2026 Atmos Environ

Convergence of mineral dust and biomass-burning aerosols over East Asia and the western North Pacific

METADATA ONLY. Metadata only: no abstract was available at harvest from any route tried. The title describes convergence of mineral dust and biomass-burning aerosol over East Asia and the western North Pacific, presumably from observation and transport analysis; the instruments, period and metrics cannot be stated. Limitation: not appraisable. Relevance: medium in principle - mixed dust-smoke plumes combine coarse non-spherical and fine absorbing particles, a regime where optical sensor size and refractive-index assumptions fail together.

doi: 10.1016/j.atmosenv.2026.122444

UK Air Quality Cell PM_{10} plume back trajectory tyre fire Mexborough vulnerability 2026 Eur J Public Health (conference abstract)

Back trajectory modelled PM_{10} plumes: population vulnerability profile from an uncontrolled fire

The UK Air Quality Cell monitored PM_{10} during an uncontrolled tyre fire; 22 hourly ADMS-STAR back-trajectory plumes were spatially averaged and overlaid on Output Areas. 24,952 residents fell in 'Unhealthy for Sensitive Groups' or worse; 7,856 (31.5%) in 'Hazardous' or beyond. Exposed areas had above-median deprivation, prior ill health and lower education. The modelling suggested field monitoring under-captured peaks. Limitation: AQI bands from a 24 h index were applied to hourly plume concentrations, and exposure is residential, ignoring sheltering and evacuation; no health outcomes. Conference abstract. Relevance: high - rapid-deploy sensor arrays around industrial fires would close the peak-capture gap the authors identify.

doi: 10.1093/eurpub/ckag169.809

US school wildfire smoke trends 2026 Environ Res

Spatio-Temporal Trends in Wildfire Smoke Exposures at Schools and Childcare Centers in the United States and Implications for Child and Adolescent Health

METADATA ONLY. No abstract at harvest. Title indicates national spatiotemporal trends in smoke exposure at schools and childcare centres. Recorded for coverage; no estimate inferred. Relevance: high if the exposure surface is sensor-informed - school sensor deployments are the most common LCS siting in the US smoke literature.

doi: 10.1016/j.envres.2026.125883

Upaul Khulna $PM_{2.5}$ meteorology HYSPLIT random forest STL 2026 Research Square (preprint)

Meteorological and HYSPLIT-Derived Transport Influences on $PM_{2.5}$ Exposure in a Khulna City, Bangladesh

One year (2024) of hourly $PM_{2.5}$ at a single Khulna station, analysed with MLR, Random Forest, STL and K-means on HYSPLIT trajectories. Winter mean ~121 $\mu g\,m^{-3}$ vs ~15 in the monsoon. Temperature and pressure were top RF predictors (OOB R2 0.64); temperature and RH were negative in MLR. Winter air arrived on short north-westerly continental paths, monsoon air on long southerly maritime paths; STL residuals flagged pre-monsoon spikes (~138 $\mu g\,m^{-3}$). Limitation: one station, one year, and reanalysis meteorology at ~50 km; season is collinear with every predictor, and the 'variance explained' shares are not additive. Preprint. Relevance: low-moderate - a coastal South Asian city with almost no monitoring, where LCS would add the spatial dimension.

doi: 10.21203/rs.3.rs-11078552/v1

Vikas wavelet background $PM_{2.5}$ India MERRA-2 2026 Environ Monit Assess

Estimating background $PM_{2.5}$ concentrations using a wavelet-based convolution framework

Wavelet coherence of low-frequency components across neighbouring MERRA-2 cells yields a spatiotemporal $PM_{2.5}$ background map for India, capturing the IGP winter build-up and pre-monsoon dust in the northwest. Against a GMM-HMM background at four cities, MAD 2.2 (Kolkata) to 10.8 (Delhi) $\mu g\,m^{-3}$. Against 5th-percentile PCHIP backgrounds across the Delhi network, MAE 21.95 and RMSE 29.45 $\mu g\,m^{-3}$, improving to 14.03/17.65 at the Delhi background site. Limitation: validation is against other statistical background definitions, not tracer or model source attribution, and MERRA-2 $PM_{2.5}$ is a reconstructed aerosol product with known bias over South Asia. Relevance: high - background/increment separation is a standard post-processing step for dense LCS networks.

doi: 10.1007/s10661-026-16082-7

Vukic et al. 2026 Environ Monit Assess

Integrated assessment of road and maritime traffic impacts on urban air quality and noise in a port-city environment

Ten sites in Split, Apr-Sep 2024: 154 port and 66 road short-duration observations of $PM_{2.5}$, PM_{10}, CO, traffic counts and (road only) noise. Port means exceeded road means ($PM_{2.5}$ 14.77 vs 11.64; PM_{10} 18.25 vs 14.70 $\mu g\,m^{-3}$) but not significantly ($p = 0.22, 0.23$). Vessel count was weakly associated with PM_{10} ($R2 = 0.082$) and stayed positively associated with port $PM_{2.5}/PM_{10}$ in cluster-robust models; wind speed lowered PM. Limitation: spot measurements with uncalibrated low-cost sensors, no reference co-location, and vessel count ignores engine type, berth time and fuel - the R2 says ship numbers are a poor proxy for ship emissions. Relevance: moderate - the authors frame LCS correctly as screening-level; port sources need continuous nodes with plume-resolving time resolution, not spot visits.

doi: 10.1007/s10661-026-15986-8

Wang China $PM_{2.5}$ decade 2026 Atmos Pollut Res

From rapid decline to slower improvement: Decadal nonlinear evolution and regional predictive associations of $PM_{2.5}$ in China (2015-2024)

METADATA ONLY. No abstract at harvest. Title indicates a nonlinear trend analysis of Chinese $PM_{2.5}$ showing deceleration of the post-2015 decline and regional predictors. Recorded for coverage. Relevance: low-moderate - slower improvement means smaller year-on-year signals, which raises the bar on sensor drift control for trend detection.

doi: 10.1016/j.apr.2026.103234

Wang et al. 2026a J Environ Sci (China)

Organosulfate-driven amplification of secondary organic aerosol formation from isoprene oxidation intermediates: Coupled effects of SO_2 and relative humidity

Smog-chamber methacrolein/OH photooxidation at high (61-65%), medium (20-22%) and low (9-12%) RH with varied SO_2. Particle mass, peak number, geometric mean diameter and SOA yield all rise with initial SO_2 at every RH; with SO_2 present, SOA mass rises as RH falls, attributed to lower wall loss of hydrophilic species and less dilution of aerosol acidity. ATR-FTIR shows organosulfates and inorganic sulfate in the particle phase. Limitation: chamber concentrations and wall-loss corrections dominate yield estimates, and no yields are reported in the abstract; transfer to ambient isoprene-SO_2 mixing is assumed. Relevance: low-moderate - acid-catalysed SOA is part of why the optical-to-mass ratio of low-cost sensors drifts with season and RH.

doi: 10.1016/j.jes.2026.02.001

Wang et al. 2026d Atmos Environ

Physics-inspired inductive spatiotemporal kriging for $PM_{2.5}$ with satellite gradient constraints

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 6 Oct with no abstract. By title, this is the central methodological problem of a sparse network - interpolating to unmonitored locations - attacked inductively (so that new sensors can be added without retraining) and constrained by satellite gradients rather than by a stationarity assumption. The Everglades record in the 2 October issue measured exactly how badly ordinary kriging and IDW fail day to day; this is the proposed repair. Relevance: high, the highest of any metadata-only record in this backfill; flagged for a full-text read as soon as the abstract appears.

doi: 10.1016/j.atmosenv.2026.122398

Wang urban morphology street-scale $PM_{2.5}$ PM_{10} change points 2026 Atmos Pollut Res

Non-linear associations, model-supported change points, and seasonal variations of urban morphology with street-scale $PM_{2.5}$ and PM_{10} concentrations

METADATA ONLY. Metadata-only record: Crossref deposit with no abstract on the day of entry, and no abstract recoverable from Europe PMC or Semantic Scholar. The title reports non-linear associations and model-supported change points between urban morphology indicators and street-scale $PM_{2.5}$ and PM_{10}, with seasonal variation. Design, city, sample size, morphology metrics and effect sizes cannot be stated from the record and are not inferred here. Limitation: unassessable until the abstract or full text is available; threshold-type 'change points' in morphology-PM regressions are prone to overfitting when sites are few. Relevance: medium - street-scale morphology thresholds bear directly on how a dense sensor network should be sited across canyon geometries.

doi: 10.1016/j.apr.2026.103242

Wei $PM_{2.5}$ ozone nonlinear coupling Puyang oilfield city GAM 2015-2025 2026 Atmosphere

Long-Term Trends and Nonlinear $PM_{2.5}$-O_3 Coupling in a Typical Oilfield City of the "2 + 26" Region

Daily $PM_{2.5}$ and MDA8 O_3 in Puyang 2015-2025 with nested GAMs adding meteorology, NO_2, humidity and season interactions. Raw correlation reversed by concentration band: r 0.271 below 35 $\mu g\,m^{-3}$, -0.208 at 35-75 and -0.313 above 75. After controlling for meteorology and NO_2, the conditional partial effect of $PM_{2.5}$ on O_3 increased over most of the range, so the negative raw correlation is largely meteorological. The low-band positive coupling strengthened over time (0.190 to 0.306). Limitation: one city, observational correlations; GAM partial effects do not identify chemical mechanisms (aerosol radiative or heterogeneous HO2 uptake). Relevance: low.

doi: 10.3390/atmos17111057

Wen and Chang 2026 Environ Pollut

Estimating marine $PM_{2.5}$ concentrations and exploring the drivers over the eastern China seas using multi-source data and machine learning

BPNN, random forest and XGBoost compared under sample-, spatial- and temporal-held-out validation for $PM_{2.5}$ over the Bohai, Yellow and East China seas; XGBoost best. Estimated fields fall west-to-east and north-to-south with maxima in Bohai Bay and off Shandong, a declining trend with winter maxima, a visible 2020 drop, and SHAP attributing spring-autumn variance to AOD and winter variance to meteorology. Limitation: there are almost no marine ground monitors, so training and validation both lean on coastal land sites - held-out skill over open water is unverified and the 'marine' field is an extrapolation; no shipping-emission predictor. Relevance: moderate - the obvious use case for buoy- or vessel-mounted low-cost nodes is precisely this validation gap.

doi: 10.1016/j.envpol.2026.129276

Wu NCP brown carbon 2026 Atmos Chem Phys

Light absorption properties and composition of water-soluble brown carbon in North China Plain: implication for an enhancing role of nitrogenous organic compounds

Simultaneous winter sampling at Beijing, Tianjin, Luancheng (rural), Handan and Jinan. Abs365 at the rural site was 1.1-3.5 times the urban values, but mass absorption efficiency peaked in Jinan (1.40 m2/g). Urban Abs365 fell ~45% from 2018 to 2023. BrC absorptivity rose from clean to haze with the N:C ratio, pointing to nitrogenous organics as chromophores. Limitation: water-soluble extracts miss the methanol-soluble, more absorbing BrC, and one winter cannot separate trend from meteorology. Relevance: moderate - the same nitrogenous organics the same-day Sci Adv record ranks most toxic are the ones that absorb at 365 nm, which a multi-wavelength absorption node can see.

doi: 10.5194/acp-26-14229-2026

Xiao NCP Sichuan haze reduced-nitrogen NOC oligomerisation $PM_{2.5}$ 2026 Environ Sci Technol

Aqueous Carbonyl-Amino Aggregation Drives Atmospheric Nitrogen-Containing Oligomer Formation in China Haze

Winter-haze $PM_{2.5}$ from the North China Plain compared with the Sichuan Basin. Reduced-nitrogen NOCs were 40% of ESI+ organics. High-molecular-weight (m/z>250) Re-NOCs rose in humid versus dry haze and correlated with NH4+ and the aqueous-SOA fraction of water-soluble organics, alongside longer aliphatic chains and higher unsaturation - read as carbonyl-amino oligomerisation and cyclisation. A marker oligomer (C8H13N3O2) tracked WSON; stronger acidity and higher temperature favoured oligomerisation over cyclisation, enriching it in the Sichuan Basin. Limitation: ESI+ is semi-quantitative and ionisation-biased, number of samples is not stated, and mechanism is inferred from correlation. Relevance: low - brown-carbon NOCs add absorbing mass that optical scattering sensors misweight under humid haze.

doi: 10.1021/acs.est.6c09940

Yadav ENSO India 2026 Atmos Pollut Res

Air Quality in Flux: The Baffling ENSO Transition in India

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 9 Oct with no abstract. By title, an analysis of how an ENSO phase transition altered Indian air quality. Interannual meteorological modes are the main reason a national PM trend over a few years is not an emissions trend - the same-day Navi Mumbai record makes the station-scale version of this point, and the 10 Oct Indo-Gangetic trend record the regional one. Relevance: moderate; on the abstract-retry list.

doi: 10.1016/j.apr.2026.103231

Yingkar $PM_{2.5}$-bound PAHs Central Asia urban trends sources risk 2026 Atmos Environ

Environment-to-Health Linkages of $PM_{2.5}$-Bound Polycyclic Aromatic Hydrocarbons in Urban Cities of Central Asia: Trends, Sources, and Risk Assessment

METADATA ONLY. Title and authors only (Yingkar, Omarova, Sovetova, Alibekov et al.). No abstract was retrievable from Crossref or Semantic Scholar at harvest. The title describes trends, source apportionment and health-risk assessment of $PM_{2.5}$-bound PAHs across urban Central Asian cities - a region with sparse speciation data and heavy coal heating. Cities, sampling design, concentrations and risk estimates are not available; nothing is inferred. Relevance: medium - coal-heating PAH peaks in Almaty-type basins are where low-cost PM networks already operate without any chemical reference; retry the abstract on a later run.

doi: 10.1016/j.atmosenv.2026.122447

Yousaf nanoscale curvature hygroscopic growth respiratory deposition ultrafine 2026 Environ Pollut

Nanoscale Curvature Controls Hygroscopic Growth and Respiratory Deposition of Atmospheric Nanoparticles

HTDMA measurements of 15-100 nm NaCl particles fed into the MPPD lung-deposition model. The Kelvin effect raised deliquescence RH to 77-81% for 15-36 nm particles and suppressed growth; at 80-90% RH, growth factors were 1.95-2.25 (15 nm) and 1.99-2.36 (100 nm). Ignoring hygroscopic growth overestimated total deposition by up to 25%, with an 18% tracheobronchial difference for particles <=50 nm, and shifted larger UFPs deeper into the lung. Limitation: pure NaCl is a best-case hygroscopic aerosol; ambient UFP is organic-rich with far lower kappa, so the correction is an upper bound. Relevance: medium - the same size-dependent water uptake that biases optical sensors in humid air also changes delivered dose, so dry-size number counts understate the problem.

doi: 10.1016/j.envpol.2026.129429

Yu Paris urban plume aerosol composition optical properties airborne ground 2026 Atmos Chem Phys

Urban-to-downwind variations in aerosol composition and optical properties in the Paris urban plume from coordinated airborne and ground-based observations

Coordinated ground and airborne measurements in Paris and its downwind plume, ~2-6 h transport. The plume modestly raised 80-200 nm particle number and moderately enhanced PM_1; organics dominated PM_1 throughout, with more oxidised organics downwind. Optical properties changed sharply: higher SSA at 450 and 630 nm downwind; refractive index moved from ~1.35-1.40 + 0.03-0.08i near source to ~1.45-1.50 + 0.015-0.02i downwind; AAE rose. Limitation: limited flights under specific meteorology; plume age is estimated, not tracked. Relevance: high - optical LCS use a single assumed refractive index; a 2-6 h change of this size shifts OPC mass conversion between urban and downwind nodes.

doi: 10.5194/acp-26-15055-2026

Zhang Fusion-DeepRSM satellite $PM_{2.5}$ 2026 Environ Sci Technol

Fusion-DeepRSM for Satellite-Derived $PM_{2.5}$ Estimation and Precursor Sensitivity Prediction

Observation-constrained deep response-surface model fusing CTM runs, satellite products and ground $PM_{2.5}$. R2 0.872, RMSE 11.82 $\mu g\,m^{-3}$, and a 4.13 $\mu g\,m^{-3}$ reduction in rural overestimation, the known failure of station-trained ML. It also returns precursor-$PM_{2.5}$ response surfaces: for 2020-2023, meteorology offset part of emission-driven improvement (+1.2 vs -1.8 $\mu g\,m^{-3}$), and precursor change drove the Feb 2023 BTH rebound. Limitation: validation against stations that are themselves inputs; the rural bias correction is quantified on a CTM testbed, not against independent rural monitors - exactly where LCS would be needed. Relevance: high - a fusion architecture into which rural sensor nodes would slot as the missing constraint.

doi: 10.1021/acs.est.6c08137

Zhao $PM_{2.5}$-O_3 co-pollution episodes BTH YRD 2026 Atmos Pollut Res

Episode analysis of the characteristics and causes of $PM_{2.5}$–O_3 co-pollution in Beijing-Tianjin-Hebei and Yangtze River Delta, China

METADATA ONLY. Title and authors only (Zhao, Xu, Gao, Zhang et al.). No abstract was retrievable at harvest. The title describes an episode analysis of simultaneous $PM_{2.5}$ and O_3 exceedance in BTH and the YRD, including causes. Data sources, period and attribution method cannot be stated. Limitation: unverifiable beyond the title. Relevance: low-medium - co-pollution episodes are where PM-only alerts from sensor networks understate risk.

doi: 10.1016/j.apr.2026.103248

Zhao et al. 2026c Atmos Pollut Res

Spatial Heterogeneity and Multi-scale Statistical Associations of Urban Air Pollution Revealed by Interpretable Machine Learning in Shenzhen, China

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 5 Oct with no abstract. By title, an interpretable machine-learning treatment of intra-urban spatial heterogeneity at several scales in Shenzhen. Scale dependence of the predictor set is the question that decides how dense a network has to be before adding nodes stops buying information, so the result matters more than the method. Relevance: potentially high; on the retry list.

doi: 10.1016/j.apr.2026.103230

Zheng et al. 2026 J Environ Sci (China)

Comparative analysis of multiple health risks from different air pollution events

Online and offline measurements in Beijing (Feb-May 2019) classify haze (83% secondary), firework (31% firework) and dust (50% dust) events and compare risk metrics against clean days. Haze and firework particles deposit mainly tracheobronchially and pulmonary, dust in extrathoracic and TB regions; ultrafines are >=81% of deposited number in every event. Carcinogenic risk exceeds 1e-4 in all events (7.6, 4.2 and 1.5 times clean-day 2.3e-4); hazard index reaches 241, 100 and 37 (firework, dust, haze) vs 23 clean. $PM_{2.5}$ OP is 0.74-0.82 vs 0.47 nmol/min/m3 clean. Limitation: one spring, one site, few events per type; HI and TCR are deterministic screening indices with default toxicity values, not health outcomes. Relevance: moderate - event rankings flip by metric, which a mass-only network cannot reproduce.

doi: 10.1016/j.jes.2026.02.033

Zhou condensation sink efficiency hygroscopicity 2026 Environ Sci Technol

Condensation Sink Efficiency of Atmospheric Particles Regulated by Their Chemical Compositions and Hygroscopicity

Defines a CS efficiency coefficient (epsilon = CS_eff/CS_calc, the latter assuming irreversible uptake) for gaseous H2SO4. Epsilon depends on composition and water uptake: dry pure organics scavenge less efficiently than inorganics, high RH raises epsilon for both, and mixed particles sit between, set jointly by composition and RH; CS efficiency is lower still for HIO_3. Including epsilon improves cluster-survival and sub-3 nm formation simulations and may explain intense NPF despite high CS in megacities. Limitation: epsilon values and their uncertainty are not given in the abstract, and field transfer is 'exploratory'. Relevance: low-moderate - NPF bursts drive number, not mass, so optical LCS miss them; CPC-equipped nodes would not.

doi: 10.1021/acs.est.6c07189

de la Cruz automated forced air injection cordwood stove $PM_{2.5}$ particle size 2026 Atmos Environ

Less Mass of $PM_{2.5}$ Emissions, Smaller Particles: The Trade-Off of Automated Forced Air Injection in a Residential Cordwood Stove

METADATA ONLY. Metadata only: no abstract was available from PubMed, Europe PMC, Crossref or Semantic Scholar at harvest. The title reports that automated forced-air injection in a residential cordwood stove lowers $PM_{2.5}$ mass emissions but shifts particles to smaller sizes. No numbers, test protocol or fuel details can be summarised. Limitation: cannot be appraised from the title; the trade-off claim needs number-based metrics to evaluate. Relevance: high in principle - a mass reduction with a shift to ultrafine sizes is precisely the case where optical mass sensors, blind below ~0.3 um, would overstate the improvement.

doi: 10.1016/j.atmosenv.2026.122443

Cardiovascular & metabolic — 28 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (28 OF 28 RECORDS IN THIS SUBTOPIC)

Andreotti active fire detections circulatory hospital admissions Brazil ecological panel 2026 Cureus

Active Fire Detections and Hospital Admissions for Circulatory Diseases in Brazil: An Ecological Panel Study (2013-2024)

27 Brazilian federative units x 12 years (324 unit-years), 13.8 million circulatory admissions and 2.3 million fire detections. Correlations were null (rho -0.03 to 0.01). In negative-binomial models with state fixed effects, year effects, temperature, poverty and primary-care coverage, the rate ratio per additional 1,000 detections/100,000 km2/year was 0.998 (0.970-1.028), null in every subset and with clustered SEs. Limitation: annual state-level fire counts are far removed from short-term smoke $PM_{2.5}$ at residence; smoke from Amazon fires is exported to other states, so the exposure is misplaced as well as aggregated. The authors correctly state this is not evidence of no effect. Relevance: low - illustrates why fire counts cannot substitute for measured PM.

doi: 10.7759/cureus.118676

Brugge HEPA sham crossover cytokines visceral adiposity 2026 Am J Prev Cardiol

Pro-inflammatory cytokines moderated by visceral adiposity from a pragmatic randomized crossover trial of portable HEPA air purifiers

116 adults (mean age 41.5) randomised to 30 days HEPA then sham, or the reverse, with a 30-day washout; ultra-sensitive cytokine assays, linear mixed models. No overall IL-1beta difference. Visceral adiposity (ABSI z >=+0.229) moderated the effect (interaction p=0.001): in the 12 participants with excess VAT, HEPA cut mean IL-1beta by 22.8% (0.138 vs 0.179 pg/mL, p<0.001); IL-6 showed a similar non-significant pattern and hsCRP little change; BMI moderated less well than ABSI. Limitation: the positive finding rests on 12 people in a post-hoc-looking subgroup chosen among several adiposity metrics, and achieved PM reduction is not reported in the abstract. Relevance: high - sham-controlled filtration with measured indoor PM is where calibrated low-cost monitors define the delivered dose.

doi: 10.1016/j.ajpc.2026.101775

Cao $PM_{2.5}$ chemical components heart disease longitudinal China 123 cities 2026 BMC Cardiovasc Disord

Association between $PM_{2.5}$ and its chemical components exposure and heart disease risk: a longitudinal study from China

8,172 participants from 123 Chinese cities; Cox models with three adjustment levels plus WQS mixtures. Per 1 $\mu g\,m^{-3}$ $PM_{2.5}$: HR 1.027 (1.022-1.032); BC had the largest per-unit estimate, HR 1.381 (1.266-1.506). All components showed non-linear exposure-response; southern residents and overweight participants had stronger associations; PAF 2.63%. Limitation: per-$\mu g\,m^{-3}$ component HRs are not comparable because component ranges differ by an order of magnitude, and components are highly collinear, so 'BC largest' is a scaling artefact unless expressed per IQR. Heart disease is likely self-reported (cohort not named in the abstract). Relevance: medium - again points to absorption (BC) as the health-relevant channel.

doi: 10.1186/s12872-026-06395-3

Cerin air pollution road environment cardiometabolic walking AusDiab 2026 Toxics

Associations of Ambient Air Pollution and Road-Related Environmental Characteristics with Cardiometabolic Health: The Role of Walking

4,141 middle-aged and older adults; GAMMs with FDR control within test families. $PM_{2.5}$ associated positively with diastolic and non-linearly with systolic BP; road density with lower HDL and higher triglycerides. Higher $PM_{2.5}$ and NO_2 were associated with more walking for transport, setting up opposing pathways via physical activity. Coefficients are not reported in the abstract. Limitation: cross-sectional, so walking cannot be ordered between exposure and outcome, as the authors state; Australian $PM_{2.5}$ contrasts are narrow (single-digit $\mu g\,m^{-3}$) and track urbanity, which drives both walkability and pollution. Relevance: low-medium - co-located PM and mobility data would let LCS studies separate these pathways.

doi: 10.3390/toxics14100920

Chen Lanzhou $PM_{2.5}$ components CVD 2026 Int J Environ Health Res

Acute cardiovascular mortality risks from $PM_{2.5}$ and its chemical components in an industrial valley city: evidence from Lanzhou, China

69,669 CVD deaths, 2015-2023. $PM_{2.5}$ lag 0: excess risk 2.06% per 10 $\mu g\,m^{-3}$ (0.66-3.48%), larger in men and under-65s. qgcomp/WQS attribute risk to PAHs and Tl/Pb/Mn in traffic-dominated Chengguan, and to nitrate/chloride/sulfate and PAHs in industrial Xigu. Limitation: components are not measured but predicted by a random forest with r>0.70, so the mixture weights inherit the predictor set of that model - district contrasts may reflect which inputs drove the prediction rather than toxicity. Relevance: moderate - a valley inversion city where source-specific nodes would test exactly these district contrasts.

doi: 10.1080/09603123.2026.2745894

Children's AIRE 2026 Int J Hyg Environ Health

Ambient particulate matter exposure and children's blood pressure in rural California, 2019-2022

592 children aged 6-8 at enrolment, blood pressure measured up to five times through spring 2022, linear mixed models. An IQR increase (38 days) in prior-year days above 50 $\mu g\,m^{-3}$ daily PM_{10} was associated with +1.1 mmHg DBP (95% CI 0.8, 1.5) and +0.9 mmHg SBP (0.3, 1.5); the top quartile of short-term $PM_{2.5}$ and PM_{10} carried 5.8 (0.4, 11.2) and 5.9 (0.2, 11.6) percentage-point higher risk of elevated or hypertensive SBP against the lowest quartile. Limitation: exposure comes from regulatory monitors in a rural region where the nearest monitor can be tens of kilometres away and agricultural and wildfire sources are highly local, so non-differential misclassification is large and almost certainly biases toward the null; paediatric BP is noisy across visits. Relevance: high - this is the archetypal study whose exposure term a rural low-cost network would transform, and the 'days above threshold' metric suits an inexpensive, lower-accuracy node better than an annual mean does.

doi: 10.1016/j.ijheh.2026.114923

Cosenza et al. 2026 medRxiv (preprint)

Associations of Outdoor Air Pollution With Cause-specific Mortality in Metastatic Prostate Cancer: A cohort study

33,384 men with metastatic prostate cancer (11 US registries, diagnosed 2000-2015, followed to 2018); 1-km $PM_{2.5}$ and NO_2 at the diagnosis address; Cox and IPW cause-specific models. Per IQR $PM_{2.5}$, cardiovascular mortality HR 1.04 (1.02-1.07), with 4th/5th vs 1st quintile HR 1.18 and 1.21; ADT-treated men with high vs low $PM_{2.5}$ had HR 1.19 (1.05-1.34), interaction p = 0.051. NO_2 was associated only with prostate-cancer death in the top quintile (HR 1.07, 1.01-1.13). Limitation: one address at diagnosis, no smoking or cardiovascular history in registry data, and cause-of-death coding in metastatic cancer is unreliable; the IQR is not stated in the abstract; preprint. Relevance: low-moderate - a high-risk population whose acute exposures personal sensors could resolve.

doi: 10.64898/2026.09.29.26364034

Du Air Health Index acute myocardial infarction 325 cities 2026 Environ Sci Technol

Establishing an Air Health Index for Predicting the Risk of Acute Myocardial Infarction Onset Associated with Air Pollution and Ambient Temperature

1,207,625 AMI cases across 325 Chinese cities; excess risks from $PM_{2.5}$, NO_2, O_3 and temperature aggregated into a normalised AHI. Per IQR, AHI associated with +5.97% (5.32-6.63%) AMI incidence vs +3.33% (2.93-3.72%) for AQHI and +2.04% (1.73-2.35%) for AQI; near-linear, no threshold; validated on an independent external dataset; 8,534 per 100,000 AMI cases attributed. Limitation: the index is fitted to the same outcome it is scored on, so in-sample superiority over AQI/AQHI is expected; IQR-scaled comparisons across indices with different distributions are not like-for-like, and 'attributable' fractions for a composite that includes temperature mix avoidable and non-avoidable risk. Relevance: high - an outcome-weighted index is a natural output layer for real-time LCS networks, provided the $PM_{2.5}$ input is calibrated.

doi: 10.1021/acs.est.6c07635

Hahad exposome heart failure older adults review 2026 Curr Heart Fail Rep

Exposomic Determinants in Heart Failure Among Older Adults - Mechanistic Links, Clinical Evidence, and Future Perspectives

Narrative review arguing for a life-course, systems-level view of heart failure in older adults in which air pollution sits alongside traffic noise, temperature extremes, metals, artificial light at night and social environment. Mechanisms proposed to converge on oxidative stress, mitochondrial dysfunction, inflammaging, impaired autophagy, epigenetic change and senescence, lowering the threshold for HF onset and decompensation. No new data or pooled estimates. Limitation: the mechanistic convergence is asserted from separate literatures; co-exposures (PM, noise, heat) are spatially correlated, and the review does not address how to separate them. Relevance: low - multi-exposure co-located sensing (PM, noise, temperature) is what the exposome framing would need.

doi: 10.1007/s11897-026-00781-w

Huang $PM_{2.5}$ coronary artery dilatation Kawasaki disease case-crossover 2026 Eur J Pediatr

Association between PM(2.5) and coronary artery dilatation with Kawasaki disease in China

847 children with Kawasaki disease, 2017-2021; case-crossover with a distributed-lag Poisson GLM on coronary Z score. Risk of coronary artery dilatation rose at lags 2-4, peaking at lag 3 with RR 1.11 (1.02-1.21); the abstract does not state the $PM_{2.5}$ increment. Mediation via CRP, ALT and haemoglobin. Limitation: a case-crossover needs an acute event with a datable onset, yet CAD is detected at echocardiography whose timing follows admission and IVIG, so the referent-window logic is shaky; the 'mediators' are markers of disease severity measured after onset, and the highlights' claim of a 'population-based cohort' contradicts the design. Relevance: low - city-monitor assignment, no sensing implication.

doi: 10.1007/s00431-026-07527-9

Huang gut microbiota mediation air pollution metabolic syndrome 2026 iMetaOmics

The mediating roles of gut microbiota in the associations of air pollution and meteorological factors with metabolic syndrome

Gut microbial community clusters were derived and related to metabolic syndrome; health-associated clusters were dominated by Bacteroides and Prevotella. Long-term NO_2, $PM_{2.5}$ and temperature were associated with microbiota composition, and Megamonas and Megasphaera statistically mediated their associations with MetS. The abstract reports no sample size, effect estimates, mediated proportions or confidence intervals. Limitation: mediation from a single stool sample assumes exposure precedes the microbiome state and that diet, a strong common cause of both microbiota and MetS, is adequately adjusted - neither can be checked from the record. Relevance: low - mechanistic hypothesis generation; exposure resolution is not the limiting factor here.

doi: 10.1002/imo2.70099

Lee CRCS-K post-discharge SO_2 stroke recurrence PM null 2026 J Stroke

Post-Discharge Ambient Sulfur Dioxide Exposure and Stroke Recurrence After Acute Ischemic Stroke

27,340 acute ischaemic stroke patients (2014-2021), district-level IDW-interpolated pollutant means over the 3 months after discharge; cause-specific Cox with death as competing risk, cluster-robust SEs, splines. 764 recurrences, 542 deaths over a median 364.8 days. PM_{10} and $PM_{2.5}$ showed no association; only SO_2 did (overall spline p=0.024), with Q4 vs Q1 HR 1.19 (0.94-1.50), stronger in age >=75 and women (interaction p 0.051, 0.062). Limitation: district IDW from sparse monitors gives Berkson-type error that biases PM toward the null, and SO_2 concentrates in harbour cities (Incheon, Ulsan, Busan) - a place effect cannot be excluded. Relevance: moderate - a null PM result built on interpolation is exactly where denser networks change the answer.

doi: 10.5853/jos.2026.03265

MESA incident AF time-varying pollution 2026 Heart Rhythm

Association of Air Pollution Exposure and Risk of Incident Atrial Fibrillation: The Multi-Ethnic Study of Atherosclerosis

6,618 MESA participants (mean age 62), 16.5 y median follow-up, 1,055 incident AF events. Exposure came from the MESA Air hierarchical spatiotemporal models, which assimilate cohort-specific snapshot monitoring, entered as time-varying 2-y averages. Per IQR of NO_2 (13.5 ppb) AF hazard rose 29% (HR 1.29, 95% CI 1.00-1.67); $PM_{2.5}$, NOx and O_3 were weaker and directionally positive, with no estimates in the abstract. Limitation: AF ascertainment from hospitalisation and study ECGs misses paroxysmal disease, and the NO_2 CI touches the null; the $PM_{2.5}$ contrast within six metros is narrow. Relevance: high methodologically - this is the cohort where supplemental community monitoring was built to cut exposure error, and $PM_{2.5}$ still came out null-leaning.

doi: 10.1016/j.hrthm.2026.10.013

MESA long-term air pollution suPAR null 2026 Environ Epidemiol

Long-term ambient air pollution is not associated with circulating soluble urokinase plasminogen activator receptor

5,072 MESA participants (2000-2002 enrolment), ELISA suPAR, multivariable linear regression. No pollutant - $PM_{2.5}$, NO_2, NOx, O_3 or light-absorbing carbon - was associated with suPAR; the $PM_{2.5}$ CI excluded effects of 2% or more per 5 $\mu g\,m^{-3}$. With two prior null cohorts, the authors conclude suPAR is not a meaningful mediator of the PM-CVD link. Limitation: single cross-sectional suPAR measure, and the US contrast of the early 2000s is narrow relative to high-exposure settings, so a null at 5 $\mu g\,m^{-3}$ does not transfer to them. Relevance: low-moderate - a well-powered null worth citing against biomarker-mediation claims; the exposure model, not sensors, sets the contrast.

doi: 10.1097/ee9.0000000000000532

Musa outdoor air pollution cardiovascular health school children Dar es Salaam 2026 BMC Public Health

Effects of outdoor air pollution on the cardiovascular health of school-aged children residing near polluted areas and the efficacy of mitigation measures in Dar es Salaam, Tanzania

Community cross-sectional study, December 2024-July 2025, of 653 school-aged children living in areas classed as highly polluted by proximity to industrial zones, major roads and waste-burning sites. Single oscillometric BP reading; 19.9% had elevated BP. In log-binomial models, prior respiratory disease carried ARR 8.81 (6.12-12.67) for cardiovascular symptoms. Awareness of mitigation policy was mixed. Limitation: no pollutant was measured and there is no unexposed comparison group, so the title's causal framing is not supported; one-visit BP overstates prevalence in children. Relevance: high as a gap marker - urban East Africa has almost no PM monitoring, and this design shows what health studies look like without it.

doi: 10.1186/s12889-026-28148-9

Rao $PM_{2.5}$ components cardiorenal multistate trajectory 2026 Int J Cardiol

Associations of $PM_{2.5}$ components with dynamic and accelerated progression of different kidney function states and subsequent cardiovascular events

307,639 participants, 1,058,210 person-years, 23,948 CVD cases and 7,763 CVD deaths; multistate and accelerated failure time models over eGFR categories 1-4. Per IQR, transition from category 2 to 3 had HR 1.273 (1.257-1.289) for sulfate, 1.304 (1.290-1.317) for nitrate and 1.280 (1.267-1.294) for ammonium, larger than for OM or BC. AFT: predicted 2-to-3 transition time 1,727 days at 4.2 vs 869 days at 7.0 $\mu g\,m^{-3}$ nitrate. Limitation: HRs near 1.3 per IQR with CIs this narrow, for highly collinear secondary inorganic components, signal spatial confounding by region more than component-specific toxicity; single-pollutant component models cannot separate them. Relevance: moderate - component attribution needs speciation that optical LCS lack.

doi: 10.1016/j.ijcard.2026.134999

Rivera-Mariani et al. 2026 medRxiv (preprint)

Inflammatory Burden, Structural Exposure, and Racial Disparities in Incident Heart Failure: Evidence from the Multi-Ethnic Study of Atherosclerosis

6,785 MESA participants followed ~16 years (485 HF events). IL-6, D-dimer, CRP and fibrinogen each predicted HF (HR 1.13-1.20 per SD); a composite index gave HR 1.25 (1.12-1.39); biomarker adjustment reduced the Black-White log-hazard gap by 57.5%. Associations were robust to $PM_{2.5}$/NO_2 co-adjustment in a geocoded subcohort. Limitation for this watch: air pollution is a sensitivity covariate only - no PM effect estimate is reported, and 'structural exposure' is inferred from a null interaction, not measured; single baseline biomarkers; preprint. Relevance: low - admitted because it tests whether PM explains part of an inflammatory pathway and reports that it does not change the estimates.

doi: 10.64898/2026.09.28.26364238

Sun long-term air pollution lipid profiles Shanxi repeated measures 2026 Front Public Health

Long-term exposure to ambient air pollution and lipid profiles: a longitudinal study in a heavily industrialized region of northern China

Repeated-measures study of 9,040 adults with 19,113 lipid measurements from a health-screening programme, 2019-2023. Per 10 $\mu g\,m^{-3}$ $PM_{2.5}$: TG +1.85%, TC +1.70%, LDL-C +1.81%, HDL-C -1.55% (linear mixed models); PM_{10}, NO_2, SO_2 and O_3 in the same direction, CO weaker; age modified the HDL-C association after FDR correction. Limitation: a health-screening population self-selects; CHAP is itself trained on the sparse Shanxi monitor network that underrepresents coal-industry hotspots, and with five correlated pollutants the '$PM_{2.5}$ most pronounced' ranking is not identified from single-pollutant models. Lipids are sensitive to fasting status, not mentioned. Relevance: moderate - industrial-region exposure surfaces are where LCS densification would most change CHAP errors.

doi: 10.3389/fpubh.2026.1950282

UKB arrhythmia NMR metabolome mediation 2026 Ecotoxicol Environ Saf

Large-scale plasma metabolic profiling identifies signatures and biomarkers associated with air pollution and arrhythmias: A prospective cohort study from UK Biobank

449,068 UK Biobank participants free of arrhythmia, median follow-up 16.3 y, 47,381 incident cases. All four pollutants were associated with arrhythmia, HR 1.015-1.257 per 10 $\mu g\,m^{-3}$ (range across pollutants; CIs not in the abstract). An elastic-net signature from 170 NMR metabolites mediated 11.2% of the $PM_{2.5}$ association (PM_{10} 5.2%, NO_2 7.4%, NOx 9.8%); lipid, ketone-body and amino-acid pathways recurred, ketone bodies most for intraventricular block. Limitation: single-baseline 2010 LUR exposure and single-time metabolomics make mediation cross-sectional in disguise; ketone bodies track fasting and diabetes status. Relevance: low-moderate for sensing - another UKB LUR result whose exposure contrast is too smooth to need dense monitoring.

doi: 10.1016/j.ecoenv.2026.120919

UKB women deprivation $PM_{2.5}$-NO_2 fatty liver 2026 Front Endocrinol (Lausanne)

Socioeconomic disadvantage, air pollution, and metabolic health among women: analyses of UK Biobank and US data

231,525 UK Biobank women: Townsend deprivation predicted FLI (beta 1.114, 95% CI 1.069-1.159) and FLI>=60 (OR 1.080, 1.076-1.084). Age-adjusted pollution associations vanished once deprivation entered the model (FLI beta -0.068, -0.137 to 0.001; FLI>=60 OR 0.999, 0.993-1.004). Across 46 US jurisdictions, +1 SD poverty meant +1.38 points female obesity after $PM_{2.5}$ adjustment. Limitation: adjusting for an area deprivation index that partly lies on the pollution pathway can over-adjust; a 2010 LUR score and state-level US $PM_{2.5}$ are coarse. The null is contextual, not causal. Relevance: low-moderate - shows how easily SES absorbs smooth LUR contrasts; finer exposure would be needed to separate them.

doi: 10.3389/fendo.2026.1948066

Wan $PM_{2.5}$ mortality urban rural elderly Beijing cohort 2026 Zhonghua Liu Xing Bing Xue Za Zhi

Association between fine particulate matter exposure and all-cause and cardiovascular mortality in urban and rural elderly population in Beijing

Beijing Elderly Comprehensive Health Cohort, n=3,472. Per 1 $\mu g\,m^{-3}$ $PM_{2.5}$: all-cause mortality HR 1.053 (1.026-1.080), CVD mortality HR 1.046 (1.001-1.100). Outdoor physical activity modified the association (P-interaction 0.022): HR 1.087 per $\mu g\,m^{-3}$ without outdoor activity vs 1.026 with it. Limitation: a 5% hazard increase per single $\mu g\,m^{-3}$ is roughly an order of magnitude above the pooled long-term estimates. In a city-scale cohort the $PM_{2.5}$ contrast is mostly urban vs rural, so residual confounding by healthcare access and SES is the likely driver. The physical-activity 'offset' is also confounded by baseline health. Relevance: low - single-city satellite exposure.

doi: 10.3760/cma.j.cn112338-20251204-00870

West Florida temperature x $PM_{2.5}$ CVD admissions 2026 J Am Heart Assoc

Joint Impacts of Ambient Air Pollution and Temperature on Cardiovascular Disease in West Florida

BayCare Health System admissions 2017-2024 analysed with distributed-lag GAMs and tensor-product temperature x $PM_{2.5}$ surfaces, copollutant-adjusted. Temperature dominated: MI risk in adults >=65 was lower at warmer days (RR75/25 0.92, 0.89-0.95), heart failure was U-shaped with heat risk in younger and Black patients (RR75/25 1.05, 1.02-1.09). $PM_{2.5}$ was weak; the largest signal was stroke in Black patients, RR75/25 1.02 (0.96-1.08). Limitation: a single-system admissions series assigns one area-wide monitor value to a coastal region with low, compressed $PM_{2.5}$, so near-null PM results partly reflect a small exposure contrast; subgroup multiplicity is unaddressed. Relevance: moderate - low-concentration subtropical settings are where denser networks would sharpen the contrast.

doi: 10.1161/jaha.126.049988

Wongcharoen short-term $PM_{2.5}$ MACE high-risk population nationwide case-crossover 2026 BMJ Open

Associations between short-term PM(2.5) exposure and major cardiovascular events in high-risk population: a nationwide case-crossover study

METADATA ONLY. Title and authors only (Wongcharoen et al.); no abstract indexed at harvest. A nationwide time-stratified case-crossover of short-term $PM_{2.5}$ and major adverse cardiovascular events restricted to a 'high-risk population' (presumably prior CVD, diabetes or hypertension). Country, n, lag structure and effect size cannot be stated. Limitation: unverifiable beyond the title; case-crossover in a pre-selected high-risk group estimates a conditional acute effect that is not comparable with general-population estimates. Relevance: low-moderate - short-lag acute effects are what real-time LCS alerts are meant to act on.

doi: 10.1136/bmjopen-2026-120476

Yadav PM pollution-accelerated cardiovascular aging PACA framework 2026 Cardiovasc Toxicol

Particulate Matter Exposure and Accelerated Cardiovascular Aging: Mechanistic Evidence, Biomarker Integration, and the PACA Framework

A conceptual review that proposes the 'Pollution-Accelerated Cardiovascular Aging' (PACA) framework to separate transient toxicity from a real shift in cardiovascular ageing trajectory. It collects established pathways (mitochondrial and redox dysfunction, epigenetic change, senescence/SASP, proteostasis, endothelial and autonomic dysfunction, the pulmonary-neural-cardiac axis) and adds a Temporal-Evidence Matrix that weighs persistence after exposure stops and change relative to age-referenced comparators. Limitation: hypothesis-generating with no new data or systematic search. Without validated ageing clocks for the vasculature the matrix is hard to operationalise. Relevance: low - longitudinal personal exposure data are the missing input the framework asks for.

doi: 10.1007/s12012-026-10209-5

Zhang short-term air pollution stroke incidence Guiyang karst city 2026 Front Public Health

Time-series analysis of the impact of short-term exposure to atmospheric pollutants on stroke risk among residents in Guiyang City, 2016-2021

Guiyang 2016-2021, 11,517 incident strokes (5.25/day), quasi-Poisson GAM with single-day and cumulative lags, controlling trend, weather, day-of-week and holidays. Per 10 $\mu g\,m^{-3}$ at the maximal lag: $PM_{2.5}$ RR 1.026 (1.005-1.048), PM_{10} 1.015 (1.001-1.030); SO_2 1.075, NO_2 1.057, CO 1.346 per mg/m3. Women and adults >=65 carried more of the risk; under-65s responded only to NO_2 and PM_{10}. Limitation: single-pollutant models with the lag chosen at the maximum inflate the PM coefficients, co-pollutant collinearity is unaddressed, and the stroke series does not split ischaemic from haemorrhagic. Relevance: moderate - a low-concentration karst basin is where fixed-site averaging hides terrain-driven gradients that a dense low-cost network would resolve.

doi: 10.3389/fpubh.2026.1777772

Zhang wildfire smoke waves long-term $PM_{2.5}$ cardiovascular hospitalisation Medicare 2026 Environ Res Health

Impact of wildfire smoke waves on the association between long-term exposure to fine particulate matter and cardiovascular hospitalizations among older adults in the United States

65.2 million Medicare beneficiaries aged >=65, 2017-2022, ZIP-year quasi-Poisson. Raising two-year $PM_{2.5}$ from 5 to 9 $\mu g\,m^{-3}$ (WHO guideline to US standard) gave overall CVD RR 1.09 (1.09-1.10) in ZIP-years without smoke waves but 1.04 (1.03-1.04) with them (ratio of RRs 1.05, 1.04-1.06); the pattern held below 9 $\mu g\,m^{-3}$ and was stronger in men, urban and western residents. Limitation: the attenuation is what one expects if smoke spikes inflate the two-year mean with a less toxic-per-mass or differently distributed exposure, so this is an exposure-metric artefact as much as a biological finding; ZIP-level assignment and 10 km smoke grids blur the contrast. Relevance: high - long-term means hide episodic peaks that only high-frequency network data resolve.

doi: 10.1088/2752-5309/aeb0c3

Zhao interpretable machine learning multi-window $PM_{2.5}$ hypertension CHNS Beijing 2026 Toxics

An Interpretable Machine Learning Framework Integrating Multi-Window Environmental Exposure for Hypertension Risk Assessment

Five classifiers (GLM, Lasso, tree, RF, XGBoost) with SHAP and partial-dependence layers on 2,523 participant-visits from the CHNS Beijing subsample; person-level and year-based splits; BP excluded from predictors. Test AUC 0.69-0.77, Brier 0.17-0.23. The 60-day $PM_{2.5}$ window was the top exposure feature and added ~0.02-0.03 AUC (p=0.03-0.05). Adjusted window-specific analyses showed inverse $PM_{2.5}$-hypertension associations, stronger for longer windows. Limitation: a ~0.02 AUC gain at borderline p with inverse associations indicates that the $PM_{2.5}$ windows act as calendar/season proxies, not causal exposure; feature importance is not effect estimation, and the CHNS visit spacing is irregular. Relevance: low.

doi: 10.3390/toxics14100871

Zhu source-specific $PM_{2.5}$ atrial fibrillation Medicare cohort 2026 Environ Health (ACS)

Source-Specific $PM_{2.5}$ Exposure and Atrial Fibrillation Risk in U.S. Older Adults: A Nationwide Cohort Study

26.3 million Medicare beneficiaries, 2000-2019, 171.6 M person-years, 4.44 M incident AF. Per IQR of source-specific $PM_{2.5}$: secondary sulfate HR 1.056 (1.053-1.058), heavy oil/coal combustion 1.048 (1.046-1.050), secondary nitrate 1.032 (1.030-1.034), vehicle 1.029 (1.028-1.030); biomass burning and dust modest, SOA null overall but positive in the West. Limitation: source profiles come from sparse speciation monitors interpolated by inverse distance over 100 km, so source contrasts are largely regional contrasts; sulfate and nitrate fields are smooth and the per-IQR scaling differs by source, so 'sulfate is most toxic' is not a per-microgram statement. Claims-based AF under-ascertains paroxysmal AF. Relevance: high - speciation, not mass, is what optical LCS cannot give; source-resolved risk raises the value of co-located filter or ACSM nodes.

doi: 10.1021/envhealth.5c00638

Neuro / mental health — 24 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (24 OF 24 RECORDS IN THIS SUBTOPIC)

ABCD air pollution 2026 Schizophr Bull

The Relationship between Air Pollution Exposure, Genetic Liability for Schizophrenia, and Brain Outcomes in the ABCD Study

1,580 adolescents with genotype, age 9-10 exposure and structural plus resting-state MRI at 13-14; linear mixed-effects models with a schizophrenia polygenic score as moderator. Higher $PM_{2.5}$ was associated with smaller right frontal pole surface area (p-FDR 0.039); the PRS interactions were non-monotonic - among low-PRS individuals, higher $PM_{2.5}$ went with larger bilateral caudate volumes, while low-PRS/medium-$PM_{2.5}$ showed smaller left caudate than medium/medium. Higher NO_2 was associated with stronger negative connectivity between networks. Limitation: a single surviving FDR-corrected main effect across a very large imaging search space, and interaction patterns that do not order monotonically in either PRS or exposure, is the signature of noise - the authors' own 'first test' framing is the right reading. Relevance: moderate - the exposure is a modelled annual surface at the residence of adolescents who move around a city all day.

doi: 10.1093/schbul/sbag138

ALFA+ TRAP CSF GFAP 2026 Alzheimers Dement

Air pollution is associated with sex-specific longitudinal changes in astrocytic and neuroinflammatory processes linked to cognitive decline in Alzheimer's disease at-risk individuals

226 cognitively unimpaired middle-aged adults at elevated AD risk with baseline and 3-year CSF. Higher traffic-related exposure associated with longitudinal increases in GFAP and IL-6, mainly in women; GFAP partially mediated associations with worsening executive and attentional performance. Limitation: no effect sizes in the abstract; n=226 split by sex across four correlated pollutants and many biomarkers is a multiplicity problem, and single-address LUR estimates ignore time-activity. Relevance: moderate - CSF longitudinal biology is rare, but this is hypothesis-generating at this n.

doi: 10.1002/alz.71837

Al-Atram early-life air pollution child neurodevelopment critical review 2026 J Xenobiot

Early Life Ambient Air Pollution and Child Neurodevelopment: Critical Review of Cohort Evidence

Narrative critical review (purposive selection, PubMed through 30 Aug 2026) weighing null findings, exposure assessment, confounding and causal interpretation. European and US multicohort studies show small associations with selected outcomes alongside substantial nulls; Canadian and Korean cohorts show pollutant- and age-specific heterogeneity. Birth weight and prematurity require total vs conditional effect distinctions; placental and imaging data support plausibility but not mediation. The authors conclude the evidence does not make pollution a determinative cause of ASD or ADHD. Limitation: no protocol, duplicate screening or risk-of-bias scoring, so selection is the authors' own, and it predates today's Medicaid wildfire-ASD cohort. Relevance: low.

doi: 10.3390/jox16060191

Baird woodsmoke CD4 T cell brain influx saturated fat diet mice 2026 Neurotoxicology

CD4+ T Cell Activation and Peripheral Immune Cell Influx into the Brain Following Wildfire Smoke Exposure is Modulated by a Saturated Fat Diet

Female C57BL/6 mice exposed to filtered air or woodsmoke, with or without a coconut-oil saturated-fat diet, sampled 1, 14 and 28 days after exposure by high-dimensional flow cytometry. Woodsmoke produced a time- and diet-dependent influx of activated CD4 T cells (CD44+, RORgamma-, FOXP3-) into brain, and LFA-1, VCAM-1 and ICAM-1 peaked one day after the last exposure; taurine and glycerolipid metabolism were upregulated. Limitation: 500 $\mu g\,m^{-3}$ is an extreme-episode dose, females only, and effect sizes are not in the abstract. Relevance: low for sensing - a mechanistic anchor for smoke-neuroinflammation epidemiology.

doi: 10.1016/j.neuro.2026.103589

Batisse et al. 2026 Epidemiology

Long-term exposure to ultrafine particles and mortality from neurodegenerative diseases: a population-based cohort study

2.1 million adults in the Canadian Census Health and Environment Cohorts, 2001-2019, 19 million person-years and 20,560 neurodegenerative deaths. Per 10,000 particles/cm3 of UFP number, dementia mortality HR 1.22 (95% CI 1.17-1.27), strengthened after adjustment for UFP size; positive for every dementia subtype, largest for vascular dementia. Models were run with and without inverse-probability censoring weights for competing events and adjusted for co-pollutants. Limitation: postal-code exposure from a land-use-regression UFP surface, no mobility or indoor time, and cause of death on the death certificate under-ascribes dementia; UFP number and NO_2 share the traffic gradient, so co-pollutant adjustment cannot fully separate them. Relevance: high - UFP is unregulated and unmeasured by every optical low-cost node, which sees mass, not number; this is the strongest current argument for CPC-class instruments in dense networks.

doi: 10.1097/ede.0000000000002054

Bloomberg et al. 2026 medRxiv (preprint)

Daily Exceedance of World Health Organization Air Pollutant Guidelines and Cognitive Aging

English Longitudinal Study of Ageing, 12,235 adults 50+ (waves 1-10, 2002-2023). Exposure is the number of days above WHO daily guideline values in the year before baseline, modelled at the residence, adjusted for the annual mean. NO_2 exceedance days (75th percentile vs median) predicted faster 14-year decline: memory -0.09 SD (-0.17 to -0.01), fluency -0.10 SD (-0.19 to 0.00). $PM_{2.5}$ and PM_{10} exceedance days were not associated with decline or baseline cognition. Limitation: exceedance count and annual mean are highly collinear, so the 'peak beyond mean' effect is fragile; one exposure year; attrition in ageing cohorts is cognition-dependent; preprint. Relevance: moderate - a null for PM peak frequency is exactly what high-time-resolution networks are positioned to retest.

doi: 10.64898/2026.09.23.26363787

Byun ADRD readmission 2026 Environ Health Perspect

$PM_{2.5}$ and 30-Day Readmission in Medicare Beneficiaries with Alzheimer's Disease and Related Dementias: Effect Modification by Neighborhood Green and Blue Space

Nationwide time-stratified case-crossover over 6,098,836 ADRD index admissions, 759,128 (12.4%) readmitted within 30 days. Per IQR (7.2 $\mu g\,m^{-3}$) of lag 0-2 $PM_{2.5}$: OR 1.010 (1.006-1.015), stronger for readmissions after day 7. Landsat- and Street-View-derived green and blue space did not conclusively modify the association. Limitation: ZIP-level modelled $PM_{2.5}$ for a largely homebound or institutionalised population is a poor proxy for the air actually breathed, which biases toward the null and makes the null modification uninterpretable. Relevance: moderate - a 1% per-IQR effect on a 6-million-person denominator is the scale at which indoor sensing in care settings would change the exposure estimate more than the model does.

doi: 10.1021/ehp.6c00147

Han et al. 2026 Int J Epidemiol

Air pollution and criminal behavior: a nationwide case-crossover analysis in China

Time-stratified case-crossover of 483,718 crime events in China (2013-2019) with conditional logistic regression per pollutant and weighted-quantile-sum for the mixture. Per IQR on the event day, crime rose 1.66% for $PM_{2.5}$, 1.88% PM_{10}, 4.03% NO_2, 2.03% CO, 2.81% SO_2 and 1.30% O_3; WQS index +3.22% per unit. Associations were stronger in the cold season, the north and lower-SES regions, and for violent crime and theft. Limitation: every pollutant is 'positive', which is what shared confounding by weather, daylight and outdoor activity patterns looks like; exposure is assigned at a coarse scale to events of unknown offender residence; no CIs in the abstract. Relevance: low - a behavioural endpoint with acute timing, where personal or hyperlocal exposure would separate PM from co-pollutants.

doi: 10.1093/ije/dyag214

Holm CHAPS Fresno neurobehavior BDNF children air pollution 2026 Front Public Health

Neurobehavioral assessments and biomarkers in school children who are highly exposed to air pollution

Children's Health and Air Pollution Study, n=193 aged 7-18. Third-grade test scores correlated with NIH Toolbox total and crystallized composites (r 0.55-0.60) and with BDNF (r 0.35). First-trimester traffic pollutants were associated with aggressive and rule-breaking behaviour; first-year exposure to all pollutants with lower BDNF, window around 5-6 months. Limitation: 193 children, many pollutants, windows and outcomes, and no effect sizes or multiple-testing control in the abstract - the window findings are exploratory. Pollutants are highly collinear in Fresno, so 'all pollutants' cannot attribute the BDNF signal to PM or EC. Relevance: medium - EC and PAH contrasts again point to absorption channels rather than mass.

doi: 10.3389/fpubh.2026.1970714

Jin wildfire carbonaceous ADRD 2026 medRxiv (preprint)

Carbonaceous Components Underlying Wildfire Smoke $PM_{2.5}$-Associated Alzheimer's Disease and Related Dementias in Older US Adults

39.7 million fee-for-service Medicare beneficiaries, 2005-2019, 8.4 million incident ADRD cases; stratified Cox. Per 1 $\mu g\,m^{-3}$, smoke $PM_{2.5}$ HR 1.029 (1.024-1.035) for ADRD and 1.015 (1.005-1.026) for AD; the smoke carbonaceous fraction HR 1.090 (1.077-1.103) for ADRD. Per IQR the non-smoke association is larger (ADRD 1.048 vs 1.009) because smoke exposure contrast is narrow. Attributable: 11,142 ADRD cases/year. Limitation: preprint; annual smoke partitioning is model-derived and its error is not propagated, and claims-based incident dementia carries diagnostic-intensity confounding that tracks rurality and fire-prone regions. Relevance: high - argues for composition-resolved smoke metrics that optical LCS cannot supply without co-located carbon measurement.

doi: 10.64898/2026.10.08.26364863

Kalkbrenner traffic air pollution ADHD Denmark critical windows 2026 Environ Res

Traffic-related air pollution links with attention deficit hyperactivity disorder in Denmark: Developmental periods of susceptibility and sex-specific associations

Danish singletons born 1981-2008: 21,860 ADHD cases and a 32,041-person random subcohort. Exposure in each developmental period was mutually adjusted for the other periods to separate windows. The second year of life dominated: EC >1.13 vs <0.49 $\mu g\,m^{-3}$ HR 1.24 (1.12-1.36), not attenuated by $PM_{2.5}$ adjustment; females HR 1.44 (1.21-1.72) vs males 1.18 (1.00-1.40). $PM_{2.5}$ gave the same window and sex pattern. Polygenic scores for ADHD were adjusted. Limitation: period-specific exposures at the same address are highly collinear, and a window identified by mutual adjustment is sensitive to differential measurement error between periods. Diagnosis-based ADHD in females is also subject to referral bias that varies with urbanicity. Relevance: medium - EC, not $PM_{2.5}$ mass, carries the signal, which argues for absorption (BC) channels in dense networks.

doi: 10.1016/j.envres.2026.125900

Khan Pakistan neurovascular review 2026 Curr Neurovasc Res

Air Pollution and Neurovascular Health in Pakistan: An Emerging Public Health Crisis in South Asia - A Narrative Review

Narrative synthesis of mechanisms - systemic inflammation, oxidative stress, endothelial dysfunction, BBB disruption, microglial activation, prothrombotic change - linking particulate exposure to ischaemic and haemorrhagic stroke and small-vessel disease, with the observation that Pakistan has no longitudinal, biomarker or exposure-linked stroke surveillance data. Limitation: no search strategy, no new data, no quantitative synthesis; the gap statement is the contribution. Relevance: low-moderate - a call for registry-linked exposure that presupposes the monitoring network Pakistan does not have.

doi: 10.2174/0115672026460820260929113419

Li built environment sleep duration air pollution mediation Yinzhou cohort 2026 BMC Public Health

Association between built environment and sleep duration mediated by air pollution: a cohort study in China

Yinzhou cohort, 18,891 adults, baseline 2015-2017 and follow-up 2021-2023; quantile g-computation for a built-environment mixture and SEM for mediation. Per quartile of the mixture: +0.057 h/year sleep (0.039-0.075). Per IQR, PM_{10}, $PM_{2.5}$ and NO_2 were associated with -0.027, -0.048 and -0.071 h sleep. $PM_{2.5}$ mediated up to 21.5% of the green-space effect; NO_2 up to 45.3% of the walkability effect. Limitation: self-reported sleep, and SEM mediation needs no unmeasured mediator-outcome confounding - noise and light at night co-vary with traffic pollution and are only partly modelled. Relevance: low-moderate - mediation of this kind needs better spatial exposure, not more covariates.

doi: 10.1186/s12889-026-28717-y

$PM_{2.5}$ and Mn biomarkers 2026 Environ Pollut

Differential associations of $PM_{2.5}$ and particulate manganese with blood-based amyloid and tau biomarkers

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 7 Oct with no abstract. By title, $PM_{2.5}$ mass and a specific metal constituent are contrasted against blood-based amyloid and tau - the preclinical Alzheimer biomarkers. Separating a constituent from total mass against a mechanistic biomarker is precisely the design the UFP dementia-mortality cohort (5 October) and the nitrate lung-cancer cohort (3 October) both point toward. Relevance: high; on the retry list.

doi: 10.1016/j.envpol.2026.129304

Papatheodorou prenatal wildfire $PM_{2.5}$ ASD ADHD Medicaid 2026 JAMA Pediatr

Prenatal Exposure to Wildfire $PM_{2.5}$ and Risk of ASD and ADHD in Children Enrolled in Medicaid

Nationwide Medicaid (MAX) cohort of 893,212 mother-child pairs with stable ZIP code in pregnancy; 8,357 ASD and 22,302 ADHD diagnoses (cumulative incidence to age 8: 1.9% and 10.4%). Cox models with distributed-lag nonlinear terms. Per IQR (0.75 $\mu g\,m^{-3}$) of cumulative smoke $PM_{2.5}$: ASD HR 1.30 (1.07-1.59); ADHD HR 1.03 (0.91-1.17). The ASD window sat in mid to late gestation. Limitation: a 30% hazard increase per 0.75 $\mu g\,m^{-3}$ is very steep for a ZIP-level smoke fraction; smoke years and regions coincide with secular and regional shifts in ASD claims coding, and claims-based diagnosis in Medicaid depends on service access. Relevance: high - smoke-specific PM at sub-$\mu g\,m^{-3}$ contrasts is exactly where dense sensor networks add resolution over 1 km model fields.

doi: 10.1001/jamapediatrics.2026.5045

Santoso air pollution noise greenness salivary cortisol BAMSE 2026 J Expo Sci Environ Epidemiol

Air pollution, road traffic noise, and greenness in relation to salivary cortisol in young adults

BAMSE birth cohort, n=1,148 at age 24; linear regression of salivary cortisol on prior-year modelled pollutants, noise and NDVI. No overall associations. In post hoc male subgroups, PM_{10} was associated with higher morning cortisol (beta 0.058, 0.003-0.112) as were NOx and noise, with greenness inverse; females showed opposite trends. Limitation: the only signals are exploratory sex-by-season subgroup contrasts after a null main analysis, with lower CIs at ~0; single-day cortisol is highly variable. Treat as null. Relevance: low - low-contrast Stockholm exposures modelled at the address.

doi: 10.1038/s41370-026-00995-z

Shanghai prenatal UFP postpartum depression 2026 Environ Pollut

Prenatal exposure to ultrafine particles and positive screening for postpartum depression: a prospective pregnancy cohort in Shanghai, China

2,647 pregnant women, two centres; 20.4% screened positive at 6-10 weeks postpartum. Per IQR, first-trimester UFP OR 1.252 (1.065-1.471) and whole-pregnancy UFP OR 1.104 (1.047-1.164); both survived adjustment for $PM_{2.5}$, which had no stable independent association. Stronger with male infants. Limitation: EPDS is a screen, not a diagnosis, and the UFP surface's spatial resolution and validation are not stated - UFP gradients decay over ~100-300 m from roads, so residential misclassification is likely large and non-differential only if traffic proximity is unrelated to deprivation, which in Shanghai it is not. Relevance: high - a second UFP-specific neuro signal this month that $PM_{2.5}$ mass does not carry, and that optical nodes cannot measure.

doi: 10.1016/j.envpol.2026.129330

Tang source PM dementia 2026 Eur Arch Psychiatry Clin Neurosci

Associations of long-term exposure to source-specific $PM_{2.5}$, lifestyle, and genetic susceptibility with dementia risk: a prospective cohort study

Cox models with time-varying source-specific $PM_{2.5}$; median follow-up 11.9 years, 5,648 incident dementia cases. Per IQR: domestic HR 1.05 (1.01-1.08), road transport 1.12 (1.04-1.21), other transport 1.12 (1.06-1.19), industry 1.07 (1.03-1.11). High exposure plus high polygenic risk gave a 236-352% excess versus the lowest-lowest stratum. Limitation: source fields come from a dispersion model whose sources are spatially collinear, and per-IQR scaling differs by source, so the ranking of transport over domestic is not a per-ug comparison; the joint-effect excess mostly restates genetic risk; the abstract does not name the cohort or country. Relevance: moderate - source attribution at residence is what a speciated or BC-capable node network could provide directly.

doi: 10.1007/s00406-026-02391-y

Wang air pollution adolescent anxiety depression Sichuan cross-sectional 2026 BMC Public Health

Air pollution and mental health in adolescents: a cross-sectional study in Sichuan, China

71,364 school students aged 10-19, surveyed over two winters (2022-23, 2023-24). Per IQR, PM_{10} anxiety OR 1.29 (1.22-1.36) and O_3 1.24 (1.19-1.29), with similar depression associations; a weighted combined score rose stepwise. $PM_{2.5}$ and SO_2 were inversely associated. Limitation: exposure was assigned from the IP address of the online survey, i.e. a point-in-time location of the device, and the PM_{10}-positive/$PM_{2.5}$-inverse split from highly correlated pollutants is the signature of collinearity and spatial confounding rather than a fraction-specific effect. Relevance: low - coarse exposure assignment.

doi: 10.1186/s12889-026-28370-5

Wang et al. 2026c Environ Pollut

Long-Term Air Pollution Exposure and Brain Structural Decline: A Longitudinal Analysis

4,276 middle-aged and older adults with repeated brain MRI; global, grey- and white-matter volumes and white-matter hyperintensity burden regressed on modelled residential $PM_{2.5}$ and PM_{10} in fully adjusted multivariable models. Higher long-term particulate exposure was associated with accelerated global and regional grey-matter loss and greater WMH progression, which the authors read as both a neurodegenerative and a cerebrovascular pathway. Limitation: the abstract gives no effect sizes, no confidence intervals, no exposure contrast and does not name the cohort or country - unusual for a quantitative imaging analysis, and it makes the result unassessable as reported; repeat-MRI subsamples are also selected for survival and compliance. Relevance: moderate - brain structure is the mediator between the UFP dementia-mortality signal and clinical disease, but this record cannot be entered as an effect estimate.

doi: 10.1016/j.envpol.2026.129269

Wang multi-omics Mendelian randomization $PM_{2.5}$ depression 2026 Research Square (preprint)

Multi-omics Mendelian Randomization Reveals the Causal Association and Mediating Mechanisms of $PM_{2.5}$ with Depression: A Systematic Study Integrating Transcriptomics, Metabolomics, and Radiomics

LD-score regression found genetic correlation between '$PM_{2.5}$ exposure' and MDD (rg 0.334, P 1.9e-11); two-sample MR gave OR 1.27 (1.07-1.51) for MDD. Two-step MR identified five blood genes, six brain-region genes and two blood fatty-acid ratios as mediators; no mediation through CSF metabolites or imaging traits. Limitation: a GWAS of residential $PM_{2.5}$ indexes where people live, so its instruments capture SES, urbanicity and migration; the genetic correlation itself shows horizontal pleiotropy with MDD is likely, which violates the exclusion restriction. The OR has no exposure unit. Preprint. Relevance: low - not an exposure-measurement result.

doi: 10.21203/rs.3.rs-10994413/v1

Yang heat anomaly wildfire smoke $PM_{2.5}$ incident depression All of Us 2026 J Affect Disord

Heat anomaly, wildfire smoke, social-material vulnerability, and incident depression in a national EHR-linked cohort: A preregistered secondary analysis

Preregistered Cox analysis in All of Us: 179,218 adults free of recorded depression, 5,484 incident cases over a mean 371 days. Exposures were the 30-day pre-index PRISM maximum-temperature anomaly, Stanford ECHO smoke $PM_{2.5}$ and a social-material vulnerability (SMV) composite. Heat anomaly: HR 1.06 (1.03-1.10) and 1.08 (1.04-1.11) per SD; SMV: HR 1.36 (1.33-1.40) per SD. Mean smoke $PM_{2.5}$ showed no association in the primary model; secondary smoke metrics were heterogeneous by metric and follow-up window. Limitation: exposure is assigned at the 3-digit ZIP level and frozen at baseline, so smoke-episode contrast is heavily smoothed and a null is weak evidence of no effect; EHR ascertainment depends on care access. Relevance: high - this is the misclassification problem that residence-scale sensor-corrected smoke $PM_{2.5}$ exists to fix.

doi: 10.1016/j.jad.2026.122663

Zhang CLHLS $PM_{2.5}$ components cognitive disorders lifestyle mediation 2026 J Glob Health

Association between long-term ambient air pollution exposure and cognitive disorders among older adults in China: the mediating role of lifestyle

Chinese Longitudinal Healthy Longevity Study; Cox, Bayesian WQS and mediation models. Per 10 $\mu g\,m^{-3}$, $PM_{2.5}$ HR 1.005 (1.001-1.009), sulphate 1.030 (1.003-1.056), nitrate 1.020 (1.003-1.037), ammonium 1.033 (1.005-1.062), O_3 1.006; mixture OR 1.028 (1.006-1.040) with O_3 carrying the largest WQS weight. Favourable vs unfavourable lifestyle HR 0.488 (0.336-0.710); lifestyle mediated 8.4-9.5% of the pollutant associations. Limitation: an HR of 1.005 per 10 $\mu g\,m^{-3}$ is at the floor of detectability and the abstract gives neither n, case count nor how cognitive disorder was ascertained (MMSE cut-offs in CLHLS are education-sensitive); 'mediation' by lifestyle that is itself a confounder of residential exposure is not identifiable. Relevance: low - gridded component surfaces, no sensor component.

doi: 10.7189/jogh.16.04307

Zhou short-term indoor $PM_{2.5}$ cognition HRV randomized crossover 2026 iScience

Short-term indoor fine particulate matter exposure impedes cognitive performance and autonomic nervous system in adults

Randomised single-blind crossover in 90 working-age adults, two stable chamber conditions set by an air purifier with real or sham filter: 1.70 vs 31.22 $\mu g\,m^{-3}$ $PM_{2.5}$ from candle burning. The high condition lowered heart rate variability and attention and memory scores, but not coordination or reasoning; reduced HRV partly mediated some cognitive declines. Effect sizes with CIs are not given in the abstract. Limitation: candle smoke carries VOCs, CO, odour and visible haze, so blinding and the PM-specific attribution are both weak; multiple cognitive domains invite selective significance. Relevance: medium - controlled indoor steps of this size are what consumer PM monitors can verify in occupied rooms.

doi: 10.1016/j.isci.2026.117904

Respiratory & allergic — 40 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (40 OF 40 RECORDS IN THIS SUBTOPIC)

Argersinger air pollution chronic rhinosinusitis review 2026 Curr Opin Otolaryngol Head Neck Surg

The link between air pollution and chronic rhinosinusitis

A short review of recent work on $PM_{2.5}$ and chronic rhinosinusitis. $PM_{2.5}$ is associated with CRS prevalence and severity and with worse disease-specific and quality-of-life outcomes after endoscopic sinus surgery. In nasal epithelial cells from CRS patients it worsens remodelling and barrier dysfunction and reduces ciliogenesis. Traffic, industrial, wildfire and pollen-related PM contribute independently. Limitation: non-systematic, no pooled estimates. Most cited associations are cross-sectional or claims-based with modelled residential exposure, and CRS phenotype (with or without polyps) is rarely separated. Relevance: low.

doi: 10.1097/moo.0000000000001164

Dublin air pollution fungal spores respiratory admissions time series 2026 Eur J Public Health (conference abstract)

Short-term effects of air pollution and aeroallergens on respiratory admissions

Dublin daily HIPE admissions with NO_2, $PM_{2.5}$, PM_1, fungal spores and meteorology; negative-binomial time series, lags 0-7. NO_2 was consistently associated with asthma, COPD and all respiratory admissions (~3-5% per unit, $p < 0.01$); particulate effects were 'outcome-specific' and not quantified in the abstract; spores showed no independent short-term effect. Limitation: per-unit NO_2 effects of 3-5% are implausibly large per ppb or $\mu g\,m^{-3}$ and suggest an unstated scale; study years and PM results are missing; conference abstract. Relevance: low-moderate - Dublin's PM_1 series is unusual, and spore counts pair naturally with optical sensor coarse channels.

doi: 10.1093/eurpub/ckag169.1252

Gonehgar sand dust storms asthma Asia systematic review 2026 J Asthma

Geographic Patterns of Asthma Morbidity: A Systematic Review of Sand and Dust Storm (SDS) Impacts in Asia

Thirty-three studies to May 2023 (13 cohort, 12 panel), mostly Japan, Korea and the Middle East. Nearly all reported positive associations between SDS days and asthma exacerbations, admissions or ED visits, peaking at 1-5 day lags, with larger effects in children and older adults and stronger effects in East Asia than the Middle East. No pooled estimate. Limitation: the search is 29 months old, English-only, and SDS exposure is defined inconsistently (event days vs PM_{10} thresholds), which precludes meta-analysis and may explain the East-West contrast. Relevance: moderate - coarse-mode dust is where optical LCS under-read most, so SDS health evidence built on dust-flagged sensor data would need a coarse-fraction correction.

doi: 10.1080/02770903.2026.2751845

Guangdong COPD surveillance PM_1 $PM_{2.5}$ PM_{10} PAF 2026 Fundamental Research

Association between chronic respiratory diseases and the attributable risk of long-term exposure to atmospheric particulate matters among middle-aged and elderly Chinese population

COPD surveillance sample of 7,371 adults >=40 y (CRD 19.4%, COPD 10.0%). Each 10 $\mu g\,m^{-3}$ of PM was associated with higher prevalence of every CRD outcome, more so in rural residents and biomass users. PAF of CRD: PM_1 19.09% (95% CI 12.41-25.02), $PM_{2.5}$ 18.98% (12.90-24.90), PM_{10} 21.40% (13.84-27.67); COPD highest. Limitation: cross-sectional prevalence, so PAF here is causal only by assumption; $PM_1/PM_{2.5}/PM_{10}$ come from the same satellite-ML product and are near-collinear, so the three PAFs are one estimate three times; biomass fuel confounds rural contrasts. Relevance: low-moderate - PM_1 adds nothing distinct when derived from the same model.

doi: 10.1016/j.fmre.2025.04.013

Guerra-Ruiz cytokines volcanic emissions Cumbre Vieja ASHES 2026 Ecotoxicol Environ Saf

Systemic cytokine signatures associated with volcanic emisions exposure and respiratory disease after the cumbre vieja eruption-ashes project.

Cross-sectional ASHES sub-study after the 2021 Tajogaite eruption: 102 adults (66 general population, 13 occupationally exposed, 23 asthmatic). Workers had the longest exposure (median 10 h vs 6 h) and higher IL-2, IL-7, IL-17A and IL-17F; asthmatics showed broader TH1/TH2/TH17 activation, and IL-4 correlated with exposure time in asthmatics (r 0.79). Limitation: exposure is hours of presence, not measured PM or SO_2, and 13 workers is too few for a stable comparison; 48 markers tested without stated correction. Relevance: medium - the eruption had dense monitoring; linking those PM records to individuals would turn this into an exposure-response study.

doi: 10.1016/j.ecoenv.2026.120950

Harb idiopathic subglottic stenosis geographic clustering ecological 2026 Laryngoscope Investig Otolaryngol

Geographic Clustering of Idiopathic Subglottic Stenosis in North America: An Ecological Cohort Study

4,450 members of an online iSGS support group (99.3% female) mapped to state/province and scaled by census White female populations. Prevalence clustered in dry, high, warm areas: Wyoming 18.1, Utah 17.4, Colorado 13.9 per million vs New Hampshire 1.2; Alberta 85.4 vs Quebec 2.9. $PM_{2.5}$ was not independently associated with prevalence but modified the elevation association. Limitation: the numerator is self-selected membership of an online group, so 'prevalence' reflects diagnosis, awareness and referral patterns; state-level ecology cannot separate climate from health-system access. Level 4 evidence by the authors' own grading. Relevance: low - a null for PM at state scale, hypothesis-generating only.

doi: 10.1002/lio2.70607

Howlett-Downing and Wichmann 2026 Int J Environ Health Res

Triangulating health risks of $PM_{2.5}$ and trace elements in Pretoria: evidence from health risk assessment, case-crossover, and concentration-response function analyses

Three methods applied to the same secondary dataset: hazard quotients, a case-crossover analysis and concentration-response functions. Hazard quotients exceeded unity in adults, children and infants; the case-crossover OR was 1.027 (95% CI 1.006-1.049) per 10 $\mu g\,m^{-3}$ $PM_{2.5}$. Meeting the South African standard would avert 158 admissions annually, rising to 342 and 748 under the WHO daily and annual guidelines; element-specific functions give 110-608. Limitation: the three methods are not independent evidence - they re-use one exposure series from a small number of sampling days, and summing element-specific attributable cases double-counts the mass they are part of; hazard quotients are screening-level and not comparable to epidemiological risk. Relevance: moderate - Southern African monitoring is thin, and a saturation low-cost network would resolve the within-city contrast this design cannot.

doi: 10.1080/09603123.2026.2733112

ISVOLCAN Tajogaite volcano La Palma lung function obstructive pattern 2026 Eur J Public Health (conference abstract)

Lung function after Tajogaite volcano eruption in La Palma: ISVOLCAN study

Random population sample of 604 adults (mean age 50; 62% women) with spirometry a median 424 days after the 85-day eruption. Obstruction prevalence 28.8%. Western (more exposed) residence: OR 1.64 (1.01-2.67); 1.74 (1.03-2.93) without prior respiratory disease. Intermediate SO_2 exposure OR 1.98 (1.18-3.34). Limitation: a fixed FEV1/FVC <70% cut-off over-calls obstruction in older adults, no pre-eruption baseline exists, and residence area is a coarse proxy that cannot separate SO_2 from volcanic PM; non-monotone SO_2 categories. Conference abstract. Relevance: moderate - volcanic events are a case for rapidly deployed PM and SO_2 sensor networks.

doi: 10.1093/eurpub/ckag169.069

Jung peaker power plant emissions ED urgent care visits Massachusetts 2026 J Clim Change Health

Individual-level health effects of living near a gas-fired "Peaker" power plant

Primary-care patients living within 30 km of a gas-fired peaker plant, 2017-2021; 2,310 ED/UC visits analysed with a case-crossover design. Exposure was a daily HyADS parcel-trajectory index (mean 1.4, SD 16, reflecting plant intermittency); models adjusted for heat index and non-peaker $PM_{2.5}$ (mean 8.5 $\mu g\,m^{-3}$). Respiratory visits OR 1.044 (1.001-1.090) per unit; non-white patients OR 1.347 (1.034-1.755). Cardiovascular, headache and pregnancy visits were null. Limitation: the exposure is unitless and ZIP-code assigned, so the OR has no concentration scale. Peaker 'on' days coincide with peak heat and grid stress. A lower CI of 1.001 across four outcome families invites multiplicity concerns. Relevance: high - intermittent point sources are exactly what fence-line LCS can time-resolve.

doi: 10.1016/j.joclim.2026.100762

Khan Gambia personal $PM_{2.5}$ pneumococcus 2026 medRxiv (preprint)

Longitudinal personal environmental exposure and Streptococcus pneumoniae colonisation in a cohort of children 5-14 years old in rural Gambia

134 children aged 5-14, 554 visits with paired personal exposure and culture. Geometric mean personal $PM_{2.5}$ 12.6 $\mu g\,m^{-3}$ (12.2-13.1), CO 0.16 ppm; colonisation 39.7%. Neither $PM_{2.5}$ nor CO was associated with colonisation after adjustment, including in within/between-child decomposition; between-child temperature was inversely associated (OR 0.74, 0.56-0.98). Limitation: personal $PM_{2.5}$ of 12.6 $\mu g\,m^{-3}$ in solid-fuel households is implausibly low for wood-smoke exposure and suggests the monitor or its wear compliance under-captured cooking peaks; a null under that exposure contrast is uninformative. Relevance: high - a clear case where personal-monitor validation, not the epidemiology, decides the result.

doi: 10.64898/2026.10.08.26364987

Kim interstitial lung disease changing climate Personal View 2026 Lancet Respir Med

Interstitial lung diseases in a changing climate: emerging risks and opportunities to improve care

A Lancet Respiratory Medicine Personal View on climate-related exposures in interstitial lung disease. Air pollution is the established risk factor. Extreme heat, viral illness, damp housing, pollen, disasters and desertification are framed as emerging threats. Wildfire smoke, occupational ILD under climate change and disaster preparedness are named as the main gaps. Limitation: an opinion synthesis with no quantitative evidence grading. ILD cohorts are small and heterogeneous, so most climate-ILD links are extrapolated from COPD and asthma. Relevance: low-medium - wildfire-smoke alerting for ILD patients is a concrete use case for hyperlocal sensing.

doi: 10.1016/s2213-2600(26)00252-3

Kinzunga Kinshasa $PM_{2.5}$ respiratory 2026 Research Square (preprint)

Acute Respiratory Impacts of Fine Particulate Matter ($PM_{2.5}$) in Kinshasa: Epidemiological Evidence and Pulmonological Implications

13,049 adult admissions at one hospital, 2018-2022; mean daily $PM_{2.5}$ 46-50 $\mu g\,m^{-3}$. Per 10 $\mu g\,m^{-3}$ over lag 0-5, adjusted OR 1.01 (1.00-1.03), strongest at 35-64 years and in Tshangu and Funa. Limitation: a single-hospital catchment against a city-wide exposure series, and an estimate whose lower bound sits on the null; the sensor and model fusion step is described but its validation against any reference in Kinshasa is not. Relevance: high - one of the first African health analyses whose exposure series is built on low-cost sensors, which makes the uncorrected sensor error the main threat to the inference.

doi: 10.21203/rs.3.rs-11301519/v1

Ko-CHENS 36 months 2026 Allergy Asthma Immunol Res

Exposure to Environmental Pollutants and the Risk of Respiratory Illnesses in 36-Month-Old Children: The Korean Children's Environmental Health Study

4,059 children with measured indoor pollutants at a 36-month home visit, blood lead, mercury and cadmium, urinary endocrine-disruptor metabolites and questionnaire outcomes, analysed by quantile g-computation. Allergic rhinitis was most prevalent; children with respiratory illness had more tobacco-smoke exposure and pet ownership, and higher dust-mite levels with recurrent wheezing and pneumonia. The mixture was positively associated with allergic rhinitis (driven by lead and phthalates) and with pneumonia (phthalates). Limitation: the indoor PM measurement is a single home visit standing in for three years of exposure, and in a mixture model dominated by biomarkers with month-long half-lives, that one spot measurement is the weakest term - which is a likely reason PM does not drive any of the mixture associations. Relevance: high - a continuously logging indoor node would replace the single visit, and this cohort shows what the single visit costs.

doi: 10.4168/aair.2026.18.5.668

Kobza respiratory ambulatory care air quality Upper Silesia 2026 Int J Occup Med Environ Health

The assessment of respiratory diseases in ambulatory care attributable to air quality in the Upper Silesian agglomeration

Five-year ecological analysis correlating daily mean concentrations from about 50 monitoring stations with National Health Fund outpatient incidence for seven respiratory ICD codes across eight regions of the Upper Silesian voivodeship. The strongest signal was for SO_2, not PM: tau 0.99 and rho 0.9 in regions 7 and 8, rho 0.78 in regions 1 and 4. Limitation: rank correlations on aggregated series without control for season, influenza, temperature or autocorrelation - seasonality alone would generate these coefficients, so the results carry no causal or PM-specific information. Relevance: low - included because $PM_{2.5}/PM_{10}$ were analysed; the coal-heating region is a classic case where denser residential PM sensing would sharpen exposure contrast.

doi: 10.13075/ijomeh.1896.02875

Kusumaningtiar school children $PM_{2.5}$ PM_{10} health risk systematic review 2026 Research Square (preprint)

Environmental Health Risks in School Children Due to Air Pollution Exposure: Systematic Literature Review

PRISMA review (PROSPERO CRD420261282406) of 12 studies published 2015-2025 on school $PM_{2.5}$ and PM_{10}, combining environmental health risk assessment, epidemiology, MPPD dosimetry and STELLA modelling. Several studies reported risk quotients >1; one Bandung study found inverse $PM_{2.5}$-lung function associations; dosimetry indicated higher airway deposition in children. Limitation: 12 heterogeneous studies with mixed methods and no quantitative synthesis; RQ>1 depends on default intake assumptions rather than measured outcomes. Preprint. Relevance: medium - underlines the need for standardised school-level monitoring that low-cost networks could provide.

doi: 10.21203/rs.3.rs-11094408/v1

Li size-specific PM lung function China Pulmonary Health Study 2026 Environ Health (ACS)

Associations between Size-Specific Particulate Matter and Lung Function: Results from the China Pulmonary Health Study

50,991 adults >=20 y from the China Pulmonary Health Study; multivariable linear models on long-term size-resolved PM. Per IQR PM_1: FEV1 -36.84 mL (-45.91 to -27.76), FVC -34.63 mL (-45.48 to -23.79); large-airway decrements grew as particle size fell, whereas the coarse fraction $PM_{2.5}$-10 gave the largest small-airway (FEF75, FEF25-75) decrements. Limitation: size fractions are derived from correlated modelled surfaces (PM_1 and $PM_{2.5}$ share most of their mass), and the abstract reports no multi-fraction model, so 'size-specific' ranking may reflect differential exposure-model error rather than biology; cross-sectional. Relevance: high - optical LCS report PM_1 and coarse bins natively, and this is evidence that those channels carry distinct health signal worth calibrating, not discarding.

doi: 10.1021/envhealth.5c00676

Li spatiotemporal causal machine learning $PM_{2.5}$ asthma Colorado counties 2026 GeoHealth

A Spatiotemporal Causal Machine Learning Framework for Quantifying the Impact of $PM_{2.5}$ on Asthma at the Community Level

Colorado county-level monthly asthma hospitalisations and ED visits, 2011-2019, modelled with a spatiotemporal representation-learning network plus a causal-inference layer that estimates counterfactual outcomes under hypothetical $PM_{2.5}$ reductions. Estimated effects were heterogeneous, larger in communities with compounded environmental and social stressors, with wider uncertainty in data-sparse counties. No effect size per $\mu g\,m^{-3}$ is given in the abstract. Limitation: county-month aggregation with monthly $PM_{2.5}$ is ecological; 'causal' here rests on unconfoundedness given learned covariates, which a neural model cannot verify, and sparse rural counties are exactly where exposure error peaks. Relevance: moderate - the model's uncertainty is concentrated where monitors are absent, an explicit argument for sensor densification in rural Colorado.

doi: 10.1029/2025gh001735

Lovric ragweed sensitisation PM_{10} pollen model comparison Croatia ATOPICA 2026 Children (Basel)

Outdoor Environmental Risk Factors for Paediatric Ragweed Sensitisation: A Formal Model Comparison

Re-analysis of 3,520 Croatian children from ATOPICA within EDIAQI; logistic models compared by AIC, adjusted for age and parental atopy. Sensitisation prevalence 10.8%. PM_{10} and categorical pollen area fit equally well (delta AIC 1.7) because both encode the same inland-coastal contrast; NO_2 added nothing. Transitional woodland-shrub aOR 0.93 (0.90-0.97) survived Bonferroni; rural residence aOR 0.69 (0.50-0.96). Limitation: the authors state it directly - area-level exposure cannot separate particulate pollution from pollen, so a PM contribution can be neither confirmed nor excluded. Relevance: moderate - an explicit case where only sub-regional, co-located PM and pollen measurement could break the collinearity.

doi: 10.3390/children13101366

Luglio air pollution pediatric asthma hospitalizations Louisiana vulnerable groups 2026 Environ Epidemiol

Associations between air pollution and pediatric asthma hospitalizations across vulnerable groups in Louisiana

28,499 asthma hospitalizations in children in Louisiana 2010-2019, time-stratified case-crossover with DLNM to lag 3. Cumulative lag 0-3: $PM_{2.5}$ OR 1.079 (1.052-1.107), O_3 1.082 (1.050-1.116), NO_2 1.071 (1.034-1.111), with increments as defined in the paper. In multipollutant models O_3 and NO_2 attenuated while $PM_{2.5}$ persisted. Effect modification by race and Medicaid status was not significant, although lag patterns differed (lags 1-2 Black children, lag 3 White children). Limitation: exposure assigned from area-level estimates in a state with sparse monitoring outside the industrial corridor, so differential misclassification by race/rurality could mask modification; NO_2 covers only 2010-2016. Relevance: medium - higher-burden groups sit where reference coverage is thinnest.

doi: 10.1097/ee9.0000000000000544

Meng traffic noise air pollution lung function randomised crossover 2026 Environ Res

Independent and interactive effects of traffic-related noise and air pollution on lung function: A randomized crossover study in healthy young adults.

Randomised crossover with four exposure conditions (noise and air pollution separated by intervention) in 51 healthy young adults beside a busy road at evening rush hour; pollutants and noise measured each session, mixed models adjusted for age, sex and BMI. Per 10 $\mu g\,m^{-3}$ PM_{10}, FEF75 fell 0.215 L/s; per 10 $\mu g\,m^{-3}$ O_3, 0.224 L/s. Noise independently reduced dynamic ventilatory indices, and exploratory interaction terms suggested noise amplified PM_1- and $PM_{2.5}$-associated FEV1 and small-airway decrements. No CIs in the abstract. Limitation: the four interventions are not described in the abstract and exposure conditions cannot be blinded, and pollutant contrasts within a single rush-hour session are small and collinear, so the interaction findings are hypothesis-generating. Relevance: medium - co-located acoustic and PM nodes are needed to separate the two stressors.

doi: 10.1016/j.envres.2026.125963

Morales El Paso port-of-entry $PM_{2.5}$ immigration COPD census tracts 2026 J Immigr Minor Health

Breathing at the Border: $PM_{2.5}$ Exposure, Immigration, and COPD Across Census Tracts Close to Ports-of-Entry

El Paso census tracts; NASA satellite-derived $PM_{2.5}$ merged with CDC tract COPD prevalence and ACS social vulnerability; GLMs with distance to the port-of-entry. $PM_{2.5}$ and percent foreign-born non-citizens were positively associated with COPD prevalence; tracts within three miles of a port had higher prevalence, and the $PM_{2.5}$ effect was stratified by proximity. No coefficients or CIs in the abstract. Limitation: CDC PLACES prevalence is itself small-area modelled from BRFSS demographics, so regressing it on demographic covariates is partly circular; smoking is not addressed; satellite $PM_{2.5}$ at ~1 km cannot resolve truck-idling gradients at bridge queues. Relevance: high - port-of-entry diesel queues are a textbook case for dense LCS siting.

doi: 10.1007/s10903-026-02022-7

Oliveira air pollution HRV lung function schoolchildren Cubatao 2026 Epidemiol Serv Saude

Effect of air pollution on heart rate variability and pulmonary function in schoolchildren: a cross-sectional study, Cubatao, Brazil, 2017.

Convenience sample of 77 schoolchildren in the Cubatao industrial zone, with pollutant levels from the state agency's monitors. Associations reported in the abstract are for gases: SO_2 and HFnu -0.68 (-1.27 to -0.11) daily, -1.29 (-2.21 to -0.38) over three days; O_3 and HFnu -0.32 (-0.62 to -0.02); NO_2 with 0.23% lower FEV1 and 0.30% lower FEF25-75 (no CI given). No PM association is reported in the abstract. Limitation: 77 children, one fixed-site monitor and a single visit leave essentially no exposure contrast across subjects - the estimates rest on day-to-day variation within a cross-sectional design, with multiple pollutants and lags tested. Relevance: low - illustrates the exposure-assignment ceiling of a single regulatory monitor.

doi: 10.1590/s2237-96222026v35e20250368.en

$PM_{2.5}$ constituents TB contacts 2026 Ann Am Thorac Soc

Differential effects of $PM_{2.5}$ constituents on tuberculosis infection and disease progression in 40,379 student close contacts

METADATA ONLY. No abstract at harvest. Title indicates a constituent-resolved analysis of $PM_{2.5}$ and latent infection and progression among 40,379 student close contacts of TB cases. Recorded for coverage; no estimate inferred. Relevance: moderate - a large contact cohort is an unusually clean design for an infection endpoint.

doi: 10.1093/annalsats/aaoag325

Pedersen early-life air pollution lung function COPSAC 2026 BMJ Open Respir Res

Impact of air pollution exposure in early life on lung function development in the COPSAC cohorts

Two Copenhagen birth cohorts (n=846) with repeated spirometry and plethysmography from age 3 to 18 years; linear mixed models. Per IQR (2.2 $\mu g\,m^{-3}$) of postnatal $PM_{2.5}$: FEV1 at age 6 -2.70% (-4.27 to -1.10) and specific airway resistance +7.30% (1.87-13.07). The deficit and the resistance increase persisted in growth trajectories through childhood, stronger in males. Limitation: COPSAC is an asthma-enriched cohort (COPSAC2000 enrolled children of asthmatic mothers), so effect sizes may not transfer to the general population; a 2.2 $\mu g\,m^{-3}$ IQR in Copenhagen is small and partly a proxy for urban residence. Relevance: medium - effects at very low contrast argue for address-level precision rather than municipal means.

doi: 10.1136/bmjresp-2025-003297

Perez-Garcia nasal EWAS respiratory biomarkers Mexican American CHAMACOS 2026 medRxiv

Nasal Epigenome-Wide Association Study of Respiratory and Allergic Biomarkers in Mexican American Young Adults

Preprint. Nasal EWAS (EPICv2, 804,220 CpGs) in 354 CHAMACOS young adults aged 22, replicated in 540 Project Viva adolescents aged 13. Forty-nine CpGs were associated with FeNO and 557 DMRs with respiratory outcomes (FDR<0.05); 26 CpGs and 103 DMRs replicated (e.g. LRRFIP1, EPX, IL13). Ten FeNO-related CpGs were associated with prenatal and/or past-year $PM_{2.5}$ or O_3 at nominal p<0.05 only. Limitation: the air-pollution step is a secondary look-up without multiple-testing control, so the $PM_{2.5}$ link is hypothesis-generating; cross-sectional nasal sampling cannot order exposure and methylation. Relevance: low-medium - an agricultural-valley cohort where PM sources are local and poorly served by regulatory monitors.

doi: 10.64898/2026.10.27.26366640

Poland air pollution COPD asthma emergency admissions DLNM heating season 2026 Eur J Public Health (conference abstract)

Air pollution and respiratory admissions in Poland: a nationwide DLNM analysis of COPD and asthma

National registry: 625,597 COPD and 153,563 asthma emergency admissions (32,893 paediatric), DLNM lag 0-10, stratified by heating season. Associations appeared only in the heating season: COPD RR per IQR $PM_{2.5}$ 1.044, PM_{10} 1.039, SO_2 1.055; lead 1.072, benzo[a]anthracene 1.057. Adult asthma stronger ($PM_{2.5}$ 1.100). In children, PM was not significant; arsenic (1.124) and ozone (1.115) were. NO_2 null throughout. No CIs in the abstract. Limitation: component and pollutant RRs come from single-pollutant models on collinear coal-combustion tracers, so 'lead' and 'PAH' effects cannot be separated from $PM_{2.5}$; conference abstract. Relevance: moderate - heating-season coal signatures are where speciation-capable nodes add value over mass.

doi: 10.1093/eurpub/ckag169.807

Sao Paulo paediatric 2026 Front Pediatr

Climate and air pollution are associated with pediatric respiratory hospitalizations in Sao Paulo, Brazil

Paediatric emergency visits and admissions recorded in 2022, geocoded, with exposures assigned by inverse-distance weighting and multivariable logistic models for two outcomes. Higher ozone (OR 1.012, 1.005-1.019), higher relative humidity (1.016, 1.005-1.027) and spring were associated with greater odds that a respiratory encounter was a bronchiolitis admission; lower temperature was associated with the unspecified-ARI outcome (0.887, 0.793-0.993 per unit). Limitation: the outcome is conditional on presenting - these are odds of diagnosis given an encounter, not risk of disease, so the design cannot speak to incidence; IDW over a sparse monitor network in a metropolitan area of 21 million is the same interpolation problem the 2 October Everglades record quantified. Relevance: moderate - no PM estimate reaches significance here, which is itself informative about what IDW exposure does to a signal.

doi: 10.3389/fped.2026.1899313

Shi air pollution pulmonary tuberculosis incidence China GTWR 2026 Front Public Health

Analysis of influencing factors of atmospheric pollution on pulmonary tuberculosis incidence in China from 2014 to 2020

Provincial panel 2014-2020 (31 units), spatial autocorrelation, space-time scanning and geographically and temporally weighted regression (R2 0.889), with 1-5-year moving-average exposure windows. High-burden clusters persisted in Xinjiang, Xizang and Qinghai (RR 3.56). $PM_{2.5}$ coefficients ran 'east-negative, west-positive'; SO_2, NO_2 and CO were jointly positive in the north-east; O_3 mostly negative. Limitation: 217 province-years and locally varying coefficients amount to a flexible fit of an ecological surface; the sign flip of $PM_{2.5}$ across China shows confounding by development and TB control, not toxicity. Relevance: low.

doi: 10.3389/fpubh.2026.1951817

Sittitoon Thailand haze $PM_{2.5}$ symptoms inequality 2026 Front Public Health

Health inequalities in $PM_{2.5}$-related symptoms during Thailand's haze season: the role of education, sociodemographic, lifestyle, protective behaviors, and environmental factors

3,916 adults interviewed January-April 2025, mixed-effects logistic models with latent-class protective-behaviour phenotypes. Symptom prevalence: respiratory 80.7%, eye 52.8%, skin 48.7%, neurological 33.1%. Higher provincial $PM_{2.5}$ was independently associated with respiratory and dermatological symptoms; non-compliers (48.6%) and the neutral group had higher odds of respiratory and neurological symptoms than compliers (33.4%). Female sex, younger age, comorbidity, alcohol, urban residence and lack of village-health-volunteer contact also raised risk. Limitation: one satellite value per province per month is assigned to every respondent, so the $PM_{2.5}$ contrast is between-province and confounded with region; self-reported symptoms during a publicised haze season invite reporting bias; odds ratios are not given in the abstract. Relevance: high - northern Thailand has dense low-cost networks that could replace provincial satellite means with district-level exposure.

doi: 10.3389/fpubh.2026.1899785

Taiwan greenness pollution child asthma 2026 J Asthma Allergy

Association of Urban Greenness and Air Pollution on Asthma Visits Among Children and Adolescents in Industrial and Non-Industrial Cities in Taiwan

METADATA ONLY. No abstract at harvest. Title indicates an analysis of greenness and air pollution with paediatric asthma visits contrasting industrial and non-industrial Taiwanese cities. Recorded for coverage. Relevance: low-moderate.

doi: 10.2147/jaa.s614875

Trachalaki IPF $PM_{2.5}$ 2026 Ann Am Thorac Soc

Are Short-Term Increases in $PM_{2.5}$ Linked to Respiratory Hospitalizations in IPF?

METADATA ONLY. No abstract in PubMed, Europe PMC or Semantic Scholar at harvest; the title format suggests a research letter on short-term $PM_{2.5}$ and respiratory admissions in idiopathic pulmonary fibrosis. Recorded for coverage only - no result can be summarised, and none is inferred. Relevance: moderate if positive, because IPF is a small, identifiable cohort for which personal or home-sensor exposure is feasible.

doi: 10.1093/annalsats/aaoag317

Tullis cystic fibrosis CRS olfaction pollutants 2026 Int Forum Allergy Rhinol

Environmental Exposures, Chronic Sinusitis, and Olfaction in People With Cystic Fibrosis

80 adults with CF and chronic rhinosinusitis enrolled 2019-2023 at US rhinology centres. Ozone correlated with worse SNOT-22 (Rs 0.34, p=0.002), QOD-NS (Rs 0.30) and smell identification (Rs -0.30); $PM_{2.5}$, PM_{10}, CO, SO_2 and NO_2 showed no significant associations. Limitation: unadjusted Spearman correlations across six pollutants and several outcomes in 80 people, with exposure from the nearest regulatory monitor to a ZIP centroid - the PM null is uninformative rather than reassuring. Relevance: moderate - a small clinical cohort is the case where residence-level sensor data would most reduce exposure misclassification.

doi: 10.1002/alr.70204

Wang Guangdong $PM_{2.5}$ oxidative potential DTT childhood allergy 2026 Environ Sci Technol

Associations of $PM_{2.5}$ and Its Oxidative Potential with Childhood Allergy-Related Symptoms: A Cross-Sectional Study in Guangdong Province, China

131,346 children aged 6-18 in six Guangdong cities. $PM_{2.5}$ mass and DTT oxidative potential were positively associated with all three symptom outcomes; mass-normalised OP (DTTm) gave the most consistent associations, lower AIC/BIC and the greatest stability in two-pollutant models. Mixture analyses implicated several constituents; OP plausibly carried part of the Co, OC, Cu and Hg associations, the largest indirect share 36.2% for OC and rhinoconjunctivitis via DTTm. Limitation: cross-sectional symptom questionnaires; OP is presumably modelled or assigned from few filter sites, and the abstract reports no effect sizes, so the model-fit comparison cannot be weighed against exposure-error differences between metrics. Relevance: high - OP is not measurable by optical LCS, which bounds what a mass-only network can say about allergy.

doi: 10.1021/acs.est.6c05462

Wang air pollutants sleep apnea polysomnography architecture Shenzhen 2026 Environ Res

Associations of multiple ambient air pollutants and sleep apnea, sleep architecture and nocturnal oxygenation in adult patients with or without obstructive sleep apnea (OSA)

Retrospective polysomnography study of 4,049 adults in Shenzhen. Over a 0-180 day lag window, $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO and diurnal temperature range were associated with greater OSA severity. $PM_{2.5}$ and PM_{10} were associated with 1.71% and 1.32% of total sleep time more N1 and 1.51% and 1.32% less N3 sleep (increments as reported). O_3 and DTR showed the clearest associations with oxygenation; $PM_{2.5}$ had a rising exposure-response. Limitation: the lag window was chosen post hoc, which inflates false-positive risk across many pollutant-outcome pairs; a clinic-referred sample means referral itself may track season and pollution. CIs are not given in the abstract. Relevance: low - city-level exposure.

doi: 10.1016/j.envres.2026.126044

Wang climate air pollutants influenza 48 countries convergent cross mapping 2026 Glob Health Res Policy

Delayed and nonlinear associations and predictive causal signals between climate factors, air pollutants, and influenza epidemics in 48 countries

48 countries, 2022-2024; FluNet weekly positivity against national climate and pollutant series in hemisphere-specific Bayesian distributed-lag models, then convergent cross mapping (CCM) for 'predictive causal' drivers. Northern Hemisphere peak OR 1.74 (1.34-2.26) at weekly cumulative SO_2 3100 $\mu g\,m^{-3}$, lag 1.6 weeks; Southern peak for organic carbon OR 1.72 (0.65-4.58), null. CCM ranked dust $PM_{2.5}$ the leading pollutant driver (19 countries). Limitation: national-level exposure against surveillance positivity is fully ecological; three post-pandemic seasons cannot separate pollution from NPI relaxation, and CCM detects shared dynamics, not causation. Relevance: low - surveillance-scale; dust $PM_{2.5}$ is a size fraction low-cost optical sensors undercount.

doi: 10.1016/j.ghrp.2026.04.002

Wilner public safety power shutoff wildfire $PM_{2.5}$ healthcare California 2026 PLoS Med

Co-exposure to planned power outages and wildfire $PM_{2.5}$ and acute healthcare utilization in California: A time-stratified case-crossover study

Statewide California encounters for adults 20+, time-stratified case-crossover, temperature-adjusted. Severe PSPS events alone: same-day respiratory encounters OR 1.06 (1.02-1.11), stronger for COPD. Joint PSPS and +10 $\mu g\,m^{-3}$ wildfire $PM_{2.5}$ vs neither: OR 1.16 (1.10-1.22), with multiplicative interaction. Cardiovascular and psychiatric encounters null. Limitation: PSPS is circuit-level but assigned by ZIP, and outage plausibly changes the exposure itself (no power for filtration or HVAC, more time outdoors or relocation), so the interaction mixes effect modification with exposure modification. Relevance: high - outages disable mains-powered sensors and purifiers precisely during smoke; battery-backed nodes are needed to measure this window.

doi: 10.1371/journal.pmed.1005270

Yoon pollutant asthma immune mechanisms review 2026 Allergy Asthma Immunol Res

Impacts and Immune Mechanisms of Pollutant-Mediated Asthma and Airway Inflammation

Narrative review of epithelial injury, barrier disruption, oxidative stress and innate signalling (NF-kB, MAPK, PI3K/Akt, inflammasome) driving T2, non-T2 and mixed asthma phenotypes; pollutant-specific arms include PM-induced NLRP3 activation, O_3 barrier disruption, DEP-driven IL-17A and pollutant-allergen co-exposure. Acknowledges that experimental doses and compositions do not map onto real-world exposure. Limitation: non-systematic, no new data, and no dose metric that a measurement study could target. Relevance: low - background for mechanism sections.

doi: 10.4168/aair.2026.18.5.640

Yuksel COPD cigarette biomass combined exposure exacerbations Turkiye 2026 Tob Induc Dis

Clinical outcomes of COPD associated with cigarette smoke, biomass smoke, and combined exposure: A retrospective observational study

319 stable COPD patients: smoking-associated 159, biomass-associated 80, combined 80. Combined exposure presented younger (59.2 vs 61.4 and 63.5 y), with lower FEV1 (48% vs 55-56% predicted), more chronic respiratory failure (26.2% vs 17.0% and 11.2%) and, adjusted, severe exacerbations AOR 1.89 (1.22-2.96) vs smoking alone. Limitation: exposure is a retrospective chart classification with no dose for biomass, and the equal group sizes (80/80) suggest selection rather than consecutive enrolment. Pack-years differ by group and are not shown to be balanced, so 'combined' may simply mean heavier total exposure. Relevance: low - no measured particle exposure.

doi: 10.18332/tid/231161

Zhou preoperative short-term $PM_{2.5}$ postoperative pulmonary complications lung cancer surgery 2026 Am J Respir Crit Care Med

Preoperative short-term $PM_{2.5}$ Exposure and Postoperative Pulmonary Complications After Lung Cancer Surgery

Metadata only: the record (AJRCCM research letter) carried no abstract at PubMed, Crossref or Semantic Scholar on the issue date. The title reports an association of short-term $PM_{2.5}$ exposure before surgery with postoperative pulmonary complications after lung cancer resection; design, n and effect size are not available here and are not inferred. Limitation (structural): preoperative exposure windows are short, so day-of-week and seasonal surgical scheduling and hospital load are the confounders any such analysis has to defeat. Relevance: medium - a perioperative window is one where short-horizon PM forecasts could change clinical scheduling. Retry the abstract before the weekly.

doi: 10.1093/ajrccm/aamag575

Zouma et al. 2026 Toxicol Rep

Pulmonary function assessment of a community exposed to air pollution in an industrial area in Ouagadougou, Burkina Faso

146 industrial-area residents vs 146 controls (Nioko 2), Nov-Dec 2023, with ATS/ERS spirometry. Symptom prevalence was higher in the industrial area (lower airway 89% vs 64%); pre-bronchodilator FEV1 %pred and FET% were lower ($p = 0.002$); restrictive defects were more common (OR 1.70, $p = 0.035$, no CI given), especially in women. Limitation: exposure is residence only - no PM or gas measurement, unadjusted univariate OR, and occupation, biomass cooking and socioeconomic differences between districts are uncontrolled. Relevance: moderate - West African industrial districts have essentially no PM monitoring; this is the kind of study a few calibrated nodes would turn from a residence contrast into an exposure-response.

doi: 10.1016/j.toxrep.2026.102344

Reproductive & developmental — 24 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (24 OF 24 RECORDS IN THIS SUBTOPIC)

Ai air pollution IVF outcomes Chengdu retrospective cohort 2026 Int J Fertil Steril

Association between Ambient Air Pollution Exposure and Reproductive Outcomes following In Vitro Fertilization: A Retrospective Cohort Study

12,453 women with fresh embryo transfer at one Chengdu centre, 2019-2022; exposure windows defined around oocyte retrieval and transfer; GLMs adjusted for clinical, sociodemographic and paternal factors. Exposure in the 85 days before oocyte retrieval (folliculogenesis window) was associated with lower odds of biochemical pregnancy, clinical pregnancy and live birth; live birth OR 0.969 (0.954-0.984) per 10 $\mu g\,m^{-3}$ $PM_{2.5}$. Limitation: four pollutants over several overlapping windows from 4 years of gridded data raise multiplicity, and in a single-city clinic cohort the exposure contrast is mostly temporal, confounded by season and protocol changes (including COVID-era disruption). Relevance: low.

doi: 10.22074/ijfs.2026.2072188.1959

Atkin air quality excess heat pregnancy outcomes review 2026 Nurs Clin North Am

Impact of Poor Air Quality and Excess Heat on Pregnancy Outcomes.

Narrative review on combined poor air quality and excess heat in pregnancy. Argues that physiological adaptations of pregnancy raise susceptibility, that structural determinants concentrate exposure in some populations, and that nurses should educate patients, promote adaptation and advocate for climate mitigation. No quantitative synthesis in the abstract. Limitation: non-systematic; heat-pollution interaction is asserted rather than quantified. Relevance: low - clinical guidance context only.

doi: 10.1016/j.cnur.2026.06.010

Falcon-Gracia BiSC placental sncRNA air pollution maternal smoking 2026 medRxiv (preprint)

Impact of air pollution and tobacco smoke on the placental small non-coding RNA landscape

390 chorionic villus samples from the Barcelona Life Study Cohort, small-RNA sequencing; 979 sncRNA clusters (miRNA 33%, piRNA 26%, snoRNA 12%, tRF 8%). Maternal smoking was associated with miR-421 down-regulation and up-regulation of SNORD113/114 snoRNAs at the imprinted DLK1-DIO_3 locus and piR-26394. No sncRNA was associated with $PM_{2.5}$, BC or NO_2 after multiple-testing correction; several miRNAs were suggestive. Limitation: preprint; a null after FDR across ~1000 features at n=390 says little about effect absence, and pregnancy-average LUR exposure is a blunt contrast for trimester-specific placental biology. Relevance: low - a usefully reported null for PM against a positive smoking control.

doi: 10.64898/2026.10.16.26365732

Gaylord DNA methylation signature prenatal air pollution ECHO cohort 2026 Environ Epigenet

A methodology to identify a DNA methylation-based signature of prenatal air pollution exposure in the ECHO Cohort

Refines a pipeline that predicts high versus low in-utero air-pollution exposure from cord-blood DNA methylation (450K/EPIC) across multiple ECHO pregnancy cohorts. Exposures were ML-modelled at geocoded addresses; training and validation sets were matched on exposure level, site and array. Validation AUCs: $PM_{2.5}$ 0.64-0.68, NO_2 0.62-0.95, O_3 0.50-0.75, varying with region, exposure window and the high/low threshold. Limitation: the target is itself a model output, so the epigenetic classifier inherits exposure-model error, and $PM_{2.5}$ discrimination is modest; dichotomisation discards dose information. Relevance: medium - a biomarker of cumulative exposure is only as good as the exposure surface it is trained on, which is where denser monitoring would raise the ceiling.

doi: 10.1093/eep/dvag034

Ge preconception folic acid air pollution birthweight 2026 Environ Health (ACS)

Preconception Folic Acid Supplementation Associated with Attenuated Effects of Air Pollution on Birthweight Outcomes

METADATA ONLY. Title and authors only (Ge, Zhang, Liao ... Berhane, Bai et al.); no abstract retrievable from PubMed, Europe PMC, Crossref or Semantic Scholar at harvest. The title claims effect modification of the air pollution-birthweight association by preconception folic acid use. Pollutants, cohort, n and effect sizes cannot be stated. Limitation: unverifiable beyond the title; supplement use is self-reported and strongly socially patterned, so 'attenuation' by folate is exposed to confounding by SES and antenatal-care quality unless handled explicitly. Relevance: low - no sensing content evident.

doi: 10.1021/envhealth.5c00779

Guangdong CHD $PM_{2.5}$ constituents 2026 Environ Pollut

Differential Impacts of Fine Particulate Matter Constituents on Congenital Heart Disease across Critical Periconceptional Windows

Guangdong CHD Registry 2004-2022: 12,680 cases, 10,266 controls, multilevel logistic regression by exposure window. Per-IQR increments of $PM_{2.5}$ mass and constituents raised CHD odds 9-16%, strongest in the preconception window and attenuating later; BC and nitrate stayed associated during pregnancy. Conotruncal defects carried the largest ORs (1.14-1.36 per IQR). Mixture models put sulfate and BC first preconception, BC and nitrate thereafter. Limitation: constituents are modelled and mutually collinear, so 'dominant constituent' rankings are mixture-model artefacts as much as biology; registry controls and the higher-SES amplification hint at ascertainment differences. CIs absent from the abstract. Relevance: moderate - constituent-resolved windows need speciation, which optical LCS cannot supply; BC from multi-wavelength nodes is the tractable piece.

doi: 10.1016/j.envpol.2026.129345

Ha et al. 2026 Kidney Res Clin Pract

Developmental $PM_{2.5}$ exposure and stage-dependent kidney senescence

METADATA ONLY. Metadata-only record: PubMed (Korean Society of Nephrology) deposited title, authors and DOI on 1 Oct with no abstract; PubMed efetch, Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes kidney senescence after developmental $PM_{2.5}$ exposure by life stage. Neither the species nor the exposure route is recoverable. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.23876/j.krcp.26.414

Hyperemesis gravidarum PM_{10} temperature 2026 Front Public Health

Seasonal ambient temperature and particulate matter as discriminators of hyperemesis gravidarum severity: a retrospective comparative observational study

211 HG patients: temperature (AUC 0.799) and PM_{10} (AUC 0.754) discriminated severe from non-severe ketonuria; SO_2 did not. The PM_{10} association attenuated after adjustment for temperature. Limitation: hospital series with no control group, ecological exposure, and PM_{10} and temperature are seasonally collinear - the attenuation says PM_{10} was largely a season marker. Relevance: low.

doi: 10.3389/fpubh.2026.1957434

ICMR-SPIC stillbirth risk factors biomass fuel India pooled cohort 2026 Lancet Reg Health Southeast Asia

Prevalence and risk factors for stillbirths in India: secondary analysis of ICMR-Stillbirth Pooled India Cohort (ICMR-SPIC)

Ten prospective pregnancy cohorts 1994-2023 harmonised, 227,821 pregnancies, 3,935 stillbirths; weighted rate 14.8/1000 (13.4-16.3). In mixed-effects modified Poisson models, biomass fuel use aIRR 1.37 (1.21-1.57) with the largest population attributable fraction of any factor, 13.95% (9.11-18.79), ahead of nulliparity and hypertensive disorders. Limitation: biomass use is a binary household marker strongly tied to rural residence, poverty and access to obstetric care; mutual adjustment for education does not remove that, so the PAF is an upper bound, not a smoke effect. Cohorts span three decades with different ascertainment. Relevance: low - the household-air-pollution burden case, without a particle metric.

doi: 10.1016/j.lansea.2026.100841

Kalam HAPIN LPG intervention child growth stunting ages 2-5 2026 medRxiv (preprint)

Effects of a 500-day LPG cookstove and fuel intervention delivered during gestation and infancy on height-for-age (HAZ) and stunting at ages 2, 3, 4 and 5 years

HAPIN randomised trial (~3,200 pregnant women; LPG stove plus free fuel to infant age 12 months), extended follow-up of 2,169 children. The intervention cut gestational $PM_{2.5}$ from 117.5 to 80.7 $\mu g\,m^{-3}$ and infant $PM_{2.5}$ from 113.2 to 36.9 $\mu g\,m^{-3}$, but post-trial exposures converged (99.9 vs 98.7). ITT effect on repeated HAZ: beta -0.06 (-0.16 to 0.03); no effect on stunting at any age and no effect modification. Limitation: even the intervention arm stayed far above the WHO guideline during gestation, so this is a null for a partial exposure contrast, not for clean air. The exposure-response analysis that would test dose is still pending. Preprint, not peer reviewed. Relevance: medium - the trial's personal monitoring shows how far stove switching moves exposure.

doi: 10.64898/2026.10.16.26365712

Kukshal et al. 2026 Indian J Pediatr

Maternal Exposure to Air Pollution During Early Pregnancy and Congenital Heart Disease: An Exploratory Case-Control Analysis from Delhi NCR

1,556 participants (1,115 CHD cases, 441 controls) with residential exposure at conception and one month either side; multinomial regression, Bonferroni-corrected. Post-conception PM_{10} was associated with all CHD (AOR 3.36, 2.05-5.52) and VSD (AOR 4.35, 2.41-7.83); preconception PM_{10} with simple CHD (2.82, 1.69-4.71); 'poor AQI' gave AORs of 2.5-4.0. Limitation: the PM_{10} contrast (per unit or category) is not given in the abstract; cases outnumber controls 2.5:1, implying hospital controls; Delhi-NCR monitor density makes residence-level contrasts small relative to city-wide covariance; ORs of 3-4 for an environmental exposure are far above the literature and suggest selection or confounding by SES. Relevance: moderate - a case for dense sensing where between-residence contrasts are the identifying variation.

doi: 10.1007/s12098-026-06363-x

Liang CHD circulating proteins 2026 Nat Commun

A nested case-control study of air pollution and congenital heart defects via circulating proteins

Maternal periconception $PM_{2.5}$, PM_{10} and NO_2 were each associated with CHD, NO_2 most strongly; supervised sparse PLS identified pollutant-specific protein signatures in maternal and umbilical cord serum that tracked exposure and CHD, implicating FGF-4, VEGF-D, MSP and angiogenin. Limitation: the abstract gives no effect sizes or n; cord serum is sampled at delivery, months after cardiogenesis, so a cord-protein signature cannot be the mechanism of a first-trimester malformation, and supervised signature selection on the same outcome inflates association. Relevance: moderate - periconception exposure windows are short, which favours high-resolution residential estimates. Held from the 9 Oct issue.

doi: 10.1038/s41467-026-77623-9

Lin prenatal air pollution transient tachypnea newborn Taiwan 2026 Front Pediatr

Prenatal air pollution exposure and the risk of transient tachypnea of the newborn

Nationwide matched design on the Taiwan NHIRD linked to the Maternal and Child Health Database, 2004-2020: 3,091 TTN cases in first-born infants and 12,364 controls (1:4 on maternal age, sex, birth date). Exposure over the 40 weeks before birth from the nearest of 71 stations to the residential postal code. Adjusted ORs: CO 1.806 (1.312-2.487), NOx 1.010 (1.005-1.015), $PM_{2.5}$ 1.001 (1.000-1.002), the last per unit concentration with a CI touching the null. Effects concentrated in the Taipei basin. Limitation: nearest-station assignment at postal-code scale, no multipollutant model reported, and the $PM_{2.5}$ signal is marginal while CO dominates, pointing to traffic as a mixture rather than PM. Relevance: medium - traffic-gradient exposure in a dense basin is where street-level sensor networks change exposure contrast.

doi: 10.3389/fped.2026.1941575

Liu $PM_{2.5}$ components small for gestational age IVF Guangzhou 2026 BMC Public Health

Maternal exposure to PM(2.5) and its components and risk of small for gestational age in women undergoing in vitro fertilization: a propensity score-matched case-control study in Guangzhou, China.

Propensity-score-matched case-control within 1,459 IVF conceptions in Guangzhou: 166 SGA cases matched 1:1. Whole-pregnancy $PM_{2.5}$ OR 1.124 (1.015-1.245); second trimester 1.082 (1.005-1.165); third trimester 1.075 (1.002-1.154); the exposure increment is not given in the abstract. SO4, OM and BC were positively associated; mixture models ranked nitrate as the dominant driver, and no subgroup reached significance. Limitation: 332 matched women give lower CI bounds within 0.2-1.5% of the null across many component-by-window tests, with no multiplicity correction reported; the component estimates come from a modelled product whose component-specific error is unknown. Relevance: low-medium - component attribution here rests on modelled speciation that sensor networks cannot supply.

doi: 10.1186/s12889-026-28800-4

Ma et al. 2026 J Environ Manage

Maternal $PM_{2.5}$ exposure induces preeclampsia-like phenotypes in mice through placental angiogenic dysfunction and lncRNA-RUNX1 dysregulation

C57BL/6J dams given intranasal heating-season $PM_{2.5}$ through gestation developed raised blood pressure at E13.5 and E17.5, proteinuria, lower VEGF and PlGF, higher TNF-alpha, IL-6 and sFlt-1, and abnormal labyrinth architecture with impaired angiogenesis; ET-1 and NO were unchanged. Transcriptomics plus ChIP/RIP implicated lncRNAs NONMMUT087808.1 and NONMMUT044449.2 acting through RUNX1. Limitation: 3 mg/kg bolus intranasal instillation is far above any ambient-equivalent deposited dose and bypasses the nasal-pharyngeal filter, so the phenotype demonstrates sufficiency, not ambient relevance; one PM source, no filtered-air-plus-vehicle dose-response. Relevance: moderate - the sFlt-1/PlGF axis is measurable in human pregnancy and is the natural mediator to test against measured exposure.

doi: 10.1016/j.jenvman.2026.131066

Moradnia Isfahan pregnant urinary metals indoor dust AQI 2026 Environ Geochem Health

Spatial variability of heavy metals in pregnant women: linking urinary biomarkers, indoor dust contamination, and air quality in Isfahan, Iran

80 pregnant women; paired urine and vacuum dust analysed by ICP-OES. Dust metals ranked Pb>Cr>Ni>Cd (Pb mean 111.10 ug/g); urine Pb 10.52, Cr 8.30, Ni 4.14 ug/L. Urinary metals and AQI differed across urban zones ($p<0.001$); dust did not (Pb $p=0.514$). Building age correlated with dust Pb (rho 0.788). Dust and urinary profiles were uncorrelated. Limitation: n=80, spot urine without creatinine adjustment stated, and the only airborne metric is a zone-level AQI - no PM mass or PM-bound metal was measured, so 'air quality' enters only as an ecological covariate. Relevance: low - illustrates the gap that personal or indoor PM monitoring with filter speciation would fill.

doi: 10.1007/s10653-026-03508-x

Ni Wuhan birth cohort air pollution neurodevelopment target trial 2026 Environ Sci Technol

Early Life Ambient Air Pollution, Sex Steroids and Neurodevelopment: Target Trial Emulation in a Prospective Cohort

3,575 mother-child pairs (2014-2019), cord sex steroids in 897. Most pollutants were inversely associated with Bayley scores across windows, strongest in the first postnatal year. Androstenedione mediated 5-12% of some preconception/prenatal associations with PDI, but the IPW indirect effects lost significance. Target trial emulation of China's 2013-2017 Action Plan: children in the later policy period scored 7.27 (MDI) and 6.41 (PDI) points higher. Limitation: the policy contrast is a birth-period comparison, so secular changes in Bayley administration, parental education and care confound it; per-pollutant effect sizes and CIs are not in the abstract. Relevance: moderate - the postnatal window being most sensitive argues for exposure assigned to the infant's home, where indoor low-cost monitors outperform residential-grid estimates.

doi: 10.1021/acs.est.6c09006

Pakistan cooking fuel undernutrition 2026 Int J Environ Health Res

Association between household fuel use and childhood undernutrition in Pakistan: evidence from a National survey

4,224 under-fives; 45.4% with anthropometric failure, 37.7% stunted. Polluting vs clean fuels: CIAF OR 1.16 (1.04-1.31), stunting 1.16 (1.03-1.31), wasting 1.13 (1.00-1.29), underweight 1.17 (1.04-1.32); dung strongest. Limitation: fuel type is a wealth marker as much as an exposure, and with no measured PM the residual socioeconomic confounding is plausibly the size of the effect; cross-sectional. Relevance: moderate - the standard HAP proxy, and a reminder of why personal-monitor substudies inside DHS-type surveys are needed.

doi: 10.1080/09603123.2026.2749055

Placental angiogenic biomarkers review 2026 Reprod Toxicol

Exposure to environmental toxicants during pregnancy and placental angiogenic biomarkers: An integrative review

Whittemore-Knafl integrative review: 19 human studies after screening. Air pollutants predominantly showed an anti-angiogenic pattern - higher sFlt-1 or sFlt-1/PlGF and lower PlGF - as did PFAS, Cd and Cu; 14 of 19 studies reported at least one association in that direction. Limitation: 19 heterogeneous studies across four toxicant classes, no pooling or bias assessment, and 'at least one association' across multiple biomarkers and windows is a low bar. Relevance: low-moderate - plausible mechanism for PM-preeclampsia links, not evidence of it.

doi: 10.1016/j.reprotox.2026.109377

Qiang maternal PM macrosomia 2026 Chin Med J (Engl)

Interaction of maternal $PM_{2.5}$ exposure and psychosocial state on newborn macrosomia: A nationwide population-based study

METADATA ONLY. Metadata-only record: the publisher deposited title, authors and DOI on 10 Oct with no abstract, and the retry chain recovered none. By title, a nationwide study of the interaction between maternal $PM_{2.5}$ exposure and maternal psychosocial state on newborn macrosomia. Interaction claims between a modelled environmental exposure and a self-reported psychosocial measure are the design most prone to residual confounding by socioeconomic position, and macrosomia is a thresholded birthweight outcome whose prevalence is sensitive to that. Relevance: moderate - birth cohorts are where personal rather than residential exposure measurement would change the estimate most. On the abstract-retry list.

doi: 10.1097/cm9.0000000000004256

Scotland prenatal $PM_{2.5}$ infant amoxicillin birthweight mediation 2026 Eur J Public Health (conference abstract)

Prenatal $PM_{2.5}$ and Infant Antibiotics: Birth-Weight Mediation via Causal Analysis

All Scottish singleton births 2009-2018, n=498,123, linked registries; g-computation mediation comparing observed $PM_{2.5}$ with a counterfactual of 5 $\mu g\,m^{-3}$. Controlled direct effect OR 1.12 (1.10-1.14); interventional direct OR 1.12 (1.11-1.14); indirect via birthweight 0.99 (0.99-1.01). Limitation: exposure IQR is only 6-8 $\mu g\,m^{-3}$, so a 12% excess over a ~2 $\mu g\,m^{-3}$ counterfactual shift is implausibly large and points to residual area-level confounding (deprivation, prescribing culture); amoxicillin dispensing is a weak infection proxy. Preliminary conference abstract. Relevance: low-moderate - fine-scale exposure contrast is too narrow for postcode annual means.

doi: 10.1093/eurpub/ckag169.955

Song et al. 2026 Environ Res

Role of Neighborhood Vulnerability and Opportunity in the Association Between Prenatal Air Pollution and Autism-Related Traits

8,598 mother-child pairs from 35 ECHO sites; pregnancy and trimester $PM_{2.5}$, NO_2 and O_3 at residence; Social Responsiveness Scale analysed by GEE, stratified by census-tract SVI and Child Opportunity Index quintiles, with interaction $p < 0.10$ as 'suggestive'. O_3 interacted with SVI ($p = 0.07$) and COI ($p = 0.01$): beta 0.05 (-0.005 to 0.10) in lowest SVI and 0.04 (0.00 to 0.08) in highest COI. Trimester $PM_{2.5}$ also showed interactions, but no stratum estimate was significant. Limitation: a liberal interaction threshold across many strata and trimesters invites false positives, and the O_3 signal appears in the advantaged strata, contrary to the hypothesis. Relevance: low - the $PM_{2.5}$ finding is null within strata.

doi: 10.1016/j.envres.2026.125777

Valuvunina $PM_{2.5}$ PurpleAir quarry low birth weight Kinshasa DRC 2026 J Toxicol

Atmospheric Pollution by Fine Particulate Matter ($PM_{2.5}$) and Low Birth Weight: A Cross-Sectional Study in Binza Ozone Health Zone, Democratic Republic of the Congo, 2022

355 mothers delivering in 2021 at six facilities near the Carrigres quarry. LBW prevalence 12.96% (9.7-16.9). PurpleAir PA-II-SD sensors, corrected with a Kinshasa BAM-derived equation, gave area $PM_{2.5}$ 54.6-102.5 $\mu g\,m^{-3}$. Residence <500 m from the quarry: aOR 4.12 (1.98-8.57); area $PM_{2.5}$ >75 $\mu g\,m^{-3}$: aOR 1.18 (0.59-2.36). Limitation: $PM_{2.5}$ was measured for 48-72 h in 2022 and assigned to births in 2021, i.e. after the pregnancies, at the health-area level, so the null $PM_{2.5}$ result says nothing about gestational exposure; quarry dust is coarse and poorly captured by Plantower optics. Relevance: high - a direct example of low-cost sensors used as the sole exposure metric, and of the temporal mismatch that undermines it.

doi: 10.1155/jt/7953451

Zhu prenatal PM_1 $PM_{2.5}$ PM_{10} childhood hemoglobin anemia Fujian birth cohort 2026 Ecotoxicol Environ Saf

Association of prenatal PM_1, $PM_{2.5}$, and PM_{10} exposure with early childhood hemoglobin levels and anemia: Identification of susceptibility windows

Fujian Birth Cohort paediatric subcohort, 1,235 mother-child pairs. Per 1 $\mu g\,m^{-3}$ over the whole pregnancy, child hemoglobin fell by 1.02 (0.53-1.52) for PM_1, 0.87 (0.46-1.28) for $PM_{2.5}$ and 0.37 (0.19-0.56) for PM_{10} (units as reported). First-trimester anemia ORs per unit were 1.23, 1.16 and 1.09; DLMs placed the hemoglobin window in early-mid pregnancy and the anemia window in the first trimester. Limitation: per-microgram effects of this size imply implausibly large decrements across the cohort's range; PM_1, $PM_{2.5}$ and PM_{10} are near-collinear satellite products, so the size-fraction ranking is not interpretable. Postnatal exposure and diet are not stated. Relevance: low - satellite-derived exposures only.

doi: 10.1016/j.ecoenv.2026.120980

Mechanistic toxicology — 58 records

Carried from the daily issues — full summaries (58 of 58 records in this subtopic)

Aged traffic particles 2026 Environ Sci Technol

Aging of Urban Traffic Fine Particles Modulates Toxicity Mechanisms and Oxidative Stress Responses

Real cold-start and urban emissions sampled in a parking garage, aged in an oxidation flow reactor. Ageing raised SOA mass up to 6.9-fold, O:C from 0.26 to 0.79, and both intrinsic and volume-normalised oxidative potential, while peroxide-sensitive DCF activity fell. Aged extracts raised cellular ROS in lung and liver cells and produced greater oxidative-stress and DNA-damage responses; fresh extracts instead suppressed cellular respiration, especially in HepG2, with distinct cell-painting morphological profiles. Limitation: extracts, not whole aerosol, so the soluble fraction is tested and the particle is not; the divergence between acellular DCF and cellular ROS shows how assay-dependent 'oxidative potential' remains. Relevance: high - oxidative potential is the leading candidate for a health-relevant metric a sensor network could one day report, and this is a direct demonstration that it changes downwind of the source.

doi: 10.1021/acs.est.6c06323

Amato mobile air-liquid interface toxicity Barcelona harbor urban 2026 Environ Sci Pollut Res Int

Assessment of air pollution toxicity in urban environments in Barcelona, Spain, using a mobile air-liquid interface system

A mobile air-liquid interface system exposed A549 cells directly to ambient air at a harbour site and an urban background site, alongside co-located particle and gas monitoring. Partial least squares explained up to 80% of the variance in ALI toxicity from air-quality parameters. Gas-phase pollutants, ozone above all, contributed more than particles. Limitation: LDH release in a single tumour-derived cell line is a crude endpoint. With two sites and highly collinear predictors, PLS loadings cannot separate ozone from photochemical-age covariates. n and exposure durations are not given in the abstract. Relevance: medium - pairing in-situ bioassays with multipollutant monitors is the template for testing whether mass-based LCS metrics track toxicity.

doi: 10.1007/s11356-026-38306-6

Awasthi air pollution oxidative stress inflammation molecular mechanisms review 2026 J Xenobiot

Molecular Mechanisms Linking Air Pollution-Induced Oxidative Stress to Inflammation and Human Disease

Broad review of ROS/RNS generation by PM, NOx, O_3, SO_2 and VOCs and downstream Nrf2, NF-kB and MAPK signalling, mitochondrial dysfunction, ER stress and epigenetic reprogramming, mapped to respiratory, cardiovascular, neurological, metabolic and carcinogenic outcomes; also covers biomarkers, susceptible windows and antioxidant-based interventions. No quantitative synthesis. Limitation: pollutants with very different chemistry are pooled under one oxidative-stress narrative, and in vitro dose relevance is not appraised; nothing here is new relative to the established mechanistic literature. Relevance: low - background for oxidative-potential (OP) metrics, which are the measurable bridge to sensing.

doi: 10.3390/jox16060185

Chen $PM_{2.5}$ mitochondria-associated membranes mouse liver 2026 Am J Physiol Cell Physiol

Exposure to airborne $PM_{2.5}$ suppresses mitochondrial associated membranes and MAM-associated metabolic regulators in mouse liver

Mice inhaling environmentally relevant $PM_{2.5}$ showed fewer mitochondria-associated membranes and fewer MAM-forming mitochondria in liver (TEM). In hepatocytes, $PM_{2.5}$ suppressed basal and glucagon-induced MAM formation (BiFC sensor); macrophage particle uptake was blocked by antioxidants or calcium chelation. Mechanism: repression of CREBH/PPARalpha programs for the mitochondrial unfolded protein response and fatty-acid oxidation. Limitation: in vitro particle doses and the inhaled dose to liver are not reconciled, and the translocation route (systemic particles vs lung-derived mediators) is assumed rather than shown. Sample sizes are not in the abstract. Relevance: low - mechanism for PM-metabolic associations.

doi: 10.1152/ajpcell.00103.2026

Chen $PM_{2.5}$ mixtures Parkinson disease INPP5D SHIP1 PI3K-AKT multi-omics 2026 J Xenobiot

Neurotoxicity of $PM_{2.5}$-Bound Multi-Pollutant Mixtures in Parkinson's Disease: Exploring the INPP5D/SHIP1-PI3K-AKT Axis via an Integrative Multi-Omics Study

Two-sample MR on public GWAS suggested a positive genetic association of $PM_{2.5}$ with PD; network toxicology intersected $PM_{2.5}$-component targets, PD risk genes and differential transcripts (195 genes), and LASSO/RF picked INPP5D (SHIP1) as hub. Docking and SH-SY5Y experiments with SRM 1648a showed SHIP1 up-regulation, PI3K-AKT suppression, inflammation and apoptosis, rescued by INPP5D knockdown. Limitation: MR for an environmental exposure rests on GWAS of residential $PM_{2.5}$, whose instruments mostly encode where people live (pleiotropy); docking affinities are not evidence. MR estimates are not reported in the abstract, nor is the SRM dose. Relevance: low.

doi: 10.3390/jox16060184

Chen pirfenidone fine particulate matter EMT mitochondrial fission autophagy 2026 Respir Res

Pirfenidone reduces fine particulate matter-induced epithelial to mesenchymal transition in lung epithelial cells through regulating mitochondrial fission and autophagy

A549 cells exposed to 100 ug/mL fine PM lost E-cadherin and gained vimentin, alpha-SMA, TGF-beta1, Snail and Slug, turned spindle-shaped and migrated more; PM increased mitochondrial fission and LC3B/p62 accumulation, i.e. impaired autophagic flux. Intratracheal PM in mice reproduced EMT markers, inflammatory infiltration and alveolar wall thickening. Pirfenidone reversed these effects in vitro and reduced collagen deposition in vivo. Limitation: 100 ug/mL on a cell monolayer and bolus instillation are far above realistic deposited doses, and A549 is a carcinoma line; the drug result says nothing about population exposure. Relevance: low - mechanism for PM-associated fibrosis; no measurement content.

doi: 10.1186/s12931-026-03804-1

Chen toluene soot toxicity 2026 Atmos Environ

Oxidative toxicity and cytotoxicity of toluene combustion soot particles mediated by their microstructure

METADATA ONLY. No abstract retrievable at harvest. From the title: soot generated from toluene combustion, with oxidative potential and cytotoxicity attributed to particle microstructure (likely graphitisation/nanostructure order). Recorded for coverage; no result is inferred. Relevance: low-moderate - if microstructure rather than mass sets toxicity, mass-based optical sensing misses a toxicity axis entirely.

doi: 10.1016/j.atmosenv.2026.122407

Comis Turin oxidative potential 2026 Atmosphere

Oxidative Potential and Elemental Fingerprints of Urban and Suburban Particulate Matter in Turin

32 PM_{10} filters (urban traffic vs rural background, 2020-2021, min and max days of four months). Urban OP-AA 0.12-0.60 vs suburban 0.10-0.38 nmol AA/min/m3; Cu was the most active single metal. Total and phosphate-buffer bioaccessible metal fractions were compared. Limitation: filters were selected at monthly concentration extremes, so OP-mass relationships are inflated by design, and n=32 cannot support source attribution of OP. Relevance: low-moderate - OP remains unmeasurable by optical nodes; the Cu dominance points to brake-wear tracers as the practical proxy.

doi: 10.3390/atmos17101000

Do $PM_{2.5}$ superior cervical ganglion calcium channels satellite glia 2026 Int J Mol Sci

$PM_{2.5}$ Exposure Disrupts Sympathetic Neurotransmission by Impairing Voltage-Gated Ca2+ Channels and Remodeling Satellite Glia in the Superior Cervical Ganglion

Rats received intracerebroventricular $PM_{2.5}$; isolated superior cervical ganglia were also exposed directly. ICV $PM_{2.5}$ raised hypothalamic TNF-alpha and IL-6. Direct exposure cut medium norepinephrine without changing tissue content; depolarisation-evoked Ca2+ responses fell after either route, and patch clamp showed lower VGCC current density without Cav alpha1 mRNA change. GFAP and Kir4.1 rose (satellite glia). Limitation: intracerebroventricular injection bypasses the lung and does not model inhalation dose or translocation; particle source and dose relevance are not given in the abstract. Relevance: low.

doi: 10.3390/ijms27209184

Donohoe ultrafine amorphous silica epithelial stress pro-fibrotic 2026 Part Fibre Toxicol

Ultrafine amorphous silica induces chronic epithelial cell stress and pro-fibrotic reprogramming in a particle size- and exposure duration-dependent manner

METADATA ONLY. Title and authors only (Donohoe, Selvamani, Dharwal, O'Neil et al.). No abstract was retrievable at harvest. The title reports that ultrafine amorphous silica causes chronic epithelial cell stress and pro-fibrotic reprogramming that depends on particle size and exposure duration - relevant because amorphous silica is usually treated as low-hazard relative to crystalline silica. Model system, doses and endpoints are not available; nothing is inferred. Relevance: low - toxicology of a particle class not resolved by mass sensors.

doi: 10.1186/s12989-026-00710-8

EPFR size-resolved PM 2026 Environ Toxicol Chem

Size-Dependent Characteristics, Exposure Risk, and Formation Mechanisms of Environmentally Persistent Free Radicals in Atmospheric Particulate Matter

EPFRs were carbon-centred radicals adjacent to oxygen groups; $PM_{2.5}$ had the highest mass-normalised concentration (8.05e16 spins/g) and g-factor (2.00337). Cigarette-equivalent exposure was 0.326/day for $PM_{2.5}$ and 0.346/day for PM_{10}. Laboratory surrogates showed smaller particles generate more EPFRs owing to larger specific surface. Limitation: the cigarette-equivalent model is a communication device without dosimetric basis, and site/season details are absent from the abstract. Relevance: low-moderate - another surface-chemistry toxicity axis that mass sensing cannot see.

doi: 10.1093/etojnl/vgag274

Edwards macrophage superoxide SOA biomass plastic burning PM 2026 ACS Omega

Superoxide Release from Macrophages upon Exposure to Secondary Organic Aerosols, Biomass, and Plastic Burning Particulate Matter

Macrophages exposed to PM suspensions, superoxide followed for 4 h by chemiluminescence plus EPR; an activation threshold is defined as the dose of a clear superoxide rise. Limonene SOA gave the largest release (~32 uM) but the highest threshold (300 ug/mL); naphthalene and toluene SOA gave 2.5-7 uM with thresholds of 0.02-0.1 ug/mL. Biomass PM had no clear threshold. Plastic-burning PM ranged widely: PET highest output (~12 uM), PS/LDPE/HDPE thresholds ~0.01 ug/mL, PVC and PET 50-100 ug/mL. Limitation: suspension doses in ug/mL do not map onto deposited alveolar doses, and a single cell line under submerged exposure omits surfactant and epithelial interaction. Relevance: moderate - per-mass oxidative potency varies by four orders of magnitude across sources, which mass-only sensors cannot register.

doi: 10.1021/acsomega.6c06562

Env epigenetics review 2026 Front Epigenet Epigenom

Environmental epigenetic modifiers in neurodegenerative diseases: a systematic review of epigenomic mechanisms and therapeutic targets

A PRISMA-structured review of DNA methylation, histone modification and non-coding RNA changes linking environmental neurotoxicants - particulate matter among them - to neurodegeneration, with emphasis on developmental windows and life-course exposure. Limitation: PRISMA structure without quantitative synthesis, risk-of-bias assessment or any pooling, over a literature in which publication bias toward positive methylation findings is severe; particulate matter is treated interchangeably with metals and solvents despite incomparable exposure metrics. Relevance: low as evidence. Entered because the epigenetic mediator claim keeps appearing beside the dementia-mortality results in this backfill and it should be visible how thin the review base under it is.

doi: 10.3389/freae.2026.1833533

Ferreira Vicente et al. 2026 Expert Rev Respir Med

Pulmonary black carbon accumulation: mechanisms, health impacts, and environmental sources

Narrative review (searches to June 2026) of BC sources, physicochemistry, exposure assessment, biodistribution and tissue retention, with emphasis on tissue-based biomonitoring and label-free optical detection, and of respiratory, cardiovascular, neurological, developmental, metabolic and immune associations. The argument: tissue BC load could bridge external exposure estimates and biologically effective dose. Limitation: narrative, no quantitative synthesis or risk-of-bias assessment; tissue BC integrates an unknown exposure history, so it is a cumulative dose marker, not a validation of any particular exposure model. Relevance: low-moderate - internal BC dose is a possible ground truth for personal aethalometer and network-derived BC exposure.

doi: 10.1080/17476348.2026.2740205

Ganse diesel exhaust particles cardiac metabolomics glycolytic shift rats 2026 Biomolecules

Diesel Exhaust Particles Exposure Disrupts Cardiac Bioenergetics and Induces a Shift Towards Glycolytic Metabolism in Rats

Sprague-Dawley rats instilled with DEP or saline three times weekly for one or three weeks. At one week, 17 of 22 myocardial metabolites changed: succinate, acetate, formate, 3-hydroxybutyrate, ATP and ADP fell; glucose and lactate rose, consistent with reduced oxidative metabolism and a glycolytic shift, plus nucleotide degradation. Most changes resolved by three weeks; plasma changes were modest. Limitation: intratracheal bolus dosing of a single reference DEP does not reproduce inhaled dose rate, group sizes are not stated, and transient changes that resolve under continued exposure question chronic relevance. Relevance: low.

doi: 10.3390/biom16101501

Gao et al. 2026 Carcinogenesis

Air pollution and early-onset lung cancer: from environmental carcinogenesis to early intervention strategies

Narrative review of mechanisms linking air pollution to early-onset lung cancer: genomic instability (DNA damage, telomere dysfunction, mitochondrial impairment, mutational signatures), epigenetic dysregulation, a tumour-promoting niche through CXCL13 and IL-1beta, immune evasion via PD-L1 and CD47-SIRPalpha, and respiratory-gut microbiome disruption, closing with candidate biomarkers and chemopreventive targets. Limitation: no systematic search or evidence grading, the 99%-above-guideline and >14%-of-lung-cancer-deaths framing is quoted rather than re-derived, and the biomarkers are called 'validated' on the strength of single studies. Relevance: low as evidence, moderate as a map of which mediators a cohort with measured personal exposure could actually test.

doi: 10.1093/carcin/bgag070

Gautam diesel exhaust miR-133b-3p CTGF Hippo ECM remodeling mouse lung 2026 Environ Mol Mutagen

Diesel Exhaust-Induced miR-133b-3p Downregulation and CTGF-Linked Hippo Pathway Inactivation Promote ECM Remodeling and Inflammation in Mouse Lungs

C57BL/6 mice received diesel exhaust 30 min/day, 5 days/week for 8 weeks versus filtered air. Microarrays showed miR-133b-3p down more than 2-fold; CTGF was confirmed as a direct target in A549 cells (mimic reduced CTGF mRNA p=0.0054, protein p=0.0001). In vivo, exhaust raised CTGF, lowered P-YAP/YAP (Hippo inactivation), increased BALF IL-18, IL-1beta and GM-CSF, raised MMP-3/7/12 and lowered LOXL1/4, consistent with ECM remodelling. Limitation: exhaust dose (PM mass, NO_2) is not reported in the abstract, so particle and gas effects cannot be separated; the CTGF link is correlative in vivo. Relevance: low - traffic-exhaust mechanism; supports particle number or black carbon over mass as the near-road metric.

doi: 10.1002/em.70088

Gut-liver axis pollutants MASLD review 2026 Int J Mol Sci

The Role of the Gut-Liver Axis in the Development of Liver Diseases Induced by Selected Environmental Pollutants: A Narrative Review

Narrative review covering heavy metals, PM and endocrine disruptors (chiefly BPA) as drivers of gut dysbiosis and barrier failure, with LPS translocation into the portal circulation activating hepatic TLR4/NF-kB, oxidative stress and steatosis; EDCs additionally act on hepatic nuclear receptors. Limitation: PM is one of three pollutant classes and the review does not separate inhaled from ingested particle routes or give dose context, so the PM-specific gut-liver claim rests on a thin, mostly rodent literature. Relevance: low for sensing - flags an extrapulmonary endpoint that cohort studies with dense exposure surfaces could test.

doi: 10.3390/ijms27198581

He FOXM1 lncRNA COPD-R biomass $PM_{2.5}$ pulmonary inflammation rat 2026 Ecotoxicol Environ Saf

FOXM1/LncRNA COPD-R regulates pulmonary inflammation in chronic obstructive pulmonary disease induced by biomass-related $PM_{2.5}$

Rats exposed to biomass-fuel smoke for seven months developed COPD-like disease; lung RNA-seq found 312 differentially expressed lncRNAs (294 up). lncRNA COPD-R rose from 0.15 to 1.05 (relative expression, $p<0.01$). In rat L2 alveolar cells, biomass-related $PM_{2.5}$ increased COPD-R; knockdown reduced PM-induced CCL2, IL-17F, MMP-9 and TIMP-1, and ChIP/luciferase assays placed FOXM1 upstream as its transcriptional activator, with rescue experiments consistent with a FOXM1-COPD-R axis. Limitation: whole-smoke exposure in vivo but $PM_{2.5}$ only in vitro, dose not reported in the abstract, and a rat lncRNA whose human orthologue is not established. Relevance: low - mechanistic support for household biomass PM, where the exposure gap is measurement in homes.

doi: 10.1016/j.ecoenv.2026.120952

Houston UFP PAH deposition 2026 Air Qual Atmos Health

PAH-associated health risk from ultrafine particles: an urban field study

Three Houston sites: UFP respiratory deposition measured directly with the Mobile Aerosol Lung Deposition Apparatus, combined with size-segregated PAH analysis to estimate deposited PAH mass. Both carcinogenic and non-carcinogenic risks were within acceptable levels. Limitation: no numbers in the abstract; PAH risk is one narrow axis of UFP toxicity, so 'acceptable' does not translate to UFP being benign. Relevance: moderate - measured deposited dose rather than ambient concentration is the right exposure metric and is rarely available.

doi: 10.1007/s11869-026-02118-x

Inhaled polystyrene microplastics pulmonary fibrosis review 2026 Front Immunol

Polystyrene-associated pulmonary fibrosis: a summary based on the available evidence and prospects

Reviews sources, deposition and pulmonary effects of inhaled polystyrene microplastics and proposes, as a working model, convergence on coupled Wnt/beta-catenin and TGF-beta1/Smad2/3 signalling driving EMT, myofibroblast differentiation and matrix remodelling. Each element is graded as direct PS-MP evidence, extrapolation from other particles, or canonical fibrosis biology - an unusually honest structure. Limitation, stated by the authors: no study tests the full cascade in one inhalation model; non-physiological doses and pristine monodisperse spheres dominate. Relevance: low-moderate - airborne microplastics sit inside optical PM size bins but are invisible as a class to LCS; exposure assessment remains filter-and-spectroscopy work.

doi: 10.3389/fimmu.2026.1953398

Jiang carbon nanoparticle NSCLC progression ARNT2 ARL14 2026 Ecotoxicol Environ Saf

Carbon nanoparticle exposure promotes the progression of non-small cell lung cancer via the ARNT2/ARL14 axis: Findings from in vivo and in vitro models

Mouse CNP-induced fibrosis model (200 ug/kg every other day, 90 days) identified ARNT2 by transcriptomics. In xenografts with continued CNP exposure, tumour volume at day 36 rose (A549 194 to 308 mm3; A427 245 to 417 mm3) along with Ki-67/PCNA and N-cadherin/vimentin. ARNT2 knockdown reversed these changes. In vitro, 2 ug/mL for 48 h increased proliferation, migration (up to 3.6-fold), invasion, stemness markers and IL-6 (up to 13.7-fold), and reduced apoptosis. Limitation: carbon black nanoparticles stand in for 'the shared core' of $PM_{2.5}$, coal dust and smoke, but lack their organic and metal fractions. Instillation doses and in-vitro concentrations far exceed lung deposition from ambient air. Group sizes are not reported. Relevance: low.

doi: 10.1016/j.ecoenv.2026.120951

Jin et al. 2026 Front Immunol

ANKRD22 contributes to NF-kB-associated inflammatory responses in alveolar type II epithelial cells during $PM_{2.5}$-triggered airway inflammation

$PM_{2.5}$ raised ANKRD22 in alveolar type II cells in a mouse airway-inflammation model and in A549 cells. AT-II-specific ANKRD22 knockout mice had less BALF leukocytosis and neutrophilia, histological inflammation and IL-6/CXCL1; siRNA knockdown in A549 lowered IL6, IL8, IL1B, CXCL1 and CXCL2. $PM_{2.5}$ increased p50, phospho-IKKb, IkBa and RELA, and the non-canonical p52/RELB arm, all attenuated by ANKRD22 loss. Limitation: particle source, dose and dosimetry are not given in the abstract, A549 is a poor AT-II surrogate, and the knockout readouts are acute. Relevance: low - mechanism only; no exposure metric that a monitoring network could map onto.

doi: 10.3389/fimmu.2026.1894350

Kim et al. 2026 J Pharmacopuncture

Acorus gramineus Solander Preserves Corneal Epithelial Barrier Integrity against $PM_{2.5}$-Induced EMT-Like Remodeling

Human corneal epithelial cells grown as 3D spheroids were exposed to $PM_{2.5}$ with or without an Acorus gramineus root extract. $PM_{2.5}$ destabilised spheroids without marked cytotoxicity, lowered transepithelial resistance, disrupted ZO-1, ZO-3 and occludin, induced EMT-like markers (N-cadherin, vimentin, MMPs) and cytokines, with TGF-b/Smad activation; the extract reversed these. Limitation: particle source and concentration are not stated in the abstract, no in vivo or tear-film model, and the therapeutic arm is a crude herbal extract. Relevance: low - the ocular surface is an underexamined direct-contact target, but nothing here quantifies dose.

doi: 10.3831/kpi.2026.29.3.314

Kou Xing Su San Sang Xing Tang $PM_{2.5}$ lung injury rats 2026 Pak J Pharm Sci

Xing Su San and Sang Xing Tang attenuate $PM_{2.5}$-induced lung injury and inflammatory marker expression in rats

40 male Wistar rats in five groups of 8. $PM_{2.5}$ at 500 $\mu g\,m^{-3}$ raised lung RAGE mRNA 2.8-fold (warm-dry) and 2.5-fold (cool-dry), NF-kB p65 2.4- and 2.2-fold, and S100A9-positive cells to 3.45% and 3.13% vs 0.82% in controls (all $P<0.05$). The two herbal formulas, given by gavage, partly reversed these markers in their matched climate model (e.g. RAGE 1.6-fold with Sang Xing Tang). Limitation: 500 $\mu g\,m^{-3}$ is an episode-level dose, n=8 per group, no histology scoring or blinding reported, and formula composition is not standardised. Relevance: low - a temperature-humidity x PM interaction in toxicology, not monitoring.

doi: 10.36721/pjps/2026.40.15845.1

Lee et al. 2026b Cell Biol Int

Curcumin-Loaded Nanospheres Attenuate Fine Dust (PM_{10})-Induced Cellular Toxicity by Suppressing Reactive Oxygen Species-Dependent Signaling in Human Keratinocytes

PM_{10} reduced HaCaT viability and raised ROS, ERK, p38 and NF-kB activity, shifted Bax/Bcl-2, cleaved caspase-3 and processed IL-1beta; curcumin-loaded nanospheres at 1e-4 ug/mL restored viability and blunted each step, as did NAC and pathway inhibitors. Limitation: this is a formulation paper - the PM_{10} arm is a generic oxidative-stress stimulus with no composition, dose-response or endotoxin control, and an effect at 1e-4 ug/mL with none at higher unformulated doses is a non-monotonic claim that needs replication before it means anything. Relevance: low - logged for completeness of the dermal-endpoint strand, not as mechanism.

doi: 10.1002/cbin.70222

Liu $PM_{2.5}$ IL-17 microglia neuroinflammation 2026 Environ Health (ACS)

IL-17 Signaling Pathway: A Potential Signaling Pathway Mediating Neuroinflammation Induced by $PM_{2.5}$

BV2 microglia exposed to $PM_{2.5}$ lost viability and raised inflammatory cytokines; microarray DEGs enriched for inflammatory GO terms and the KEGG IL-17 pathway with downstream NF-kB and MAPK. An anti-IL-17A antibody reversed cytokine release, microglial activation and behavioural deficits (implying an in vivo arm). Limitation: the abstract gives no $PM_{2.5}$ source, composition or dose, so environmental relevance cannot be judged; BV2 is an immortalised murine line with an exaggerated inflammatory phenotype, and IL-17 is usually T-cell/gamma-delta derived, so microglial IL-17 signalling needs cell-type attribution. Relevance: low - mechanistic only.

doi: 10.1021/envhealth.6c00080

Liu Streptococcus D19T fermentation supernatant asthmatic mice $PM_{2.5}$ 2026 Environ Int

Protective effect and mechanism of Streptococcus D19T fermentation supernatant in asthmatic mice exposed to $PM_{2.5}$

METADATA ONLY. Title and authors only (Liu, Sun, Wang, Jin et al.); no abstract retrievable at harvest. The title describes a probiotic-derived fermentation supernatant protecting asthmatic mice against $PM_{2.5}$-aggravated airway disease. Dose, model and outcomes cannot be stated. Limitation: unverifiable beyond the title; $PM_{2.5}$ aggravation models usually use instilled boluses far above ambient deposition. Relevance: low.

doi: 10.1016/j.envint.2026.110584

Liu cerium dioxide scavenging environmentally persistent free radicals $PM_{2.5}$ filtration review 2026 Int J Mol Sci

Cerium Dioxide-Based Scavenging of Environmentally Persistent Free Radicals in Fine Particulate Matter: A Prospective Review

A hypothesis-stating review, explicitly not a validated result. EPFRs persist on $PM_{2.5}$ for days to months and drive hydroxyl-radical generation by Fenton-like cycling after deposition. The authors propose migrating CeO2 Ce3+/Ce4+ redox cycling, an established antioxidant mechanism, into ambient-temperature air filters to quench EPFRs on captured particles, and argue that high-temperature catalytic uses of CeO2 are not a precedent. They list the open questions: direct quenching vs ROS interception, humidity, airflow, contact time, and CeO2 fate. Limitation: no data; the only cited support is gas-phase ROS scavenging in cigarette smoke. Relevance: low - filter chemistry, not measurement.

doi: 10.3390/ijms27209232

Liu et al. 2026c Environ Sci Technol

Bulk $PM_{2.5}$ Masks Oxidative Potential and Reactive Oxygen Species Responses by Metal-Organic Complexation in PM_1 and PM_1-2.5 Mixtures

Roadside PM_1 and PM_1-2.5 were tested separately and mixed. Mixing is antagonistic on every endpoint - interaction factor 0.65-0.81 for ascorbate OP, 0.80-0.93 for DTT OP, 0.10-0.88 for cellular ROS and 0.43-0.78 for mitochondrial superoxide - whereas metal-free biomass-burning PM is strictly additive. Cu2+-humic acid mixtures reproduce the antagonism (IF 0.22-0.95), and EPR shows complexed Cu forming nonlinearly with the HA:Cu ratio, explaining why studies disagree. Limitation: one roadside site, extract-based assays, and Cu-HA as the single model complex; whether in vivo lining fluid reproduces the complexation is untested. Relevance: high - bulk-$PM_{2.5}$ OP, the metric proposed as a toxicity-weighted complement to mass, understates the oxidative burden of its size fractions; size-resolved sampling is needed to interpret it.

doi: 10.1021/acs.est.6c09837

Liu et al. 2026d Environ Sci Technol

Metabolite-Associated Inflammation during Combined Exposure to Cooking Oil Fumes and Atopic Dermatitis: A Multi-Organ Spatial Metabolomics Study

A mouse model of atopic dermatitis with cooking-oil-fume co-exposure, profiled by untargeted and spatial metabolomics with in situ immunofluorescence across lung, blood, heart and kidney. AD+COF gave the largest rises in IL-6 and TNF-a and the largest metabolite shifts: succinate accumulation, xanthine and uric-acid depletion, glucose perturbation and spatial lipid remodelling, correlated across organs. Limitation, stated by the authors: no tracing, receptor blockade or mitochondrial assays, so inter-organ signalling is associative, and the high-concentration regimen limits environmental extrapolation; COF particle size and mass are not given in the abstract. Relevance: low - indoor cooking PM is a large exposure class, but this design cannot inform a dose metric.

doi: 10.1021/acs.est.6c11745

Lung-on-a-chip pollutant toxicity review 2026 Front Bioeng Biotechnol

Microfluidic lung-on-a-chip for evaluating inflammation and toxicity from environmental pollutants

Review of microfluidic lung-on-a-chip platforms - air-liquid interface, cyclic stretch, perfused endothelium - for pollutant toxicology, covering PM, pathogens, VOCs and emerging nanomaterials, with advantages in biomimicry and real-time readout and the obstacles of chip standardisation, operational complexity and subsystem integration. Limitation: no quantitative synthesis; dose delivery (deposited mass per area) is the variable that decides translatability and is not addressed in the abstract. Relevance: low - only connects to monitoring if chips are dosed at measured ambient deposition rates.

doi: 10.3389/fbioe.2026.1942916

Melatonin DPM2.5 2026 Ecotoxicol Environ Saf

Melatonin protects keratinocytes from diesel-derived $PM_{2.5}$-induced damage by suppressing JNK/p38 MAPK signaling and caspase activation

Diesel $PM_{2.5}$ raised intracellular ROS, damaged macromolecules, disrupted mitochondrial membrane potential and drove apoptosis in HaCaT cells; melatonin scavenged radicals, restored viability and membrane potential, partly restored Bcl-2 against raised Bax and Bim, and suppressed caspase-9 and -3 and JNK/p38 MAPK activation, with a hairless-mouse arm. Limitation: the same template as the curcumin-nanosphere record on 3 October - a protective-agent paper in which the PM arm is a generic oxidant stimulus, with no composition, no dose-response against a deposited-dose estimate and no endotoxin control; diesel PM is a reference material, not an ambient aerosol. Relevance: low as mechanism; logged to keep the dermal-endpoint strand complete and because the DPM2.5 reference material is at least characterised, unlike most such papers.

doi: 10.1016/j.ecoenv.2026.120879

Mountain West wood smoke 2026 Part Fibre Toxicol

Household wood smoke exposure, lung black carbon deposition, and genomic instability in U.S. Mountain West residents

METADATA ONLY. Metadata-only record: Crossref deposited title, authors and DOI on 7 Oct; neither PubMed, Europe PMC nor Semantic Scholar held an abstract, and Scholar Gateway is still refusing. By title this is an unusually direct chain - a household exposure, a measured internal dose (lung black carbon deposition) and a genomic endpoint in the same human subjects. Internal dose is what every ambient-monitoring epidemiology study substitutes a model for, so a paper that measures it is worth the retry. Relevance: high, the highest of the day's metadata-only records; flagged for full-text read.

doi: 10.1186/s12989-026-00702-8

Mouth-throat replica deposition 2026 J Aerosol Sci

The influence of model selection and flow rate on aerosol transport in replica models of the mouth and throat: An experimental study

METADATA ONLY. No abstract at harvest. Title indicates an experimental comparison of mouth-throat replica geometries and flow rates for aerosol transport. Recorded for coverage. Relevance: low-moderate - dosimetry inputs.

doi: 10.1016/j.jaerosci.2026.106906

Nored lithium-ion battery fire soot PM pulmonary toxicity mice fluoride 2026 Environ Res

Pulmonary Toxicity of Particulate-Phase Combustion Byproducts from Lithium-Ion Battery Fires

Soot (10 fire tests) and smoke PM (2 tests) from lithium-ion battery thermal runaway were size-filtered to <10 um and given to mice (100 ug). Soot had more carbon (37% vs 18%), PM more anions (12% vs 6%), inorganic elements similar (23-28%); particulate fluoride reached ~5% of mass. Only soot altered ventilatory timing, peak expiratory flow and relaxation time; both raised lung inflammation to levels comparable with other combustion sources. Fluoride, Al, Fe, Ni and Li were associated with toxicity. Limitation: a single 100 ug bolus (aspiration/instillation) is not inhalation dosimetry, two smoke tests is a thin basis, and component-toxicity associations across so few samples are exploratory. Relevance: low - toxicology; flags fluoride-bearing PM as a new indoor/urban fire hazard.

doi: 10.1016/j.envres.2026.126023

Onoda et al. 2026 Part Fibre Toxicol

Gene-environment interaction in neurodevelopment: Setd5 haploinsufficiency and prenatal carbon nanoparticle exposure synergistically induce ASD-like phenotypes

Pregnant C57BL/6N dams received intranasal carbon black nanoparticles (95 ug/kg) or water on GD5 and GD9; offspring were genotyped wild-type or Setd5+/- and analysed by sex. Only male Setd5+/- offspring exposed to CBNP showed a higher proportion with ASD-like behaviour (reduced social-novelty exploration, fixation on a familiar mouse) with selective loss of parvalbumin interneurons; either factor alone had minimal effect. Limitation: a single dose by intranasal bolus, a model particle without the metals or organics of ambient PM, and 'proportion of individuals' outcomes in small groups; group sizes are not in the abstract. Relevance: low-moderate - a gene-by-exposure synergy implies population risk is concentrated in susceptible genotypes that a mean exposure-response hides.

doi: 10.1186/s12989-026-00705-5

$PM_{2.5}$ HCC miR-25 2026 Transl Cancer Res

$PM_{2.5}$ promotes migration of hepatocellular carcinoma by targeting miR-25

$PM_{2.5}$ lowered E-cadherin, raised vimentin and miR-25 and increased MHCC97H migration; miR-25 mimics amplified and inhibitors attenuated the effect; mouse models showed enhanced tumour growth and invasion. Limitation: single cell line, unstated doses, and a single-miRNA mechanism for a pleiotropic exposure. Relevance: low.

doi: 10.21037/tcr-2026-0728

$PM_{2.5}$ NLRP3 inflammasome lung cancer review 2026 Int J Mol Sci

Possible Links Between $PM_{2.5}$-Induced NLRP3 Inflammasome Activation and Its Role in Lung Cancer Progression and Drug Resistance

Narrative review positioning $PM_{2.5}$-driven ROS as the trigger for NLRP3 inflammasome assembly, caspase-1 activation and IL-1beta/IL-18 maturation, sustaining chronic pulmonary inflammation in lung carcinogenesis. It then hypothesises that NLRP3 signalling contributes to lung-cancer drug resistance and proposes NLRP3 as a prognostic marker and target, noting NLRP3 inhibitors are in trials for other inflammatory diseases but none in oncology. Limitation: the drug-resistance link is explicitly hypothetical and assembled from heterogeneous in vitro doses; no systematic search or grading. Relevance: low for sensing - no exposure metric beyond mass $PM_{2.5}$.

doi: 10.3390/ijms27198931

$PM_{2.5}$ thyroid vitamin C rats 2026 Ecotoxicol Environ Saf

Vitamin C attenuates $PM_{2.5}$-induced thyroid dysfunction and mitophagy dysregulation by rebalancing ER-mitochondrial Ca2+ dynamics

Female rats and FRTL5 thyrocytes: $PM_{2.5}$ impaired thyroid function and histology, triggered ER stress, remodelled MAM proteins (IP3R1, GRP75, VDAC1), caused mitochondrial Ca2+ overload, ROS, depolarisation, PINK1/Parkin changes, GPX4 loss and Fe2+ accumulation; 4-PBA, 2-APB, Mdivi-1 and vitamin C each mitigated damage. Limitation: doses not given in the abstract and likely supra-environmental. Relevance: low - mechanism for a thyroid signal that the epidemiology (13 Oct fire-PM thyroid cancer) has not yet tied to function.

doi: 10.1016/j.ecoenv.2026.120920

Ruan gut microbiota dose-time trajectories $PM_{2.5}$ PFOS mouse 2026 Environ Pollut

Time-resolved profiling reveals nonlinear dose-time trajectories of gut microbiota in $PM_{2.5}$ and PFOS exposure models

Two mouse models ($PM_{2.5}$ inhalation; PFOS in drinking water) sampled repeatedly across dose levels and exposure stages, with 16S sequencing and generalised additive models for dose-time structure. Controls showed clear temporal succession, which the authors stress must be modelled before attributing change to exposure. In the $PM_{2.5}$ model, restructuring peaked at early-to-intermediate stages, with nonlinear trajectories in Lactobacillus, Prevotella and the Firmicutes/Bacteroidetes ratio; PFOS effects were later and longer. No effect sizes with CIs are reported in the abstract. Limitation: functional pathways are inferred, not measured; doses and particle source are not given in the abstract, and cage effects dominate small murine microbiome studies. Relevance: low - mechanistic, but the time-varying response argues against single-timepoint toxicology.

doi: 10.1016/j.envpol.2026.129383

Saveleva et al. 2026 Mol Neurobiol

Urban PM0.2 Affects Astrocytes by Inducing Xenobiotic and Oxidative Stress Responses

Primary astrocytes exposed to collected urban PM0.2 at 50 ug/ml for 24 h: 2,217 mRNAs altered, with strong induction of xenobiotic metabolism (CYP genes, Ahrr), Nrf2 oxidative stress responses and glutathione detoxification (GST superfamily, Nqo1); 354 proteins nominally changed, 142 concordant with mRNA. Reanalysis of published data shows microglia go pro-inflammatory while astrocytes go detoxification/antioxidant. Limitation: one dose, one time point, 50 ug/ml is orders above any plausible brain dose, no particle-free extract control for organics vs core; proteomics is labelled exploratory. Relevance: low-moderate - reinforces that ultrafine/quasi-ultrafine number, not $PM_{2.5}$ mass, is the metric for neuro endpoints, which most LCS do not resolve.

doi: 10.1007/s12035-026-06232-w

Sharma et al. 2026 Aerosol Sci Technol

Characterization of carbonaceous particles generated from three flame modes using a single burner to enable properties-based exposure studies

METADATA ONLY. Metadata-only record: Taylor & Francis deposited title, authors and DOI on 6 Oct with no abstract. By title, a single burner operated in three flame modes to generate carbonaceous particles of controlled and differing properties, explicitly to support properties-based rather than mass-based exposure studies. That framing is the open problem behind every toxicology record in this backfill: dose expressed as mass hides composition, and a generator that varies one property at a time is how it gets separated. Relevance: high; on the retry list.

doi: 10.1080/02786826.2026.2737177

Sun respirable mining dust natural killer cell dysfunction 2026 Environ Sci Technol

Respirable Particulate Matter Induces Natural Killer Cell Dysfunction via Oxidative Stress-Mediated Lysosomal and Mitochondrial Injury: Implications for Mineral-Rich Mining Dust Exposure.

In-vitro exposure of NK cells to respirable mineral dusts with a K562 target-cell challenge. Injury was dose- and composition-dependent: apoptosis/necrosis, shifted cytokines, altered perforin/granzyme B function and impaired killing. Silica- and silicate-rich dusts were worst; iron-bearing and carbonaceous dusts produced distinct oxidative patterns. Mitochondrial depolarisation, lower mitochondrial mass and lysosomal deacidification converged across dusts; ascorbic acid partially restored organelle function and IFN-gamma. Doses and their relation to inhaled lung burden are not given in the abstract. Limitation: an isolated cell line in submerged culture, no lung-burden dosimetry, so the proposed adverse outcome pathway to lung cancer is inferential. Relevance: low - composition (silica fraction) matters, which mass sensors cannot resolve. Note: this DOI was held from the 3 Oct issue as unauthored and is now carried here.

doi: 10.1021/acs.est.6c07515

Tampere SOA toxicity 2026 Environ Res

From primary to secondary aerosol: chemical transformation and toxicity of gasoline vehicle exhaust under contrasting driving conditions

A Euro 6 plug-in hybrid with a gasoline particulate filter run over a mild RDE cycle and a highly dynamic Combined cycle, cold and hot, with diluted exhaust aged in the Tampere Secondary Aerosol Reactor and delivered to A549 cells through a mobile air-liquid-interface system, fresh and aged, whole exhaust and gas-phase only. Ageing consistently shifted aerosol composition and toxicity relative to primary emissions. Limitation: one vehicle, one fuel, and an OFR delivers days of equivalent ageing in minutes at OH concentrations far above ambient, which over-produces highly oxygenated products; ALI dose at the cell surface is rarely quantified in the same units as an inhaled dose. Relevance: high - a GPF-equipped modern vehicle emitting little primary PM and substantial secondary aerosol is the case in which a tailpipe-mass regulatory metric and a downwind sensor network disagree by construction.

doi: 10.1016/j.envres.2026.125881

Tang CPS-II plasma proteome air pollution lung cancer ever-smokers 2026 Environ Sci Technol

Analysis of the Potential Mediating Role of Plasma Proteins in the Association Between Air Pollution Exposure and Lung Cancer Risk Among Ever-Smokers

95 matched case-control pairs of ever-smokers with pre-diagnostic plasma proteomics from the CPS-II Nutrition Cohort. Seventeen proteins associated with both exposures and lung cancer (CX3CL1, ADAM15, CXCL17 among them, $p<0.05$); high-dimensional mediation (HIMA) flagged six, including CX3CL1 and CXCL17 (Bonferroni $p<0.05$). Limitation: 190 people against a high-dimensional proteome, and the mediation signals disappeared when cases diagnosed within a year of blood draw were excluded - consistent with reverse causation by subclinical tumour. The abstract does not name the pollutants' exposure model. Pathways (chemokine signalling, immune modulation) are hypothesis-generating only. Relevance: low - biomarker discovery, no sensing implication.

doi: 10.1021/acs.est.6c07407

Time-restricted feeding $PM_{2.5}$ adipose metabolism 2026 J Nutr Biochem

Metabolic Reprogramming by Time-Restricted Feeding Confers Resistance to $PM_{2.5}$-Induced Metabolic Disturbances in Adipose Tissue

METADATA ONLY. No abstract at harvest. Title indicates an experimental study in which time-restricted feeding attenuated $PM_{2.5}$-induced metabolic disturbance in adipose tissue. Recorded for coverage; no estimate inferred. Relevance: low for sensing - a dietary effect modifier, of interest only if human chrono-nutrition data are later paired with personal exposure.

doi: 10.1016/j.jnutbio.2026.110521

Vocal fold fibroblasts 2026 Laryngoscope

Effects of Fine Particulate Matter ($PM_{2.5}$) in Cultured Human Vocal Fold Fibroblasts

METADATA ONLY. Metadata-only record: Wiley deposited title, authors and DOI on 7 Oct with no abstract. By title, an in-vitro exposure of human vocal fold fibroblasts to $PM_{2.5}$ - an unusual target tissue, and a plausible one given that the larynx sits in the deposition path of coarse and fine particles and that occupational voice disorders cluster in dusty trades. Relevance: low-moderate; on the retry list.

doi: 10.1002/lary.70976

Wang seasonal PM size constituents oxylipin panel Beijing 2026 Environ Sci Technol

Winter and Summer Differences in Oxylipin Responses to Ambient Particles: Roles of Sizes and Constituents

Panel of 123 adults in Beijing with repeated visits in winter and summer; 27 $PM_{2.5}$ mass, size and constituent metrics against five lipoxygenase-derived oxylipins. Associations peaked at lag 0-1 d; across metrics, oxylipin increases ranged 3.3-114.2%, most robust for UFP, accumulation-mode number, OC and chloride. UFP showed an exposure-response only in summer: per IQR, +12.9 to +97.7% in summer vs -7.7 to +18.8% in winter; accumulation-mode and OC effects were larger in winter. No CIs in the abstract. Limitation: 27 metrics x 5 biomarkers x 2 seasons x several lags is a large test space, and the season contrast also changes time outdoors and ventilation. Relevance: high - size-resolved number, not mass, carries the summer signal, which mass-only optical sensors cannot report.

doi: 10.1021/acs.est.6c08647

Yadav et al. 2026 Environ Pollut

The dual role of atmospheric HULIS in inducing climate forcing and oxidative stress in the northwestern Himalayas

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 6 Oct with no abstract. By title, HULIS measured for both light absorption (brown-carbon forcing) and oxidative stress, in a high-altitude Himalayan setting. HULIS is the fraction that links the optical property a low-cost sensor responds to with the oxidative potential that toxicology implicates, so a single study reporting both is more useful than two reporting one each. Relevance: high; on the retry list.

doi: 10.1016/j.envpol.2026.129300

Yoon PRL-3 ezrin airway epithelium particulate matter 2026 Respir Res

PRL-3-sensitive cytoskeletal signaling mediates early airway epithelial responses to particulate matter

Phosphoproteomics of human bronchial epithelial cells under non-cytotoxic PM exposure, with pharmacological and genetic PRL-3 (PTP4A3) suppression and an acute mouse airway exposure model. PM rapidly and transiently upregulated PRL-3 and relocalised it to the membrane, with PRL-3-sensitive ezrin dephosphorylation in vitro and in vivo. Blocking PRL-3 restored ezrin phosphorylation and attenuated EMT-associated changes and MUC5AC upregulation, independent of Akt, ERK and mTOR, and reproducibly across UPM, DEP and PM_{10}. Limitation: doses are described as non-lethal but not related to deposited human doses; acute exposure says nothing about chronic remodelling. Relevance: low for sensing - a candidate early-effect pathway shared across PM types, i.e. not composition-specific.

doi: 10.1186/s12931-026-03834-9

Yu alveolar organoid $PM_{2.5}$ 2026 Front Toxicol

Differential toxicity of water- and DMSO-extractable $PM_{2.5}$ fractions in human alveolar organoids

Human alveolar organoids exposed to extracted $PM_{2.5}$ showed dose- and time-dependent morphological damage, more severe for the DMSO-extractable (organic) fraction, with ROS rise, cell death and a sustained fall in AT2 cells plus a small AT1 increase suggesting AT2-to-AT1 transdifferentiation. Limitation: doses are not reported in deposited-mass terms, so relevance to ambient exposure is unknowable, and extraction solvents select chemistry rather than particles. Relevance: low - model development. Held from the 9 Oct issue.

doi: 10.3389/ftox.2026.1943932

Zakari PM_1 microbial profiles lung gut microbiome mice 2026 J Hazard Mater

Associations of particulate matter-associated microbial profiles with lung and intestinal microbiomes in healthy mice

Mice were housed under ambient PM_1, HEPA-filtered air or clean air; PM_1 was characterised for ions, metals, carbon fractions, oxidative potential and its own 16S microbiome (Proteobacteria-dominated). Lung diversity barely changed while the gut microbiome shifted: PM_1 reduced Actinobacteriota and Bacteroidota and raised Proteobacteria versus both control arms. Eleven families and six genera overlapped among PM_1, lung and gut communities in the PM_1 group. Limitation: group sizes and exposure concentrations are not given in the abstract, gases co-vary with PM_1, and taxon overlap does not show transfer from air to gut. Relevance: low - airborne microbial load is a PM component no optical sensor resolves.

doi: 10.1016/j.jhazmat.2026.143978

Zendi brown carbon wood pyrolysis tracers cytotoxicity A549 2026 Chem Res Toxicol

Fine Brown Carbon Aerosol Emitted by Wood Pyrolysis - New Harmful Tracers and Cytotoxicity in Human Lung Cells

Smouldering-regime BrC was generated by 350 C wood pyrolysis, extracted in three solvents and characterised by UHPLC-ESI-ToF-MS non-target analysis (776-1,132 molecular features per extract). Water extracts were dominated by lignin phenols and aromatic acids; methanol and acetone pulled less oxidised terpenoid and sterol-like species. Cytotoxicity in A549 cells ranked water < methanol < acetone, consistent with lipophilic, membrane-partitioning and redox-active compounds; named contributors to baseline narcosis included 3-hydroxybenzoic, salicylic and p-coumaric acids, benzene-1,2,4-triol and coniferylaldehyde. Limitation: solvent extracts at unreported dose-equivalence to inhaled mass, a single carcinoma line, and pyrolysis rather than flaming or aged smoke. Relevance: low-medium - supports BrC (UV-absorbing) channels as a toxicity-weighted metric.

doi: 10.1021/acs.chemrestox.6c00273

Zeng RON toxicity 2026 Sci Adv

Reduced organic nitrogen compounds dominate the toxicity of $PM_{2.5}$ organics following internal exposure

Nontargeted high-resolution MS of urban $PM_{2.5}$ organics is combined with protein-adductomics to define the fraction that actually reaches the circulation, then scored with multi-endpoint toxicity prediction. Reduced organic nitrogen (RON: amines, amides, N-heterocycles) carries 39% of the predicted multi-organ toxicity of the internally exposed fraction against 23% for the oxygenated organics the field usually targets, and binds human serum albumin strongly enough to explain systemic distribution. Limitation: the toxicity shares are model predictions on annotated features, not measured dose-response; nontargeted annotation confidence and ionisation bias toward basic nitrogen both inflate RON's apparent share. Relevance: high - the toxic fraction is invisible to optical mass sensing, an argument for adding composition proxies (e.g. N-containing BrC absorption) to networks.

doi: 10.1126/sciadv.aea8654

Zhang et al. 2026c Free Radic Biol Med

$PM_{2.5}$ exposure impairs BMSC function and dysregulates mitochondrial oxidative stress: multi-omics insights into potential osteoporotic mechanisms

Rat BMSCs exposed to $PM_{2.5}$ lost viability, cytoskeletal integrity, proliferation and migration, with suppressed osteogenesis via beta-catenin/Runx2, raised ROS and disturbed mitochondrial homeostasis. Transcriptomics gave 2,372 DEGs (911 up / 1,461 down) and metabolomics 171 DEMs, converging on cAMP signalling; forskolin rescued viability, ROS, membrane potential, migration and differentiation. Limitation: a single pooled $PM_{2.5}$ sample of unstated composition at unstated dose, applied directly to a cell type that in vivo sits behind the alveolar barrier - the rescue is strong internal evidence for the pathway but says nothing about delivered dose at the marrow. Relevance: low for sensing, moderate for the bone-outcome literature that epidemiology has begun to report.

doi: 10.1016/j.freeradbiomed.2026.10.004

Zhao coal dust macrophage extracellular vesicles thyroid cancer miR-205-5p 2026 Front Immunol

Coal dust-stimulated macrophage-derived extracellular vesicles promote thyroid cancer cell phenotypes involving miR-205-5p, CAMK1G, and NF-kB-associated signaling

THP-1 macrophages exposed to characterised coal dust released extracellular vesicles enriched in miR-205-5p; these increased TPC-1 thyroid cancer cell viability, clonogenicity, migration and invasion and reduced apoptosis. Effects ran through CAMK1G and NF-kB (p65, IkBa phosphorylation) and were attenuated by miR-205-5p inhibition or CAMK1G knockdown; coal-dust-exposed mice and humanised xenograft mice were used in vivo. Limitation: the premise - coal dust promoting thyroid cancer - has no epidemiological anchor, and macrophage dosing in vitro is far above realistic lung burdens. Relevance: low - occupational mechanism without an exposure metric.

doi: 10.3389/fimmu.2026.1856770

de Luna wildfire PM toxicity systematic review 2026 J Hazard Mater

Systematic review on the toxicity of wildfire emitted particulate matter: State of science, knowledge gaps and way forward

46 in vitro and in vivo studies, 2005-2025 (epidemiology excluded). Only 42% of in vitro and 27% of in vivo studies used PM actually collected during wildfires; the rest used surrogate burns. In vitro work concentrates on lung (50%) and macrophages (17%); inhalation studies on lung (46%), heart (15%) and brain (15%). Mechanisms reported are mainly inflammation, oxidative stress and DNA damage, and toxicity varied with dose, size, fuel and burn condition. Gaps: incomplete physicochemical characterisation, unrealistic doses, few chronic or toxicokinetic studies, no work on mixtures with urban pollution. Limitation: no quantitative synthesis or risk-of-bias grading, so heterogeneity is described, not explained. Relevance: medium - the field cannot yet say which smoke property a sensor should track beyond mass.

doi: 10.1016/j.jhazmat.2026.143985

Occupational & indoor — 50 records

Carried from the daily issues — full summaries (50 of 50 records in this subtopic)

Aljabri et al. 2026 Indoor Air

Indoor Air Quality and Exposure Characterization in Operational Waterpipe Cafes: Field Measurements of Particulate and Volatile Pollutants in Al-Madinah Al-Munawwarah

Direct measurement in five operating waterpipe cafes on weekdays and weekends: mean weekday $PM_{2.5}$ 285.5 +/- 149 $\mu g\,m^{-3}$ and PM_{10} 301.6 +/- 156 $\mu g\,m^{-3}$, weekend TVOC 0.50 mg/m3, formaldehyde below short-term reference values. Variability tracked smoking intensity and ventilation. The $PM_{2.5}/PM_{10}$ ratio near 0.95 is the signature of a pure combustion source with no coarse fraction. Limitation: instrument make, calibration and whether an optical sensor was humidity- or composition-corrected are not stated - charcoal and tobacco smoke is exactly where optical PM monitors read high; five venues, no personal sampling, no worker shift-length exposure. Relevance: high for occupational indoor monitoring, where a factor-of-two optical bias still leaves concentrations an order of magnitude above guideline.

doi: 10.1155/ina/8494753

Alquwayi respirable crystalline silica occupational exposure Saudi Arabia review 2026 Front Public Health

Potential risk of occupational exposure to respirable crystalline silica in Saudi Arabia: a comprehensive review of the current status and future challenges

Single-author review integrating literature, labour and mining statistics, mineralogy and regulation through a 'silica hazard triangle' (workforce, materials, activities). Concludes that mining, construction, cement, glass, engineered stone and photovoltaic manufacturing expansion plus arid-climate dust raise RCS exposure potential, while measured exposure, epidemiology and surveillance data are scarce. Limitation: no new measurements and no quantitative exposure estimates - the conclusion is an inference from sector size, not from data. Relevance: low-moderate - optical LCS cannot speciate silica, so RCS monitoring needs gravimetric/FTIR sampling regardless.

doi: 10.3389/fpubh.2026.1918362

Arif Agra footwear cluster ventilation IAQ tiers 2026 Research Square (preprint)

Ventilation, indoor air quality and thermal comfort in India's leather footwear manufacturing cluster: a tier-resolved field study

Agra footwear cluster, three enterprise tiers (formal factory to cottage unit), 378 worker-sessions over four seasons and 147 days, breathing-zone measurements paired with ASHRAE 55 ballots. All pollutants rose linearly from large to cottage scale as air velocity fell from 0.80 to 0.16 m/s; velocity explained 79% of between-tier pollutant variance. No session anywhere met the WHO $PM_{2.5}$ guideline; 91.3% exceeded the ACGIH formaldehyde TLV. Limitation: unreviewed preprint; PM instrument and calibration are not described in the abstract, and session means against an annual guideline overstate comparability. Relevance: moderate - informal-sector workplaces with no monitoring are an obvious deployment niche for wearable or wall-mounted low-cost PM nodes.

doi: 10.21203/rs.3.rs-10780874/v1

Arif Agra leather footwear facilities indoor $PM_{2.5}$ CO_2 formaldehyde 2026 Research Square (preprint)

Indoor Air Quality and Thermal Environment Across Three Leather Footwear Facilities in Agra India

Three leather-footwear facilities (cottage, small, organised), 27 months, 608 sessions. Facility dominated variance for every parameter (partial eta2 0.10-0.87); cottage vs organised means $PM_{2.5}$ 2.0-fold, CO_2 1.9-, formaldehyde 3.0- and VOC 3.5-fold higher. Season dominated PM and thermal variables, tracking ventilation; indoor/outdoor $PM_{2.5}$ ratios 0.48-1.00 across facilities; CO_2 above the ASHRAE threshold in 99.5% of cottage sessions vs 0% organised. Limitation: preprint; one facility per tier, so scale is confounded with building; instruments and their calibration are not stated, and spot sessions are not shift-weighted personal exposure. Relevance: moderate - informal-sector workplaces are where portable LCS are the only feasible monitor.

doi: 10.21203/rs.3.rs-10962722/v1

Blackley et al. 2026 Ann Work Expo Health

Aerosol mitigation efficiency of high-volume suction systems during aerosol-generating dental procedures with patients

Five commercial high-volume suction systems - standard HVE, three intraoral systems and one modified HVE - measured during real crown preparations (n=26) and ultrasonic scaling (n=25), with direct-reading instruments near the dentist and in the hallway. Procedure-to-background concentration ratios were modelled with covariates (suction flow rate, procedure duration, tooth location, scaler type and quadrant); a system counted as effective only when the 95% lower confidence limit of the ratio was <=1, i.e. indistinguishable from background. Limitation: unblinded single-site evaluation with modest per-system n, patients differ across procedures, and the short-duration ratio design cannot speak to a full clinical day. Relevance: high - this is the design a sensing group should copy. The estimand is 'did the control return the air to background', tested against the instrument's own noise, rather than a percentage reduction quoted without a comparison.

doi: 10.1093/annweh/wxag081

Blanco-Alegre ironing indoor ultrafine particles ventilation eBC 2026 Atmosphere

Ironing as a Source of Indoor Aerosol Particles: Effects of Ventilation and Ironing Conditions

Residential measurements with a conventional steam iron and a steam iron with a portable boiler, at low and higher ventilation. PNC rose by about two orders of magnitude during ironing, mostly 20-80 nm, to a maximum of 7.12e5 cm^{-3} (conventional iron, low ventilation). Emission rates 8.1e11-1.5e12 particles/min, highest for the boiler iron at low ventilation; modes shifted to larger sizes after ironing; eBC changed little. Limitation: one dwelling, few repetitions, and the source mechanism (water additives, fabric, soleplate heating) is not separated; no mass concentration is reported, so health relevance is by number only. Relevance: medium - an ultrafine source that optical home sensors will undercount almost entirely.

doi: 10.3390/atmos17111063

Charres et al. 2026 Atmos Pollut Res

Insights into the organic and toxicological characteristics of PM_{10} from indoor and outdoor settings in a school environment

Metadata only. Metadata-only record: Elsevier deposited title, authors and DOI on 2 Oct with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes organic and toxicological characterisation of indoor and outdoor PM_{10} at a school; the author list (Cipoli, Vicente, Evtyugina) is the Aveiro aerosol group, but this watch does not infer location or method from authorship. Relevance: potentially high - indoor/outdoor school PM composition is what classroom low-cost sensors cannot report. On the retry list.

doi: 10.1016/j.apr.2026.103223

Chen dormitory Stokes 2026 Build Environ

Size-resolved aerosol transport and exposure risk in university dormitories driven by Stokes number topology and source-flow coupling

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 9 Oct with no abstract. By title, size-resolved aerosol transport and exposure risk in dormitories organised around Stokes number and source-flow coupling - i.e. the regime where inertia, not the mean flow, decides where particles land, which is why a single room-average concentration misrepresents exposure for coarse particles. Relevance: moderate - the argument for size-resolved rather than $PM_{2.5}$-mass indoor nodes. On the abstract-retry list.

doi: 10.1016/j.buildenv.2026.115373

Creosote railroad switch PAH workers 2026 Ann Work Expo Health

Assessing PAH exposure during construction work of railroad switches using creosote-impregnated ties

Workers assembling railroad switches from creosote ties: breathing-zone sum of 16 PAHs 120,000-280,000 ng/m3 on a single sampled shift; skin pyrene 0.31-140 ng/cm2, highest on palms and calves; urinary 1-HP rose across the week (1.9-17 umol/mol creatinine), most above the UK guidance value by day 4. A PBTK model attributed uptake mainly to the dermal route, and pyrene under protective clothing implies fabric penetration. Limitation: one air-sampling shift, small crew, and the particle/vapour split is not reported, so the inhalation share rests on the model. Relevance: low for ambient sensing - a reminder that biomarker-based particulate PAH exposure is not reducible to air concentration.

doi: 10.1093/annweh/wxag088

Dental scaling bioaerosols 2026 J Hosp Infect

Divergent bacterial signatures in dental clinic bioaerosols: a real-field investigation

Ultrasonic scaling caused a large surge in sub-micron particle counts, but culturable bacteria concentrated in the 1.1-3.3 um stages. Airborne bacterial communities differed from PPE (ANOSIM R=0.633, p=0.001) and were indistinguishable from non-touch surfaces; oral pathogens were rare in air. Limitation: single clinic and procedure type, small sample count not given. Relevance: moderate - a clean demonstration that particle number from an optical counter is not a proxy for biological load; mirrors the 4 Oct suction study.

doi: 10.1016/j.jhin.2026.10.003

Du classroom CO_2 proxy PM VOC ventilation types 2026 Indoor Environments

CO_2 as a proxy for PM and VOCs in classrooms across ventilation types and activities

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. The title tests whether CO_2 tracks PM and VOCs in classrooms across ventilation types and activities - i.e. whether a cheap NDIR sensor can stand in for particle and VOC monitoring. Number of classrooms, ventilation categories, PM instrumentation and agreement statistics cannot be stated. Limitation: unverifiable beyond the title; physically, CO_2 tracks occupant-generated bioeffluents and air exchange, whereas PM indoors also depends on outdoor infiltration, resuspension and filtration, so decoupling under mechanical ventilation is expected. Relevance: high - directly asks whether school CO_2 monitor rollouts can be read as PM exposure surveillance.

doi: 10.1016/j.indenv.2026.100200

Fan wildfire smoke indoor $PM_{2.5}$ absenteeism Oklahoma Head Start OK-AIR 2026 J Expo Sci Environ Epidemiol

Wildfire smoke, indoor and outdoor $PM_{2.5}$, and sick-related absenteeism in Oklahoma Head Start early care and education centers

Cohort 1 of the OK-AIR cluster-randomised 2x2 factorial trial (HEPA, upper-room UVGI): 5 centres, 20 classrooms, 204 children aged 0-5, 2023-2024. On smoke-impacted days, the intervention period (pooled arms) was associated with 29.0% lower indoor $PM_{2.5}$ (22.8-34.7%) after adjustment for outdoor $PM_{2.5}$. Each +100 in the 7-day mean of regional FIRMS fire detections: sick-related absence OR 1.106 (1.018-1.201). Limitation: the period contrast pools all arms, so it is not a HEPA effect, and smoke days cluster in Feb-Mar, confounded by respiratory-virus season; fire counts are a regional proxy, not exposure. Relevance: high - classroom sensor plus EPA outdoor pairing is a deployable I/O design for childcare networks.

doi: 10.1038/s41370-026-00993-1

Geng grain dust simulator manikin respirator workplace protection factor 2026 J Agric Saf Health

Using a Grain Dust Simulator with a Manikin System to Evaluate Performances of Commonly Used Respirators

A grain dust simulator combined with a breathing manikin estimated SWPFs for four common respiratory protection devices at four respirable dust levels (5, 10, 15, 20 mg/m3) and two inhalation flows (30 and 85 L/min), benchmarked against real-time dust in on-farm grain bins. The platform was repeatable and separated device performance; the nuisance dust mask is not recommended for grain bins. SWPF values are not in the abstract. Limitation: manikin faces do not reproduce human fit variability or movement, so SWPF overstates real protection for poorly fitted FFRs; a static simulator lacks the burst exposures of bin entry. Relevance: low-medium - inside/outside real-time dust monitors are the field version of this test.

doi: 10.13031/jash.16485

Global IAQ open standards framework 2026 Build Environ

A unified framework for global indoor air quality: transitioning from guidelines to actionable open regulatory standards

METADATA ONLY. No abstract at harvest. Title indicates a proposal to move indoor air quality from guideline values to open, enforceable standards. Recorded for coverage. Relevance: moderate - enforceable indoor standards would make low-cost indoor PM monitors compliance instruments, with the performance-certification burden that implies.

doi: 10.1016/j.buildenv.2026.115384

Hunter e-cigarette aerosol early post-emission particle decay controlled conditions 2026 Sci Total Environ

Early post-emission particle decay dynamics of E-cigarette aerosols under controlled conditions

METADATA ONLY. Title and authors only (Hunter, Jacoby, Ferdousi-Rokib, Asa-Awuku et al.). No abstract was retrievable at harvest. The title describes the first seconds-to-minutes decay of e-cigarette aerosol under controlled conditions - evaporation of volatile droplets that shrinks particles rapidly after exhalation or emission. Instruments, conditions and decay constants are not available; nothing is inferred. Relevance: medium - e-cigarette droplets produce large, short-lived spikes on indoor optical PM sensors that mass-based readings misstate because the particles are liquid and evaporating; useful for indoor network event classification.

doi: 10.1016/j.scitotenv.2026.182379

Indoor NPF from outdoor O_3 2026 Environ Sci Technol

Elevated Outdoor O_3 Can Sustain New Particle Formation and Growth Indoors

In an authentic office in a high-O_3 region, mechanical ventilation supplied enough O_3 that indoor new particle formation was VOC-limited, not O_3-limited; limonene addition drove growth at 20-100 nm/h, one to two orders above outdoor rates, with meaningful mass from grown particles. Limitation: one building, injected limonene rather than real product use, and no occupancy. Relevance: high - indoor UFP and secondary mass from cleaning products are invisible to $PM_{2.5}$-only indoor sensors until particles grow past ~300 nm.

doi: 10.1021/acs.est.6c07592

K-HEALTHinAIR multicountry indoor air homes schools low-cost sensors 2026 Eur J Public Health (conference abstract)

Multicountry evaluation of indoor air in homes and schools: results from K-HEALTHinAIR project

Low-cost sensors in homes and schools across four countries, 2025. Home $PM_{2.5}$ was elevated with smoking (93.9 $\mu g\,m^{-3}$), air conditioning (34.8), gas cooking (29.5) and Dutch district heating (19.4). School $PM_{2.5}$ was higher in Poland (22.0) than Austria (9.5). $PM_{2.5}$ exceedance reached 56% of days in Austrian wood-heated homes and 36% in Spanish smokers' homes; CO_2 exceedance was 41% in Austrian schools. Limitation: sensor calibration and RH correction are not described, so cross-country contrasts may partly be instrument and humidity artefacts; sampling of homes is unstated. Conference abstract. Relevance: high - a multi-country indoor LCS deployment whose QA is the open question.

doi: 10.1093/eurpub/ckag169.339

Kazakhstan PRISm 2026 Front Public Health

Occupational associations of preserved ratio impaired spirometry (PRISm) in a population-based study

2,275 adults from the general population with spirometry and a lifetime job-exposure matrix. PRISm prevalence was 12% (95% CI 10.7-13.3), higher in women (14%) than men (10%). Occupational airborne exposure including VGDF was not associated with PRISm, as ever-exposure or cumulative years; BMI was (adjusted OR 1.05 per unit). Limitation: a job-exposure matrix built for European job titles applied to Kazakh work histories misclassifies heavily and will null out a real association; cross-sectional design cannot separate PRISm from healthy-worker selection out of dusty jobs. Relevance: moderate - a clean negative in a heavily industrial country, and a direct argument for measured rather than matrix-assigned occupational exposure.

doi: 10.3389/fpubh.2026.1921126

Ketsakorn and Chaiyadej 2026 PLoS One

Probabilistic health risk assessment of respirable dust exposure among stone carvers in Thailand using average daily dose and Monte Carlo simulation

Personal sampling across mining, cutting, splitting, knocking and carving, with a 10,000-iteration Monte Carlo ILCR using US EPA exposure parameters. Mean respirable dust was below both the Thai OEL and the TLV everywhere, highest in carving at 0.051 mg/m3; ILCR stayed under 1e-6, highest in carving at 3.3e-7. Limitation: respirable dust mass is the wrong metric for a silica hazard - no crystalline silica fraction is reported, and an EPA inhalation-unit-risk framework applied to a mixed mineral dust produces a number with no established exposure-response behind it; small sample, one season. Relevance: low-moderate - the task-level contrast is exactly what a wearable logging monitor resolves and a single shift-average sample does not.

doi: 10.1371/journal.pone.0359706

Kim pig building air cleaner TSP $PM_{2.5}$ productivity 2026 J Anim Sci Technol

Association between indoor air pollutants and pig productivity index by on-site application of air cleaner

One commercial pig building, control vs air-cleaner-treated room, monitored six months across rearing stages. Air-cleaner reduction efficiency rose with stage, 5-10% at farrowing to 30-45% at fattening; treated rooms showed better weight gain, feed conversion, activity index and mortality. Weight gain and activity correlated with TSP and $PM_{2.5}$ (and airborne fungi). Limitation: one room per arm, no randomisation or replication - room effects are inseparable from treatment; the cleaner technology and PM instrumentation are not stated. Relevance: low - livestock housing is a large agricultural PM source and an occupational exposure for farm workers, but this design supports neither inference.

doi: 10.5187/jast.2500087

Konisky et al. 2026 Build Environ

Evaluating Indoor Air Pollution in Urban, Low-Income, Multi-Family Buildings: A Case Study from Indianapolis

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 1 Oct with no abstract; PubMed efetch, Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes an indoor air-pollution evaluation in low-income multi-family buildings in an Indianapolis case study. Directly relevant to low-cost indoor sensor deployments in under-monitored housing. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.buildenv.2026.115356

Kulmala determination deposition rates indoor particle dynamics 2026 Aerosol Sci Technol

Determination of deposition rates from indoor particle dynamics

METADATA ONLY. Title and authors only (Kulmala I., Siilin, Hakala, Ehder-Gahm); no abstract retrievable at harvest. The title describes deriving particle deposition rates from indoor concentration dynamics. Method, size range and validation cannot be stated. Limitation: unverifiable beyond the title; decay-based deposition estimates are confounded by air-exchange and mixing assumptions unless these are measured independently. Relevance: medium - deposition rates are an input for indoor/outdoor ratio models used to interpret paired indoor and outdoor low-cost sensors.

doi: 10.1080/02786826.2026.2754776

Kuye ultrafine particle respiratory deposition dose wood stoves UK homes 2026 Atmosphere

Respiratory Deposition Dose of Ultrafine Particles from Improved Wood Stoves and Open Fireplaces

Exploratory measurements in four UK homes (Ecodesign stove, multifuel Ecodesign stove x2, open fireplace), 11-12 winter days each, 1-min PNC combined with age/sex/activity ventilation rates and the ICRP model. Daily inhaled dose 0.45-16.0 x10^9 particles/kg/day sitting and 2.37-73.2 x10^9 at moderate activity (4.5-5.3-fold higher). Child scenarios had 2.7-3.8x the adult body-mass-normalised dose; household mean deposition rate 1.50-5.99 x10^8 particles/min. Limitation: n=4 homes that differ simultaneously in appliance, fuel, room, ventilation and behaviour, so stove-type effects cannot be identified; doses are modelled from concentration, not measured. Relevance: medium - residential UFP is invisible to mass sensors; indoor PNC nodes are the relevant instrument.

doi: 10.3390/atmos17111039

LiFePO4 workers cardiac autonomic 2026 Occup Environ Med

Association between occupational LiFePO4 exposure and cardiac autonomic function: a cross-sectional study of cathode material production workers

METADATA ONLY. No abstract at harvest. Title indicates a cross-sectional study of battery-cathode workers' LiFePO4 particle exposure and cardiac autonomic function. Recorded for coverage. Relevance: moderate - battery-material dust is an emerging occupational particle class with little exposure data.

doi: 10.1136/oemed-2025-110695

Mahmoud filtration risk 2026 Build Environ

The link between aerosol dispersion, infection risk and air filtration

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 9 Oct with no abstract. By title, a link between indoor aerosol dispersion, filtration and infection risk - the Wells-Riley family of models, whose weak point is that a well-mixed room assumption and a measured clean-air delivery rate rarely both hold. Relevance: moderate - filtration intervention studies in this corpus (the 4 Oct AFRI-c HEPA null among them) keep failing on exposure verification, which is a monitoring problem before it is a filtration problem. On the abstract-retry list.

doi: 10.1016/j.buildenv.2026.115378

Martin simplified method portable air cleaner CADR low-cost sensors 2026 MethodsX

A simplified method for evaluating fine particulate matter removal performance of portable air cleaners

Methods paper for community-run CADR testing: release a short-lived nebuliser aerosol in a sealed room, log $PM_{2.5}$ decay with and without the cleaner on low-cost sensors, and compute CADR from the decay-rate difference. Residential low-cost-sensor CADRs ran ~26% below laboratory values from research-grade instruments - a conservative bias. Of ten units, a four-filter Corsi-Rosenthal box performed best among DIY builds; a fan shroud added ~11%. Across 135 models DIY units cost far less per CADR unit. Limitation: the 26% gap mixes sensor size-response with room mixing and leakage, and is not decomposed; nebuliser aerosol is not wildfire smoke. Relevance: high - a practical protocol in which the low-cost sensor is the reference, with its bias quantified.

doi: 10.1016/j.mex.2026.104216

Matuskova wood stove phase-resolved particle-bound PAHs 2026 Environ Res

Characterisation of particle-bound PAHs emitted from a residential wood-burning stove and their dependence on combustion phases

Three temperate wood species burned in a standard residential stove, aerosol split by cascade impactor across five combustion phases. PM peaked at ignition and flaming; PAHs followed for spruce and black locust, while beech peaked later (interval 4), coinciding with low flue-gas oxygen. Size distributions were monomodal at ~0.1 um aerodynamic diameter with PM_1 70-89% of mass. Hardwoods emitted mainly LMW PAHs; spruce gave an initial HMW burst, shifting to LMW during burnout, when PAHs moved towards coarser particles. Limitation: one stove under nominal operation; real-world overloading and damp fuel dominate emissions. Relevance: moderate - an ~0.1 um mode sits below optical LCS detection efficiency, so wood-smoke PM is under-counted by size-blind calibrations.

doi: 10.1016/j.envres.2026.125931

Mohammed mobile HEPA-UV air decontamination operating theatre 2026 Infect Prev Pract

Evaluation of a mobile surgical air decontamination device in reducing airborne microbial load in the operating theatre of a tertiary care hospital.

Before-after evaluation of a mobile HEPA + UV decontamination unit in one operating theatre over 15 days (July 2025). Mean total particle count fell from 6,836 to 4,072 particles/m3 (-40.4%), significant in the 0.3 and 0.5 um channels; CFU fell from 34.5 to 15.2 CFU/m3 (P 0.003). Limitation: one theatre, short before-after design with no control period or blinding to activity level, and no clinical outcome (surgical-site infection). Relevance: low-medium - shows optical counters as the performance metric for portable cleaners in clinical rooms.

doi: 10.1016/j.infpip.2026.100583

Nahian Siegel in-situ testing in-duct air cleaners residential 2026 Build Environ

In-situ testing of in-duct air cleaners in residential buildings

METADATA ONLY. Title and authors only (Nahian, Siegel); no abstract retrievable at harvest from Crossref, Europe PMC or Semantic Scholar. The title describes in-situ testing of in-duct air cleaners (filters or electronic cleaners in forced-air systems) in occupied homes, presumably measuring particle removal under real operation. Country, number of homes, test protocol and removal metrics cannot be stated. Limitation: unverifiable beyond the title; in-situ effectiveness depends on HVAC runtime fraction, which lab single-pass efficiency ignores. Relevance: moderate - indoor LCS networks are the practical tool for verifying such devices in homes.

doi: 10.1016/j.buildenv.2026.115494

Neonatal air cleaners 2026 Build Environ

Portable air cleaners in neonatal facilities: Hybrid experimental-CFD assessment of different configurations

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 8 Oct with no abstract. By title, a combined experimental and CFD assessment of portable air cleaner configurations in neonatal facilities. Placement, not device rating, is what decides delivered clean-air rate in a real room, and the AFRI-c care-home null (4 October) is the clinical consequence of getting it wrong unmeasured. A neonatal unit is the setting where the dose argument is strongest. Relevance: high; on the retry list.

doi: 10.1016/j.buildenv.2026.115381

Ng particulate matter respiratory deposition inhalation dose biomass burning homes indigenous communities 2026 Atmos Environ

Particulate Matter Exposure, Respiratory Deposition and Inhalation Dosage Assessment in Biomass Burning Homes in Rural Indigenous Communities

METADATA ONLY. Title and authors only (Ng, Monalisa, Balachandran, Hoque et al.); day-0 Elsevier deposit with no abstract at harvest. The title describes household PM measurement in biomass-burning homes in rural indigenous communities with respiratory deposition and inhaled-dose modelling. Instruments, n and concentrations cannot be stated. Limitation: unverifiable beyond the title; deposition models (e.g. MPPD) are sensitive to the assumed size distribution, which optical monitors in smoke-laden kitchens report poorly. Relevance: medium - household biomass smoke is a key use case for low-cost PM monitors.

doi: 10.1016/j.atmosenv.2026.122448

Office air purifier field CADR 2026 Build Environ

Air purifier performance in an occupied office: Field CADR and effectiveness from a reconstructed device-off condition

METADATA ONLY. No abstract at harvest. Title indicates field estimation of purifier clean-air delivery rate in an occupied office using a reconstructed device-off counterfactual. Recorded for coverage. Relevance: high if the counterfactual is sensor-derived - field CADR is how purifier-trial exposure contrasts should be verified.

doi: 10.1016/j.buildenv.2026.115390

Othman et al. 2026 Indoor Air

Impact of Urban Land Use on Indoor Air Quality: Quantifying Ambient Particle Infiltration and Asthma Risk in Naturally Ventilated School Buildings

Ten primary schools stratified by land use (high-traffic, industrial, residential, mixed industrial/high-traffic, mixed industrial/port) with paired indoor-outdoor $PM_{2.5}$ and PM_1. Indoor maxima were at mixed industrial/port schools (24 h $PM_{2.5}$ 35.33 +/- 9.25, PM_1 23.79 +/- 5.87 $\mu g\,m^{-3}$); outdoor maxima at high-traffic schools (34.50 +/- 20.64 and 23.78 +/- 13.09); land-use class differences were significant ($p<0.05$) and the mean infiltration factor exceeded 0.6. Limitation: ten buildings across five classes is two per class, so the land-use contrast is confounded with building; Finf estimated from indoor-outdoor ratios without an independent air-change measurement absorbs any indoor source into the infiltration term. Relevance: high - Finf > 0.6 in naturally ventilated classrooms means outdoor network data is a usable proxy for children's exposure in this climate, which is the assumption most school studies make without testing.

doi: 10.1155/ina/5050009

Parthiban et al. 2026 Cureus

Ocular Surface Disorders in Traffic Police Officers in the Union Territory of Puducherry, India: A Cross-Sectional Study

106 traffic police examined with OSDI, Schirmer, TBUT, tear meniscus height and conjunctival impression cytology: Nelson grade 1-3 metaplasia in 83%, dry-eye symptoms in 78%; grade associated with years of field duty, OSDI and TBUT (chi-square $p<0.05$). Limitation: no unexposed comparison group, no exposure measurement, UV and pollution are inseparable, and years of service is collinear with age; the authors' 'clear exposure-outcome relationship' is not supported by this design. Relevance: low - a case for wearable PM monitoring in traffic-police cohorts, which would make this question answerable.

doi: 10.7759/cureus.115533

Patras primary school classroom PM airborne viruses ventilation 2026 Eur J Public Health (conference abstract)

Dual burden of PM and airborne viruses in primary schools: A pilot study in Patras, Greece

March 2026 sampling in 28 classrooms of seven primary schools. Mean $PM_{2.5}$ 37.8 and PM_{10} 109 $\mu g\,m^{-3}$; every classroom exceeded the WHO 24 h guideline, and at least one respiratory virus was detected in every room (SARS-CoV-2 most often). Frequent natural ventilation was associated with lower PM ($p < 0.033$, Cliff's delta >0.58) and less viral RNA. $PM_{2.5}/PM_{10}$ 0.35 points to resuspension. Limitation: short occupied-hours sampling compared against a 24 h guideline overstates exceedance, and one month in one city; instrument type unstated. Conference abstract. Relevance: moderate - coarse-dominated classroom PM is where optical LCS under-read most.

doi: 10.1093/eurpub/ckag169.1262

Pigakis cement dust small airway dysfunction CPET 2026 Int Arch Occup Environ Health

Early respiratory effects of long-term cement dust exposure: association with exercise capacity in asymptomatic workers

500 non-smoking, non-atopic cement workers aged >=35 with >=10 years' exposure underwent spirometry, lung volumes, impulse oscillometry and CPET. A subset had small airway dysfunction despite normal spirometry; they had 19.4% lower VO2 peak (29 vs 36 mL/kg/min) and lower breathing reserve (34% vs 43%). Limitation: no unexposed comparator and no measured dust concentrations, so this shows SAD is associated with exercise limitation, not that cement dust causes either; the SAD prevalence is not given. Relevance: low-moderate - occupational dust surveillance still relies on tenure, which personal PM monitors could replace.

doi: 10.1007/s00420-026-02247-x

Portable air purifiers airborne bacteria AMR offices schools 2026 Front Public Health

Response of culturable bacteria and their resistance to air purification across compartments

Offices in Valencia and classrooms in Dublin sampled before and after portable air purifier deployment; 341 isolates, MALDI-TOF identification, susceptibility testing and resistance-gene PCR. Office airborne isolates fell from 41 to 5; school airborne isolates fell from 102 to 47 and resistant prevalence from 43% to 21%. Surface and human-associated communities shifted less, and human-associated samples became Staphylococcus-dominated with more non-susceptible isolates. Limitation: single pre/post snapshots without control rooms or particle counts, culture-only recovery, small isolate numbers. Relevance: low-moderate - purifier trials rarely log bioaerosol and PM together; co-located OPCs would let isolate counts be normalised to removed particle number.

doi: 10.3389/fpubh.2026.1945592

Rees et al. 2026 NIHR Open Res

Lessons learned from using high efficiency particulate air (HEPA) filtration units in care homes: interviews with care home staff, residents and relatives from the AFRI-c trial

Reflexive thematic analysis of interviews with 25 staff, 20 residents and 12 relatives inside the AFRI-c pragmatic cluster RCT, whose main result was that HEPA units did not reduce respiratory infections. Units became normalised but were judged unsuitable for some residents; lockable settings and wall fixing were asked for; cost was seen as prohibitive sector-wide, raising an equity concern that filtration would reach only residents who can pay. Maintenance burden was minor. Limitation: qualitative, post hoc, and conducted among people who already knew the trial was running; no measurement of unit runtime, filter loading or achieved air-change rate - which is the missing variable that would explain a null. Relevance: high - a filtration null without exposure verification is the standing weakness of this trial literature, and a logging low-cost PM node in each room is the cheap fix.

doi: 10.3310/nihropenres.13993.1

Rodriguez et al. 2026 Indoor Air

Integrated Assessment of Indoor Air Pollution and Associated Health Risks in a Hemodialysis Unit

One hemodialysis treatment room monitored for a year: ~1,400 time-resolved readings of CO_2, O_3 and PM0.3/2.5/10 with simultaneous outdoor measurements, plus 12 monthly VOC campaigns. Indoor CO_2 often exceeded the Spanish threshold (indoor-outdoor difference >350 ppm, peaks >1,100 ppm), indicating poor ventilation. $PM_{2.5}$, PM_{10} and O_3 had I/O ratios below 1 (predominantly outdoor origin) and were generally under WHO guidelines with occasional peaks; most VOCs had I/O >1, and benzene and ethylbenzene cancer risks were below 1e-6. Limitation: one room, instrument type and PM calibration are not given in the abstract, and 'occasional peaks' are not quantified. Relevance: moderate - dialysis patients spend ~12 h/week indoors in one room; I/O ratios tell a facility whether filtration or ventilation is the lever.

doi: 10.1155/ina/4283416

Saborit gas cooking indoor air quality households 7 European countries 2026 Indoor Environments

Characterization of the impact of gas cooking on indoor air quality in households across 7 European countries

Metadata only: no abstract at Crossref, Europe PMC or Semantic Scholar on the issue date. The title reports household measurements of gas-cooking impacts on indoor air quality across seven European countries; which pollutants (NO_2, $PM_{2.5}$, ultrafine particles) were measured, the number of homes and the instrumentation are not available here and are not inferred. Limitation (structural): cooking PM is dominated by the food and method rather than the fuel, so attributing particle increments to gas requires an electric comparison arm. Relevance: medium - multi-country household campaigns of this type typically rely on low-cost PM and NO_2 sensors. Retry the abstract before the weekly.

doi: 10.1016/j.indenv.2026.100202

Simkovich LPG stove RCT quality of life 2026 Indoor Air

Cooking With Liquefied Petroleum Gas or Biomass and Quality of Life: Results From a Randomized Controlled Trial

180 women primary cooks aged 25-64 randomised to an LPG stove, 12 months of free fuel and behavioural reinforcement vs usual cooking; 179 completed. ITT at 12 months: SF-36 mental 48.3 vs 50.9 ($p=0.59$), physical 53.1 vs 47.6 ($p=0.08$), EQ-5D VAS 67.0 vs 71.7 ($p=0.05$); no personal $PM_{2.5}$-QOL exposure-response. More intervention women felt financially better off (42% vs 13%) and reported higher income (33% vs 9%), both $p<0.001$. Limitation: QOL is a secondary outcome in a trial powered for something else, unblinded, and the fuel subsidy itself transfers income, so the financial result is partly mechanical. Relevance: moderate - personal $PM_{2.5}$ monitoring was available but the QOL signal was absent; intervention trials remain the strongest design for testing whether measured reductions matter.

doi: 10.1155/ina/9936046

Stange SEM-Raman inhalable particles indoor artificial turf infill 2026 Research Square (preprint)

Correlative SEM–Raman Microscopy of Inhalable Particles Sampled during Indoor Football on Artificial Turf with Crumb Rubber and Olive-Stone Infills

Air sampling during matches in two Norwegian indoor halls, one with crumb-rubber infill and one with olive-stone/quartz infill, analysed by correlative SEM, AI particle classification and micro-Raman. Both infills released airborne particles. Olive-stone particles appeared indoors only; tyre rubber appeared indoors and outdoors, implicating traffic as well as infill. PM_{10} was 58.4 $\mu g\,m^{-3}$ in the olive-stone hall vs 6.0 $\mu g\,m^{-3}$ with crumb rubber, and fungal spores were found with the organic infill. Limitation: one hall per infill, so the tenfold PM_{10} difference is confounded by hall ventilation, occupancy and cleaning. Preprint. Relevance: low-medium - substituting a 'sustainable' material changed composition and raised mass.

doi: 10.21203/rs.3.rs-11040166/v1

Stefaniak recycled plastic bag HDPE filament 3-D printing emissions 2026 J Chem Health Saf

Recycling Plastic Bags into Filament: Consideration of Emissions during Melting, Extrusion, and 3-D Printing.

Waste HDPE plastic bags from South Africa and the US were blended with virgin HDPE, extruded into filament and 3-D printed, with particle- and gas-phase emissions measured through each stage. Filament extrusion emitted mostly 1.2-5.2 nm particles; printing emitted mostly 5.6-560 nm particles, with particle yields (no./g processed) 2-3 orders of magnitude higher than extrusion. Printing also released acetone (1.4-39.4 ug/g printed) and formaldehyde (9.9-16.1 ug/g). Mechanical properties matched virgin HDPE. Limitation: yields are per mass processed in a controlled enclosure, not personal breathing-zone concentrations, and no uncertainty is attached to the order-of-magnitude contrast; feedstock contamination of bag plastic was not characterised. Relevance: low-medium - nearly all the number emission sits below the ~300 nm optical cut of consumer PM sensors, so maker-space monitoring needs a CPC or diffusion charger, not a PMS-class node.

doi: 10.1021/acs.chas.5c00138

Suh residential occupant activities airborne microplastic emissions 2026 Build Environ

Microplastics emissions from occupant activities in residential spaces: Experimental quantification and relative importance analysis of particle generation factors

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. The title reports experimental quantification of microplastic particle emissions from occupant activities in residential spaces and a relative-importance analysis of generation factors (activity type, textiles, surfaces presumably). Size range, identification method (FTIR, Raman, pyrolysis GC-MS), emission rates and whether the inhalable fraction was measured cannot be stated. Limitation: unverifiable beyond the title; microplastic counts are method-bound and seldom expressed as PM mass. Relevance: moderate - indoor activity-resolved particle sources are what household LCS see as unexplained spikes; polymer fibres also scatter differently from combustion aerosol.

doi: 10.1016/j.buildenv.2026.115399

Uberlandia street vendors $PM_{2.5}$ micronuclei panel 2026 Int J Hyg Environ Health

Urban air pollution and genotoxic damage in street vendors: a longitudinal repeated-measures panel study

238 men over four campaigns spanning rainy and dry seasons: 116 outdoor street vendors in the city centre vs 122 indoor formal vendors in a reference area. PM concentrations and MN frequency were both consistently higher in the centre. Poisson GEE: per 1 $\mu g\,m^{-3}$ $PM_{2.5}$, MN IRR 1.024 (1.013-1.034) rainy and 1.026 (1.019-1.034) dry; no season interaction (p=0.378). Limitation: exposure is fixed-point area monitoring, not personal, and is collinear with occupation group, so the $PM_{2.5}$ slope partly encodes street vs indoor work (traffic gases, PAHs, sun, smoking patterns). Relevance: moderate - seven fixed points is exactly the density at which a small low-cost array could convert an ecological contrast into personal-scale exposure.

doi: 10.1016/j.ijheh.2026.114932

Walter neighbourhood air exchange inhaled dose children 2026 J Expo Sci Environ Epidemiol

Neighborhood-scale disparities in air exchange rates and inhaled pollution dose among school-aged children

METADATA ONLY. No abstract at harvest. Title indicates neighbourhood-scale modelling of housing air exchange rates and the resulting inhaled pollutant dose among school-aged children, framed as a disparity analysis (Northeastern University group). Recorded for coverage; no estimate inferred. Relevance: high if infiltration is parameterised from data - indoor/outdoor LCS pairs measure exactly this ratio.

doi: 10.1038/s41370-026-00989-x

Wang electret HVAC filter charge neutralisation wildfire smoke 2026 Environ Sci Technol

Charge-Driven Degradation of Residential Electret HVAC Filters under Wildfire Smoke Exposure.

Laboratory flaming and smouldering wood smoke, field sampling at prescribed burns and controlled soot challenges on residential electret HVAC filters. Filtration efficiency collapsed at particle loadings far below those that raise pressure drop; the driver was soot-rich particles neutralising electret charge irreversibly, not mechanical clogging or organic-aerosol deposition. Built into a dynamic indoor model driven by EPA monitor data for a regional wildfire event, effective filter life fell from months to days at moderate outdoor $PM_{2.5}$, with protection varying by building and ventilation. No efficiency numbers are given in the abstract. Limitation: few filter products and smoke types; charge loss in the field depends on soot fraction, which varies by fire stage. Relevance: high - indoor/outdoor LCS pairs can detect the efficiency collapse that pressure-based replacement rules miss.

doi: 10.1021/acs.est.6c03046

Zargarian spent coffee grounds MOF filtration membrane $PM_{2.5}$ bioaerosol photothermal 2026 Small

Spent Coffee Grounds-Derived Micro-Nano-Architectured Air Filtration Membranes for Multiwavelength Light-Triggered Bioaerosol Inactivation

Single-step co-electrospinning of MOF-177/polyacrylonitrile nanofibres with electrosprayed ZIF-8-decorated spent coffee ground microparticles. At 4.5x the NIOSH respirator test face velocity the membrane captured 99.7% of $PM_{2.5}$ at about half the pressure drop of a MOF-free control and removed 83.4% of ammonia; photothermal heating >80 C under simulated sunlight or NIR restored microbial suppression after three days of office air. Limitation: one challenge aerosol and short loading; no size-resolved efficiency at the MPPS, no long-term loading curve, and MOF stability under humidity is not shown. Relevance: low - a filtration material, not a measurement; at most a candidate inlet pre-filter for bioaerosol-sensitive sensor housings.

doi: 10.1002/smll.76528

Zhang WWTP bioaerosol 2026 Environ Res

Comprehensive Characterization of Bioaerosol Microbial Communities, ARG, and Occupational Exposure Risks in WWTP and Comprehensive Landfill

Paired sampling at a wastewater plant and an integrated solid-waste facility with 16S sequencing, HT-qPCR, QMRA and ML. Culturable bacteria were 3.4-fold higher at the landfill (1,726 vs 502 CFU/m3) and so was the fine biogenic particle share (27% vs 15%), but airborne ARG/MGE copies were an order of magnitude higher at the WWTP (5,462 vs 584 copies/m3), giving it the larger estimated worker inhalation dose. Limitation: two facilities, short campaigns, and culturable counts undercount viable-but-nonculturable cells; ARG copies per m3 do not establish viable resistant organisms. Relevance: low-moderate - optical sensors see biogenic particles as generic mass; the case for fluorescence-capable occupational monitors.

doi: 10.1016/j.envres.2026.125888

Zhao et al. 2026d Build Environ

Behavior-resolved bioaerosol exposure modeling reveals scale-dependent occupational risk in pediatric wards

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 5 Oct with no abstract. By title, a bioaerosol exposure model in paediatric wards resolved by occupant behaviour, reporting scale-dependent occupational risk. Behaviour-resolved models are the class of work that decides whether a fixed room monitor can stand in for a worker's exposure at all, and the paediatric ward is a setting where that substitution is routinely assumed. Relevance: potentially high; on the retry list.

doi: 10.1016/j.buildenv.2026.115364

Burden, policy & mitigation — 64 records

Carried from the daily issues — full summaries (64 of 64 records in this subtopic)

Abdella Italy urban-rural 2026 Atmos Environ

Cleaning the air or shifting the burden? Characterizing urban-rural disparities in air pollution-attributable mortality across Italy

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 10 Oct with no abstract. By title, an attributable-mortality analysis asking whether Italian air-quality gains have been distributed or merely displaced from urban to rural populations. Rural exposure is systematically the worse-measured half of any national network, so a rural-urban disparity estimate inherits the monitoring asymmetry it is trying to describe - the 8 October US block-level record showed spatial aggregation attenuating exactly this kind of disparity. Relevance: moderate-high; on the abstract-retry list.

doi: 10.1016/j.atmosenv.2026.122420

Abu Dhabi PM burden 2008-2022 2026 Risk Anal

Ambient Air Pollution Health Impacts in High-Income Countries, Part II: A Case Study in the United Arab Emirates

Attributable deaths from anthropogenic PM and ozone in Abu Dhabi rose from 231 (90% UI 59-427) in 2008 to 1,249 (507-2,169) in 2019-2022, despite air quality holding roughly steady while population doubled. The increase reflects population growth, better health records and, mainly, steeper concentration-response functions. Limitation: a fivefold change driven largely by swapping CRFs is a change in assumptions, not in burden; the 'anthropogenic' split in a dust-dominated desert regime is itself model-dependent. Relevance: moderate - shows how far CRF choice swamps measured trends, which bounds what better monitoring alone can change.

doi: 10.1111/risa.70368

Air-health web atlases review Italy 2026 Epidemiol Prev

PNC project 'Atlante Aria e Salute': exploratory web review and quality assessment of open-access atlases on the health effects of air pollution

Protocol-driven review of public air-and-health atlases (Sep-Dec 2024): 1,000 websites screened, 15 potentially relevant, 8 included (3 governmental, 3 academic, 2 foundations). All report $PM_{2.5}$ and/or PM_{10}; NO_2 and O_3 are less common; all but one use annual averages, six give dose-response estimates (mostly all-cause long-term mortality), only two cause-specific. Assessed for interactivity, Flesch readability, WCAG accessibility and transparency. Limitation: eight sites is a thin sample and search-engine screening is not reproducible. Relevance: moderate - public health-impact maps ignore sub-annual and sub-city variation, exactly what dense networks measure.

doi: 10.19191/ep26.4-5.s1.a1026.089

Atlante Aria e Salute Delphi co-creation 2026 Epidemiol Prev

Co-creation as a decision-making tool in the PNC 'Atlante Aria e Salute' project to define the contents of an integrated atlas

Two Delphi rounds (16 then 22 experts and regional stakeholders) plus plenaries to fix the Italian national air-and-health atlas: data sources (ISTAT cause-specific mortality, hospital discharges), territorial disaggregation, pollutant-outcome pairs graded on causal certainty, public-health relevance and routine-data accuracy, attributable-fraction methods and policy counterfactuals. Limitation: process paper; the abstract is truncated before the final outcome list, and no exposure model is specified. Relevance: low-moderate - a national open-data atlas is a natural consumer of fused monitor/LCS exposure surfaces if its resolution is fine enough.

doi: 10.19191/ep26.4-5.s1.a952.090

BC outpatient risk 2026 Eco Environ Health

Black carbon as a major contributor to ambient $PM_{2.5}$-related cause-specific outpatient risks: implications for early warning and health benefits evaluation of air quality improvement

Over 212 million daily outpatient records, 2013-2017, with conditional logistic regression per city pooled by random effects and three complementary models testing constituent independence. Black carbon carried the strongest independent associations, and at lag0 rather than the lag0-1 typical of admissions: per IQR (1.8 $\mu g\,m^{-3}$) increases ranged from 1.33% (0.93-1.74) for chronic kidney disease to 3.89% (3.30-4.49) for COPD, attributable fractions 2.38% to 6.12%. Both $PM_{2.5}$- and BC-attributable burdens fell substantially after the 2013 clean-air programme. Limitation: constituents come from a modelled reanalysis and are collinear; outpatient visits are supply-sensitive, and the lag0 finding could reflect care-seeking on visibly bad days rather than a faster biological response. Relevance: high - BC is measurable by low-cost aethalometry, and a same-day constituent signal is the one an operational early-warning network could actually act on.

doi: 10.1016/j.eehl.2026.100283

BEST-COST all-cause vs disease-specific exposure-response functions burden $PM_{2.5}$ NO_2 2026 Eur J Public Health (conference abstract)

Effect of all-cause versus disease-specific exposure-response functions on air pollution burden

BEST-COST burden estimates (healthiar R package) in five countries, comparing WHO all-cause mortality ERFs with disease-specific ERFs. Disease-specific YLL were 2-4x lower (ratio 0.26-0.52; median 0.42 for $PM_{2.5}$, 0.31 for NO_2). Country rankings were stable (Spearman rho >=0.90), but absolute agreement was moderate for $PM_{2.5}$ (ICC 0.46) and poor for NO_2 (0.15). Limitation: five countries is too few for inference (Wilcoxon p=0.0625); the gap partly reflects causes of death excluded from the disease-specific set, which is a substantive question, not a method artefact. Conference abstract. Relevance: moderate - ERF choice, not exposure error, dominates burden uncertainty.

doi: 10.1093/eurpub/ckag169.810

Beveridge HeatRisk AQI cardiovascular action algorithm 2026 Am J Prev Cardiol

HeatRisk and air quality risk recommendations: A narrative review and action algorithm for the care of people at risk for and living with cardiovascular disease

PubMed-based narrative review (2000-2025) prioritising systematic reviews, mapped onto CDC, EPA and NWS guidance to build a clinical counselling algorithm for heat and poor air quality in people with or at risk of CVD. It summarises excess risk for heart failure, ischaemic heart disease, stroke and MI, and high-risk groups (pregnancy, older age, certain medications, occupational exposure), and proposes action at a risk score >=3. Limitation: non-systematic and the scoring algorithm is not validated against outcomes. Relevance: moderate - AQI-triggered clinical advice is only as good as the local $PM_{2.5}$ behind it; sensor-informed AQI (AirNow Fire and Smoke map) is the practical feed.

doi: 10.1016/j.ajpc.2026.101704

Bhattarai India crop burning inventories 2026 Atmos Environ

Air Quality Impacts of Post-monsoon Crop Residue Burning in Northern India Based on Regional and Global Emission Inventories

METADATA ONLY. No abstract at harvest. Title indicates a comparison of regional and global emission inventories for post-monsoon crop-residue burning and the resulting air-quality impact in northern India. Recorded for coverage. Relevance: moderate - inventory spread is the dominant uncertainty in Indo-Gangetic smoke attribution, and dense sensors are the observational constraint that could arbitrate between inventories.

doi: 10.1016/j.atmosenv.2026.122423

Bong et al. 2026 medRxiv (preprint)

Delayed Coal Plant Retirements in the Era of Data Centers Carry a Measurable Mortality Cost

Of 50 US coal plants scheduled (Jan 2024) to retire before 2035, 20 had later or removed dates by Jan 2026. Projecting plant-attributable $PM_{2.5}$ and mortality among Medicare beneficiaries for 2026-2035, the revised schedule adds 1,256 deaths (1,136-1,376), 3.1 times the burden of the original schedule. Eighteen of the 20 plants sit in balancing authorities with hyperscale data-centre demand, and several federal orders and utility filings cite data-centre growth. Limitation: the interval reflects concentration-response uncertainty only, not dispersion-model or future-emission uncertainty; operating hours of delayed plants are assumed; attribution to data centres is circumstantial, not causal; preprint. Relevance: moderate - coal-sulfate $PM_{2.5}$ is regional and secondary, so it is a background term networks see as a baseline, not as a local plume.

doi: 10.64898/2026.09.29.26364336

Brunori Rome urban vineyard grapevine leaf PM interception 2026 Plants (Basel)

Plant-Atmosphere Interactions in Urban Vineyards of Rome: A Digital Phenotyping Approach for the Functional Selection of Grapevine Cultivars

Urban vineyard of 170 cultivars at Rome's Botanical Garden, 2025 season: 54 cultivars screened by leaf micromorphology, five assessed in depth (anatomy, chlorophyll, reflectance indices, SEM-counted PM). After ~20 days without heavy rain, leaf-surface PM ranged from 52,024 (Barbera) to 77,462 particles/mm2 (Maturano), comparable to some urban tree species; adaxial deposition was over twice abaxial. Limitation: particle counts, not mass or size-resolved deposition flux; five cultivars, one site, one accumulation window, and no ambient PM measurement to convert counts into a removal rate. Relevance: low - green-infrastructure mitigation claims need co-located ambient PM, which a low-cost sensor at canopy height would supply cheaply.

doi: 10.3390/plants15193014

Canada wildfire smoke guidance 2026 Can J Public Health

Public health guidance about wildfire smoke: Four targets for improvement

Commentary proposing four distinct targets for improving Canadian wildfire-smoke guidance: a stronger evidence base (subpopulations, long-term effects, protective interventions), ethically deliberated recommendations, evaluation of guidance effectiveness beyond self-report, and a process-and-content standard for authoritative guidance. Limitation: no data. Relevance: moderate - the evaluation target presupposes measured indoor and personal exposure during smoke events, which is where low-cost sensors already dominate the evidence.

doi: 10.17269/s41997-026-01267-x

Cantu air quality across the lifespan nursing response review 2026 Nurs Clin North Am

Breathing Unequal Air: Global and US Air Quality Across the Lifespan and the Nursing Response.

Narrative review for nurses of global and US air quality trends and health effects across the lifespan, citing ~8 million deaths/year attributable to air pollution, with $PM_{2.5}$, O_3 and NO_2 as the principal pollutants. Covers cardiopulmonary, neurocognitive and perinatal outcomes, nursing practice and policy, and argues for integrating air-quality awareness into nursing curricula. No new data. Limitation: non-systematic; no synthesis of effect sizes. Relevance: low - possible audience for patient-facing use of public sensor maps.

doi: 10.1016/j.cnur.2026.06.007

China BC burden 2026 Nat Commun

The Hidden Health and Climate Burden of Black Carbon under China's Dual Carbon Strategy

A localized BC inventory and revised risk function put over 540,000 premature deaths attributable to black carbon in China in 2015 - more than twice previous estimates - while the CMIP6 inventory is found to overestimate BC radiative forcing by 45%. Projections suggest demographic ageing may offset the health gains from falling emissions. Limitation: a doubled burden rests on a 'revised health risk function' whose derivation is the entire result and cannot be checked from the abstract; BC-specific concentration-response functions are confounded with co-emitted species in every source study they come from, so attributing 540,000 deaths to BC rather than to combustion mixture is a strong claim. Relevance: high - BC is measurable by low-cost aethalometry, and the inventory-vs-measurement gap this paper quantifies is what a dense BC network would close.

doi: 10.1038/s41467-026-77632-8

East Asia $PM_{2.5}$ O_3 warming levels 2026 Environ Pollut

Uneven Future $PM_{2.5}$ and O_3 responses to Emission Reductions under Different Warming Levels in East Asia

METADATA ONLY. No abstract at harvest. Title indicates climate-penalty-modulated $PM_{2.5}$ and O_3 responses to emission cuts in East Asia. Recorded for coverage. Relevance: low.

doi: 10.1016/j.envpol.2026.129354

GBD 2023 G20 stroke air pollution temperature 2026 Thromb Haemost

Estimates, temporal evolution, and contributors of the stroke burden due to air pollution and non-optimal temperature across G20 nations, 1990-2023

GBD 2023, G20 nations. Age-standardised stroke mortality attributable to air pollution plus non-optimal temperature fell from 52.81 to 17.46/100,000 (1990-2023; EAPC -3.44%, 95% CI -3.57 to -3.30); DALY rate EAPC -3.29%. The decline is driven by household air pollution; ambient PM and heat burdens rose or stalled, highest in low/middle-SDI members, and ARIMA projections to 2040 keep them flat. Limitation: GBD re-tabulation with joint attribution of two risk classes; the ARIMA projection adds no exposure information. Relevance: low - confirms that ambient PM, not household fuel, now carries the residual stroke burden.

doi: 10.1055/a-2977-5555

Global atmospheric Pb 2000-2019 2026 Environ Pollut

Global modelling of atmospheric lead (Pb) concentrations and health risks during 2000-2019

METADATA ONLY. No abstract at harvest. Title indicates global modelling of atmospheric Pb and associated risk over two decades. Recorded for coverage. Relevance: low.

doi: 10.1016/j.envpol.2026.129346

Global heatwave-fire smoke coexposure 2026 Sci Adv

Rising global population coexposure to heatwaves and landscape fire-sourced air pollution

Global 2000-2019: 55.9% of the population lived where heatwave-smoke coexposure increased; 916 million people/year had at least one coexposure day, rising by 287 million per decade; mean coexposure days per person rose from 1.0 to 1.7. Of the 5.5 billion person-day/decade increase, 70.5% was intensity and 29.5% population growth; low-HDI countries carried the most. Limitation: the fire-PM product is model-derived with no stated threshold sensitivity, and 'coexposure day' is a binary that discards concentration. Relevance: moderate - joint heat-smoke days are also when optical sensors are most biased (low RH, fresh smoke), so the exposure most in need of measurement is the hardest to measure cheaply.

doi: 10.1126/sciadv.aea9749

Grzesczyk-Nojszewska Poland air pollution interventions public attitudes survey 2026 Int J Occup Med Environ Health

Public attitudes toward personal and population interventions to reduce air pollution exposure: a 2025 nationwide cross-sectional survey in Poland

CAWI survey, May 2025, quota sample of 1,092 adults. 76.4% saw air pollution as requiring public action (women aOR 1.47, 1.11-1.96; age >=60 aOR 2.49, 1.59-3.91). Commonest protective behaviours: closing windows during episodes (44.2%) and avoiding outdoor time (36.5%). Fines for burning prohibited materials in home furnaces had 70.2% support. Limitation: quota online panels over-represent the engaged; no link to local concentrations, and stated behaviours are unverified. Relevance: medium - window closing is the dominant action, so indoor/outdoor paired sensing is what tells people whether it works.

doi: 10.13075/ijomeh.1896.02831

Gujral Dominici Memphis datacenter $PM_{2.5}$ 2026 Preprints.org (preprint)

Behind-the-Meter Versus Grid Electricity for AI Datacenters: Magnitude, Geography, and Equity of $PM_{2.5}$-Attributable Mortality in Memphis

Two datacenters 12 km apart. At a common 240 MW, grid-supplied C1 carried 10.83 attributable deaths/y (95% interval 7.83-13.84) vs 3.67 (2.65-4.68) for behind-the-meter C2; at real/permitted capacities the order reversed, 11.35 (6.48-17.29) for C2 vs 8.10 (4.60-12.38) for C1. 94% of C1 deaths fell >250 km away; 82% of C2 deaths within 50 km, concentrated in the low-income, high people-of-colour stratum. Limitation: preprint; permitted capacity is not operating capacity, and on-site turbine emission factors and the reduced-form dispersion model are the whole result. Relevance: high - a near-source burden that a local LCS network around the site could verify directly.

doi: 10.20944/preprints202610.0819.v1

Guo lung cancer burden 2026 Chin Med J Pulm Crit Care Med

Trends in lung cancer burden, risk factors, and disease projections in China and the United States, 1990-2023

GBD 2023 data with joinpoint regression, three-factor Das-Gupta decomposition and Bayesian age-period-cohort projection to 2050, comparing lung cancer burden and its attributable risks - smoking, air pollution, occupational exposure, metabolic and dietary factors - between China and the United States. Limitation: GBD attribution for air pollution is a modelled counterfactual built on exposure surfaces and borrowed concentration-response functions, so the pollution share is an assumption propagated forward, not an observation; the abstract reports no attributable fractions. Relevance: low-moderate - useful as the burden framing that the measurement side of this corpus feeds, but it adds no exposure information of its own.

doi: 10.1016/j.pccm.2026.08.006

He ozone-dominant air pollution risk ageing inequality China counties 2026 Ecotoxicol Environ Saf

A nationwide effect-based assessment of ozone-dominant air pollution risk patterns and ageing-related inequality in China

Interpretable machine learning was fit to cancer data from 813 Chinese counties (2018-2019) to get non-linear long-term exposure-response shapes for cold-season $PM_{2.5}$ and warm-season O_3. Model-based thresholds then anchored exceedance and excess-hazard indices projected over 2017-2023. $PM_{2.5}$ pressure fell while O_3 pressure rose, with hotspots in Sichuan-Chongqing and the Yangtze River Delta that coincide with ageing. Limitation: two years of county cancer data cannot identify a dose-response threshold. ML-derived 'thresholds' in an ecological panel are model artefacts confounded by region, and the abstract gives no effect sizes or CIs. Relevance: low - county-level modelled exposure.

doi: 10.1016/j.ecoenv.2026.120945

Idowu medical waste incinerator airborne heavy metals Monte Carlo Nigeria 2026 Research Square (preprint)

Monte Carlo-Based Probabilistic Health Risk Assessment of Airborne Heavy Metals from a Medical Waste Incinerator in Southwestern Nigeria

Particulate matter collected on passive glass slides around a medical-waste incinerator for two months; ten metals analysed, converted to airborne concentrations, and propagated through 10,000 Monte Carlo draws. Median HQ ~400, 95th percentile 850; exceedance probability near 100%; ILCR above benchmarks. Limitation: passive deposition slides measure settled mass, not airborne concentration - the conversion to C_air requires an assumed deposition velocity that dominates the result, and an HQ of 400 almost certainly reflects that conversion rather than real inhalation dose. Preprint. Relevance: moderate as a cautionary example - fence-line LCS or low-volume samplers would have given a defensible air concentration.

doi: 10.21203/rs.3.rs-11381994/v1

Imman caregivers $PM_{2.5}$ prevention preschool children northern Thailand qualitative 2026 Child Health Nurs Res

Caregivers' experiences of $PM_{2.5}$ prevention for preschool children in Thailand: a qualitative study with focus group interviews

Focus groups plus in-depth interviews with 20 primary caregivers (14 women) of preschool children in a rural northern Thai community subject to recurrent haze; content analysis (Graneheim and Lundman). Themes: barriers of awareness and responsibility against pervasive local sources; individual prevention driven by visible symptoms rather than air-quality information; and a shift toward collective action and calls for structural change (waste-burning control). Limitation: one community, 20 participants, no link to measured exposure or behaviour change. Relevance: medium - caregivers act on symptoms, not data; local sensor displays only help if they reach this audience in an actionable form.

doi: 10.4094/chnr.2025.047

Indo-Gangetic BC 2026 Atmos Pollut Res

Long-term decline in Black Carbon Aerosol and Associated Health Risk over north-western Indo-Gangetic Plain

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 8 Oct with no abstract. By title, a long-term decline in black carbon over the north-western Indo-Gangetic Plain with an attached health-risk estimate. A declining BC trend in one of the most polluted regions on earth is a strong claim that depends entirely on aethalometer calibration and filter-loading correction being stable across the record - the commonest source of spurious long-term trends in BC. The same day carries two independent BC burden records (China, 540,000 deaths; the 274-city outpatient series on 7 October), so the trend direction matters. Relevance: high; on the retry list.

doi: 10.1016/j.apr.2026.103227

Isfahan synbiotic omega-3 RCT 2026 J Res Med Sci

Effects of synbiotic versus omega-3 supplementation on total antioxidant capacity in adolescents exposed to ambient air pollution: A randomized double-blind controlled trial

70 healthy adolescents aged 10-18, 8 weeks synbiotic (n=38) vs omega-3 (n=32). TAC rose within both arms (839.8 to 903.4 and 898.6 to 953.3 umol/L) with no between-group difference (p=0.75). Limitation: active-comparator trial with no placebo arm, so within-arm rises cannot be separated from regression to the mean or seasonal change; exposure is not measured, so 'exposed to air pollution' is residence in Isfahan. Relevance: low.

doi: 10.4103/jrms.jrms_753_25

Izmir maritime 2026 Atmos Environ

Quantifying Maritime Contributions to Urban Air Quality in Izmir Bay Using Tier 3 Emission Inventory and AERMOD

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 8 Oct with no abstract. By title, a bottom-up Tier 3 (movement-based) ship emission inventory coupled to AERMOD for a port city. Port and shipping contributions are among the hardest urban sources to attribute from a land-based network, because the source is offshore, mobile and upwind only part of the time - the same problem the Geoje Island AMS record (6 October) attacked by measurement rather than by model. Relevance: moderate-high; on the retry list.

doi: 10.1016/j.atmosenv.2026.122410

Jiang PLA melt-blown nonwoven filtration durability humidity airflow 2026 Langmuir

Service Durability of PLA Melt-Blown Nonwovens: Roles of Humidity and Airflow Velocity in Degradation and Filtration Performance

Full-factorial service-condition study (humidity, airflow velocity, PM loading, natural microbes) on PLA melt-blown filter media. Degradation stays early-stage, first-order, k up to 3.88e-2 /h, surface-erosion dominated. Humidity drives degradation (90% of short-term water-contact-angle variance; 77% of yield-strength and 91% of pressure-drop variance long-term); airflow velocity governs dynamic filtration via particle-media contact. Limitation: the abstract gives no filtration efficiency numbers by particle size, so it is unclear whether MPPS performance falls within a realistic mask or HVAC service life. Relevance: low - mitigation materials; humid-climate filter life matters for purifier intervention trials.

doi: 10.1021/acs.langmuir.6c03984

K-HEALTHinAIR indoor $PM_{2.5}$ COPD attributable burden Spanish households 2026 Eur J Public Health (conference abstract)

Attributable Burden of COPD from Indoor $PM_{2.5}$ Exposure in Spanish Households, 2025

Comparative risk assessment combining measured household $PM_{2.5}$ (K-HEALTHinAIR), GBD 2023 Spanish rates and published exposure-response functions. Mean attributable COPD burden ~116 thousand DALYs; by household type, homes with smokers 327 thousand, air-conditioning for heating 183 thousand, no climate control 192 thousand, gas cooking 119 thousand. Limitation: applying ambient-derived ERFs to indoor $PM_{2.5}$ of different composition (tobacco, cooking) and scaling household-type estimates to national totals as if every home were that type; the subgroup figures are not additive and no uncertainty is given. Conference abstract. Relevance: moderate - burden estimates driven by low-cost indoor measurements.

doi: 10.1093/eurpub/ckag169.1053

Landscape fire smoke health action plans review 2026 Public Health Chall

Health Management Planning for Landscape Fire Smoke Events: A Literature Review and Suggested Human-Centred Framework for Resource Development

Reviews smoke-event health management plans for at-risk individuals in Australia, Canada and the US and finds only four resources, two of them dedicated landscape-fire-smoke action plans. Proposes a design-thinking, human-centred framework for producing tailored plans for priority populations and clinicians. Limitation: a non-systematic search of grey literature; the framework is a process proposal with no evaluation of uptake or exposure reduction. Relevance: moderate - action plans hinge on local, timely $PM_{2.5}$ triggers, which is where corrected LCS (e.g. AirNow Fire and Smoke Map) carry operational weight.

doi: 10.1002/puh2.70428

Lee et al. 2026c ACS Omega

Exhaust Emission Characteristics of Gasoline Engines with Secondary Air Systems and Electrically Heated Catalysts in Cold-Start WLTC Mode

A 1.6 L GDI engine with warm-up and underfloor catalysts, an electrically heated catalyst (EHC) and secondary air injection (SAI) was run over the cold WLTC phase, including a -7 C coolant case. EHC + SAI over the first 80 s cut THC 69.1%, CO 65.7% and NOx 80.6% for 26.7 Wh; best strategy was EHC pre-heat 50 s before start, held 30 s after. SAI without EHC raised CO. PN rose sharply in the cold case but optimised EHC-SAI coordination cut it by >30%. Limitation: one engine, one calibration, dynamometer cycle rather than real driving, and PN is reported only as a relative reduction - no absolute PN, size distribution or sub-23 nm fraction, which is where cold-start GDI particles sit. Relevance: moderate - cold-start PN is the largest remaining tailpipe contribution in urban low-cost-network data, and sub-23 nm particles are invisible to the optical nodes watching for it.

doi: 10.1021/acsomega.6c05980

Lee type 2 diabetes burden Asia GBD 2023 particulate matter leading risk 2026 Diabet Med

The evolving burden of type 2 diabetes mellitus in Asia, 1990-2023: A Global Burden of Disease Study 2023

GBD 2023 secondary analysis, 34 Asian countries in five regions, 1990-2023. Age-standardised T2DM prevalence and DALY rates rose almost everywhere, most in South Asia (prevalence +66.5%, DALYs +75.6%); High-income Asia Pacific DALYs fell 2.4% (UI crosses zero). Particulate matter pollution was the leading attributable risk factor in every Asian region, though low-fruit diet led in South Asia (157 DALYs/100,000). Limitation: attribution inherits the GBD $PM_{2.5}$-diabetes exposure-response curve and satellite-model exposure surfaces; it is not new evidence of causality, and the abstract gives no PM-attributable numbers. Relevance: low - burden framing; the exposure layer is exactly the satellite-fusion product low-cost networks are used to correct in Asia.

doi: 10.1111/dme.70421

Liang emergency haze controls causal ML synthetic control Shijiazhuang 2026 Environ Sci Technol

Evaluating Emergency Emission Controls During an Unprecedented Regional Haze Event Using Causal Machine Learning

Three-step causal evaluation (meteorological normalisation, augmented synthetic control, ML prediction) of Shijiazhuang's 45-day 'Sharp Sword' campaign in 2016. Local emission strength fell 4.3% ($PM_{2.5}$), 4.8% (PM_{10}), 15.8% (SO_2), 5.8% (NO_2), 15.3% (CO); concentrations fell 42.1 $\mu g\,m^{-3}$ (14.8%) for $PM_{2.5}$ and 47.4 $\mu g\,m^{-3}$ (12.4%) for PM_{10}. Independent PMF showed vehicle, steelmaking and coal contributions 4.1%, 11.0% and 21.7% lower than in 2017. $PM_{2.5}$ still averaged 243.2 $\mu g\,m^{-3}$ during the campaign. Limitation: donor-city synthetic controls share the regional haze and possibly spillover controls, biasing the effect toward zero; no uncertainty intervals in the abstract. Relevance: medium - a transferable counterfactual framework for evaluating interventions from network data.

doi: 10.1021/acs.est.6c12991

Lu $PM_{2.5}$ longevity country panel specification analysis 2026 Front Public Health

Paradoxical $PM_{2.5}$-longevity estimates in global country panels are sensitive to differential trend control: a specification analysis, 2000-2023

4,406 country-years, two-way fixed effects with GDP, health spending and urbanisation. The conventional model gives a paradoxical +0.406 y life expectancy per 10 $\mu g\,m^{-3}$ $PM_{2.5}$ (0.144-0.668); economy-specific linear trends remove it (-0.023 y, -0.175 to 0.129), as do first differences (-0.006 y, -0.085 to 0.073). A future-exposure lead (+2 y) was positive alone (0.296, 0.042-0.549) but attenuated with current exposure. Limitation: the trend-adjusted estimates are null rather than negative, so the paper diagnoses bias without recovering the expected harm; national averaging discards almost all exposure contrast. Relevance: moderate - a methods warning for any ecological $PM_{2.5}$ panel, including those fed by sensor-network or satellite exposure surfaces.

doi: 10.3389/fpubh.2026.1936760

Lu kitchen air purifier health-economic outcomes 338 Chinese cities 2026 Environ Int

Heterogeneous health-economic outcomes of kitchen air purifier interventions: A 338-city study in urban China

METADATA ONLY. Title and authors only (Lu, Yao, Jiang, Fu et al.). No abstract was retrievable at harvest. The title reports a 338-city assessment of heterogeneous health-economic outcomes of kitchen air-purifier interventions in urban China. Exposure model, health endpoints, cost basis and effect sizes are not available; nothing is inferred. Relevance: moderate - cooking-generated PM is the largest indoor source a low-cost sensor sees, and any intervention-scale cost-effectiveness estimate depends on indoor concentration inputs such sensors could supply. Retry abstract first.

doi: 10.1016/j.envint.2026.110587

Ma TRWP China 2026 Nat Commun

Spatiotemporal dynamics of tire wear particles reveal escalating emission inequalities across China's road network

A road-segment-resolved national inventory fuses harmonised field measurements, ML and traffic activity: China's roads emitted 425 kt of TRWP-derived $PM_{2.5}$ in 2023, ~80% from modelled vehicle-induced resuspension. TRWP's share of traffic $PM_{2.5}$ is projected to rise 34-fold from 2011 to 2050, reaching 67% of traffic $PM_{2.5}$ emissions, with exposure concentrated in megacity corridors; combined mitigation cuts emissions 41%. Limitation: resuspension dominates the total and is the least constrained term - it is modelled, not measured, and its emission factor varies by an order of magnitude with road silt loading and humidity. Relevance: high - non-exhaust PM is coarse-skewed and rubber/mineral-rich, exactly the aerosol on which Plantower-class sensors misreport mass, so a growing TRWP share silently changes network calibration.

doi: 10.1038/s41467-026-77475-3

Mohamadi Isfahan $PM_{2.5}$ AirQ+ WRF-SMOKE-CMAQ health damage 2026 J Environ Health Sci Eng

Assessing the health impacts of $PM_{2.5}$ from anthropogenic and natural sources in Isfahan, Iran

Station annual $PM_{2.5}$ fed AirQ+; at a 2020 GDP of USD 2,283 per capita, annual health damage was USD 441 M (253-628 M). A year-long WRF-SMOKE-CMAQ run attributed 7.9% of damage to iron and steel, 2.2% to cement and 1.7% to brick kilns; 'environmental' (natural) particles accounted for USD 115.3 M (26%). Limitation: attributing damage linearly by modelled mass share assumes equal toxicity, and the share of dust (natural) depends on how the inventory handles anthropogenic fugitive dust. Few stations in a large city. Relevance: low - a standard AirQ+ exercise; denser monitoring would mainly tighten the exposure input.

doi: 10.1007/s40201-026-01011-0

Mukhamediyarov Kazakhstan code 2026 Atmosphere

Air Quality and Emission Trends Following the 2021 Environmental Code Reform in Kazakhstan Across Regions and Industrial Sectors

A before-and-after comparison of Kazakhstan's 2021 Environmental Code reform, which tightened standards and pushed Best Available Techniques. Reported stationary-source emissions fell roughly 5-6% between 2021 and 2024, but no statistically significant improvement in assessed ambient indicators appeared between 2021 and 2025, and episodic PM exceedances persisted in several cities. Limitation: no control region and no counterfactual, so the null is as much about the design as about the policy; reported emissions are self-declared inventory, and the ambient network is sparse enough that a 5% emission change would be undetectable even if real. Relevance: moderate - a clean example of why policy evaluation needs measurement density the regulatory network does not have; the 7 October congestion-pricing record is the version that had it.

doi: 10.3390/atmos17100993

NYC congestion pricing 2026 Environ Sci Technol Lett

Attributing $PM_{2.5}$ and NO_2 Changes One Year after New York City's Congestion Pricing Policy

The Central Business District Tolling Program began in January 2025; its first year is evaluated against 2022-2024 with difference-in-differences on NYCCAS real-time $PM_{2.5}$ and on satellite NO_2. Within the congestion relief zone, $PM_{2.5}$ fell by 1.44 $\mu g\,m^{-3}$ attributable to the policy ($p<0.001$), or 1.33 $\mu g\,m^{-3}$ allowing for spillover, while nearby non-zone sites showed a non-significant 0.50 $\mu g\,m^{-3}$ rise. Tropospheric NO_2 fell over 20% across metropolitan New York against a 2018-2024 baseline, but only 1.23% was attributable to the programme ($p=0.096$). Limitation: one treated zone and one post-period, so the parallel-trends assumption rests on three pre-years; the non-zone increase is the signal that displacement may be real but underpowered. Relevance: high - a 1.4 $\mu g\,m^{-3}$ step change across a few square kilometres is exactly the contrast a dense low-cost network resolves and a regulatory network, here a purpose-built survey, only barely does.

doi: 10.1021/acs.estlett.6c00610

Pannu et al. 2026 Research Square (preprint)

Assessment of the significance of urban vegetation in air phytoremediation: Quantitative analysis of particulate matter removal

Leaves of five morphologies were sampled every fourth day for a year, leaf-surface PM quantified in three size classes and scaled to 20 dominant species and canopy cover. Delhi's urban forest (23% of area) removed 527 +/- 301 t $PM_{2.5}$ and 7,907 +/- 1,079 t total PM - about 1.08% of annual $PM_{2.5}$ emissions in the 2018 SAFAR inventory. Directly measured removal is ~24% of the deposition-velocity estimate for $PM_{2.5}$ and ~4.6% for PM_{10}; leaf loading saturates in Nov-Jan. Fine deposits are enriched in Cr, Ni, Cu, Zn and Pb. Limitation: five morphologies extrapolated to 20 species, wash-off and resuspension partly estimated, an emissions denominator from 2018; preprint. Relevance: moderate - urban greening claims based on deposition-velocity models overstate removal four-fold.

doi: 10.21203/rs.3.rs-10839085/v1

Polwiang Bangkok Chiang Mai burning control DLNM GAM $PM_{2.5}$ regimes 2026 Atmos Pollut Res

Differential sensitivity of urban and biomass-dominated $PM_{2.5}$ regimes to burning control: A distributed-lag generalized additive model analysis of Bangkok and Chiang Mai (2021-2024)

METADATA ONLY. Title and author only; no abstract retrievable at harvest. The title states a distributed-lag GAM comparing the sensitivity of $PM_{2.5}$ to burning-control measures in an urban regime (Bangkok) and a biomass-burning-dominated regime (Chiang Mai), 2021-2024. Effect sizes, lag structure, the burning-control variable (fire counts, policy dates) and confounder set cannot be stated. Limitation: observational policy attribution in a GAM is sensitive to meteorological adjustment and transboundary smoke from Myanmar and Laos; single author. Relevance: moderate - Chiang Mai is a dense PMS-sensor setting (DustBoy network), and regime-specific response is where LCS spatial coverage tests policy effects.

doi: 10.1016/j.apr.2026.103236

Prieto Perez Huszar future central European air quality emission changes 2026 Atmos Environ

The role that changes in emissions play in future central European air-quality

METADATA ONLY. Title and authors only (Prieto Perez, Huszar, Karlicky); no abstract retrievable at harvest. The group runs regional climate-chemistry (RegCM/WRF-CAMx) scenarios; the title separates the emission-change contribution to future central European air quality from climate change. Pollutants, scenarios and horizons cannot be stated. Limitation: unverifiable beyond the title; scenario attribution depends on the emission pathway chosen. Relevance: low - policy context for the PM levels European LCS networks will measure.

doi: 10.1016/j.atmosenv.2026.122404

Quadros et al. 2026 Commun Earth Environ

Global air quality and human health impacts of growing aircraft emissions

Global aircraft emissions projected 2019-2040 and run through a chemistry transport model. 2019: 33,900 (95% CI 23,500-45,600) $PM_{2.5}$ deaths and 24,600 (15,500-34,200) O_3 deaths attributable to aviation. By 2040 these rise 58%/82% (low), 119%/148% (baseline) and 169%/205% (high scenario); the non-aviation background shifts aircraft impacts by up to 44%. Limitation: the CIs propagate concentration-response uncertainty only, not CTM, inventory (especially cruise NOx-nitrate) or grid resolution, which dominate a marginal-sector estimate; most $PM_{2.5}$ impact is secondary nitrate and sulfate far from airports. Relevance: moderate - the near-airport UFP signal low-cost networks can see is not the mass-based burden counted here.

doi: 10.1038/s43247-026-03732-4

Ranzi et al. 2026 Epidemiol Prev

Atlante Aria e Salute: methods for estimating the health impacts of air pollution and national application

Methodological description of the Air and Health Atlas, an interactive national platform. $PM_{2.5}$ and NO_2 are modelled at 1 km with random forests, converted to population-weighted exposure at municipality level against 2021 census blocks, and combined with provincial death-certificate rates through HRAPIE-2 concentration-response functions against the 5 $\mu g\,m^{-3}$ WHO counterfactual; 2019 is shown as an illustration, with pronounced north-south gradients. Limitation: population-weighted exposure at municipality scale cannot capture within-municipality contrast, the random-forest surface is validated against the same monitors that train it, and HRAPIE-2 functions are transferred from other populations. Relevance: high - this is the operational template a low-cost network would be asked to improve, and the 1 km population-weighted step is exactly where added density pays.

doi: 10.19191/ep26.4-5.s1.a1030.092

Rodriguez Bogota car-free 2026 Research Square (preprint)

Annual trends in ambient particulate matter during car free days in Bogota from 2016 to 2025

Nine annual Car-Free Day observations from the regulatory network, OLS and weighted regressions with 2,000-replicate bootstrap. PM_{10} fell significantly, slope -4.04 $\mu g\,m^{-3}$ per year (weighted bootstrap CI -6.49 to -0.81), median change -41.8%; $PM_{2.5}$ trend was -1.72 $\mu g\,m^{-3}$ per year (CI -3.17 to 0.40), not significant. Limitation: n = 9 single days with no control days, so the trend is the city's background trend plus that day's weather - it says nothing about the effect of the car-free intervention itself. Relevance: low-moderate - the right design (Car-Free Day minus matched control days, by station) is one that a dense sensor network could deliver at street scale.

doi: 10.21203/rs.3.rs-10864736/v1

Ruengdit active-flow respirator fit children Chiang Mai 2026 Front Public Health

Active-flow systems enhance respirator fit success in children: a comparative study with standard disposable models

82 children aged 5-10, modified OSHA quantitative fit test. A standard triple-fold KN95-equivalent FFR passed in 3.7%; adding a portable HEPA unit supplying 66.5 L/min positive pressure raised the pass rate to 23.2% (OR 17.0) and median fit factor from 9.5 to 39.5. Wider bizygomatic breadth (aOR 1.31) and longer menton-sellion (1.11) favoured fit; head nodding was worst. Limitation: no CI given for the OR; crossover order and fit-test learning effects are not described, and 77% still failed with the active system. Relevance: high - a median fit factor of 9.5 means adult masks give children roughly 90% leakage-limited protection at best, which matters for exposure-reduction advice issued from LCS haze alerts.

doi: 10.3389/fpubh.2026.1851331

Ruwali Lary US infant life expectancy environmental determinants machine learning 2026 Sensors (Basel)

Estimating Life Expectancy in the U.S.: Identifying Environmental Determinants and Disparities (2003 to 2019)

County-level IHME life expectancy for the under-1 age group, contiguous US 2003-2019, regressed on 33 ECMWF environmental variables with ML. Marked north-south disparity, southern counties lowest. With only five variables the model reached R2 0.81 and RMSE 1.06 years on an independent test set. Beyond PM, O_3 and CO, lower formaldehyde, sulphate and dust aerosol, more vegetation and lower wet-bulb temperature ranked as predictors of higher life expectancy. Limitation: no socioeconomic, race or healthcare covariates, so the south-north environmental gradient absorbs deprivation; feature importance is predictive, not causal, and IHME county estimates are themselves smoothed models. Relevance: moderate - a reanalysis-driven baseline that dense LCS data could sharpen.

doi: 10.3390/s26206478

Scott equine air quality index Delphi management 2026 J Am Vet Med Assoc

Experts recommend management strategies for horses during poor air quality

A three-round modified Delphi with 15 equine veterinarians, 11 of whom completed all rounds. The panel adapted the EPA AQI public messaging to equine health risk and set management, training and competition recommendations for each AQI category. Experts agreed that healthy horses as well as compromised ones can have short- and long-term effects. Limitation: 11 completers, and the recommendations rest on translational extrapolation from human data, as the authors acknowledge. There are no equine exposure-response data. Relevance: medium - venue-level LCS readings at stables and competition grounds are what would operationalise AQI-tiered decisions.

doi: 10.2460/javma.26.05.0436

Shandong integrated air-pollutant GHG inventory 2026 Atmosphere

Strategies for Co-Control of Air Pollutants and Greenhouse Gases: Insights from an Integrated Emission Inventory in Shandong Province

2024 integrated inventory for 16 Shandong prefectures, 10 air pollutants and 4 GHGs over 8 source categories. Power/heat and industry gave 86.84% of CO_2; industry dominated SO_2, CO and VOCs; mobile sources NOx; agriculture NH_3, CH4 and N2O; fugitive dust dominated primary PM. Interspecies correlations show partial synergy, so co-control cannot be assumed from shared sources. Limitation: inventory only, no uncertainty bounds in the abstract, and fugitive-dust PM is the least constrained category in any inventory. Relevance: low-moderate - fugitive dust dominance is a coarse-mode signal that $PM_{10}/PM_{2.5}$ ratios from LCS can check locally.

doi: 10.3390/atmos17101016

Sharma France net-zero benefits 2026 Environ Int

Health and economic benefits of air quality improvements in France through diverse net-zero transition scenarios by 2050

METADATA ONLY. No abstract at harvest. Title indicates a health-impact and economic assessment of air-quality co-benefits under several French net-zero pathways to 2050. Recorded for coverage; no estimate is inferred. Relevance: low - scenario work, but it fixes the concentration ranges a future network would need to resolve.

doi: 10.1016/j.envint.2026.110555

US vehicle emission burden 2026 Environ Sci Technol

Spatially Refined Analysis of Emission Burdens and Equity from Light-, Medium-, and Heavy-Duty Vehicle Traffic in the United States

Block-level on-road emission intensities for the whole United States. Medium- and heavy-duty trucks are under 11% of vehicle kilometres but 46% of NOx and 50% of $PM_{2.5}$ emission burden; truck $PM_{2.5}$ is exhaust-dominated while tyre and brake wear are nearly half of light-duty $PM_{2.5}$. People of colour face burdens averaging 40% above the national mean (Asian populations over 60%), lower-income populations 20-25% higher, with significant disparities in 85-89% of counties, rural as well as urban. Critically, spatial aggregation attenuates the measured disparity. Limitation: this is an emission-intensity metric, not a concentration or an exposure - no dispersion, no background, no infiltration - so it cannot be read as an inhaled dose. Relevance: high - the aggregation result is the quantitative case for network density: resolution is not cosmetic, it determines whether a disparity is visible at all.

doi: 10.1021/acs.est.6c12227

Urban tree bark PM retention functional traits 2026 Environ Pollut

Species-specific PM retention by urban tree bark is associated with functional traits under controlled exposure

Modified open-top chambers compared PM capture by bark, leaves and branches of 12 calibration species. Per unit area bark retained more PM than foliage in 11 of 12 (Cinnamomum camphora 364.5 vs Juniperus formosana 112.1 ug/cm2). Upscaled to whole plants, bark $PM_{2.5}$-10 retention was 1.40x that of leaves while $PM_{2.5}$ was comparable - a coarse-selective filter. Lenticel density, roughness and groove prominence predicted interspecies variation; applied to 26 Hangzhou species the spread was 4.83-fold. Limitation: retention mass under controlled exposure says nothing about net ambient concentration reduction, and independent validation was limited. Relevance: low for sensing - deposition-mitigation claims of this type need ambient gradient measurement to be credible.

doi: 10.1016/j.envpol.2026.129262

Wang carbon footprint air-pollutant emissions Beijing 2022 Winter Olympics 2026 Atmosphere

Linking Carbon Footprints to Air-Pollutant Emissions from Mega Sporting Events: Evidence from the Beijing 2022 Winter Olympics

Screening-level inventory converting the verified 713,788 t CO_2e footprint of the Beijing 2022 Winter Olympics (2016-2022) into co-emitted pollutants via Chinese pollutant-to-CO_2 ratios over eight emission carriers, with Shapley attribution and Monte Carlo/Sobol uncertainty. Construction produced 63% of CO_2e, ~70% of 1,335 t NOx and 76% of 279 t $PM_{2.5}$. Green electricity avoided 31% of no-measure emissions; offsets reduced no air pollutant. Limitation: emission-side only - ratio-based $PM_{2.5}$ with no dispersion, concentration or exposure step, so there is no health quantity, and national average ratios may misstate construction-site fugitive dust. Relevance: low - construction dust is the source class that site-perimeter low-cost PM monitoring is typically deployed to police.

doi: 10.3390/atmos17111072

Wang climate-adaptive city pilot $PM_{2.5}$ health dividend China 2026 Front Public Health

Does climate adaptation policy deliver a health dividend? Exposure, modeled health burden, and defensive capacity in China's climate-adaptive city pilots

Difference-in-differences on a 2011-2022 panel of 264 prefecture cities, 21 treated by China's 2017 climate-adaptive city pilot. The pilot is associated with $PM_{2.5}$ lower by 3.87 $\mu g\,m^{-3}$ (95% CI -6.49 to -1.25; wild-bootstrap p=0.0067); GEMM translates this into an ~8.1% lower modelled burden (-11.32 to -4.76%) - an accounting result, not observed mortality. Defensive-capacity responses (physicians, beds, insurance) are imprecise. Limitation: the estimate shrinks and loses precision in the matched sample and after pollution-convergence adjustment, so selection of already-improving cities is not excluded; 21 treated units is a small base. Relevance: low-medium - policy evaluation at city scale rests on gridded $PM_{2.5}$ whose own error is not propagated.

doi: 10.3389/fpubh.2026.1987084

Wang et al. 2026b J Environ Sci (China)

What drives long-term variations in biomass burning emissions in China?

A 2008-2023 inventory of domestic and open biomass burning with Kaya/LMDI decomposition of $PM_{2.5}$ emission change. 2008 emissions were 3.22 Mt $PM_{2.5}$, 0.50 Mt EC and 1.41 Mt OC; all species fell 50-65% by 2023. Domestic burning declined mainly through the economic factor on burned biomass (183.4% of the change) and urbanisation (54.7%); open straw burning fell through activity intensity (-115%), whose effect weakened after 2017. Limitation: LMDI partitions an identity, it does not test causation, and activity data for domestic and open burning are the weakest inputs of any Chinese inventory; no uncertainty bounds are given. Relevance: low-moderate - the declining rural smoke signal is a baseline shift any rural sensor trend analysis has to account for.

doi: 10.1016/j.jes.2026.05.013

Warminski SRC pellets boiler ESP PM CO emissions 2026 Molecules

Short-Rotation Coppice Pellets as Biofuels in a Modern Boiler with an Electrostatic Precipitator: Particulate Matter and Carbon Monoxide Emissions

Short-rotation coppice pellets (willow, poplar, black locust) versus pine and birch in a pellet boiler with an electrostatic precipitator. Species significantly affected PM and CO; the ESP cut mean PM by 52.5% (53.81 to 25.54 mg/m3) without affecting CO. Highest PM: willow without ESP (90.96 mg/m3); lowest: pine with ESP (17.51). Poplar gave low PM but the highest CO; black locust the best balance (combustion efficiency 93.20%). Limitation: one boiler, total PM only (no size distribution or PM_1 fraction, where residential-combustion mass sits), and steady-state operation that misses start-up peaks. Relevance: low - source-side mitigation; a 52% ESP removal at total-mass level likely overstates sub-micron removal.

doi: 10.3390/molecules31193443

Yang et al. 2026b Environ Res

Process-resolved characterization and plant-level inventory of multi-pollutant emissions from China's coking industry

Stack measurements at coal charging, coke-oven flue gas, dry quenching and pushing in two plants (total PM 3.12-37.72 mg/m3; sulfate 28-59% of ions; As and Cd enriched in $PM_{2.5}$; the condensable fraction carries 44.5-57.4% of trace elements in flue gas) feed a 307-plant inventory for 2012-2024. 2022 national emissions: 159.8 kt PM, 66.6 kt SO_2, 82.3 kt NOx, 838.3 kt VOCs, 4.24 kt PAHs, 707.6 t trace elements; BACT cuts PM to 15.5 kt by 2050. Limitation: two plants anchor factors for 307; fugitive emissions, flagged as a priority, are the least measured; condensable PM is operationally defined. Relevance: moderate - condensables are invisible to stack filters but appear as ambient $PM_{2.5}$ downwind, where fence-line sensor networks see them.

doi: 10.1016/j.envres.2026.125824

Yu multicountry $PM_{2.5}$ admissions 2026 Lancet Planet Health

Cause-specific hospital admissions associated with short-term $PM_{2.5}$ exposure: a multicountry time-series analysis

346.5 million admissions from 949 communities in six countries/territories, 2000-2019; quasi-Poisson time series pooled by meta-analysis. Per 10 $\mu g\,m^{-3}$, all-cause admissions RR 1.080 (1.074-1.086), with effects persisting to 21 days. Malignant neoplasms (1.108, 1.094-1.123), mental disorders (1.100) and nervous system disease (1.085) all exceed respiratory (1.067, 1.060-1.074). Attributable fraction against the WHO daily 15 $\mu g\,m^{-3}$ fell from 8.9% (2000-09) to 3.9% (2010-19). Limitation: a 21-day distributed lag on cancer admissions invites harvesting and scheduling artefacts - an elective oncology admission is not an acute PM event - and heterogeneity across countries is described as substantial but the pooled RR is reported without it. Relevance: high - the long-lag, non-respiratory signal argues for continuous rather than episodic sensing.

doi: 10.1016/j.lanplh.2026.101526

Zhang Jiaxing inventory co-control screening $PM_{2.5}$ GHG 2026 Atmosphere

Inventory-Based Co-Control Screening of Air Pollutants and Greenhouse Gases in a Typical Yangtze River Delta City

Harmonised 2024 integrated inventory plus enterprise statistics for Jiaxing; 54 non-zero source categories scored on an air-pollutant H-score (mean of $PM_{2.5}$, NOx, SO_2, VOCs, NH_3 shares) against a GWP100 CO_2-eq G-score, screened at the 70th percentile. Coupled co-control candidates: CHP, thermal power, motor vehicles; decoupled single-pollutant priorities: industrial solvents (VOCs) and paved-road dust (PM). A 75th-percentile benchmark implies 7,941 t VOC reduction among reporting enterprises. Limitation: equal-weight averaging of pollutant shares ignores toxicity and secondary PM formation potential; no ambient or health validation. Relevance: low - road-dust PM priority is the coarse fraction optical LCS undercount.

doi: 10.3390/atmos17101023

Zhang US wildfire smoke five-year mortality impulse response 2026 medRxiv (preprint)

Wildfire smoke leaves a large five-year imprint on mortality in the United States

County-month mortality and wildfire-$PM_{2.5}$ panels 2011-2023; episodic smoke pulses are used to deconvolve overlapping responses and recover an impulse-response function. Excess mortality peaks about three years after a pulse and is gone by five. Integrated over 2011-2023 this gives 1.7 million smoke-associated deaths (95% UI 1.1-2.3 million), roughly ten times what long-term ambient $PM_{2.5}$ relative risks imply; circulatory and respiratory causes contribute comparably, adults >=65 dominate, and responses were larger during COVID-19. Limitation: a three-year lag in an ecological panel is exposed to slow confounders (migration, economic shocks, drought, heat) co-varying with fire years; the order-of-magnitude excess needs independent replication before it enters burden accounting. Relevance: high - a multi-year lag makes historical smoke-$PM_{2.5}$ reconstructions, including LCS-corrected ones, the binding input.

doi: 10.64898/2026.10.15.26365428

Zhang early-onset female lung cancer burden GBD 2021 2026 Asia Pac J Clin Oncol

Global Burden and Trends of Early-Onset Tracheal, Bronchus, and Lung Cancer Among Women From 1990 to 2021.

GBD 2021 secondary analysis of early-onset (under 50) female TBL cancer, 1990-2021. Global rates declined (all EAPCs <0), but low-middle SDI regions rose (DALY EAPC 0.66) and South Asia fastest (DALY EAPC 1.42). Ambient PM was the fastest-growing attributable risk (attributable DALY rate EAPC 1.22, ASDR 1.27). Limitation: GBD attributable burden is model output that inherits the exposure surface and IER curves; in young women with few cases, estimates in low-SDI regions rest largely on modelled covariates rather than registries. Relevance: low - burden, not measurement.

doi: 10.1111/ajco.70212

Zhang et al. 2026d Atmos Pollut Res

Impact of strong biomass burning in Northeast China: air pollution, enhancement ratio among pollutants, and health risk

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 3 Oct with no abstract. By title, an episode study of strong agricultural biomass burning in Northeast China reporting enhancement ratios among co-emitted pollutants and a health-risk estimate. Enhancement ratios (dPM / dCO, for instance) are the quantity that lets a sparse network attribute an episode to burning rather than to secondary formation, so the numbers matter more than the burden claim. Relevance: moderate; on the retry list.

doi: 10.1016/j.apr.2026.103224

Zhang respirable magnetite emissions coal-fired power China 2026 Environ Sci Technol

Emissions of Respirable Magnetite from the Coal-Fired Power Sector in China: Spatial Inequalities and Mitigation Pathways

Multifactor emission model for respirable magnetite from Chinese coal-fired power: 2,828.7 (2,307.1-3,371.4) t/y, spatially concentrated by tiered emission standards. Guizhou and Sichuan supply 4% of coal power but tightening their standards alone would cut >31% of national magnetite emissions. Limitation: emission factors for magnetite specifically (as opposed to total PM or Fe) rest on few stack measurements, and the uncertainty interval looks narrow for a species-level inventory; health relevance is asserted from brain-magnetite literature rather than exposure-response data. Relevance: moderate - magnetite is invisible to optical LCS mass; magnetic or filter-based speciation would be needed to verify hotspots.

doi: 10.1021/acs.est.6c08602

Zhao et al. 2026b Atmosphere

Pollution Characteristics, Influencing Factors and Health Risk Assessments of Mercury in Atmospheric $PM_{2.5}$ from Xinxiang, China

Year-long filter sampling gives an annual mean $PM_{2.5}$-bound Hg of 0.399 ng/m3, significantly higher in winter ($p<0.001$) and in the heating than the non-heating period, positively correlated with SO_2, CO, $PM_{2.5}$, OC, EC, Cl-, Zn and Ca and negatively with temperature, O_3 and V - a coal-combustion signature. Hazard index 2.06e-3 (children) and 8.46e-4 (adults), both far below 1, with ingestion the dominant modelled pathway. Limitation: the samples are from 2015 and describe an emissions regime two major control phases out of date; an ingestion-dominant pathway for an airborne metal is an artefact of the EPA soil-oriented default parameters. Relevance: low - logged as a composition time point, not as a risk estimate.

doi: 10.3390/atmos17100974

Zhao et al. 2026e Research Square (preprint)

Thermal-oxygen evolution and atmospheric pollutant emission characteristics during the vertical propagation of underground coal fires

A scaled physical-similarity rig run from ignition to burnout with continuous monitoring at six depths. Thermal-oxygen evolution split into incubation, rapid heating, intensive reaction, rapid cooling and burnout; the upper layers burned out while the middle sustained reaction and the deepest was oxygen-limited; surface response lagged the seam. Particulate matter and light hydrocarbons were released early (CH4 and C2H6 concentrated in the first 40-60 h, at background by ~80 h) while CO_2 and CO dominated later. Limitation: a similarity-scaled rig reproduces sequence, not magnitude - emission factors from it cannot be transferred to a real coal fire, and no scaling validation is offered; preprint. Relevance: moderate - the early-PM, late-CO sequence is a usable detection signature for surface sensor arrays over suspected underground fires.

doi: 10.21203/rs.3.rs-10517753/v1

Other clinical endpoints 28 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (28 OF 28 RECORDS IN THIS SUBTOPIC)

Beijing CKD cohort 2026 Environ Pollut

Association between Long-term Exposure to Air Pollution and Incident Chronic Kidney Disease among Adults in Beijing, China

15,580 adults in the Beijing Air Pollution and Population Health Cohort, 664 incident CKD cases 2019-2025, time-varying Cox models. Per 10 $\mu g\,m^{-3}$: PM_{10} HR 1.18 (1.06-1.31) and CO 1.02 (1.01-1.03) were significant; $PM_{2.5}$ 1.22 (0.99-1.50) and NO_2 1.17 (0.96-1.43) were positive but crossed unity; O_3 was inverse, 0.77 (0.68-0.87). The composite score gave 1.51 (1.07-2.12) per IQR, with stronger associations in women, older adults and urban residents. Limitation: six years of follow-up in one city where all pollutants fell together makes single-pollutant HRs nearly uninterpretable - the inverse ozone estimate is the tell, since ozone rose as PM fell; eGFR-defined incident CKD is sensitive to a single creatinine measurement. Relevance: moderate - kidney endpoints are under-studied, but this design cannot separate the pollutants.

doi: 10.1016/j.envpol.2026.129302

Bottini antibiotic resistance Ecoli Salmonella $PM_{2.5}$ ecological 2026 One Health

Environmental, socioeconomic, and livestock-related factors associated with antibiotic resistance in Escherichia coli, Shigella spp., and Salmonella enterica

397,103 isolates from human, animal and environmental sources (NCBI Pathogen Detection, 2003-2018) in mixed-effects negative binomial models. For E. coli/Shigella, $PM_{2.5}$ was associated with beta-lactam resistance (PRR 1.39, 1.11-1.74), alongside low income and population density; for S. enterica, $PM_{2.5}$ and tetracycline resistance PRR 1.65 (0.98-2.76), with warm climate and poultry density. Limitation: national $PM_{2.5}$ is a marker of industrialisation, sanitation and antibiotic use, and sequencing effort varies by country; the association is not interpretable as a PM effect. Relevance: low - included because $PM_{2.5}$ is reported with an estimate; the airborne-resistome hypothesis would need measured bioaerosol, not country means.

doi: 10.1016/j.onehlt.2026.101600

CLHLS indoor constituents 2026 BMC Public Health

Association between exposure to indoor $PM_{2.5}$ constituents and mortality risk among Chinese older adults: a nationwide prospective cohort study

12,607 participants (mean age 86.8) followed 2008-2018, 8,643 deaths over a median 4.14 years, time-varying Cox with restricted cubic splines. Per 1 $\mu g\,m^{-3}$: $PM_{2.5}$ HR 1.003 (1.001-1.005), black carbon 1.072 (1.035-1.111), organic matter 1.009 (1.002-1.016), sulfate 1.010 (1.001-1.019); associations were stronger in rural residents, non-linear, robust to excluding hypertensive and diabetic participants, and $PM_{2.5}$ tracked the neutrophil-to-lymphocyte ratio as a candidate inflammatory mediator. Limitation: 'indoor' here is an ambient field multiplied by a modelled infiltration factor, not a measurement - and the infiltration factor is itself a function of building and season, so constituent contrasts partly encode housing type; black carbon's HR is an order of magnitude above $PM_{2.5}$'s per unit mass, which is biologically plausible and also what collinearity with a small-magnitude constituent produces. Relevance: high - this is the exact quantity a cheap indoor node measures directly, and the study had to model it.

doi: 10.1186/s12889-026-29130-1

China fire-sourced $PM_{2.5}$ cancer 2026 Environ Pollut

Extreme, Accumulated, and Lagged Exposure to Fire-Sourced $PM_{2.5}$ and Site-specific Cancer Risk in China

Fire-sourced $PM_{2.5}$ was positively associated with several site-specific cancers, with lagged and accumulated metrics showing the broadest pattern (thyroid, kidney, breast, lung, leukaemia, lymphoma) and extreme exposure the narrowest (strongest for larynx). Against all-source $PM_{2.5}$, fire $PM_{2.5}$ associations were stronger for thyroid, kidney, larynx, lymphoma and oral cancers, and stronger in the north and in low-GDP areas. Limitation: no effect sizes, n or geographic unit in the abstract; many sites x three metrics is a multiple-comparison surface, and thyroid cancer incidence is notoriously driven by screening intensity, which tracks region and income. Relevance: moderate - source-resolved smoke again outperforms mass.

doi: 10.1016/j.envpol.2026.129343

Dahl air pollution bone fractures high-risk population 2026 Environ Int

Air pollution and bone fractures in a high-risk population: differences by age and sex

METADATA ONLY. No abstract at harvest. The title indicates an analysis of ambient air pollution and fracture risk in a population at high fracture risk, stratified by age and sex. Recorded for coverage; no exposure model, design or estimate is inferred. Relevance: low-moderate - fracture is an under-studied PM endpoint, and bone-health associations have so far rested on residential annual means.

doi: 10.1016/j.envint.2026.110570

Deng ultrafine particles kidney disease ED visits New York 2026 Environ Res

A mysterious danger for kidney disease exacerbations: short-term impacts of ultrafine particles on hospital visits

New York State discharge data 2013-2018 (95% of hospitals), 269,945 kidney-disease ED visits, conditional logistic regression per IQR of modelled UFP. Associations appeared at lag 1 and on every cumulative lag 0-1 to 0-6, peaking at lag 0-3: excess risk 0.74% (0.28-1.20) per IQR. Larger in adults >65, non-African American patients, hot and dry days, and fall/winter. Limitation: UFP is a CTM product, not a measurement; its spatial error is large and likely correlated with $PM_{2.5}$ and NO_2, and the abstract reports no co-pollutant adjustment, so 'UFP-specific' toxicity is not isolated. The effect per IQR is small. Relevance: high - a state-scale UFP health signal built entirely on modelled number concentration is the case for CPC or OPC-extended nodes to validate it.

doi: 10.1016/j.envres.2026.126022

Duan landscape fire-sourced $PM_{2.5}$ kidney function eGFR children China 2026 Environ Res

Landscape fire-sourced $PM_{2.5}$ and kidney function in children: evidence from a national cross-sectional study in China

China National Nutrition and Health Surveillance of Children and Lactating Women 2016-2018, n=12,102 aged 5-18. Per IQR (0.505 $\mu g\,m^{-3}$) of fire-sourced $PM_{2.5}$: eGFR -0.83 mL/min/1.73 m2 (-1.03 to -0.62); middle-school students -1.46 (-1.83 to -1.09) vs primary -0.59 (-0.84 to -0.34). Associations were stronger with low NDVI and low vitamin or carbohydrate intake. Limitation: an IQR of 0.5 $\mu g\,m^{-3}$ is a tiny contrast for an effect this size, which points to residual spatial confounding (fire-PM tracks rural, agricultural regions); creatinine-based eGFR in children depends on muscle mass and hydration, which also vary by region. Cross-sectional. Relevance: low - source-specific PM from a chemical-transport model, no sensing.

doi: 10.1016/j.envres.2026.125978

Hefei conjunctivitis 2026 Front Public Health

Ambient air pollutants and outpatient visits for conjunctivitis in Hefei, China: a time-series study of hospital-based data during 2014-2023

73,043 conjunctivitis outpatient records, 2014-2023, in a time-series design with stratification by sex, age, season and pandemic period. All five pollutants were positively associated with daily visits, strongest generally at lag0; SO_2 carried the largest lag-0 estimate, then NO_2, with pollutant-specific cumulative lag patterns. Limitation: no effect sizes, units or intervals appear in the abstract, so nothing can be entered; SO_2 ranking first in a city where SO_2 has fallen to a few $\mu g\,m^{-3}$ suggests a marker for combustion-season conditions rather than a conjunctival effect, and single-city time series cannot separate co-varying pollutants. Relevance: low-moderate - ocular surface endpoints are plausible for coarse and irritant exposure, and a record of the claim is worth keeping even though this analysis does not establish it.

doi: 10.3389/fpubh.2026.1926405

Hoang $PM_{2.5}$ minimal residual disease childhood ALL REDIAL 2026 JNCI Cancer Spectr

Fine particulate matter and minimal residual disease in childhood acute lymphoblastic leukemia from the REDIAL consortium

2,125 children with ALL; mean daily residential $PM_{2.5}$ during induction in quartiles; logistic regression. In B-ALL, positive end-of-induction MRD: Q2 aOR 1.56 (1.08-2.24), Q3 1.51 (1.05-2.17), Q4 1.19 (n.s.). Restricted to NCI high-risk B-ALL, aORs rose to 1.73-2.28 across Q2-Q4. Limitation: a non-monotone quartile pattern with the top quartile null is not a dose-response; census-tract $PM_{2.5}$ over a ~4-week induction is tightly tied to treatment centre and ethnicity, which REDIAL was built to study and which also predicts MRD. Relevance: low-medium - induction-window exposure is short enough that a home monitor would be the right instrument.

doi: 10.1093/jncics/pkag116

Hu air pollution built environment lung cancer recurrence after surgery China 2026 Research Square (preprint)

Ambient Air Pollution, Residential Built Environment, and Lung Cancer Recurrence After Curative-Intent Surgery: A Retrospective Cohort Study in Three Chinese Cities

542 adults after curative resection at three tertiary hospitals (2019-2026); 60 recurrences, median follow-up 27.3 months. Site-stratified meta-analysis, per within-site SD of first-year post-diagnosis exposure: $PM_{2.5}$ HR 1.41 (1.08-1.84), PM_{10} 1.38 (1.10-1.73), NO_2 1.67, CO 1.51, SO_2 1.59; O_3 null. A co-exposure component gave HR 1.51 (1.20-1.89); green-space diversity HR 0.59/SD. Limitation: 60 events across six pollutants, mixtures and built-environment terms is badly underpowered for adjustment; within-site SD contrasts are small and residence tracks stage, histology and access to follow-up imaging. Preprint. Relevance: low.

doi: 10.21203/rs.3.rs-11311605/v1

Jiang air pollution meteorology hand foot mouth disease Shenzhen DLNM 2026 Front Public Health

Ambient air pollution and meteorological factors associated with hand, foot, and mouth disease risk in Shenzhen, China: a time-series study

Shenzhen daily HFMD counts 2013-2022 with six pollutants and meteorology in DLNMs, including pre/post EV71 vaccine and COVID-19 periods. $PM_{2.5}$ and PM_{10} showed risk peaks at low concentrations and 'protective' effects at extreme highs; SO_2 and CO raised cumulative risk at extremes; temperature showed a dual-peak pattern and high humidity raised risk. No effect sizes with CIs in the abstract. Limitation: HFMD is a contact-transmitted enteroviral disease with strong seasonality driven by school terms; non-monotone, sign-flipping PM curves are the signature of residual seasonal confounding and DLNM over-flexibility, not biology. Relevance: low.

doi: 10.3389/fpubh.2026.1920424

Li ambient air pollution gastric cancer incidence 204 countries ecological 2026 Curr Med Sci

Ambient Air Pollution and Gastric Cancer Incidence: A Multinational Ecological Analysis (1990-2020)

Negative-binomial fixed-effects panel over 204 countries, 1990-2020, from GBD 2021. Per unit $PM_{2.5}$: +0.3% (0.1-0.5%) in age-standardised gastric cancer incidence; NO_2 +1.8% (1.2-2.4%), O_3 +0.6% (0.3-0.9%). Splines show nonlinearity; $PM_{2.5}$ lag window 3-4 years; male and high-middle-SDI strata most affected. Limitation: both exposure and outcome are GBD modelled quantities that share covariates (SDI), and a 3-4 year lag for a cancer with a decades-long Helicobacter-driven course is biologically implausible; country fixed effects do not remove the H. pylori decline that tracks development. The authors concede ecological status and call for individual-level validation. Relevance: low.

doi: 10.1007/s11596-026-00255-z

Lima et al. 2026 medRxiv (preprint)

Epigenetic Exposure Signatures Distinguish Early-Onset from Later-Onset Colorectal Cancer in a Racially Diverse US Cohort

Applies published methylation risk scores (lifestyle, air pollution, pesticides) to tumour tissue from 146 DANCE colorectal-cancer cases (29 early-onset, 69 intermediate, 48 late) and relates them to age at onset, adjusting for race, geography, Area Deprivation Index and a multi-exposure index. Higher odds of early onset were associated with MRS for 2,4-D, heptachlor, coarse PM, Mediterranean diet and smoking. Limitation: decisive - MRS were trained on blood, applied here to tumour DNA, whose methylome is reshaped by the cancer itself; 29 early-onset cases; signatures are proxies validated against observed exposure only by Spearman correlation; no effect sizes in the abstract; preprint. Relevance: low - a coarse-PM methylation proxy is not an exposure measurement.

doi: 10.64898/2026.09.28.26363311

Lin $PM_{2.5}$ frailty trajectories CHARLS 2026 J Occup Environ Med

The association of particulate matter 2.5 exposure with frailty trajectories in middle-aged and older Chinese adults and the potential modification by lifestyle and socioeconomic status

10,659 CHARLS participants; deficit-accumulation frailty index, three GBTM trajectories (stable low 64%, increasing to frailty 29%, continuously worsening 7%). Per 10 $\mu g\,m^{-3}$ $PM_{2.5}$, odds of the two adverse trajectories rose 15.2% and 12.3% (ordered logistic; no CIs in the abstract). Higher SES attenuated the association (RERI -0.067); unhealthy sleep duration amplified it (RERI 0.163). Limitation: GBTM classes are model artefacts with posterior misclassification carried into a second-stage regression without uncertainty propagation; CHARLS addresses are coarse (city/county), so exposure contrast is mostly between regions with different healthcare and diet. Relevance: low - regional satellite exposure, no sensing implication.

doi: 10.1097/jom.0000000000003913

Lin air pollution cervical cancer recurrence mortality Fudan cohort 2026 Environ Health (ACS)

Long-Term Exposure to Ambient Air Pollution and Cervical Cancer Outcomes in Chinese Women: A Prospective Study on Recurrence and Mortality Risks

3,758 women with primary cervical cancer (2019-2022), median follow-up ~3 y; 217 recurrences, 113 deaths. Per IQR $PM_{2.5}$: recurrence SHR 1.93 (1.40-2.66, Fine-Gray), death HR 2.55 (1.78-3.67); PM_{10} SHR 1.65 (1.13-2.42) and HR 2.21 (1.38-3.54); NO_2 and SO_2 also positive, O_3 null. Limitation: hazard ratios near 2-2.5 per IQR from 113 deaths are implausibly large for a residential long-term contrast and point to confounding by referral geography - patients from outside Shanghai (higher PM) likely differ in stage, HPV management and follow-up intensity; single-centre, short follow-up, no adjustment for distance to the hospital reported in the abstract. Relevance: low - model-based residential exposure, no sensing implication.

doi: 10.1021/envhealth.6c00011

Ma green space pollution aging 2026 Environ Health Perspect

Associations of Green Space and Air Pollution Mixture with Heterogeneous Biological Aging Patterns

11,760 adults aged 20-65 with repeated measures, accelerated longitudinal design; three BAA trajectories. Pollution mixture OR 1.16 (1.08-1.24) for accelerated-aging membership, driven mainly by O_3 and PM_{10}; NDVI OR 0.84 (0.75-0.95), 25% of which was mediated through lower pollution. Limitation: trajectories differ in level, not slope, so the design measures cross-sectional BAA differences dressed as trajectories; exposure contrast within one city is narrow and residential NDVI is strongly SES-patterned. Relevance: moderate - the narrow within-city contrast is the case for intra-urban sensor density. Held from the 9 Oct issue.

doi: 10.1021/ehp.6c00004

Northern Thailand NPC survival $PM_{2.5}$ and coarse PM 2026 J Clin Med

The Effect of Fine ($PM_{2.5}$) and Coarse (PM_{10}-2.5) Particulate Matter on Survival Outcomes in Nasopharyngeal Carcinoma Patients: A Retrospective Cohort Study in a Northern Thailand Tertiary Care Center

METADATA ONLY. No abstract at harvest. Title indicates a single-centre retrospective cohort of nasopharyngeal carcinoma patients in northern Thailand relating fine and coarse PM to survival. Recorded for coverage; no estimate inferred. Relevance: moderate - northern Thailand's biomass-burning season is a well-instrumented low-cost-sensor region, and a coarse-fraction prognosis question needs PM_{10}-2.5 that optical LCS estimate poorly.

doi: 10.3390/jcm15197367

Ran $PM_{2.5}$ MASLD proteins 2026 J Adv Res

Association of protein signature of fine particulate matter with MASLD: the underlying pathways and Mendelian randomization

52,190 participants free of metabolic dysfunction-associated steatotic liver disease at baseline, 624 incident cases over a median 11.8 years. $PM_{2.5}$ gave HR 1.19 (1.10-1.30) and a 700-protein elastic-net signature of $PM_{2.5}$ gave HR 1.19 (1.08-1.31), the proteins enriched in NF-kB signalling and cytokine-receptor interaction; Mendelian randomisation supported CD28, CSF3 and HAVCR2 as causally associated with MASLD, and the signature mediated 15% (2.7-37.7%) of the $PM_{2.5}$-MASLD association. Limitation: a 700-protein signature trained on the same cohort in which it is then tested will absorb variance from everything $PM_{2.5}$ correlates with - smoking, adiposity, deprivation - so the mediated fraction is an upper bound, and its CI nearly spans an order of magnitude; the MR arm addresses protein-to-MASLD causality, not exposure-to-protein. Exposure is an annual mean from an interpolated surface, which is the weakest term in the chain. Relevance: moderate - liver endpoints are under-studied, and 15% mediation at best leaves the mechanism mostly open.

doi: 10.1016/j.jare.2026.10.021

SEER wildfire $PM_{2.5}$ head and neck cancer 2026 Laryngoscope

Wildfire Air Pollution Exposure and Head and Neck Cancer Incidence

SEER 2016-2022, county-level DLNM with 0-10-year lags, adjusted for tobacco, alcohol and demographics with county random effects. Cumulative RR per additional smoke-exceedance day 1.06 (1.04-1.09); any annual exposure 1.41 (1.24-1.60); per percentage-point of days >5 $\mu g\,m^{-3}$ 1.23 (1.13-1.33). Larynx largest (1.13, 1.07-1.19 per day), oropharynx 1.09. Limitation: ecological county-level exposure and covariates, a ten-year lag fitted on seven years of incidence, and wildfire-exposed western counties differ in screening and HPV epidemiology; the p16 stratification is the right control but its result is truncated in the abstract. Relevance: moderate-high - a second smoke-cancer signal this week (cf. 13 Oct China), both favouring upper aerodigestive sites.

doi: 10.1002/lary.70874

Sandal et al. 2026 Clin Kidney J

Disasters and hazard exposure in end-stage kidney disease: a scoping review of patient outcomes

PRISMA scoping review: 10,040 screened, 30 studies, ~1.07 million dialysis patients and 64,531 transplant recipients. Poor air quality was associated with 5-139% higher mortality; extreme heat 9-45%; 23-59% of patients had dialysis disruption in disasters. Limitation: scoping design, no pooling or risk-of-bias grading, and 'poor air quality' spans wildfire smoke, $PM_{2.5}$ increments and AQI categories, so the 5-139% range is not an effect size. Relevance: low-moderate - dialysis patients are an identifiable, routinely contacted population where sensor-triggered alerts could plausibly be tested.

doi: 10.1093/ckj/sfag318

Slovakia $PM_{2.5}$ lung cancer mortality gender lag ecological 2026 Eur J Public Health (conference abstract)

$PM_{2.5}$ exposure and lung cancer mortality in Slovakia: gender differences and delayed effects

National time-series ecology, 2010-2023: population-weighted $PM_{2.5}$ fell from 24.4 to 15.2 $\mu g\,m^{-3}$ while lung-cancer mortality fell from 30.7 to 25.3 per 100,000 (AAPC -1.5%), driven by men (56.8 to 42.6; AAPC -2.2%); female rates were flat (12.6). Correlation with $PM_{2.5}$ r=0.95 overall and 0.97 in men, none in women; lags suggested emerging female associations. Limitation: 14 annual points of two monotonic trends correlate trivially; male smoking decline over the same period explains the male-only pattern, and the outcome is a GBD model output. Conference abstract. Relevance: low.

doi: 10.1093/eurpub/ckag169.1260

Tian et al. 2026 Hum Genet

Annual average exposure to ambient air pollution, polygenic risk, and incidence of major chronic diseases: a large-scale prospective cohort study

313,534 UK Biobank participants with LUR annual pollutant estimates and weighted polygenic risk scores for each of five disease groups; Cox models with RERI and AP for additive interaction. $PM_{2.5}$ carried the strongest association, HR 1.22 per 5 $\mu g\,m^{-3}$ (95% CI 1.19-1.26); high genetic risk acted independently (diabetes HR 1.77 per SD of PRS), with significant additive interaction between high PRS and high $PM_{2.5}$ for neurodegenerative disease (RERI 0.14) and diabetes (RERI 0.21). Limitation: UK Biobank is a healthy-volunteer cohort with a narrow $PM_{2.5}$ range, the LUR surface is a single baseline-era estimate carried forward, and five outcomes with four pollutants invites selective emphasis - the interaction terms are modest and not corrected for multiplicity. Relevance: moderate - gene-environment interaction raises the value of resolving exposure well at the individual level, which is the case for personal rather than ambient monitoring.

doi: 10.1007/s00439-026-02875-w

Wang $PM_{2.5}$ lung cancer nonsmokers narrative review IL-1beta EGFR screening 2026 J Toxicol Environ Health B Crit Rev

Environmental $PM_{2.5}$ exposure associated with lung cancer in nonsmokers: a narrative review of high-risk groups, mechanisms, and implications for screening

Narrative review of cohort, meta-analytic and toxicological evidence on $PM_{2.5}$ and lung cancer in never-smokers (<100 lifetime cigarettes), weighted to East Asian women. Emphasises macrophage IL-1beta signalling promoting EGFR-mutant adenocarcinoma, and lists high-exposure microenvironments: incense burning, cooking and solid fuel, subway/bus/motorbike commuting and professional driving, firefighting, construction, informal e-waste recycling. Argues for cumulative $PM_{2.5}$ in risk-prediction models and screening of exposed never-smokers. Limitation: non-systematic; no new quantitative synthesis, and the microenvironment list is not dose-ranked. Relevance: medium - cumulative personal $PM_{2.5}$ as a screening criterion would need exactly the personal and microenvironment data wearable sensors produce.

doi: 10.1080/10937404.2026.2747089

Wen landscape fire $PM_{2.5}$ renal hospitalisations six-country time-series 2026 Environ Sci Technol

Landscape Fire-Sourced $PM_{2.5}$ and Cause-Specific Renal Hospitalizations: A Multicountry Two-Stage Time-Series Analysis of 12.5 Million Admissions

12.5 million renal admissions across 1020 locations in six countries, 2000-2019; two-stage time-series with fire-sourced $PM_{2.5}$ separated from the non-fire remainder. Per 10 $\mu g\,m^{-3}$ cumulative lag 0-5, RR 1.011 (1.007-1.015) for all renal admissions, strongest for acute kidney injury; associations larger for fire-sourced than non-fire $PM_{2.5}$, larger in warmer areas and graded by GDP per capita. Attributable burden 1.38 admissions per 100,000 per year. Limitation: the fire/non-fire split comes from a chemical-transport-plus-ML partition, not measured tracers, so 'fire PM is more toxic' partly inherits the partition's error structure; heat co-occurs with smoke and dehydration drives AKI, and residual confounding by temperature is the obvious threat. Relevance: moderate - smoke-episode exposure is where LCS density most improves over sparse FRM/FEM coverage.

doi: 10.1021/acs.est.6c04677

Xia et al. 2026 J Hazard Mater

Long-term exposure to fine particulate matter constituents and lung cancer risk in Chinese never-smokers

262,785 never-smokers in the China Kadoorie Biobank, 2,646 incident lung cancers; time-dependent Cox models per IQR of each constituent, quantile g-computation for the mixture, and a polygenic risk score for interaction. HR per IQR: nitrate 1.35 (1.23-1.48), ammonium 1.22 (1.13-1.31), chloride 1.20 (1.12-1.28); mixture HR 1.13 (1.07-1.19), dominated by NO_3- and NH4+; additive and multiplicative NO_3- x PRS interaction. Limitation: constituents come from one satellite/ML reanalysis in which nitrate and ammonium are strongly collinear and partly inherited from total-mass predictors, so 'nitrate drives the risk' may be 'the best-resolved component wins'; second-hand smoke and cooking fuel only partly captured. Relevance: high - constituent-specific risk is the argument for composition-capable nodes (nitrate is semi-volatile and biased low on heated optical sensors).

doi: 10.1016/j.jhazmat.2026.143759

Yang and Liu 2026 medRxiv (preprint)

No evidence for large causal effects of six air pollutants on non-suppurative otitis media: a two-sample Mendelian randomization study

Two-sample MR with UK Biobank pollutant GWAS and FinnGen R13 OME (12,397 cases, 464,237 controls); IVW primary, full pleiotropy battery, positive (asthma, rhinitis, CRS) and negative (suppurative OM) controls. No pollutant survived FDR; the nominal PM_{10} OR 1.56 (0.95-2.55) reversed with the 88-SNP set (0.64, 0.24-1.69). Minimum detectable ORs 1.69-32.1, so moderate effects are not excluded - the authors say so. Limitation: 'genetic instruments for ambient PM' index residential and behavioural traits, not exposure; null MR here is uninformative by construction, and MVMR F $<$ 10. Relevance: low - included as a properly bounded null, not as evidence of absence.

doi: 10.64898/2026.09.30.26364391

Zhang air pollution cataract meta-analysis 2026 Front Med (Lausanne)

Air pollution exposure and cataract outcomes: a systematic review and pollutant-specific meta-analysis

Fifteen observational studies, seven pooled with REML random effects (search to 10 Aug 2026; ROBINS-E). $PM_{2.5}$: HR 1.036 (0.988-1.086) per 1 $\mu g\,m^{-3}$, I2 99.7%; NO_2: HR 1.041 (1.013-1.070) per 10 $\mu g\,m^{-3}$, I2 86%; black carbon per IQR: HR 1.010 (0.983-1.038), I2 33%. Highest-vs-lowest quartile was positive for NO_2 (HR 1.094, 1.057-1.133) and null for $PM_{2.5}$ and PM_{10}; Knapp-Hartung widened the NO_2 interval. Limitation: with I2 near 100% the pooled $PM_{2.5}$ point estimate is not interpretable as a common effect - it averages incompatible exposure models and outcome definitions across too few studies. Relevance: low-moderate - the $NO_2 > PM_{2.5}$ ordering points at traffic proximity, which a dense network resolves better than satellite $PM_{2.5}$.

doi: 10.3389/fmed.2026.1969759

Zhong air pollution years of life lost cancer Pudong Shanghai time series 2026 PLoS One

The short-term impact of ambient air pollution on years of life lost from cancer in Pudong New Area, Shanghai: A time series study

Daily time series of cancer deaths and YLL in Pudong, 2013-2021, with DLNM (linear for YLL, quasi-Poisson for deaths). Per 10 $\mu g\,m^{-3}$ at lag 0-1: $PM_{2.5}$ +2.22 YLL (0.71-3.74), PM_{10} +1.04 (0.17-1.91); SO_2, NO_2 and O_3 also positive. Effects were marginally larger in men, younger and more educated people. Limitation: short-term harvesting of cancer deaths is mostly mortality displacement in terminally ill patients, and single-pollutant models leave the $PM_{2.5}$ estimate inseparable from co-pollutants; YLL per day is driven by a few young deaths, so the scale is unstable. Relevance: low - single-district monitor-average exposure.

doi: 10.1371/journal.pone.0360049

Pooled corpus register — October 2026

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
ABCD air pollution	The Relationship between Air Pollution Exposure, Genetic Liability for Schizophrenia, and Brain Outcomes in the ABCD Study	*Schizophr Bull*	Prospective cohort	Modelled annual $PM_{2.5}$, NO2 and O3 at age 9-10	USA (ABCD Study)
ALFA+ TRAP CSF GFAP	Air pollution is associated with sex-specific longitudinal changes in astrocytic and neuroinflammatory processes linked to cognitive decline in Alzheimer's disease at-risk individuals	*Alzheimers Dement*	Prospective cohort	LUR NO2, PM_{10}, $PM_{2.5}$, $PM_{2.5}$ absorbance	Spain (Barcelona, ALFA+)
Al-Atram early-life air pollution child neurodevelopment critical review	Early Life Ambient Air Pollution and Child Neurodevelopment: Critical Review of Cohort Evidence	*J Xenobiot*	Critical review	Prenatal and early-childhood $PM_{2.5}$, PM_{10}, NO2	Global / multi-region
Baird woodsmoke CD4 T cell brain influx saturated fat diet mice	CD4+ T Cell Activation and Peripheral Immune Cell Influx into the Brain Following Wildfire Smoke Exposure is Modulated by a Saturated Fat Diet	*Neurotoxicology*	Animal exposure study	Woodsmoke at 0.5 mg/m3, 4 h every other day for 14 days	Not applicable (laboratory)
Batisse et al.	Long-term exposure to ultrafine particles and mortality from neurodegenerative diseases: a population-based cohort study	*Epidemiology*	Retrospective cohort	High-resolution UFP number concentration and UFP size at residential postal code, 3-year moving average	Canada (Montreal and Toronto)
Bloomberg et al.	Daily Exceedance of World Health Organization Air Pollutant Guidelines and Cognitive Aging	*medRxiv (preprint)*	Prospective cohort	Annual count of days above WHO daily guideline for $PM_{2.5}$, PM_{10}, NO2	United Kingdom (England, ELSA)
Byun ADRD readmission	$PM_{2.5}$ and 30-Day Readmission in Medicare Beneficiaries with Alzheimer's Disease and Related Dementias: Effect Modification by Neighborhood Green and Blue Space	*Environ Health Perspect*	Time-stratified case-crossover	Daily modelled $PM_{2.5}$ at residential ZIP Code, lag 0-2	United States (contiguous, Medicare 2000-2016)
Han et al.	Air pollution and criminal behavior: a nationwide case-crossover analysis in China	*Int J Epidemiol*	Time-stratified case-crossover	Daily $PM_{2.5}$, PM_{10}, NO2, CO, SO2, O3 (six pollutants, WQS mixture)	China (national)
Holm CHAPS Fresno neurobehavior BDNF children air pollution	Neurobehavioral assessments and biomarkers in school children who are highly exposed to air pollution	*Front Public Health*	Prospective birth cohort + DLNM	Daily prenatal and first-year $PM_{2.5}$, PM_{10}, O3, NO2, NOx, PAH456 and EC	USA (Fresno, California)
Jin wildfire carbonaceous ADRD	Carbonaceous Components Underlying Wildfire Smoke $PM_{2.5}$-Associated Alzheimer's Disease and Related Dementias in Older US Adults	*medRxiv (preprint)*	Retrospective cohort	Annual smoke vs non-smoke $PM_{2.5}$ and smoke EC/OC (high-resolution model)	United States (Medicare, nationwide)
Kalkbrenner traffic air pollution ADHD Denmark critical windows	Traffic-related air pollution links with attention deficit hyperactivity disorder in Denmark: Developmental periods of susceptibility and sex-specific associations	*Environ Res*	Case-cohort study	Daily address-level $PM_{2.5}$ and elemental carbon (DEHM/UBM/AirGIS), pre-conception to age 12	Denmark (national registers)
Khan Pakistan neurovascular review	Air Pollution and Neurovascular Health in Pakistan: An Emerging Public Health Crisis in South Asia - A Narrative Review	*Curr Neurovasc Res*	Narrative review	$PM_{2.5}$, PM_{10}, UFP (literature)	Pakistan
Li built environment sleep duration air pollution mediation Yinzhou cohort	Association between built environment and sleep duration mediated by air pollution: a cohort study in China	*BMC Public Health*	Prospective cohort	Residential PM_{10}, $PM_{2.5}$, NO2 (modelled annual) as mediators	China (Yinzhou, Ningbo)
$PM_{2.5}$ and Mn biomarkers	Differential associations of $PM_{2.5}$ and particulate manganese with blood-based amyloid and tau biomarkers	*Environ Pollut*	Metadata only (no abstract)	Metadata only - $PM_{2.5}$ and particulate manganese against blood amyloid and tau biomarkers (by title)	Not recoverable from metadata

First author	Focus	Journal	Design	Metric	Region
Papatheodorou prenatal wildfire $PM_{2.5}$ ASD ADHD Medicaid	Prenatal Exposure to Wildfire $PM_{2.5}$ and Risk of ASD and ADHD in Children Enrolled in Medicaid	*JAMA Pediatr*	Retrospective cohort + DLNM	Weekly wildfire-smoke-attributable $PM_{2.5}$ (1 km), aggregated to residential ZIP code across pregnancy	USA (nationwide Medicaid, 2007-2014)
Santoso air pollution noise greenness salivary cortisol BAMSE	Air pollution, road traffic noise, and greenness in relation to salivary cortisol in young adults	*J Expo Sci Environ Epidemiol*	Cross-sectional (birth cohort)	Address-level $PM_{2.5}$, PM_{10}, black carbon, NOx (dispersion models), prior year	Sweden (Stockholm, BAMSE)
Shanghai prenatal UFP postpartum depression	Prenatal exposure to ultrafine particles and positive screening for postpartum depression: a prospective pregnancy cohort in Shanghai, China	*Environ Pollut*	Prospective cohort	Residential trimester-specific UFP number and $PM_{2.5}$ (modelled)	China (Shanghai)
Tang source PM dementia	Associations of long-term exposure to source-specific $PM_{2.5}$, lifestyle, and genetic susceptibility with dementia risk: a prospective cohort study	*Eur Arch Psychiatry Clin Neurosci*	Prospective cohort	Source-specific $PM_{2.5}$ (domestic, road transport, other transport, industry), time-varying	Not stated (large prospective cohort)
Wang air pollution adolescent anxiety depression Sichuan cross-sectional	Air pollution and mental health in adolescents: a cross-sectional study in Sichuan, China	*BMC Public Health*	School-based cross-sectional	PM_{10}, $PM_{2.5}$, O3, CO, SO2, NO2 assigned via IP-geolocated survey location; winter periods	China (Sichuan)
Wang et al.	Long-Term Air Pollution Exposure and Brain Structural Decline: A Longitudinal Analysis	*Environ Pollut*	Prospective cohort	Residential $PM_{2.5}$ and PM_{10} from high-resolution spatiotemporal models	Not stated in abstract (middle-aged and older adult imaging cohort)
Wang multi-omics Mendelian randomization $PM_{2.5}$ depression	Multi-omics Mendelian Randomization Reveals the Causal Association and Mediating Mechanisms of $PM_{2.5}$ with Depression: A Systematic Study Integrating Transcriptomics, Metabolomics, and Radiomics	*Research Square (preprint)*	Two-sample Mendelian randomisation	Genetically predicted $PM_{2.5}$ exposure (GWAS instruments)	Global / multi-region (GWAS summary data)
Yang heat anomaly wildfire smoke $PM_{2.5}$ incident depression All of Us	Heat anomaly, wildfire smoke, social-material vulnerability, and incident depression in a national EHR-linked cohort: A preregistered secondary analysis	*J Affect Disord*	Prospective cohort (registry linkage)	30-day mean wildfire-smoke $PM_{2.5}$ (Stanford ECHO model) at ZIP3	USA (national, All of Us)
Zhang CLHLS $PM_{2.5}$ components cognitive disorders lifestyle mediation	Association between long-term ambient air pollution exposure and cognitive disorders among older adults in China: the mediating role of lifestyle	*J Glob Health*	Prospective cohort	Long-term $PM_{2.5}$, sulphate, nitrate, ammonium, O3 (per 10 $\mu g/m^3$)	China (CLHLS, multi-province)
Zhou short-term indoor $PM_{2.5}$ cognition HRV randomized crossover	Short-term indoor fine particulate matter exposure impedes cognitive performance and autonomic nervous system in adults	*iScience*	Randomised crossover trial	Candle-generated indoor $PM_{2.5}$, 1.70 vs 31.22 $\mu g/m^3$ (purifier vs sham filter)	UK (laboratory)
Cardiovascular & metabolic					
Andreotti active fire detections circulatory hospital admissions Brazil ecological panel	Active Fire Detections and Hospital Admissions for Circulatory Diseases in Brazil: An Ecological Panel Study	*Cureus*	Ecological panel, fixed effects	Annual satellite active-fire detection density (INPE) as smoke proxy	Brazil (27 federative units)
Brugge HEPA sham crossover cytokines visceral adiposity	Pro-inflammatory cytokines moderated by visceral adiposity from a pragmatic randomized crossover trial of portable HEPA air purifiers	*Am J Prev Cardiol*	Randomised crossover trial	Indoor PM reduction by portable HEPA vs sham filtration (30-day periods)	Not stated in abstract
Cao $PM_{2.5}$ chemical components heart disease longitudinal China 123 cities	Association between $PM_{2.5}$ and its chemical components exposure and heart disease risk: a longitudinal study from China	*BMC Cardiovasc Disord*	Prospective cohort	$PM_{2.5}$ and components (SO4, NO3, NH4, OM, BC), gridded	China (123 cities)
Cerin air pollution road environment cardiometabolic walking AusDiab	Associations of Ambient Air Pollution and Road-Related Environmental Characteristics with Cardiometabolic Health: The Role of Walking	*Toxics*	Cross-sectional (cohort baseline)	Residential $PM_{2.5}$ and NO2 (modelled); road density, distance to nearest busy road	Australia (AusDiab wave 3)

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Chen Lanzhou $PM_{2.5}$ components CVD	Acute cardiovascular mortality risks from $PM_{2.5}$ and its chemical components in an industrial valley city: evidence from Lanzhou, China	*Int J Environ Health Res*	Time-series (GAM) with mixture models	Daily $PM_{2.5}$; random-forest-predicted components (PAHs, metals, ions)	China (Lanzhou)
Children's AIRE	Ambient particulate matter exposure and children's blood pressure in rural California, 2019-2022	*Int J Hyg Environ Health*	Prospective cohort	12-month and 7-day $PM_{2.5}$ and PM_{10} from regulatory monitors linked to residence	USA (rural California)
Cosenza et al.	Associations of Outdoor Air Pollution With Cause-specific Mortality in Metastatic Prostate Cancer: A cohort study	*medRxiv (preprint)*	Retrospective cohort	Residential $PM_{2.5}$ and NO2 at diagnosis, 1-km exposure models	USA (11 state registries)
Du Air Health Index acute myocardial infarction 325 cities	Establishing an Air Health Index for Predicting the Risk of Acute Myocardial Infarction Onset Associated with Air Pollution and Ambient Temperature	*Environ Sci Technol*	Health impact model	Daily $PM_{2.5}$ (with NO2, O3, temperature) from high-resolution surfaces	China (325 cities)
Hahad exposome heart failure older adults review	Exposomic Determinants in Heart Failure Among Older Adults - Mechanistic Links, Clinical Evidence, and Future Perspectives	*Curr Heart Fail Rep*	Narrative review	Air pollution (PM) within a multi-exposure exposome (noise, heat, metals, light at night, greenness)	Global / multi-region
Huang $PM_{2.5}$ coronary artery dilatation Kawasaki disease case-crossover	Association between PM(2.5) and coronary artery dilatation with Kawasaki disease in China	*Eur J Pediatr*	Case-crossover	Short-term $PM_{2.5}$ (distributed lag 0-7 d)	China (single paediatric centre)
Huang gut microbiota mediation air pollution metabolic syndrome	The mediating roles of gut microbiota in the associations of air pollution and meteorological factors with metabolic syndrome	*iMetaOmics*	Microbiome mediation analysis	Long-term residential $PM_{2.5}$ and NO2; ambient temperature	China (cohort location not stated)
Lee CRCS-K post-discharge SO2 stroke recurrence PM null	Post-Discharge Ambient Sulfur Dioxide Exposure and Stroke Recurrence After Acute Ischemic Stroke	*J Stroke*	Prospective cohort (registry linkage)	3-month post-discharge district PM_{10}, $PM_{2.5}$, NO2, SO2, CO, O3 (IDW)	South Korea (CRCS-K-NIH registry)
MESA incident AF time-varying pollution	Association of Air Pollution Exposure and Risk of Incident Atrial Fibrillation: The Multi-Ethnic Study of Atherosclerosis	*Heart Rhythm*	Prospective cohort	Spatiotemporal-model $PM_{2.5}$, NO2, NOx, O3 at residence; 2-y rolling means updated every 30 d	United States (6 MESA metros)
MESA long-term air pollution suPAR null	Long-term ambient air pollution is not associated with circulating soluble urokinase plasminogen activator receptor	*Environ Epidemiol*	Observational - cross-sectional	MESA Air spatiotemporal $PM_{2.5}$, NO2, NOx, O3, light-absorbing carbon (long-term residential)	United States (six MESA metros)
Musa outdoor air pollution cardiovascular health school children Dar es Salaam	Effects of outdoor air pollution on the cardiovascular health of school-aged children residing near polluted areas and the efficacy of mitigation measures in Dar es Salaam, Tanzania	*BMC Public Health*	Cross-sectional survey	None measured; residence near industry, major roads or waste burning used as proxy	Tanzania (Dar es Salaam)
Rao $PM_{2.5}$ components cardiorenal multistate trajectory	Associations of $PM_{2.5}$ components with dynamic and accelerated progression of different kidney function states and subsequent cardiovascular events	*Int J Cardiol*	Prospective cohort	2-year $PM_{2.5}$ and components: sulfate, nitrate, ammonium, organic matter, black carbon	Not stated in abstract (cohort of 307,639)
Rivera-Mariani et al.	Inflammatory Burden, Structural Exposure, and Racial Disparities in Incident Heart Failure: Evidence from the Multi-Ethnic Study of Atherosclerosis	*medRxiv (preprint)*	Prospective cohort	$PM_{2.5}$ and NO2 as co-adjustment in a geocoded subcohort (not the exposure of interest)	USA (MESA)
Sun long-term air pollution lipid profiles Shanxi repeated measures	Long-term exposure to ambient air pollution and lipid profiles: a longitudinal study in a heavily industrialized region of northern China	*Front Public Health*	Prospective cohort	3-y mean $PM_{2.5}$, PM_{10} (ChinaHighAirPollutants, geocoded residence)	China (Shanxi Province)

First author	Focus	Journal	Design	Metric	Region
UKB arrhythmia NMR metabolome mediation	Large-scale plasma metabolic profiling identifies signatures and biomarkers associated with air pollution and arrhythmias: A prospective cohort study from UK Biobank	*Ecotoxicol Environ Saf*	Prospective cohort	LUR $PM_{2.5}$, PM_{10}, NO2, NOx at baseline residence	United Kingdom (UK Biobank)
UKB women deprivation $PM_{2.5}$-NO2 fatty liver	Socioeconomic disadvantage, air pollution, and metabolic health among women: analyses of UK Biobank and US data	*Front Endocrinol (Lausanne)*	Observational - cross-sectional	2010 residential $PM_{2.5}$-NO2 score (UK); state $PM_{2.5}$ (US)	United Kingdom (UK Biobank) / United States
Wan $PM_{2.5}$ mortality urban rural elderly Beijing cohort	Association between fine particulate matter exposure and all-cause and cardiovascular mortality in urban and rural elderly population in Beijing	*Zhonghua Liu Xing Bing Xue Za Zhi*	Prospective cohort	Satellite-gridded annual $PM_{2.5}$ at residence (2010-2014 mean 79.8 $\mu g/m^3$)	China (Beijing urban and rural)
West Florida temperature x $PM_{2.5}$ CVD admissions	Joint Impacts of Ambient Air Pollution and Temperature on Cardiovascular Disease in West Florida	*J Am Heart Assoc*	Time-series	Daily network-monitor $PM_{2.5}$ and temperature, DLNM lags 0-7 with tensor-product interaction	United States (West Florida)
Wongcharoen short-term $PM_{2.5}$ MACE high-risk population nationwide case-crossover	Associations between short-term PM(2.5) exposure and major cardiovascular events in high-risk population: a nationwide case-crossover study	*BMJ Open*	Metadata only (no abstract)	Short-term $PM_{2.5}$	Thailand (inferred from author group; not stated in record)
Yadav PM pollution-accelerated cardiovascular aging PACA framework	Particulate Matter Exposure and Accelerated Cardiovascular Aging: Mechanistic Evidence, Biomarker Integration, and the PACA Framework	*Cardiovasc Toxicol*	Mechanistic review	$PM_{2.5}$ and ultrafine particles (literature)	Global / multi-region (review)
Zhang short-term air pollution stroke incidence Guiyang karst city	Time-series analysis of the impact of short-term exposure to atmospheric pollutants on stroke risk among residents in Guiyang City, 2016-2021	*Front Public Health*	Time-series (generalized additive model)	Daily fixed-site $PM_{2.5}$ and PM_{10} (city mean), with SO2, NO2, O3, CO	China (Guiyang)
Zhang wildfire smoke waves long-term $PM_{2.5}$ cardiovascular hospitalisation Medicare	Impact of wildfire smoke waves on the association between long-term exposure to fine particulate matter and cardiovascular hospitalizations among older adults in the United States	*Environ Res Health*	Nationwide cohort	Two-year mean $PM_{2.5}$ (1 km ML model) and smoke $PM_{2.5}$ (10 km); smoke wave = >=2 d with smoke $PM_{2.5}$ >10 $\mu g/m^3$	USA (contiguous)
Zhao interpretable machine learning multi-window $PM_{2.5}$ hypertension CHNS Beijing	An Interpretable Machine Learning Framework Integrating Multi-Window Environmental Exposure for Hypertension Risk Assessment	*Toxics*	Gradient-boosting model + SHAP interpretability	Cumulative $PM_{2.5}$ over 0, 7, 15, 30 and 60 days	China (Beijing, CHNS subsample)
Zhu source-specific $PM_{2.5}$ atrial fibrillation Medicare cohort	Source-Specific $PM_{2.5}$ Exposure and Atrial Fibrillation Risk in U.S. Older Adults: A Nationwide Cohort Study	*Environ Health (ACS)*	Retrospective cohort	Source-apportioned $PM_{2.5}$ (Multilinear Engine on CSN/IMPROVE speciation; IDW to ZIP code, 100 km)	USA (nationwide Medicare)
Reproductive & developmental					
Ai air pollution IVF outcomes Chengdu retrospective cohort	Association between Ambient Air Pollution Exposure and Reproductive Outcomes following In Vitro Fertilization: A Retrospective Cohort Study	*Int J Fertil Steril*	Retrospective cohort	CHAP 1 km $PM_{2.5}$, PM_{10}, O3, NO2 at residence by IVF stage	China (Chengdu)
Atkin air quality excess heat pregnancy outcomes review	Impact of Poor Air Quality and Excess Heat on Pregnancy Outcomes.	*Nurs Clin North Am*	Narrative review	Ambient air pollution and heat (reviewed)	Not stated in record
Falcon-Gracia BiSC placental sncRNA air pollution maternal smoking	Impact of air pollution and tobacco smoke on the placental small non-coding RNA landscape	*medRxiv (preprint)*	Cross-sectional (birth cohort)	Prenatal $PM_{2.5}$, black carbon, NO2 (hybrid dispersion-LUR)	Spain (Barcelona, BiSC)
Gaylord DNA methylation signature prenatal air pollution ECHO cohort	A methodology to identify a DNA methylation-based signature of prenatal air pollution exposure in the ECHO Cohort	*Environ Epigenet*	Cross-sectional (birth cohort)	Prenatal $PM_{2.5}$, NO2, 8-h max O3 from geocoded ML exposure models	USA (multi-site ECHO cohorts)
Ge preconception folic acid air pollution birthweight	Preconception Folic Acid Supplementation Associated with Attenuated Effects of Air Pollution on Birthweight Outcomes	*Environ Health (ACS)*	Metadata only (no abstract)	Ambient air pollution (metrics not stated in record)	China (inferred from author group; not stated in record)

First author	Focus	Journal	Design	Metric	Region
Guangdong CHD $PM_{2.5}$ constituents	Differential Impacts of Fine Particulate Matter Constituents on Congenital Heart Disease across Critical Periconceptional Windows	*Environ Pollut*	Case-control study	Residential $PM_{2.5}$ mass and constituents (sulfate, nitrate, ammonium, BC, OM) by periconceptional window	China (Guangdong)
Ha et al.	Developmental $PM_{2.5}$ exposure and stage-dependent kidney senescence	*Kidney Res Clin Pract*	Metadata only (no abstract)	Metadata only - developmental $PM_{2.5}$ exposure (by title)	Not recoverable from metadata
Hyperemesis gravidarum PM_{10} temperature	Seasonal ambient temperature and particulate matter as discriminators of hyperemesis gravidarum severity: a retrospective comparative observational study	*Front Public Health*	Observational - cross-sectional	Ambient PM_{10}, SO2 and temperature (regional)	Turkiye
ICMR-SPIC stillbirth risk factors biomass fuel India pooled cohort	Prevalence and risk factors for stillbirths in India: secondary analysis of ICMR-Stillbirth Pooled India Cohort	*Lancet Reg Health Southeast Asia*	Prospective cohort	Household biomass fuel use (questionnaire); no PM measurement	India (ten pooled cohorts)
Kalam HAPIN LPG intervention child growth stunting ages 2-5	Effects of a 500-day LPG cookstove and fuel intervention delivered during gestation and infancy on height-for-age (HAZ) and stunting at ages 2, 3, 4 and 5 years	*medRxiv (preprint)*	Trial / intervention	Personal $PM_{2.5}$, CO and black carbon (gestation, infancy, post-trial)	Guatemala, India and Rwanda (HAPIN 2.0 follow-up)
Kukshal et al.	Maternal Exposure to Air Pollution During Early Pregnancy and Congenital Heart Disease: An Exploratory Case-Control Analysis from Delhi NCR	*Indian J Pediatr*	Case-control	Periconceptional $PM_{2.5}$, PM_{10}, NO2, CO, SO2, O3, AQI at geocoded residence	India (Delhi NCR)
Liang CHD circulating proteins	A nested case-control study of air pollution and congenital heart defects via circulating proteins	*Nat Commun*	Nested case-control	Periconception $PM_{2.5}$, PM_{10}, NO2; maternal and cord serum proteomics	China
Lin prenatal air pollution transient tachypnea newborn Taiwan	Prenatal air pollution exposure and the risk of transient tachypnea of the newborn	*Front Pediatr*	Case-cohort study	40-week prenatal $PM_{2.5}$, CO, NOx from nearest of 71 monitoring stations (postal code)	Taiwan (national)
Liu $PM_{2.5}$ components small for gestational age IVF Guangzhou	Maternal exposure to PM(2.5) and its components and risk of small for gestational age in women undergoing in vitro fertilization: a propensity score-matched case-control study in Guangzhou, China.	*BMC Public Health*	Case-control study	Trimester-averaged $PM_{2.5}$ and SO4, NO3, NH4, OM, BC (gridded estimates at residence)	China (Guangzhou)
Ma et al.	Maternal $PM_{2.5}$ exposure induces preeclampsia-like phenotypes in mice through placental angiogenic dysfunction and lncRNA-RUNX1 dysregulation	*J Environ Manage*	Animal model	Heating-season northern-China $PM_{2.5}$, intranasal 3 mg/kg every other day, GD0.5-17.5	China (laboratory; Taiyuan-type heating-season PM)
Moradnia Isfahan pregnant urinary metals indoor dust AQI	Spatial variability of heavy metals in pregnant women: linking urinary biomarkers, indoor dust contamination, and air quality in Isfahan, Iran	*Environ Geochem Health*	Cross-sectional environmental survey	Zone-level Air Quality Index; indoor settled dust metals (not airborne PM)	Iran (Isfahan)
Ni Wuhan birth cohort air pollution neurodevelopment target trial	Early Life Ambient Air Pollution, Sex Steroids and Neurodevelopment: Target Trial Emulation in a Prospective Cohort	*Environ Sci Technol*	Prospective birth cohort	Residential ambient pollutants (PM and gases) from preconception to age 2	China (Wuhan)
Pakistan cooking fuel undernutrition	Association between household fuel use and childhood undernutrition in Pakistan: evidence from a National survey	*Int J Environ Health Res*	Observational - cross-sectional	Household air pollution proxied by self-reported primary cooking fuel	Pakistan (national, PDHS 2017-18)
Placental angiogenic biomarkers review	Exposure to environmental toxicants during pregnancy and placental angiogenic biomarkers: An integrative review	*Reprod Toxicol*	Integrative review	$PM_{2.5}$, PM_{10}, NO2, O3 among other toxicants	Global / multi-region
Qiang maternal PM macrosomia	Interaction of maternal $PM_{2.5}$ exposure and psychosocial state on newborn macrosomia: A nationwide population-based study	*Chin Med J (Engl)*	Metadata only (no abstract)	Metadata only - maternal $PM_{2.5}$ exposure with psychosocial state, nationwide (by title)	China (nationwide)

First author	Focus	Journal	Design	Metric	Region
Scotland prenatal $PM_{2.5}$ infant amoxicillin birthweight mediation	Prenatal $PM_{2.5}$ and Infant Antibiotics: Birth-Weight Mediation via Causal Analysis	*Eur J Public Health* (*conference abstract*)	Prospective cohort (registry linkage)	Annual postcode $PM_{2.5}$, pregnancy-month weighted (mean 7 $\mu g/m^3$)	United Kingdom (Scotland, 2009-2018)
Song et al.	Role of Neighborhood Vulnerability and Opportunity in the Association Between Prenatal Air Pollution and Autism-Related Traits	*Environ Res*	Prospective birth cohort	Prenatal $PM_{2.5}$, NO2, O3 at residence (pregnancy and trimester means)	USA (ECHO, 35 sites)
Valuvunina $PM_{2.5}$ PurpleAir quarry low birth weight Kinshasa DRC	Atmospheric Pollution by Fine Particulate Matter ($PM_{2.5}$) and Low Birth Weight: A Cross-Sectional Study in Binza Ozone Health Zone, Democratic Republic of the Congo, 2022	*J Toxicol*	Cross-sectional survey	Calibrated PurpleAir PA-II-SD $PM_{2.5}$ (48-72 h per health area; 0.52 x raw + 3.2 vs BAM)	Democratic Republic of the Congo (Kinshasa)
Zhu prenatal PM1 $PM_{2.5}$ PM_{10} childhood hemoglobin anemia Fujian birth cohort	Association of prenatal PM1, $PM_{2.5}$, and PM_{10} exposure with early childhood hemoglobin levels and anemia: Identification of susceptibility windows	*Ecotoxicol Environ Saf*	Prospective birth cohort	Trimester-specific PM1, $PM_{2.5}$, PM_{10} (ChinaHighAirPollutants)	China (Fujian)
Respiratory & allergic					
Argersinger air pollution chronic rhinosinusitis review	The link between air pollution and chronic rhinosinusitis	*Curr Opin Otolaryngol Head Neck Surg*	Narrative review	$PM_{2.5}$ (traffic, industrial, wildfire, pollen sources)	Global / multi-region (review)
Dublin air pollution fungal spores respiratory admissions time series	Short-term effects of air pollution and aeroallergens on respiratory admissions	*Eur J Public Health* (*conference abstract*)	Time-series	Daily NO2, $PM_{2.5}$, PM1 (EPA Ireland); fungal spore counts	Ireland (Dublin)
Gonehgar sand dust storms asthma Asia systematic review	Geographic Patterns of Asthma Morbidity: A Systematic Review of Sand and Dust Storm (SDS) Impacts in Asia	*J Asthma*	Systematic review	Dust-event PM_{10} and $PM_{2.5}$ (study-specific definitions)	Asia (Japan, Korea, Middle East)
Guangdong COPD surveillance PM1 $PM_{2.5}$ PM_{10} PAF	Association between chronic respiratory diseases and the attributable risk of long-term exposure to atmospheric particulate matters among middle-aged and elderly Chinese population	*Fundamental Research*	Observational - cross-sectional	722-day mean PM1, $PM_{2.5}$, PM_{10} at 1 km (residential address, 2013-2019)	China (Guangdong)
Guerra-Ruiz cytokines volcanic emissions Cumbre Vieja ASHES	Systemic cytokine signatures associated with volcanic emisions exposure and respiratory disease after the cumbre vieja eruption-ashes project.	*Ecotoxicol Environ Saf*	Observational - cross-sectional	Volcanic gas and particulate exposure (hours of exposure, group membership)	Spain (La Palma)
Harb idiopathic subglottic stenosis geographic clustering ecological	Geographic Clustering of Idiopathic Subglottic Stenosis in North America: An Ecological Cohort Study	*Laryngoscope Investig Otolaryngol*	Ecological	State/province $PM_{2.5}$ alongside temperature, humidity and elevation	North America (USA and Canada)
Howlett-Downing and Wichmann	Triangulating health risks of $PM_{2.5}$ and trace elements in Pretoria: evidence from health risk assessment, case-crossover, and concentration-response function analyses	*Int J Environ Health Res*	Time-stratified case-crossover	Sampled $PM_{2.5}$ mass and $PM_{2.5}$-bound trace elements, Apr 2017 - Feb 2020	South Africa (Pretoria)
ISVOLCAN Tajogaite volcano La Palma lung function obstructive pattern	Lung function after Tajogaite volcano eruption in La Palma: ISVOLCAN study	*Eur J Public Health* (*conference abstract*)	Cross-sectional (cohort baseline)	Residence area during the 2021 eruption; SO2 and PM exposure categories	Spain (La Palma, Canary Islands)
Jung peaker power plant emissions ED urgent care visits Massachusetts	Individual-level health effects of living near a gas-fired "Peaker" power plant	*J Clim Change Health*	Case-crossover	HyADS trajectory-derived peaker-plant exposure index (unitless); background $PM_{2.5}$ adjusted	USA (Massachusetts)
Khan Gambia personal $PM_{2.5}$ pneumococcus	Longitudinal personal environmental exposure and Streptococcus pneumoniae colonisation in a cohort of children 5-14 years old in rural Gambia	*medRxiv* (*preprint*)	Prospective cohort	Personal 24-h $PM_{2.5}$ and CO monitors, repeated (8 visits)	Gambia (rural)
Kim interstitial lung disease changing climate Personal View	Interstitial lung diseases in a changing climate: emerging risks and opportunities to improve care	*Lancet Respir Med*	Narrative review	Air pollution and wildfire smoke among climate exposures (literature)	Global / multi-region (review)

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Kinzunga Kinshasa $PM_{2.5}$ respiratory	Acute Respiratory Impacts of Fine Particulate Matter ($PM_{2.5}$) in Kinshasa: Epidemiological Evidence and Pulmonological Implications	*Research Square (preprint)*	Time-stratified case-crossover	Daily $PM_{2.5}$ from PurpleAir and QuantAQ Modulair sensors plus bias-corrected GEOS-CF	Democratic Republic of the Congo (Kinshasa)
Ko-CHENS 36 months	Exposure to Environmental Pollutants and the Risk of Respiratory Illnesses in 36-Month-Old Children: The Korean Children's Environmental Health Study	*Allergy Asthma Immunol Res*	Prospective birth cohort (mixtures)	Home-visit indoor PM_{10} and $PM_{2.5}$ with VOCs and dust-mite allergen; blood metals and urinary metabolites	South Korea (Ko-CHENS)
Kobza respiratory ambulatory care air quality Upper Silesia	The assessment of respiratory diseases in ambulatory care attributable to air quality in the Upper Silesian agglomeration	*Int J Occup Med Environ Health*	Ecological	Daily $PM_{2.5}$, PM_{10}, SO2, NOx, Pb, benzene from 50 regional stations	Poland (Upper Silesia)
Kusumaningtiar school children $PM_{2.5}$ PM_{10} health risk systematic review	Environmental Health Risks in School Children Due to Air Pollution Exposure: Systematic Literature Review	*Research Square (preprint)*	Systematic review (PRISMA)	Indoor and outdoor school $PM_{2.5}$ and PM_{10}	Global / multi-region
Li size-specific PM lung function China Pulmonary Health Study	Associations between Size-Specific Particulate Matter and Lung Function: Results from the China Pulmonary Health Study	*Environ Health (ACS)*	Observational - cross-sectional	Long-term PM1, PM1-2.5, $PM_{2.5}$, $PM_{2.5}$-10, PM_{10} (per IQR)	China (national, CPH Study)
Li spatiotemporal causal machine learning $PM_{2.5}$ asthma Colorado counties	A Spatiotemporal Causal Machine Learning Framework for Quantifying the Impact of $PM_{2.5}$ on Asthma at the Community Level	*GeoHealth*	Causal ML on monitoring records	County-month $PM_{2.5}$ (exposure surface not stated in abstract)	USA (Colorado)
Lovric ragweed sensitisation PM_{10} pollen model comparison Croatia ATOPICA	Outdoor Environmental Risk Factors for Paediatric Ragweed Sensitisation: A Formal Model Comparison	*Children (Basel)*	School-based cross-sectional	Area-level PM_{10} and NO2; regional ragweed pollen; CORINE land cover (5 km)	Croatia (ATOPICA / EDIAQI)
Luglio air pollution pediatric asthma hospitalizations Louisiana vulnerable groups	Associations between air pollution and pediatric asthma hospitalizations across vulnerable groups in Louisiana	*Environ Epidemiol*	Time-stratified case-crossover	Daily $PM_{2.5}$, O3 and NO2 (NO2 2010-2016 only)	USA (Louisiana)
Meng traffic noise air pollution lung function randomised crossover	Independent and interactive effects of traffic-related noise and air pollution on lung function: A randomized crossover study in healthy young adults.	*Environ Res*	Randomised crossover trial	Roadside PM1, $PM_{2.5}$, PM_{10}, O3, NO2 and noise during evening rush hour	China (urban roadside)
Morales El Paso port-of-entry $PM_{2.5}$ immigration COPD census tracts	Breathing at the Border: $PM_{2.5}$ Exposure, Immigration, and COPD Across Census Tracts Close to Ports-of-Entry	*J Immigr Minor Health*	Ecological / registry analysis	Census-tract $PM_{2.5}$ (NASA satellite-derived)	USA (El Paso, Texas, US-Mexico border)
Oliveira air pollution HRV lung function schoolchildren Cubatao	Effect of air pollution on heart rate variability and pulmonary function in schoolchildren: a cross-sectional study, Cubatao, Brazil, 2017.	*Epidemiol Serv Saude*	School-based cross-sectional	Regulatory monitor concentrations (CETESB), daily and 2-3 day averages	Brazil (Cubatao, Sao Paulo State)
$PM_{2.5}$ constituents TB contacts	Differential effects of $PM_{2.5}$ constituents on tuberculosis infection and disease progression in 40,379 student close contacts	*Ann Am Thorac Soc*	Metadata only (no abstract)	$PM_{2.5}$ constituents	Not stated
Pedersen early-life air pollution lung function COPSAC	Impact of air pollution exposure in early life on lung function development in the COPSAC cohorts	*BMJ Open Respir Res*	Prospective birth cohort	Modelled multi-source $PM_{2.5}$ and NO2 at home address (DEHM/UBM/AirGIS-type chain)	Denmark (COPSAC2000 and COPSAC2010)
Perez-Garcia nasal EWAS respiratory biomarkers Mexican American CHAMACOS	Nasal Epigenome-Wide Association Study of Respiratory and Allergic Biomarkers in Mexican American Young Adults	*medRxiv*	Cross-sectional (cohort baseline)	Prenatal and past-year $PM_{2.5}$ and O3 (residential)	USA (Salinas Valley, California, CHAMACOS; Project Viva replication)
Poland air pollution COPD asthma emergency admissions DLNM heating season	Air pollution and respiratory admissions in Poland: a nationwide DLNM analysis of COPD and asthma	*Eur J Public Health (conference abstract)*	Distributed-lag time-series	PM_{10}, $PM_{2.5}$, NO2, SO2, PAHs and metals in PM_{10} (national monitoring)	Poland (nationwide, 2012-2019)

First author	Focus	Journal	Design	Metric	Region
Sao Paulo paediatric	Climate and air pollution are associated with pediatric respiratory hospitalizations in Sao Paulo, Brazil	*Front Pediatr*	Observational - cross-sectional	IDW-interpolated pollutant and meteorological fields at geocoded residence, 2022	Brazil (Metropolitan Sao Paulo)
Shi air pollution pulmonary tuberculosis incidence China GTWR	Analysis of influencing factors of atmospheric pollution on pulmonary tuberculosis incidence in China from 2014 to 2020	*Front Public Health*	Ecological / spatial	Provincial annual $PM_{2.5}$, SO2, NO2, CO, O3	China (31 provinces)
Sittitoon Thailand haze $PM_{2.5}$ symptoms inequality	Health inequalities in $PM_{2.5}$-related symptoms during Thailand's haze season: the role of education, sociodemographic, lifestyle, protective behaviors, and environmental factors	*Front Public Health*	Cross-sectional survey	Provincial monthly $PM_{2.5}$ from satellite-based products	Thailand (nationwide)
Taiwan greenness pollution child asthma	Association of Urban Greenness and Air Pollution on Asthma Visits Among Children and Adolescents in Industrial and Non-Industrial Cities in Taiwan	*J Asthma Allergy*	Metadata only (no abstract)	Ambient air pollution (unspecified) and greenness	Taiwan
Trachalaki IPF $PM_{2.5}$	Are Short-Term Increases in $PM_{2.5}$ Linked to Respiratory Hospitalizations in IPF?	*Ann Am Thorac Soc*	Metadata only (no abstract)	Short-term $PM_{2.5}$	Not stated
Tullis cystic fibrosis CRS olfaction pollutants	Environmental Exposures, Chronic Sinusitis, and Olfaction in People With Cystic Fibrosis	*Int Forum Allergy Rhinol*	Cross-sectional (cohort baseline)	3-y mean $PM_{2.5}$, PM_{10} (and gases) from nearest EPA monitor to residential ZIP	United States (multicentre)
Wang Guangdong $PM_{2.5}$ oxidative potential DTT childhood allergy	Associations of $PM_{2.5}$ and Its Oxidative Potential with Childhood Allergy-Related Symptoms: A Cross-Sectional Study in Guangdong Province, China	*Environ Sci Technol*	School-based cross-sectional	$PM_{2.5}$ mass, oxidative potential (DTTv, DTTm), metals, carbonaceous, ions	China (Guangdong, six cities)
Wang air pollutants sleep apnea polysomnography architecture Shenzhen	Associations of multiple ambient air pollutants and sleep apnea, sleep architecture and nocturnal oxygenation in adult patients with or without obstructive sleep apnea	*Environ Res*	Retrospective cohort	$PM_{2.5}$, PM_{10}, $PM_{2.5}$ inorganic components, NO2, SO2, CO, O3; lag 0-180 d	China (Shenzhen)
Wang climate air pollutants influenza 48 countries convergent cross mapping	Delayed and nonlinear associations and predictive causal signals between climate factors, air pollutants, and influenza epidemics in 48 countries	*Glob Health Res Policy*	Bayesian spatiotemporal model	Weekly cumulative SO2, organic carbon and dust $PM_{2.5}$ (gridded; source not stated in abstract)	Global (48 countries)
Wilner public safety power shutoff wildfire $PM_{2.5}$ healthcare California	Co-exposure to planned power outages and wildfire $PM_{2.5}$ and acute healthcare utilization in California: A time-stratified case-crossover study	*PLoS Med*	Time-stratified case-crossover	ZIP-level daily wildfire $PM_{2.5}$ (per 10 $\mu g/m^3$) jointly with public safety power shutoffs	USA (California, 2013-2019)
Yoon pollutant asthma immune mechanisms review	Impacts and Immune Mechanisms of Pollutant-Mediated Asthma and Airway Inflammation	*Allergy Asthma Immunol Res*	Narrative review	PM, DEP, O3, NO2, SO2, VOCs, bioaerosols	Not stated
Yuksel COPD cigarette biomass combined exposure exacerbations Turkiye	Clinical outcomes of COPD associated with cigarette smoke, biomass smoke, and combined exposure: A retrospective observational study	*Tob Induc Dis*	Retrospective clinical series	Household biomass smoke (fuel type and duration from records); no PM measurement	Turkiye (multicentre)
Zhou preoperative short-term $PM_{2.5}$ postoperative pulmonary complications lung cancer surgery	Preoperative short-term $PM_{2.5}$ Exposure and Postoperative Pulmonary Complications After Lung Cancer Surgery	*Am J Respir Crit Care Med*	Retrospective cohort	Short-term preoperative $PM_{2.5}$ (metadata only)	China
Zouma et al.	Pulmonary function assessment of a community exposed to air pollution in an industrial area in Ouagadougou, Burkina Faso	*Toxicol Rep*	Observational - cross-sectional	Residence in an industrial area vs control district (no PM measured in abstract)	Burkina Faso (Ouagadougou)
Mechanistic toxicology					
Aged traffic particles	Aging of Urban Traffic Fine Particles Modulates Toxicity Mechanisms and Oxidative Stress Responses	*Environ Sci Technol*	Chamber / laboratory	Parking-garage aerosol (cold-start plus urban), fresh and OFR-aged; oxidative potential, SOA mass, O:C	Not stated in abstract (parking-garage sampling)

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Amato mobile air-liquid interface toxicity Barcelona harbor urban	Assessment of air pollution toxicity in urban environments in Barcelona, Spain, using a mobile air-liquid interface system	*Environ Sci Pollut Res Int*	Exposure assessment + in vitro	Co-measured particle and gas concentrations; whole-air ALI exposure	Spain (Barcelona harbour and urban background)
Awasthi air pollution oxidative stress inflammation molecular mechanisms review	Molecular Mechanisms Linking Air Pollution-Induced Oxidative Stress to Inflammation and Human Disease	*J Xenobiot*	Mechanistic review	PM, NOx, O3, SO2, VOCs (generic)	Global / multi-region
Chen $PM_{2.5}$ mitochondria-associated membranes mouse liver	Exposure to airborne $PM_{2.5}$ suppresses mitochondrial associated membranes and MAM-associated metabolic regulators in mouse liver	*Am J Physiol Cell Physiol*	Animal exposure study + in vitro receptor overexpression	Concentrated ambient $PM_{2.5}$ inhalation (mouse); $PM_{2.5}$ particles on hepatocytes/macrophages	USA (laboratory)
Chen $PM_{2.5}$ mixtures Parkinson disease INPP5D SHIP1 PI3K-AKT multi-omics	Neurotoxicity of $PM_{2.5}$-Bound Multi-Pollutant Mixtures in Parkinson's Disease: Exploring the INPP5D/SHIP1-PI3K-AKT Axis via an Integrative Multi-Omics Study	*J Xenobiot*	Mendelian randomisation + TWAS	GWAS-instrumented $PM_{2.5}$; NIST SRM 1648a urban PM in SH-SY5Y cells	Not stated (public GWAS; laboratory)
Chen pirfenidone fine particulate matter EMT mitochondrial fission autophagy	Pirfenidone reduces fine particulate matter-induced epithelial to mesenchymal transition in lung epithelial cells through regulating mitochondrial fission and autophagy	*Respir Res*	Animal + in vitro	Fine PM (standard reference material type not stated), 100 ug/mL in vitro; intratracheal in mice	Laboratory (no field site)
Chen toluene soot toxicity	Oxidative toxicity and cytotoxicity of toluene combustion soot particles mediated by their microstructure	*Atmos Environ*	Metadata only (no abstract)	Laboratory toluene-flame soot	Laboratory
Comis Turin oxidative potential	Oxidative Potential and Elemental Fingerprints of Urban and Suburban Particulate Matter in Turin	*Atmosphere*	Chamber / laboratory	PM_{10} oxidative potential (ascorbic acid assay); total and bioaccessible metals	Italy (Turin)
Do $PM_{2.5}$ superior cervical ganglion calcium channels satellite glia	$PM_{2.5}$ Exposure Disrupts Sympathetic Neurotransmission by Impairing Voltage-Gated Ca2+ Channels and Remodeling Satellite Glia in the Superior Cervical Ganglion	*Int J Mol Sci*	Animal + in vitro	Collected $PM_{2.5}$, intracerebroventricular and ex vivo direct exposure	Korea (laboratory)
Donohoe ultrafine amorphous silica epithelial stress pro-fibrotic	Ultrafine amorphous silica induces chronic epithelial cell stress and pro-fibrotic reprogramming in a particle size- and exposure duration-dependent manner	*Part Fibre Toxicol*	Metadata only (no abstract)	Ultrafine amorphous silica particles	Not stated in record
EPFR size-resolved PM	Size-Dependent Characteristics, Exposure Risk, and Formation Mechanisms of Environmentally Persistent Free Radicals in Atmospheric Particulate Matter	*Environ Toxicol Chem*	Field measurement	EPFRs (EPR) in $PM_{2.5}$, PM_{10}, TSP; SiO2-catechol-ZnO simulations	China
Edwards macrophage superoxide SOA biomass plastic burning PM	Superoxide Release from Macrophages upon Exposure to Secondary Organic Aerosols, Biomass, and Plastic Burning Particulate Matter	*ACS Omega*	In vitro exposure	Laboratory SOA (limonene, toluene, naphthalene), biomass- and plastic-burning PM suspensions	Not applicable (laboratory)
Env epigenetics review	Environmental epigenetic modifiers in neurodegenerative diseases: a systematic review of epigenomic mechanisms and therapeutic targets	*Front Epigenet Epigenom*	Systematic review (PRISMA)	Particulate matter among heavy metals, gaseous pollutants and synthetic neurotoxicants (review)	Global
Ferreira Vicente et al.	Pulmonary black carbon accumulation: mechanisms, health impacts, and environmental sources	*Expert Rev Respir Med*	Narrative review	Black carbon: sources, tissue retention, label-free optical detection	Not stated (review)
Ganse diesel exhaust particles cardiac metabolomics glycolytic shift rats	Diesel Exhaust Particles Exposure Disrupts Cardiac Bioenergetics and Induces a Shift Towards Glycolytic Metabolism in Rats	*Biomolecules*	Animal model	Diesel exhaust particles, intratracheal instillation 3x/week for 1 or 3 weeks	Spain (laboratory)
Gao et al.	Air pollution and early-onset lung cancer: from environmental carcinogenesis to early intervention strategies	*Carcinogenesis*	Narrative review	Ambient and household $PM_{2.5}$/PM_{10} with PAH, metal and VOC components (review)	Global

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Gautam diesel exhaust miR-133b-3p CTGF Hippo ECM remodeling mouse lung	Diesel Exhaust-Induced miR-133b-3p Downregulation and CTGF-Linked Hippo Pathway Inactivation Promote ECM Remodeling and Inflammation in Mouse Lungs	*Environ Mol Mutagen*	Animal exposure study	Whole diesel exhaust, 30 min/day, 5 d/week, 8 weeks vs filtered air	India (inferred from author group; laboratory)
Gut-liver axis pollutants MASLD review	The Role of the Gut-Liver Axis in the Development of Liver Diseases Induced by Selected Environmental Pollutants: A Narrative Review	*Int J Mol Sci*	Narrative review	PM among heavy metals and EDCs (toxicological literature)	Not applicable
He FOXM1 lncRNA COPD-R biomass $PM_{2.5}$ pulmonary inflammation rat	FOXM1/LncRNA COPD-R regulates pulmonary inflammation in chronic obstructive pulmonary disease induced by biomass-related $PM_{2.5}$	*Ecotoxicol Environ Saf*	Animal + in vitro	Biomass-fuel smoke (rats, 7 months) and biomass-related $PM_{2.5}$ (L2 cells)	China (inferred from author group; laboratory)
Houston UFP PAH deposition	PAH-associated health risk from ultrafine particles: an urban field study	*Air Qual Atmos Health*	Field measurement	UFP respiratory deposition (MALDA) and size-segregated UFP-bound PAHs	United States (Greater Houston)
Inhaled polystyrene microplastics pulmonary fibrosis review	Polystyrene-associated pulmonary fibrosis: a summary based on the available evidence and prospects	*Front Immunol*	Narrative review	Airborne polystyrene micro/nanoplastics (inhalation)	Not applicable
Jiang carbon nanoparticle NSCLC progression ARNT2 ARL14	Carbon nanoparticle exposure promotes the progression of non-small cell lung cancer via the ARNT2/ARL14 axis: Findings from in vivo and in vitro models	*Ecotoxicol Environ Saf*	Animal + in vitro	Carbon nanoparticles (intranasal 15-200 ug/kg; in vitro 2 ug/mL)	China (laboratory)
Jin et al.	ANKRD22 contributes to NF-kB-associated inflammatory responses in alveolar type II epithelial cells during $PM_{2.5}$-triggered airway inflammation	*Front Immunol*	Animal + in vitro	$PM_{2.5}$ instillation (mouse) and A549 exposure; dose not in abstract	China (laboratory)
Kim et al.	Acorus gramineus Solander Preserves Corneal Epithelial Barrier Integrity against $PM_{2.5}$-Induced EMT-Like Remodeling	*J Pharmacopuncture*	In vitro	$PM_{2.5}$ on human corneal epithelial spheroids; dose not in abstract	South Korea (laboratory)
Kou Xing Su San Sang Xing Tang $PM_{2.5}$ lung injury rats	Xing Su San and Sang Xing Tang attenuate $PM_{2.5}$-induced lung injury and inflammatory marker expression in rats	*Pak J Pharm Sci*	Animal exposure study	$PM_{2.5}$ inhalation 500 $\mu g/m^3$, 3 h/day, 28 days, under warm-dry or cool-dry conditions	China (laboratory)
Lee et al.	Curcumin-Loaded Nanospheres Attenuate Fine Dust (PM_{10})-Induced Cellular Toxicity by Suppressing Reactive Oxygen Species-Dependent Signaling in Human Keratinocytes	*Cell Biol Int*	In vitro exposure	PM_{10} applied to HaCaT keratinocytes; curcumin nanospheres as the test article	South Korea (laboratory)
Liu $PM_{2.5}$ IL-17 microglia neuroinflammation	IL-17 Signaling Pathway: A Potential Signaling Pathway Mediating Neuroinflammation Induced by $PM_{2.5}$	*Environ Health (ACS)*	Animal + in vitro	Collected $PM_{2.5}$ suspension (dose not stated in abstract)	Laboratory
Liu Streptococcus D19T fermentation supernatant asthmatic mice $PM_{2.5}$	Protective effect and mechanism of Streptococcus D19T fermentation supernatant in asthmatic mice exposed to $PM_{2.5}$	*Environ Int*	Metadata only (no abstract)	$PM_{2.5}$ exposure in an asthmatic mouse model	Laboratory (not stated)
Liu cerium dioxide scavenging environmentally persistent free radicals $PM_{2.5}$ filtration review	Cerium Dioxide-Based Scavenging of Environmentally Persistent Free Radicals in Fine Particulate Matter: A Prospective Review	*Int J Mol Sci*	Narrative review	EPFRs on $PM_{2.5}$ surfaces; CeO2-functionalised filter media (proposed)	Not stated (review)
Liu et al.	Bulk $PM_{2.5}$ Masks Oxidative Potential and Reactive Oxygen Species Responses by Metal-Organic Complexation in PM1 and PM1-2.5 Mixtures	*Environ Sci Technol*	Chamber / laboratory	Roadside PM1 and PM1-2.5 extracts; OP (AA, DTT), cellular ROS, mitochondrial superoxide	China (roadside site)
Liu et al.	Metabolite-Associated Inflammation during Combined Exposure to Cooking Oil Fumes and Atopic Dermatitis: A Multi-Organ Spatial Metabolomics Study	*Environ Sci Technol*	Animal model	Cooking oil fume inhalation (controlled high-concentration regimen)	China (laboratory)
Lung-on-a-chip pollutant toxicity review	Microfluidic lung-on-a-chip for evaluating inflammation and toxicity from environmental pollutants	*Front Bioeng Biotechnol*	Narrative review	Atmospheric PM, pathogens, VOCs, aromatic hydrocarbons, micro/nanomaterials (in vitro)	Laboratory

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Melatonin DPM$_{2.5}$	Melatonin protects keratinocytes from diesel-derived PM$_{2.5}$-induced damage by suppressing JNK/p38 MAPK signaling and caspase activation	*Ecotoxicol Environ Saf*	Animal + in vitro	Diesel-derived PM$_{2.5}$ applied to HaCaT keratinocytes and to HR-1 hairless mice	South Korea (laboratory)
Mountain West wood smoke	Household wood smoke exposure, lung black carbon deposition, and genomic instability in U.S. Mountain West residents	*Part Fibre Toxicol*	Metadata only (no abstract)	Metadata only - household wood smoke, lung black carbon deposition and genomic instability (by title)	USA (Mountain West)
Mouth-throat replica deposition	The influence of model selection and flow rate on aerosol transport in replica models of the mouth and throat: An experimental study	*J Aerosol Sci*	Metadata only (no abstract)	Aerosol deposition in mouth-throat replicas	Laboratory
Nored lithium-ion battery fire soot PM pulmonary toxicity mice fluoride	Pulmonary Toxicity of Particulate-Phase Combustion Byproducts from Lithium-Ion Battery Fires	*Environ Res*	Animal exposure study	LIB thermal-runaway soot and smoke PM (<10 um), chemical composition	USA (fire test facility; mouse exposures)
Onoda et al.	Gene-environment interaction in neurodevelopment: Setd5 haploinsufficiency and prenatal carbon nanoparticle exposure synergistically induce ASD-like phenotypes	*Part Fibre Toxicol*	Animal model	Carbon black nanoparticles, intranasal 95 ug/kg on gestational days 5 and 9	Japan (laboratory)
PM$_{2.5}$ HCC miR-25	PM$_{2.5}$ promotes migration of hepatocellular carcinoma by targeting miR-25	*Transl Cancer Res*	Animal + in vitro	PM$_{2.5}$ exposure of MHCC97H cells and mouse tumour models	Laboratory
PM$_{2.5}$ NLRP3 inflammasome lung cancer review	Possible Links Between PM$_{2.5}$-Induced NLRP3 Inflammasome Activation and Its Role in Lung Cancer Progression and Drug Resistance	*Int J Mol Sci*	Narrative review	PM$_{2.5}$ (toxicological literature)	Not applicable
PM$_{2.5}$ thyroid vitamin C rats	Vitamin C attenuates PM$_{2.5}$-induced thyroid dysfunction and mitophagy dysregulation by rebalancing ER-mitochondrial Ca2+ dynamics	*Ecotoxicol Environ Saf*	Animal + in vitro	PM$_{2.5}$ instillation (rats) and FRTL5 thyrocyte exposure	Laboratory
Ruan gut microbiota dose-time trajectories PM$_{2.5}$ PFOS mouse	Time-resolved profiling reveals nonlinear dose-time trajectories of gut microbiota in PM$_{2.5}$ and PFOS exposure models	*Environ Pollut*	Animal exposure study	PM$_{2.5}$ inhalation at multiple dose levels (separate PFOS drinking-water model)	Not stated
Saveleva et al.	Urban PM0.2 Affects Astrocytes by Inducing Xenobiotic and Oxidative Stress Responses	*Mol Neurobiol*	In vitro exposure	Urban ambient PM0.2 (aerodynamic diameter <0.2 um), 50 ug/ml, 24 h	Not stated (laboratory)
Sharma et al.	Characterization of carbonaceous particles generated from three flame modes using a single burner to enable properties-based exposure studies	*Aerosol Sci Technol*	Metadata only (no abstract)	Metadata only - carbonaceous particles from three flame modes on one burner (by title)	Not recoverable from metadata
Sun respirable mining dust natural killer cell dysfunction	Respirable Particulate Matter Induces Natural Killer Cell Dysfunction via Oxidative Stress-Mediated Lysosomal and Mitochondrial Injury: Implications for Mineral-Rich Mining Dust Exposure.	*Environ Sci Technol*	In vitro	Compositionally distinct respirable PM (silica/silicate, iron-bearing, carbonaceous mine dusts)	Not stated in record
Tampere SOA toxicity	From primary to secondary aerosol: chemical transformation and toxicity of gasoline vehicle exhaust under contrasting driving conditions	*Environ Res*	Chamber / laboratory	Primary and photochemically aged gasoline exhaust; PAHs, nitro- and oxy-PAHs, water-soluble elements, SP-AMS	Finland (Tampere, chassis dynamometer)
Tang CPS-II plasma proteome air pollution lung cancer ever-smokers	Analysis of the Potential Mediating Role of Plasma Proteins in the Association Between Air Pollution Exposure and Lung Cancer Risk Among Ever-Smokers	*Environ Sci Technol*	Nested case-control	Six ambient pollutants and their mixture (pre-diagnosis)	USA (CPS-II Nutrition Cohort)
Time-restricted feeding PM$_{2.5}$ adipose metabolism	Metabolic Reprogramming by Time-Restricted Feeding Confers Resistance to PM$_{2.5}$-Induced Metabolic Disturbances in Adipose Tissue	*J Nutr Biochem*	Metadata only (no abstract)	PM$_{2.5}$ exposure (experimental; protocol not stated)	Not stated

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Vocal fold fibroblasts	Effects of Fine Particulate Matter ($PM_{2.5}$) in Cultured Human Vocal Fold Fibroblasts	*Laryngoscope*	Metadata only (no abstract)	Metadata only - $PM_{2.5}$ applied to cultured human vocal fold fibroblasts (by title)	Not recoverable from metadata
Wang seasonal PM size constituents oxylipin panel Beijing	Winter and Summer Differences in Oxylipin Responses to Ambient Particles: Roles of Sizes and Constituents	*Environ Sci Technol*	Panel study	27 $PM_{2.5}$ metrics: mass, UFP and accumulation-mode number, OC, chloride and other constituents	China (Beijing)
Yadav et al.	The dual role of atmospheric HULIS in inducing climate forcing and oxidative stress in the northwestern Himalayas	*Environ Pollut*	Metadata only (no abstract)	Metadata only - humic-like substances, light absorption and oxidative potential (by title)	India (northwestern Himalayas)
Yoon PRL-3 ezrin airway epithelium particulate matter	PRL-3-sensitive cytoskeletal signaling mediates early airway epithelial responses to particulate matter	*Respir Res*	Animal + in vitro	Urban PM (UPM), diesel exhaust particles, PM_{10} reference materials	Not applicable (laboratory)
Yu alveolar organoid $PM_{2.5}$	Differential toxicity of water- and DMSO-extractable $PM_{2.5}$ fractions in human alveolar organoids	*Front Toxicol*	In vitro exposure	Water- vs DMSO-extractable $PM_{2.5}$ fractions	Laboratory (human alveolar organoids)
Zakari PM1 microbial profiles lung gut microbiome mice	Associations of particulate matter-associated microbial profiles with lung and intestinal microbiomes in healthy mice	*J Hazard Mater*	Murine inhalation exposure	Ambient PM1 (ions, metals, carbon, oxidative potential, 16S) vs HEPA-filtered and clean air	Not stated in record (ambient-air exposure chamber)
Zendi brown carbon wood pyrolysis tracers cytotoxicity A549	Fine Brown Carbon Aerosol Emitted by Wood Pyrolysis - New Harmful Tracers and Cytotoxicity in Human Lung Cells	*Chem Res Toxicol*	In vitro (cell lines)	Fine brown carbon from 350 C wood pyrolysis; water, methanol and acetone extracts	Not applicable (laboratory)
Zeng RON toxicity	Reduced organic nitrogen compounds dominate the toxicity of $PM_{2.5}$ organics following internal exposure	*Sci Adv*	Chamber / laboratory	Urban $PM_{2.5}$ organic fraction: nontargeted HRMS, serum-albumin adductomics, multi-endpoint toxicity prediction	China (urban)
Zhang et al.	$PM_{2.5}$ exposure impairs BMSC function and dysregulates mitochondrial oxidative stress: multi-omics insights into potential osteoporotic mechanisms	*Free Radic Biol Med*	In vitro exposure	$PM_{2.5}$ suspension applied to rat bone-marrow mesenchymal stem cells (dose not in abstract)	China (laboratory)
Zhao coal dust macrophage extracellular vesicles thyroid cancer miR-205-5p	Coal dust-stimulated macrophage-derived extracellular vesicles promote thyroid cancer cell phenotypes involving miR-205-5p, CAMK1G, and NF-kB-associated signaling	*Front Immunol*	Experimental / toxicology	Characterised coal dust (macrophage exposure; murine exposure)	China (laboratory)
de Luna wildfire PM toxicity systematic review	Systematic review on the toxicity of wildfire emitted particulate matter: State of science, knowledge gaps and way forward	*J Hazard Mater*	Systematic review (PRISMA)	Wildfire-emitted PM (field-collected or laboratory surrogate)	Global / multi-region
Occupational & indoor					
Aljabri et al.	Indoor Air Quality and Exposure Characterization in Operational Waterpipe Cafes: Field Measurements of Particulate and Volatile Pollutants in Al-Madinah Al-Munawwarah	*Indoor Air*	Field measurement	Indoor $PM_{2.5}$ and PM_{10} with TVOC and formaldehyde, peak occupancy, 5 venues	Saudi Arabia (Al-Madinah)
Alquwayi respirable crystalline silica occupational exposure Saudi Arabia review	Potential risk of occupational exposure to respirable crystalline silica in Saudi Arabia: a comprehensive review of the current status and future challenges	*Front Public Health*	Narrative review	Respirable crystalline silica (respirable dust fraction)	Saudi Arabia
Arif Agra footwear cluster ventilation IAQ tiers	Ventilation, indoor air quality and thermal comfort in India's leather footwear manufacturing cluster: a tier-resolved field study	*Research Square* (*preprint*)	Cross-sectional (occupational)	Breathing-zone $PM_{2.5}$ with CO2, TVOC, formaldehyde, air velocity	India (Agra)

First author	Focus	Journal	Design	Metric	Region
Arif Agra leather footwear facilities indoor $PM_{2.5}$ CO2 formaldehyde	Indoor Air Quality and Thermal Environment Across Three Leather Footwear Facilities in Agra India	*Research Square* (*preprint*)	Instrument evaluation	Indoor $PM_{2.5}$/PM_{10}, I/O $PM_{2.5}$ ratio, CO2, formaldehyde, TVOC	India (Agra)
Blackley et al.	Aerosol mitigation efficiency of high-volume suction systems during aerosol-generating dental procedures with patients	*Ann Work Expo Health*	Field measurement	Direct-reading number and mass concentrations, PM4 (respirable) and PM_{10} (thoracic), with size distribution	USA (NIOSH, clinical dental operatories)
Blanco-Alegre ironing indoor ultrafine particles ventilation eBC	Ironing as a Source of Indoor Aerosol Particles: Effects of Ventilation and Ironing Conditions	*Atmosphere*	Measurement campaign	Particle number concentration and size distribution, eBC	Not stated in record
Charres et al.	Insights into the organic and toxicological characteristics of PM_{10} from indoor and outdoor settings in a school environment	*Atmos Pollut Res*	Metadata only (no abstract)	Metadata only - indoor/outdoor school PM_{10} organic speciation and toxicity (by title)	Not recoverable from metadata
Chen dormitory Stokes	Size-resolved aerosol transport and exposure risk in university dormitories driven by Stokes number topology and source-flow coupling	*Build Environ*	Metadata only (no abstract)	Metadata only - size-resolved indoor aerosol transport and exposure, Stokes-number framing (by title)	Not stated (university dormitories)
Creosote railroad switch PAH workers	Assessing PAH exposure during construction work of railroad switches using creosote-impregnated ties	*Ann Work Expo Health*	Field measurement	Full-shift breathing-zone PAH (16 EPA priority; particle + vapour), dermal tape strips, urinary 1-HP	Sweden
Dental scaling bioaerosols	Divergent bacterial signatures in dental clinic bioaerosols: a real-field investigation	*J Hosp Infect*	Field measurement	Optical particle sizer; Andersen six-stage culturable and filter bioaerosol sampling	Not stated (dental clinic)
Du classroom CO2 proxy PM VOC ventilation types	CO2 as a proxy for PM and VOCs in classrooms across ventilation types and activities	*Indoor Environments*	Metadata only (no abstract)	Classroom PM (fractions not stated) vs CO2 and VOCs	Not stated in record
Fan wildfire smoke indoor $PM_{2.5}$ absenteeism Oklahoma Head Start OK-AIR	Wildfire smoke, indoor and outdoor $PM_{2.5}$, and sick-related absenteeism in Oklahoma Head Start early care and education centers	*J Expo Sci Environ Epidemiol*	Cohort (secondary trial analysis)	Continuous classroom $PM_{2.5}$ paired with hourly EPA outdoor $PM_{2.5}$; HMS smoke days; FIRMS fire detections	USA (Oklahoma)
Geng grain dust simulator manikin respirator workplace protection factor	Using a Grain Dust Simulator with a Manikin System to Evaluate Performances of Commonly Used Respirators	*J Agric Saf Health*	Laboratory + field measurement	Respirable grain dust, 5-20 mg/m3; real-time on-farm grain-bin dust	USA (laboratory + on-farm)
Global IAQ open standards framework	A unified framework for global indoor air quality: transitioning from guidelines to actionable open regulatory standards	*Build Environ*	Metadata only (no abstract)	Indoor air quality metrics (incl. PM)	Global / multi-region
Hunter e-cigarette aerosol early post-emission particle decay controlled conditions	Early post-emission particle decay dynamics of E-cigarette aerosols under controlled conditions	*Sci Total Environ*	Metadata only (no abstract)	E-cigarette aerosol particle size/number decay	Not stated in record
Indoor NPF from outdoor O3	Elevated Outdoor O3 Can Sustain New Particle Formation and Growth Indoors	*Environ Sci Technol*	Field measurement	Indoor particle number size distributions; ventilated O3; limonene addition	Not stated (aged unoccupied office, high-O3 region)
K-HEALTHinAIR multicountry indoor air homes schools low-cost sensors	Multicountry evaluation of indoor air in homes and schools: results from K-HEALTHinAIR project	*Eur J Public Health* (*conference abstract*)	Long-term indoor monitoring campaign	Low-cost sensor CO2, PM_{10}, $PM_{2.5}$ (annual 24 h means, 2025)	Europe (Spain, Austria, the Netherlands, Poland)

First author	Focus	Journal	Design	Metric	Region
Kazakhstan PRISm	Occupational associations of preserved ratio impaired spirometry (PRISm) in a population-based study	*Front Public Health*	Observational - cross-sectional	Lifetime occupational vapour, gas, dust and fume exposure by job-exposure matrix; no PM measured	Kazakhstan (Aktobe, Karaganda, Kostanay)
Ketsakorn and Chaiyadej	Probabilistic health risk assessment of respirable dust exposure among stone carvers in Thailand using average daily dose and Monte Carlo simulation	*PLoS One*	Observational - cross-sectional	Personal respirable dust, NIOSH 7601, 8-h TWA across five task groups	Thailand
Kim pig building air cleaner TSP $PM_{2.5}$ productivity	Association between indoor air pollutants and pig productivity index by on-site application of air cleaner	*J Anim Sci Technol*	Intervention field trial (non-randomised)	TSP, $PM_{2.5}$, airborne fungi in pig housing	South Korea (commercial pig farm)
Konisky et al.	Evaluating Indoor Air Pollution in Urban, Low-Income, Multi-Family Buildings: A Case Study from Indianapolis	*Build Environ*	Metadata only (no abstract)	Metadata only - indoor air pollution in low-income housing (by title)	Not recoverable from metadata
Kulmala determination deposition rates indoor particle dynamics	Determination of deposition rates from indoor particle dynamics	*Aerosol Sci Technol*	Metadata only (no abstract)	Indoor particle concentration decay (size-resolved, presumed)	Finland (inferred from author affiliation; not stated in record)
Kuye ultrafine particle respiratory deposition dose wood stoves UK homes	Respiratory Deposition Dose of Ultrafine Particles from Improved Wood Stoves and Open Fireplaces	*Atmosphere*	Field measurement	Indoor particle number concentration (1-min) with ICRP deposition model	United Kingdom (four homes)
LiFePO4 workers cardiac autonomic	Association between occupational LiFePO4 exposure and cardiac autonomic function: a cross-sectional study of cathode material production workers	*Occup Environ Med*	Metadata only (no abstract)	Occupational LiFePO4 particle exposure	Not stated
Mahmoud filtration risk	The link between aerosol dispersion, infection risk and air filtration	*Build Environ*	Metadata only (no abstract)	Metadata only - indoor aerosol dispersion against filtration and modelled infection risk (by title)	Not stated (indoor)
Martin simplified method portable air cleaner CADR low-cost sensors	A simplified method for evaluating fine particulate matter removal performance of portable air cleaners	*MethodsX*	Field co-location + chamber	$PM_{2.5}$ decay measured with low-cost sensors vs research-grade instruments	USA (residence and laboratory)
Matuskova wood stove phase-resolved particle-bound PAHs	Characterisation of particle-bound PAHs emitted from a residential wood-burning stove and their dependence on combustion phases	*Environ Res*	Chamber emission characterisation	Cascade-impactor size-resolved PM and particle-bound PAHs (GC-MS/MS)	Central Europe (laboratory stove; beech, black locust, spruce)
Mohammed mobile HEPA-UV air decontamination operating theatre	Evaluation of a mobile surgical air decontamination device in reducing airborne microbial load in the operating theatre of a tertiary care hospital.	*Infect Prev Pract*	Intervention field trial (non-randomised)	Optical particle counts (0.3, 0.5 um and larger) and culturable CFU	India (New Delhi)
Nahian Siegel in-situ testing in-duct air cleaners residential	In-situ testing of in-duct air cleaners in residential buildings	*Build Environ*	Metadata only (no abstract)	Residential in-duct air cleaner performance (particle removal presumed)	Not stated (record metadata only)
Neonatal air cleaners	Portable air cleaners in neonatal facilities: Hybrid experimental-CFD assessment of different configurations	*Build Environ*	Metadata only (no abstract)	Metadata only - portable air cleaner placement assessed by experiment plus CFD (by title)	Not recoverable from metadata
Ng particulate matter respiratory deposition inhalation dose biomass burning homes indigenous communities	Particulate Matter Exposure, Respiratory Deposition and Inhalation Dosage Assessment in Biomass Burning Homes in Rural Indigenous Communities	*Atmos Environ*	Metadata only (no abstract)	Household PM from biomass burning; respiratory deposition / inhaled dose	India (inferred from author affiliation; not stated in record)
Office air purifier field CADR	Air purifier performance in an occupied office: Field CADR and effectiveness from a reconstructed device-off condition	*Build Environ*	Metadata only (no abstract)	Indoor PM with portable air cleaner (field CADR)	Not stated

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Othman et al.	Impact of Urban Land Use on Indoor Air Quality: Quantifying Ambient Particle Infiltration and Asthma Risk in Naturally Ventilated School Buildings	*Indoor Air*	Field measurement	Paired indoor-outdoor 24 h $PM_{2.5}$ and PM1 with infiltration factor, 10 schools across five land-use classes	Malaysia (Klang Valley)
Parthiban et al.	Ocular Surface Disorders in Traffic Police Officers in the Union Territory of Puducherry, India: A Cross-Sectional Study	*Cureus*	Observational - cross-sectional	Occupational traffic exposure (>1 year field duty); no PM measured	India (Puducherry)
Patras primary school classroom PM airborne viruses ventilation	Dual burden of PM and airborne viruses in primary schools: A pilot study in Patras, Greece	*Eur J Public Health (conference abstract)*	School-based cross-sectional	Classroom $PM_{2.5}$/PM_{10} (triplicate) plus airborne virus RT-qPCR	Greece (Patras)
Pigakis cement dust small airway dysfunction CPET	Early respiratory effects of long-term cement dust exposure: association with exercise capacity in asymptomatic workers	*Int Arch Occup Environ Health*	Cross-sectional (occupational)	Cement dust (>=10 y occupational tenure; no measured concentrations in abstract)	Not stated in abstract
Portable air purifiers airborne bacteria AMR offices schools	Response of culturable bacteria and their resistance to air purification across compartments	*Front Public Health*	Bioaerosol monitoring campaign	Culturable airborne, surface and human-associated bacteria before/after portable purifiers	Spain (Valencia) + Ireland (Dublin)
Rees et al.	Lessons learned from using high efficiency particulate air (HEPA) filtration units in care homes: interviews with care home staff, residents and relatives from the AFRI-c trial	*NIHR Open Res*	Qualitative study (within RCT)	HEPA filtration units as the intervention; no PM measured in this component	United Kingdom (care homes, AFRI-c cluster RCT)
Rodriguez et al.	Integrated Assessment of Indoor Air Pollution and Associated Health Risks in a Hemodialysis Unit	*Indoor Air*	Field measurement	Indoor/outdoor PM0.3, $PM_{2.5}$, PM_{10}, O3, CO2; monthly VOC campaigns	Spain (hemodialysis unit)
Saborit gas cooking indoor air quality households 7 European countries	Characterization of the impact of gas cooking on indoor air quality in households across 7 European countries	*Indoor Environments*	Long-term indoor monitoring campaign	Household indoor air during gas cooking (metadata only)	Europe (7 countries)
Simkovich LPG stove RCT quality of life	Cooking With Liquefied Petroleum Gas or Biomass and Quality of Life: Results From a Randomized Controlled Trial	*Indoor Air*	Randomised controlled trial	Personal $PM_{2.5}$ exposure (exposure-response analysis)	Not stated in abstract (biomass-cooking households)
Stange SEM-Raman inhalable particles indoor artificial turf infill	Correlative SEM–Raman Microscopy of Inhalable Particles Sampled during Indoor Football on Artificial Turf with Crumb Rubber and Olive-Stone Infills	*Research Square (preprint)*	Field study	PM_{10} mass; single-particle SEM + AI classification + micro-Raman of airborne dust	Norway (two indoor sports halls)
Stefaniak recycled plastic bag HDPE filament 3-D printing emissions	Recycling Plastic Bags into Filament: Consideration of Emissions during Melting, Extrusion, and 3-D Printing.	*J Chem Health Saf*	Chamber emission characterisation	Particle number size distribution (1.2-560 nm), particle yields per g plastic; acetone and formaldehyde	United States (laboratory; South African and US HDPE bags)
Suh residential occupant activities airborne microplastic emissions	Microplastics emissions from occupant activities in residential spaces: Experimental quantification and relative importance analysis of particle generation factors	*Build Environ*	Metadata only (no abstract)	Airborne microplastic particles from occupant activities	Not stated in record
Uberlandia street vendors $PM_{2.5}$ micronuclei panel	Urban air pollution and genotoxic damage in street vendors: a longitudinal repeated-measures panel study	*Int J Hyg Environ Health*	Panel study	PM1, $PM_{2.5}$, PM_{10} at seven fixed work-area monitoring points	Brazil (Uberlandia)
Walter neighbourhood air exchange inhaled dose children	Neighborhood-scale disparities in air exchange rates and inhaled pollution dose among school-aged children	*J Expo Sci Environ Epidemiol*	Metadata only (no abstract)	Residential air exchange rate and infiltrated PM / inhaled dose	United States

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Wang electret HVAC filter charge neutralisation wildfire smoke	Charge-Driven Degradation of Residential Electret HVAC Filters under Wildfire Smoke Exposure.	*Environ Sci Technol*	Materials / filtration	Flaming/smouldering wood smoke, prescribed-burn smoke and soot challenges; EPA monitor $PM_{2.5}$ for model inputs	United States (laboratory, prescribed burns, regional wildfire scenario)
Zargarian spent coffee grounds MOF filtration membrane $PM_{2.5}$ bioaerosol photothermal	Spent Coffee Grounds-Derived Micro-Nano-Architectured Air Filtration Membranes for Multiwavelength Light-Triggered Bioaerosol Inactivation	*Small*	Bench / device experiment	$PM_{2.5}$ filtration efficiency (laboratory aerosol)	Laboratory (no field site)
Zhang WWTP bioaerosol	Comprehensive Characterization of Bioaerosol Microbial Communities, ARG, and Occupational Exposure Risks in WWTP and Comprehensive Landfill	*Environ Res*	Field measurement	Culturable bioaerosol (CFU/m3), fine biogenic particle fraction, airborne ARG/MGE copies	China (central)
Zhao et al.	Behavior-resolved bioaerosol exposure modeling reveals scale-dependent occupational risk in pediatric wards	*Build Environ*	Metadata only (no abstract)	Metadata only - behaviour-resolved bioaerosol exposure model in hospital wards (by title)	China (by journal and author affiliation; not asserted)
Sensing, forecasting & instrumentation					
Aakko-Saksa et al.	Aggregated non-methane organic and inorganic gases as indicators of aged particulate matter induced by traffic	*Environ Sci: Atmos*	Metadata only (no abstract)	Metadata only - aggregate NMVOC as proxy for aged traffic PM (by deposit teaser)	Not recoverable from metadata
Ai optical trapping Raman single carbonaceous aerosol smoke	In-Situ Single-Particle Sensing of Carbonaceous Aerosols in Freshly Generated Smoke Using Optical Trapping-Raman Spectroscopy	*Sensors (Basel)*	Laboratory instrument development	Single airborne smoke particles (several um, 10 pg), Raman spectra	Not stated (laboratory)
Bachour WRF-Chem AOD post-correction hot desert	Statistical post-correction of WRF-Chem aerosol optical depth under hot-desert conditions	*Atmos Pollut Res*	Metadata only (no abstract)	WRF-Chem AOD vs ground photometry	Not stated
Bei UPLC-MS/MS 74 antibiotics in $PM_{2.5}$ Beijing	Optimization and application of a sensitive UPLC-MS/MS method for the quantitative determination of antibiotics in $PM_{2.5}$	*J Chromatogr A*	Analytical method validation	Filter-collected 24 h $PM_{2.5}$, 74 antibiotics by SPE + UPLC-MS/MS	China (Beijing)
Birinci Bosphorus bridge traffic ML proxy $PM_{2.5}$ NO2	Machine-Learning Proxy of Air-Quality Variability Associated with Intercontinental Bridge Traffic over the Bosphorus in Istanbul	*Atmosphere*	Machine-learning model	Hourly $PM_{2.5}$, PM_{10}, coarse fraction, NO2, NO, NOx	Turkiye (Istanbul)
CONIOPOL CL61 aerosol typing	New operational perspective to identify aerosols and hydrometeors in real-time with a pioneering algorithm	*Atmos Meas Tech*	Instrument development	Volume depolarisation ratio and attenuated backscatter (Vaisala CL61 ceilometer)	Belgium (Uccle)
Cai self-supervised stacking particulate nitrate sparse observations	A Self-supervised Stacking Framework for Estimating Particulate Nitrate in Two Chinese Urban Agglomerations under Sparse Ground Observations	*Atmos Environ*	Metadata only (no abstract)	Particulate nitrate (estimated from sparse ground observations)	China (two urban agglomerations)
Calabar ZigBee WSN	Design and Performance Evaluation of a ZigBee-Based Wireless Sensor Network for CO and $PM_{2.5}$ Air Quality Monitoring in Calabar Metropolis	*Research Square (preprint)*	Instrument development	Low-cost CO and $PM_{2.5}$ nodes on an IEEE 802.15.4 ZigBee mesh, with MATLAB models of sensing, link, energy and dispersion	Nigeria (Calabar)
Chang et al.	Wintertime vertical stratification of air pollutants and meteorological associations within an urban forest canopy in Shenyang, China	*Environ Pollut*	Long-term field measurement campaign	$PM_{2.5}$, PM_{10} with O3, NO2, SO2, CO and meteorology at 1.5, 10 and 20 m	China (Shenyang)
Chemyakin et al.	Single-scattering look-up table for remote sensing and in situ black carbon studies	*Opt Lett*	Algorithm development	Black carbon fractal aggregates, single-scattering properties 0.35-2.3 um	Not stated (computational)

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Chen et al.	Linking Aerosol Chemical Composition to Volatility: A Predictive Parameterization and Implications for Future Clean Air	*J Aerosol Sci*	Metadata only (no abstract)	Metadata only - composition-to-volatility parameterisation (by title)	Not recoverable from metadata
Chiappini WRF-Chem nudging	Regime-Dependent Effectiveness of Spectral Nudging in WRF-Chem: Meteorological and Saharan Dust Skill over the Central Mediterranean	*Atmosphere*	Chemical transport model	Modelled Saharan dust AOD550 against AERONET; meteorological RMSE from nudged vs free runs	Europe (Central Mediterranean)
Chyzhykov gravimetric filter water Karl Fischer bias	Impact of Mass Loading and Water Speciation on Precision and Artifacts in Gravimetric Measurements of Atmospheric Particulate Matter	*Research Square (preprint)*	Laboratory metrology	Gravimetric PM1, $PM_{2.5}$, PM_{10} on 47 mm glass-fibre filters; Karl Fischer coulometric filter water	Poland (Krakow steelworks; Zdzieszowice coal)
Cover solvatochromic fluorophores in situ aerosol fluorescence	Evaluation of solvatochromic fluorophores for in situ fluorescence characterization of aerosols	*Aerosol Sci Technol*	Metadata only (no abstract)	Aerosol particles (in situ fluorescence probes)	Not stated in record
Dai et al.	Cross-phase partitioning of sulfur-nitrogen ratios and aerosol mixing-state evolution based on single-particle observations	*Atmos Chem Phys*	Long-term field measurement campaign	Single-particle aerosol mass spectrometry with gas-, particle- and number-based S/N ratios	China (Yangzhou, eastern China)
Dayare Navi Mumbai	Weather, Co-Pollutants and Daily $PM_{2.5}$ in a Coastal Indian City: Multi-Station Evidence for Meteorology-Triggered Air-Quality Warnings in Navi Mumbai	*Preprints.org (preprint)*	Monitoring-record analysis	Daily $PM_{2.5}$ at five CAAQMS stations (2021-2025), hourly at two; ERA5 checked against radiosonde	India (Navi Mumbai)
Drone boundary-layer profiles Germany	Influence of irradiance and mixing layer height on the vertical trace matter distribution in the lower planetary boundary layer - drone-based investigation	*Atmos Chem Phys*	Field measurement	Drone vertical profiles of particle mass and number, CO2, O3 to 500 m	Germany (rural)
Dust lidar asymptotics	Asymptotic behavior of lidar scattering properties of absorbing dust aerosols across Rayleigh and geometrical-optics regimes: theory and implications	*Atmos Chem Phys*	Theoretical / optical modelling	Lidar ratio, depolarisation ratio, SSA vs size parameter and iron-oxide fraction	Not applicable (theory)
Epple DLR tire-wear particle chassis dynamometer	Temperature and Inflation Pressure Effects on Airborne Tire Wear Particle Measurements on a Chassis Dynamometer	*Atmosphere*	Laboratory measurement	Airborne TRWP number (MCPC 7 nm-2 um; OPS 0.3-10 um) and OPS mass proxy	Germany (DLR)
Fang LCS calibration review	Calibration of Low-Cost Particulate Matter Sensors for Exposure Assessment: A Critical Review	*Curr Environ Health Rep*	Systematic review (PRISMA)	Calibrated R2 and RMSE of low-cost PM sensors against reference instruments, laboratory vs field	Global (66 studies, 2015-2025)
Fluorescence-Raman-Mie lidar $PM_{2.5}$ profiles Beijing	Classification-constrained retrieval of $PM_{2.5}$ vertical profiles from fluorescence-Raman-Mie polarization lidar	*Atmos Meas Tech*	Remote sensing / lidar retrieval	Lidar extinction to $PM_{2.5}$, aerosol typed by depolarisation ratio and fluorescence capacity	China (Beijing)
Ghaffarpasand Pope vehicle remote sensing real-world emissions high emitters	Emerging Evidence on Real-World Exhaust Emissions and High Emitting Vehicles from Remote Sensing Observations	*Atmos Pollut Res*	Metadata only (no abstract)	On-road remote sensing of exhaust (pollutants not stated in record)	Not stated (review)
Ghajari fuzzy DL Tehran	A Novel Fuzzy Hybrid Deep Learning Architecture for Spatiotemporal $PM_{2.5}$ Prediction	*Research Square (preprint)*	Machine-learning model	Station $PM_{2.5}$ with meteorological, satellite, topographic, population and road inputs, 2014-2016	Iran (Tehran)
HMAC message authentication vs LoRa telemetry poisoning	Securing AI-Driven Systems for Environmental Sustainability in IoT Applications: A Lightweight Message-Authentication Scheme Against Telemetry Poisoning	*Sensors (Basel)*	Method / algorithm	Network $PM_{2.5}$ telemetry (public datasets) used for forecast-poisoning test	Not applicable

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Hazelwood et al.	Size-Resolved Particle Dry Deposition Velocities in a Controlled Spherical Chamber: Effects of Turbulence, Surface Complexity, and Particle Physicochemical Properties	*J Aerosol Sci*	Metadata only (no abstract)	Metadata only - size-resolved dry deposition velocity in a controlled chamber (by title)	Not recoverable from metadata
He dust-storm PM_{10} deep learning	Transport-aware spatiotemporal deep learning for accurate PM_{10} forecasts during dust storms	*Atmos Environ*	Metadata only (no abstract)	PM_{10} during dust storms	Not stated
Hu et al.	Columnar optical properties of locally emitted light-absorbing carbonaceous aerosols	*J Environ Sci (China)*	Measurement campaign	BC and BrC absorption (AE33, 370/470 nm), AOD/AAOD/SSA at 550 nm	China (Beijing, surface + mountain station)
Huang foliar particulate UV-Vis-NIR calibration transfer	Nondestructive UV-Vis-NIR assessment of foliar particulate retention in urban vegetation: cross-species transfer and calibration correction	*Environ Monit Assess*	Spectroscopic calibration-transfer study	Leaf-surface particulate mass per area (washing-filtration reference)	China (Nanjing)
Intra four co-deployed electrostatic PM monitors power plant agreement	Multi-point monitoring of PM_{10}, $PM_{2.5}$, and PM1.0 at a combined-cycle power plant: field agreement between four co-deployed electrostatic monitors	*Research Square (preprint)*	Instrument evaluation	PM_{10}, $PM_{2.5}$, PM1.0 (corona charging, impaction, Faraday-cup electrometry)	Thailand (Samut Prakan)
Kalisa East Africa passive sampler network PAHs metals	First Results from a Passive Air Monitoring Network across Five East African Countries for Polycyclic Aromatic Hydrocarbons (PAHs) and Metals	*Environ Pollut*	Passive-sampler network campaign	PUF passive samplers: 23 PAHs, 24 trace metals (total and water-soluble)	East Africa (five countries, 10 sites incl. Bukavu)
Kang et al.	Moisture-coupled aerosol filtration in moss-based porous media: enhanced particle capture and aerodynamic resistance trade-off	*Build Environ*	Metadata only (no abstract)	Metadata only - filtration efficiency vs pressure drop in moss media under humidity (by title)	South Korea (by author affiliation; not asserted)
Kara Pythagorean fuzzy $PM_{2.5}$ monitoring station placement Trabzon	A Pythagorean Fuzzy decision making framework for $PM_{2.5}$ monitoring station placement in urban areas: A case study of Trabzon, Turkiye	*J Air Waste Manag Assoc*	Delphi consensus	$PM_{2.5}$ monitoring network siting (criteria-weighted)	Turkiye (Trabzon)
Kheder Foehn generative km-scale global $PM_{2.5}$ forecast	Global probabilistic forecasting of air pollution	*arXiv (preprint)*	Deep-learning forecasting	Global kilometre-scale $PM_{2.5}$ estimates (training and target)	Global
Li runway visual range model brown carbon aerosol chromatic effects	Refined runway visual range measurement model incorporating brown carbon aerosol effects	*Opt Lett*	Theoretical / optical modelling	Brown carbon aerosol spectral extinction	Not stated in record
Loh et al.	Seasonal characteristics of submicron aerosols in the coastal atmosphere of southeastern Korea: Influence of urban-port outflow	*Mar Pollut Bull*	Long-term field measurement campaign	PM1 mass, composition and size distribution by HR-ToF-AMS, four seasons, with PMF	South Korea (Geoje Island, downwind of Busan)
MISR Research Product Algorithm 1.1 km AOD	The MISR Research Product Algorithm - producing global, self-consistent, pixel-level aerosol retrievals for the Multi-angle Imaging SpectroRadiometer	*Atmos Meas Tech*	Satellite retrieval evaluation	MISR AOD, Angstrom exponent, particle shape and SSA at 1.1 km	Global
Malik MPSS calibration	Traceable calibration of MPSS in fast-scan mode: Recommendations for PNSD measurement at urban traffic hot spots	*Aerosol Sci Technol*	Metadata only (no abstract)	Metadata only - traceable calibration of mobility particle size spectrometers in fast-scan mode (by title)	Not stated (calibration method)
Manchanda hyperlocal BC sensing Bayesian inverse modeling West Oakland	Hyperlocal sensing and inverse modeling reveal community impacts of urban air pollutant emissions	*Sci Adv*	Method / algorithm	Black carbon from dense fixed-site network (6.7 sites/km2) plus mobile mapping	USA (West Oakland, California)
Mei et al.	Synthesis of the tethered balloon system and other TRACER campaign measurements elucidates aerosol property profiles	*Atmos Chem Phys*	Long-term field measurement campaign	Vertical profiles of aerosol number concentration and size distribution; CCN inferred by kappa-Koehler	USA (Houston, Texas)

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Mercado-Velazquez low-cost sensors three-tier London calibration transferability	How Accurate Are Low-Cost Air Quality Sensors? A 10-Month Study of Three-Tier Comparison with Three Calibration Methods and Cross-Site Transferability	*Sensors (Basel)*	Calibration model comparison	Low-cost optical $PM_{2.5}$ plus MOS/electrochemical NO2, CO, O3 vs reference	United Kingdom (London, two sites 1.6 km apart)
Metris and Metris	Fungus above us: exploring the atmospheric mycobiome by aircraft	*PeerJ*	Field measurement	Airborne fungal bioaerosol, aircraft sampling 2022-2023 with remote-sensing and meteorological covariates	Europe (aircraft surveys; United Kingdom-based)
Nobin Bangladesh GNN $PM_{2.5}$	Explainable spatiotemporal Graph Neural Networks: Multi-City $PM_{2.5}$ forecasting in Bangladesh with geographic graph topology validation	*PLoS One*	Machine-learning model	Daily $PM_{2.5}$, 12-city network, 1-3 day forecasts	Bangladesh (12 cities)
Ouimette PMS5003 PurpleAir internal heating hygroscopic mass reduction	Air Flow, Heat Transfer, and Hygroscopic Aerosol Mass Reduction in the Plantower PMS5003 Sensor and PurpleAir Monitor	*Preprints.org (preprint)*	Laboratory + field measurement	PMS5003 / PurpleAir $PM_{2.5}$ vs Teledyne T640x; detection-zone T and RH	USA (Santa Rosa, California; Winston-Salem, North Carolina)
Pavlidis cold start	Measurements of Real-World Cold Start Emissions in a Parking Garage	*Environ Sci: Atmos*	Field measurement	Real-world cold-start particle and gas emissions measured in a parking garage	Not stated (urban parking garage)
Pires quantum NN	An Annealing-based Quantum Neural Network with Discrete Wavelet Transforms for Enhanced Air Pollution Nowcasting	*arXiv (preprint)*	Machine-learning model	Urban $PM_{2.5}$ time series (UK, high-density data)	United Kingdom
Pyae Tin Saw portable $PM_{2.5}$ cross-site prediction fixed station	Short-term $PM_{2.5}$ data continuity: Cross-site prediction of portable measurements using a nearby fixed monitoring station	*Environ Pollut*	Metadata only (no abstract)	Portable $PM_{2.5}$ monitor vs nearby fixed-station $PM_{2.5}$	Thailand (inferred from author affiliation; not stated in record)
Qian MAG-MISD probabilistic multi-station $PM_{2.5}$ forecasting	MAG-MISD: A Mode-Specific Adaptive Graph Framework for Multiscale Interaction and Station Dependence in Probabilistic $PM_{2.5}$ Forecasting	*Preprints.org (preprint)*	Deep-learning forecasting	Hourly station $PM_{2.5}$, 127 Yangtze River Delta monitors, 8 years	China (Yangtze River Delta)
Rawat et al.	Integrated Assessment of $PM_{2.5}$ in the Central Himalaya using Satellite, Reanalysis, and a new In-situ CUPI Sensor network	*Atmos Environ*	Metadata only (no abstract)	Metadata only - satellite + reanalysis + a new in-situ low-cost PM sensor network (by title)	India (Central Himalaya)
Ren physics-constrained UQ urban $PM_{2.5}$ prediction driving factors	Integrating Implicit Physics Constraints and Uncertainty Quantification for Urban $PM_{2.5}$ Prediction and Driving Factor Analysis	*Build Environ*	Metadata only (no abstract)	Urban $PM_{2.5}$ (prediction target)	Not stated in record
Risti Jambi PMS5003 PMS7003 vs BAM-1020	Evaluation of Low-Cost $PM_{2.5}$ Sensor Accuracy in Comparison with Government Reference Instrument	*Lingkar: J Environ Eng*	Co-location calibration campaign	Plantower PMS5003 and PMS7003 vs BAM-1020 $PM_{2.5}$	Indonesia (Jambi)
SEARO air-quality early-warning review	Prediction accuracy of available early warning systems for air pollution across WHO South-East Asia region: a systematic review	*Front Public Health*	Systematic review	Forecast $PM_{2.5}$ and other pollutants (RMSE, R2, categorical accuracy, bias)	South-East Asia (India, Bangladesh, Thailand)
Saharan fresh dust optics	Optical properties and shape-dependent retrievals of the imaginary part of the refractive index of freshly emitted Saharan dust	*Atmos Chem Phys*	Field measurement	Size-resolved SSA, AAE, asymmetry, MAE/MSE; retrieved imaginary refractive index	Morocco (Sahara)
Schafer AERONET lunar AOD assessment	An assessment of lunar photometry in AERONET	*Atmos Meas Tech*	Instrument evaluation	Night-time lunar AOD (440-1640 nm), Angstrom exponent, SDA fine/coarse AOD	Global / multi-region (492 AERONET sites)
Schmidt AOD gap filling iterative singular spectrum analysis AERONET Cuiaba	Gap filling in long-term AOD time series using Iterative Singular Spectrum Analysis at a tropical AERONET site	*Environ Monit Assess*	Method / algorithm	AERONET daily aerosol optical depth	Brazil (Cuiaba-Miranda, Mato Grosso)
Shai Nanozen RT vs gravimetric	Comparison of respirable dust concentrations between gravimetric weighing and real-time dust monitoring equipment in non-mining industries	*Ann Work Expo Health*	Field measurement	Respirable dust: Nanozen DustCount 9000 optical vs its built-in filter cassette	South Africa (Gauteng)

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Sharma Singapore aerosol type classification ground remote sensing reanalysis	Aerosol Types Classification in the Tropical Climate of Singapore by Integrating Ground-Based Remote Sensing and Reanalysis Datasets	*Aerosol Air Qual Res*	Metadata only (no abstract)	Columnar aerosol optical properties (AOD, Angstrom exponent presumably)	Singapore
Sim real-time dust monitor bias gravimetric meta-analysis	Systematic bias in real-time dust monitoring: a meta-analysis of measurement discrepancy and its implications for occupational exposure assessment.	*Ann Work Expo Health*	Systematic review + random-effects meta-analysis	Optical real-time dust monitors (DustTrak, SidePak, pDR) vs gravimetric samplers	Global / multi-region (occupational literature 2010-2025)
Spinosa volcanic Saharan dust biomass burning aerosol advections Naples	Multiple advections of volcanic, Saharan dust and biomass burning aerosols over Naples	*Atmos Res*	Metadata only (no abstract)	Columnar/ profiled aerosol type (likely lidar-based)	Italy (Naples)
Srivastav ion particle size distribution NPF Indo-Gangetic Plain	Ion and particle size distribution measurements in the Indo-Gangetic Plain: seasonal insights and challenges in new particle formation event characterization	*Atmos Chem Phys*	Long-term field measurement campaign	Air-ion (0.8-42 nm) and particle (2.5-42 nm) size distributions	India (Sonipat, Haryana)
Stutz Fairbanks inversion layers	Vertical distribution of pollution trapped by wintertime surface inversions in Fairbanks, Alaska, during the ALPACA experiment	*Atmos Chem Phys*	Field measurement	Vertical trace-gas profiles (LP-DOAS) with 1-D chemistry-transport model	United States (Fairbanks, Alaska)
Sylvester PurpleAir Philadelphia	Validation of PurpleAir $PM_{2.5}$ Sensors in Temperate Urban Environments	*ACS EST Air*	Field measurement	PurpleAir PA-II $PM_{2.5}$ (cf_1 vs atm outputs) co-located with reference; RH from PA and weather station	United States (Philadelphia)
Taiwan transboundary PM_{10} forecast	Forecasting transboundary PM_{10} pollution in northern Taiwan using upwind observations and transport timing	*Environ Pollut*	Machine-learning model	PM_{10} at 3 receptors; 9 continental upwind records; trajectory timing composite	Taiwan (northern)
Trivanovic periodic technical inspection particle number devices gasoline exhaust	Assessment of existing PN-PTI devices to measure gasoline exhaust	*J Aerosol Sci*	Metadata only (no abstract)	Solid particle number (PN) at tailpipe, PTI-grade counters	Switzerland (inferred from author group; not stated in record)
UAV rotor-wash bias in airborne pollutant sensing	UAV-Induced Measurement Bias in Air Pollution Sensing: A Physics-Informed Framework for Characterization, Sensor Placement, and Correction	*Sensors (Basel)*	UAV instrument deployment	UAV-mounted $PM_{2.5}$, PM_{10} and gas sensors vs fixed reference	Not stated
Vannini biochar passive biomonitor potentially toxic elements lichens	Wood-derived biochar as a potential sustainable alternative to lichens for monitoring airborne potentially toxic elements	*Atmos Environ*	Metadata only (no abstract)	Airborne potentially toxic elements accumulated on exposed biochar	Not stated in record
Waghmare machine learning explainable AI $PM_{2.5}$ prediction Nagpur	Machine learning models with explainable AI for $PM_{2.5}$ prediction for Urban air quality assessment	*MethodsX*	Gradient-boosting model + SHAP interpretability	Station $PM_{2.5}$ with co-pollutants and meteorology	India (Nagpur, Maharashtra)
Wang $PM_{2.5}$-bound PAH association rules machine learning forecasting Nagoya	Characterizing statistical association patterns and local forecasting of $PM_{2.5}$-bound polycyclic aromatic hydrocarbon concentrations in Nagoya, Japan	*Environ Monit Assess*	Machine-learning model	Hourly $PM_{2.5}$-bound PAHs, BC, NOx, oxidants, meteorology	Japan (Nagoya)
Washeem machine learning condensation bioaerosol sampler collection volume	Machine-Learning-Guided Prediction and Temperature Optimization for Collection Volume Control in a Condensation-Based Bioaerosol Sampler	*J Aerosol Sci*	Metadata only (no abstract)	Bioaerosol (condensation-growth liquid collection)	Not stated in record
Wildfire smoke onset detection fixed cameras	Detecting Wildfire Smoke Onset from Fixed View Camera Networks	*Sensors (Basel)*	Method / algorithm	Visible smoke plumes in fixed-camera video (no mass metric)	United States (California) + transfer sets
Wu ProNet physics-AI coupled CTM $PM_{2.5}$ O3 forecasting 338 cities	Fusing physics and AI enhances accuracy and diagnostic capabilities in air quality forecasting	*Natl Sci Rev*	Deep-learning forecasting	$PM_{2.5}$ and O3 forecasts (338 Chinese cities)	China (338 cities)
Wu source-dependent organic aerosol hygroscopicity eastern China biomass burning	Source-dependent organic aerosol hygroscopicity in eastern China: observational evidence for conditional enhancement under biomass-burning influence	*Atmos Chem Phys*	Field campaign	Humidified light scattering f(RH) with OA composition; ZSR closure for kappa_OA	China (eastern, rural site)

First author	Focus	Journal	Design	Metric	Region
Yan et al.	Baseline Determination for Threshold-Based Interpretation and Groups of Single-Particle Aerosol Fluorescence Measurements	*Atmosphere*	Laboratory sampler validation	Single-particle fluorescence (Swisen-sPoleno Jupiter), bioaerosol 10-30 um	Ireland (laboratory + ambient case studies)
Yang et al.	An NB-IoT-Based Architecture with Spatial-Statistical Analytics for Cross-Domain Air and Water Quality Monitoring in Aquaculture	*Sensors (Basel)*	Real-time sensor campaign	Low-cost PM1/$PM_{2.5}$/PM_{10} channel among multi-pollutant air nodes	Taiwan (Tunghai University facility)
Yang et al.	Development and Diagnosis of a Temporal-Context U-Net Deep Learning Surrogate Model for CMAQ Particulate Nitrate over China	*Environ Sci Technol Lett*	Machine-learning model	Hourly surface particulate nitrate, CMAQ emulation (July/December 2011-2020)	China (national)
Zhang et al.	Full field of view temporal calibration for the directional polarimetric camera onboard the GaoFen-5 (02) satellite based on a data reconstruction method	*Appl Opt*	Metadata only (no abstract)	Metadata only - on-orbit temporal calibration of a directional polarimetric camera (by title)	China (GaoFen-5 (02) mission)
Zhao spin-trapping EPR radicals atmospheric fine particulate matter cavity quality factor	Optimized spin-trapping method for radicals detection induced by atmospheric fine particulate matter: Critical roles of temporal evolution and cavity quality factor	*Aerosol Sci Technol*	Metadata only (no abstract)	$PM_{2.5}$-induced radicals (EPR spin trapping)	Laboratory (not stated)
zum Berge multicopter vibration optical particle counter accuracy	Influence of multicopter-induced vibrations on the measurement accuracy of optical particle counter	*Aerosol Sci Technol*	Metadata only (no abstract)	Optical particle counter size-resolved counts on a multicopter UAV	Not stated in record
Exposure assessment & modelling					
Accra $PM_{2.5}$ elemental composition PMF source shift e-waste burning	Shifting Dominant Air Pollution Sources in Sub-Saharan Africa's Changing Urban Environment: Analysis of Elemental Composition and Source Contributions of Ambient Fine Particulate Matter ($PM_{2.5}$) Pollution in Accra, Ghana	*Environ Health Perspect*	Intensive field campaign, PMF source apportionment	Weekly gravimetric $PM_{2.5}$ filters, ED-XRF elements, PMF	Ghana (Accra, 146 sites)
Aerosol Fe isotopes	Isotopic composition of aerosol iron from anthropogenic sources: implications for source apportionment of aerosol iron	*Atmos Chem Phys*	Source apportionment	delta-56Fe end-members for desert dust and anthropogenic aerosol iron sources	Global (source characterisation)
Akhtar et al.	Spatiotemporal characteristics of ozone and $PM_{2.5}$ pollution in Jilin City: insights from observations in 2020 and 2023	*Environ Monit Assess*	Chemical transport model	Warm-season $PM_{2.5}$ and O3, 3 km WRF-CMAQ + surface monitors, met-normalised	China (Jilin City)
Aktypis nucleation-mode particles marine atmosphere downwind urban	Frequent Appearance of Nucleation-Mode Particles in the Marine Atmosphere Downwind of an Urban Area	*Atmos Environ*	Metadata only (no abstract)	Nucleation-mode particle number size distributions	Not stated in record
Aloraini negative control PSCF CWT trajectory dust attribution Riyadh	Transport Pathways Versus Dust Sources: A Negative-Control Test of Trajectory-Statistical Attribution in Riyadh, Saudi Arabia	*Atmosphere*	Monitoring-record analysis	METAR present-weather dust codes (locally lifted vs suspension), 2016-2025	Saudi Arabia (Riyadh, Arabian Peninsula)
Alturki ensemble ML emulation GEOS-CF $PM_{2.5}$ Arabian Peninsula spatial transfer	Interpretable Ensemble Machine Learning for Statistical Emulation of Summer $PM_{2.5}$ Change in GEOS-CF over the Arabian Peninsula: How Far Does Spatially Blocked Skill Transfer?	*Atmosphere*	Machine-learning model	GEOS-CF summer $PM_{2.5}$ relative change 2020-2025 (0.25 deg); TROPOMI, ERA5-Land, CHIRPS, MODIS predictors	Arabian Peninsula
Alvi PMF trace elements PM_{10} Faisalabad health risk	PMF Resolved Source Identification & Health Risk Assessment of Trace-elements in Airborne Particulate Matter (PM_{10}) from South-Asian Megacity	*J Hazard Mater*	Source apportionment	Filter PM_{10} with ICP-OES trace elements; PMF	Pakistan (Faisalabad, Indo-Gangetic Plain)
Anjum Idaho $PM_{2.5}$ 2005-2024 wildfire vs wood-smoke regimes	Long-Term Spatiotemporal Variability of Fine Particulate Matter ($PM_{2.5}$) Across Idaho, USA, Using Statistical and Geospatial Tools	*Research Square (preprint)*	Monitoring-record analysis	EPA AQS daily $PM_{2.5}$, 32 sites (4,260 site-months)	USA (Idaho)

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Anwar Pretoria FTIR	$PM_{2.5}$ Composition and Organic Matter Sources in Pretoria, South Africa Employing Fourier Transform-Infrared Spectroscopy	*Aerosol Air Qual Res*	Metadata only (no abstract)	Metadata only - FT-IR functional-group speciation of $PM_{2.5}$ organic matter (by title)	South Africa (Pretoria)
Arora rural PAH	Seasonal PAH Exposure and Health Risks at Central European Rural Background Sites Driven by Combustion Sources and Transport Pathways	*Atmos Environ*	Metadata only (no abstract)	Metadata only - particle-bound PAH at Central European rural background sites, with risk estimates (by title)	Europe (Central, rural background)
Balabat et al.	Urban Road Dust as a Microbial Hotspot: Spatial Variation of Bacterial Communities and Metal Associations in the Philippines	*World J Microbiol Biotechnol*	Field measurement	Road dust (resuspendable coarse reservoir): 16S V3-V4 metabarcoding, trace metals, pollution indices	Philippines (Metro Manila, Cebu, Cagayan de Oro)
Baliaka Eaton urban firestorm airborne toxic metals lead Los Angeles	Fire at the boundary: Airborne toxic metals from the 2025 Los Angeles urban firestorm	*Sci Adv*	Field measurement	Real-time airborne trace elements (XRF) in PM during the Eaton fire	USA (Los Angeles, California)
Barnett-Itzhaki hourly compound heat air pollution extremes	Hourly Resolution Reveals Hidden Temporal Windows of Compound Heat-Air Pollution Extremes Masked by Daily Averaging	*Atmos Pollut Res*	Metadata only (no abstract)	Hourly air pollution (pollutant set not stated) co-occurring with heat	Not stated in record
Barthaux & Zuend	Estimating Speciated Input Concentrations for Biomass Burning Organic Aerosol Modeling	*Environ Sci: Atmos*	Metadata only (no abstract)	Metadata only - speciated BB VOC/POA model inputs (by deposit teaser)	Not recoverable from metadata
Bei air moisture particulate matter winter Guanzhong Basin	Contributions of air moisture to particulate matter pollution during wintertime in the Guanzhong Basin, China	*Atmos Pollut Res*	Metadata only (no abstract)	Wintertime particulate matter (fraction not stated in record)	China (Guanzhong Basin)
Bloise water-soluble carbon nitrogen PM_{10} Chennai street vertical	Seasonal, Spatial, and Near-Surface Vertical Variability of Water-Soluble Carbon and Nitrogen in PM_{10} in Chennai, India	*Atmosphere*	Field measurement	PM_{10} mass, WSOC, WSTN, WSIN at 3, 16 and 37 m	India (Chennai street sites)
Budapest deposition	Aerosol Deposition in the Human Respiratory System from Distinct Urban Sources Using Multiple Exposure Metrics	*Environ Sci Technol*	Exposure model (secondary data)	Source-apportioned particle number size distributions, 6-1000 nm, with lung deposition modelling	Hungary (Budapest urban background)
Carnec adjoint nodal IAQ	Adjoint-based corrective approach for nodal indoor air quality predictions	*Indoor Environments*	Metadata only (no abstract)	Metadata only - adjoint correction of nodal (multi-zone) indoor air quality models (by title)	Not stated (building model)
Chakraborty fog intensity droplet size aqueous SOA Kanpur	Fog Intensity and Droplet Size-Resolved Aqueous Secondary Organic Aerosol (aqSOA) Formation in Polluted Winter Fog: New Insights from a Combined Observational and Modeling Study	*Environ Sci Technol*	Field campaign	Size-resolved fogwater, offline HR-AMS with PMF; ambient OA	India (Kanpur)
Chaudhary India meteorology	Meteorological modulation of ground-level air quality in India based on in situ measurements	*Environ Monit Assess*	Monitoring-record analysis	PM_{10}, $PM_{2.5}$, O3, NO2, SO2 with PBL height, RH, temperature, precipitation, wind, surface pressure (2017-2024)	India (national, Indo-Gangetic Plain focus)
Chavan and Lataye	Application of Openair package in R programming and PCA: unveiling air quality dynamics and meteorological influences	*Environ Sci Pollut Res Int*	Monitoring-record analysis	2023 CAAQMS records: PM_{10}, $PM_{2.5}$ with six co-pollutants and four meteorological variables	India (Nagpur)

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Chen Chengdu WSON partitioning	Simultaneous measurements of gas- and aerosol-phase water-soluble organic nitrogen in the winter urban atmosphere of Chengdu, China	*Atmos Chem Phys*	Field measurement	Gas- and particle-phase WSON and amines; aerosol liquid water content	China (Chengdu)
Chen ambient $PM_{2.5}$ representativeness children microenvironments	Evaluating the representativeness of ambient $PM_{2.5}$ for children's exposure and source characteristics across microenvironments during school semesters	*Build Environ*	Metadata only (no abstract)	Ambient vs microenvironmental $PM_{2.5}$ (instrumentation not stated)	Not stated in record
Chen primary brown carbon China 2005-2020 trends radiative effect	Long-term evolution and effects of primary brown carbon aerosol in China	*Atmos Chem Phys*	Chemical transport modelling	Modelled primary brown carbon (internal-mixing optical model), $PM_{2.5}$	China
Cheng BTH multi-source satellite $PM_{2.5}$ stacking	Machine Learning-Based Retrieval of $PM_{2.5}$ Concentrations Using Multi-Source Satellite Data in the Beijing-Tianjin-Hebei Region of China	*Atmosphere*	Machine-learning model	Satellite AOD + LST, NDVI, tropospheric NO2 + meteorology -> surface $PM_{2.5}$	China (Beijing-Tianjin-Hebei)
Cheng LETKF WRF-Chem CALIOP component-resolved aerosol assimilation East Asia	Constraining the vertical profiles of aerosol component and heating rate over east Asia through assimilating CALIOP observations	*Atmos Chem Phys*	Chemical transport modelling	Species-resolved extinction profiles: dust, sea salt, black carbon, water-soluble	East Asia
China 20 cities building green $PM_{2.5}$	City-size-dependent interactive effects of building and green-space on $PM_{2.5}$ pollution: Evidence from 20 representative Chinese cities	*Environ Pollut*	Metadata only (no abstract)	$PM_{2.5}$ vs urban morphology	China (20 cities)
Conformal $PM_{2.5}$ mapping	Preregistered External Validation of Spatially Valid Conformal Prediction for Air-Pollution Exposure Mapping in China	*Research Square* (*preprint*)	Statistical model	Province-level population-weighted $PM_{2.5}$, 1998-2024 (n=837), with distribution-free prediction intervals	China (provincial)
DC 1 km WRF-SMOKE-CMAQ	Neighborhood-Scale O3, NO2, and $PM_{2.5}$ in Washington, D.C. from a 1 x 1 km2 WRF-SMOKE-CMAQ Framework	*ACS EST Air*	Chemical transport model	Hourly O3, NO2, $PM_{2.5}$ at 1 km (2017)	United States (Washington, D.C.)
Dayanandana CMIP6 dust optical depth MENA evaluation	Evaluation of Dust Optical Depth in CMIP6 Simulations over the Middle East-North Africa Region	*Atmos Environ*	Metadata only (no abstract)	Dust optical depth (CMIP6 models vs observations)	Multi-country (Middle East and North Africa)
Delhi-NCR COVID transitions	Time series evidence on air quality transitions under urban shock during COVID-19 shutdown in Delhi-NCR, India	*Sci Prog*	Monitoring-record analysis	Daily $PM_{2.5}$, PM_{10}, NOx, SO2, CO, O3 at 10 CAAQMS stations	India (Delhi-NCR)
Dhital on-road black carbon PM exposure Nepal route travel mode	On-road exposure to black carbon and particulate matter in Nepal: Effects of route attributes and travel modes	*J Air Waste Manag Assoc*	Mobile monitoring campaign	Equivalent BC (aethalometer), PM1, $PM_{2.5}$, PM_{10} (portable optical)	Nepal (Kathmandu Valley and intercity highways)
Fang end of uniformity spatial climatic toxicological redefinitions air quality	The End of Uniformity: Spatial, Climatic, and Toxicological Redefinitions of Air Quality	*Environ Sci Technol*	Metadata only (no abstract)	Not stated (perspective on air-quality metrics)	Not stated in record
Fazelvalipour LA microplastics	Size-Resolved Characteristics and Source Influences of Airborne Microplastics in Central Los Angeles	*Preprints.org* (*preprint*)	Field measurement	PM_{10}-associated airborne microplastics (counts/m3, morphology, size) with PM_{10} composition	United States (central Los Angeles)
Five megacities $PM_{2.5}$ O3 trends meteorology 2018-2023	Urban Air Quality Trends and Meteorological Influences Across Five Global Megacities, 2018-2023	*Atmosphere*	Monitoring-record analysis	Monitor $PM_{2.5}$, PM_{10}, O3, NO2, SO2 with temperature, SLP, wind	Global (Shanghai, Delhi, Paris, Los Angeles, Sao Paulo)
GOES-R $PM_{2.5}$ CONUS	Machine Learning Predictions of $PM_{2.5}$ and Their Applications to Exposure and Health Assessment in the CONUS	*GeoHealth*	Machine-learning model	Hourly $PM_{2.5}$ at 0.01 degrees from gap-filled GOES-R geostationary AOD plus random forest	USA (contiguous)

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Ghazaryan explainable ML annual $PM_{2.5}$ Europe Central Asia	Explainable Machine Learning for $PM_{2.5}$ Prediction and Comparative Country-Level Risk Assessment in Europe and Central Asia	*Preprints.org (preprint)*	Machine-learning model	Annual city $PM_{2.5}$ vs climatic and socio-economic indicators	Europe and Central Asia (10 cities)
Ghosh meter-scale heterogeneity new particle formation 3D model	How Microscale Flow and Vapor Transport Shape New Particle Formation and Growth.	*Environ Sci Technol*	Theory / simulation	Simulated nucleation-mode particle number and size distributions (3D-ICAM, 10 m resolution)	Not stated in record
Gray-green infrastructure $PM_{2.5}$ MSPA	Spatially Explicit Morphological Regulation of Gray-Green Infrastructure for $PM_{2.5}$ Mitigation in High-Density Cities: An Interpretable MSPA-Machine Learning Framework	*Build Environ*	Metadata only (no abstract)	$PM_{2.5}$ vs green-space morphology (MSPA)	Not stated
Hanoi $PM_{2.5}$-bound PCDD/Fs	Levels and contributing sources to $PM_{2.5}$-bound PCDD/Fs and dl-PCBs in ambient air of Hanoi	*Atmos Pollut Res*	Metadata only (no abstract)	$PM_{2.5}$-bound PCDD/Fs and dioxin-like PCBs	Vietnam (Hanoi)
Harbin ammonium $PM_{2.5}$ and precipitation sources	From atmosphere to river: Tracing the origins and deposition of ammonium in both $PM_{2.5}$ and precipitation in Harbin	*Environ Pollut*	Metadata only (no abstract)	$PM_{2.5}$ ammonium (isotopic source tracing) and wet deposition	China (Harbin)
Hayli Gubbi eruption	Short-term Atmospheric Impacts of the Hayli Gubbi Volcanic Eruption 2025	*Atmos Environ*	Metadata only (no abstract)	Metadata only - short-term atmospheric impacts of a 2025 eruption (by title)	Ethiopia (Hayli Gubbi, Afar)
He NYC-METS size-resolved organic aerosol offline AMS	Characteristics of size-resolved organic aerosols using offline aerosol mass spectrometry: NYC-METS	*Aerosol Sci Technol*	Metadata only (no abstract)	Size-resolved organic aerosol (offline AMS)	USA (New York City)
Hopke et al.	Characterization of PM Pollution Episodes in Kuwait	*Atmos Pollut Res*	Metadata only (no abstract)	Metadata only - PM episode characterisation (by title)	Kuwait
Imran Chinese cities urban air quality determinants interpretable ML 2014-2023	Spatio-temporal variation and determinants of urban air quality in Chinese cities during 2014-2023: an interpretable machine learning analysis	*J Environ Manage*	Metadata only (no abstract)	City-level air quality ($PM_{2.5}$ presumably among pollutants)	China (multi-city)
Jabbari-Hichri on-road emissions light-duty vehicles particulate gaseous pollutants	On-road emissions of light-duty vehicles: detailed analysis of particulate and gaseous pollutants	*Atmos Pollut Res*	Metadata only (no abstract)	Light-duty vehicle on-road particulate and gaseous emissions	France (inferred from author affiliation; not stated in record)
Julaha vertical aerosol chemical composition 230 m 4 m ACSM rural background	Meteorological Effects on the Vertical Variability of Aerosol Chemical Composition at a Rural Background Site	*Environ Pollut*	Field measurement	Submicron Org, SO4, NO3, NH4 (ToF-AMS at 230 m, ToF-ACSM at 4 m), eBC, OC/EC	Czech Republic (rural background tower site)
Kalkavouras ammonia sulfuric acid BVOC urban new particle formation Athens	Assessing the combined roles of ammonia, sulfuric acid, and biogenic volatile organic compounds in urban new particle formation	*Environ Pollut*	Long-term field measurement campaign	Particle number size distributions with NH3, BVOCs and proxy H2SO4	Greece (Athens, Thissio)
Kaskaoutis in situ aerosol optical properties eastern Mediterranean	In situ aerosol optical properties in the background eastern Mediterranean: seasonal variability, regional transport, and source attribution	*Atmos Environ*	Metadata only (no abstract)	In situ aerosol scattering and absorption	Not stated in record (eastern Mediterranean)
Kim SpaMixed spatial mixed models microbiome $PM_{2.5}$	Spatial mixed models for assessing environmental exposure effects on the microbiome	*Genome Med*	Statistical methods (simulation + application)	$PM_{2.5}$ (two applied studies)	Not stated (methods paper)
Kirsanov COSMO-ART Saint Petersburg UHI pollution PM_{10} feedback	Assessing Urban Pollution and Heat Island Feedback in a Northern Megacity Using the Fully Coupled COSMO-ART Model	*Atmosphere*	Chemical-transport modelling	Near-surface PM_{10}, NO, CO, O3; aerosol radiative feedback	Russia (Saint Petersburg)
Kwon seasonal $PM_{2.5}$ secondary inorganic aerosol livestock rural Ulsan	Seasonal Composition of $PM_{2.5}$ and Secondary Inorganic Aerosols at a Rural Site near Livestock Facilities in Ulsan, South Korea	*Atmosphere*	Filter sampling + composition	Weekly 24 h $PM_{2.5}$ filters, eight water-soluble ions	South Korea (Ulsan, rural)
Langford cooking dominant primary PM1 eddy covariance London Beijing Delhi	Traffic and combustion emission controls leave cooking as a dominant local source of primary PM1 in megacities	*npj Clim Atmos Sci*	Eddy covariance flux measurement, method development	Chemically resolved non-refractory PM1 fluxes (aerosol mass spectrometry + eddy covariance)	Global / multi-region (London, Beijing, Delhi)

First author	Focus	Journal	Design	Metric	Region
Li Guangzhou 3D vegetation	Scale-Dependent Effects of Three-Dimensional Urban Vegetation on Land Surface Temperature and Air Pollutants: Synergy and Trade-Off Zones in Guangzhou, China	*Atmosphere*	Statistical model	1 km gridded $PM_{2.5}$, O3, NO2; layered vegetation volume	China (Guangzhou)
Li atmospheric cadmium concentration deposition China modelling	Modeling the spatial distributions and seasonal variations of atmospheric cadmium concentration and deposition in China	*Atmos Environ*	Metadata only (no abstract)	Particle-bound atmospheric cadmium (concentration and deposition)	China (national)
Lian firework barium aerosols single-particle SPAMS Guangzhou	Evolution of firework-related barium aerosols: insights from single-particle analysis and mass concentration monitoring	*Atmos Chem Phys*	Field measurement	Online Ba mass concentration plus single-particle aerosol mass spectrometry (SPAMS)	China (Guangzhou, four sites)
Liang deep learning low-visibility events fog-haze eastern China	Classification and quantification of low-visibility events using deep learning over eastern China	*Atmos Chem Phys*	Deep-learning model	Surface $PM_{2.5}$ with geostationary satellite and meteorology	China (eastern)
Liu Beijing autumn-winter $PM_{2.5}$ oxidative potential source predictors	Autumn-winter variability and source-related predictors of $PM_{2.5}$ oxidative potential in urban Beijing	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ oxidative potential (assay not stated) and source tracers	China (Beijing)
Liu $PM_{2.5}$ sources mixing state Chengdu Kangding altitude	Altitudinal and seasonal differences in PM(2.5) sources, secondary formation, and particle mixing states in the basin-plateau transition zone of Southwest China.	*Environ Monit Assess*	Source apportionment	Filter $PM_{2.5}$ chemistry (ions, carbon) and single-particle mass spectrometry	China (Chengdu and Kangding, Sichuan)
Liu et al.	Persistent cold air pools and associated sustained haze pollution in Taiyuan Basin	*J Environ Sci (China)*	Monitoring-record analysis	Surface $PM_{2.5}$ vs basin meteorology (soundings, valley heat deficit, WRF)	China (Taiyuan Basin)
Liu et al.	Metropolitan atmospheric microplastics: Characteristics and its inputs to fluvial system	*J Environ Sci (China)*	Field measurement	Atmospheric microplastic deposition (n/m2/day), fibres and fragments	China (Chengdu)
Lu heterogeneous ammonia uptake SOA WRF-Chem North China Plain	Impacts of the heterogeneous ammonia uptake on air quality in the North China Plain	*Atmos Chem Phys*	Chemical transport modelling	WRF-Chem $PM_{2.5}$, inorganic aerosol and nitrogen-containing organics (NOCs)	China (North China Plain)
Lu street-view greenery street canyon PM concentrations	Association between urban greenery and particulate matter concentrations in street canyons: a local-scale study based on street view imagery	*Environ Sci: Atmos*	Field measurement	Street-canyon PM concentrations (metric not stated in abstract)	Not stated in abstract
Lucknow Holika Dahan $PM_{2.5}$	Impact of Holika Dahan on ambient $PM_{2.5}$ and associated health risks in an urban environment of Lucknow, India	*Environ Monit Assess*	Field measurement	Night-time $PM_{2.5}$ mass and chemistry (tracers, ATR-FTIR), HYSPLIT, AERMOD	India (Lucknow)
Luo et al.	Primary nitrated phenol emissions from heavy-duty diesel vehicles based on real-world activity data in Hunan, China	*Environ Pollut*	Modelling / inventory	Nitrated phenols (light-absorbing BrC), filter + PUF exhaust sampling	China (Hunan)
Magalhaes extreme rainfall landslides PM_{10} $PM_{2.5}$ Sao Sebastiao	Can extreme rainfall associated with landslides affect air quality? A case study in a coastal city	*Environ Monit Assess*	Monitoring-record analysis	Hourly PM_{10} and $PM_{2.5}$ (regulatory station), Dec 2021-Feb 2025	Brazil (Sao Sebastiao, Sao Paulo State)
Mai population-weighted $PM_{2.5}$ O3 China 2015-2024 preprint	Population Weighted Ambient $PM_{2.5}$ and O3 Concentrations in China from 2015 to 2024	*Research Square (preprint)*	Exposure model (secondary data)	Annual 1 km CHAP $PM_{2.5}$ and O3 weighted by WorldPop	China (national)
Makacha PRECISE personal ambient $PM_{2.5}$ monitoring framework pregnancy SSA	An integrated climate-structured personal and ambient air pollution monitoring framework embedded within a pregnancy and postpartum cohort in Sub-Saharan Africa	*Front Glob Womens Health*	Instrument evaluation	Minute-resolved personal and fixed-site $PM_{2.5}$ (wearable LCS)	Sub-Saharan Africa (Gambia, Kenya, Mozambique)
Mandujano change-point detection trend estimation long-term PM records	Assessing the Effect of Change-Point Detection Techniques on Trend Estimation in Long-Term Particulate Matter Records	*Environ Sci Atmos*	Metadata only (no abstract)	Long-term PM monitoring records	Not stated in record

First author	Focus	Journal	Design	Metric	Region
Mao et al.	Seasonal and Pollution-Level Variations in $PM_{2.5}$-Bound PFASs and Associated Health Risks in Jinan, a Typical Northern Chinese Industrial City	*Atmosphere*	Field measurement	$PM_{2.5}$-bound PFASs (17 compounds, LC-MS/MS), 163 filters	China (Jinan)
Mielnik et al.	Comparing geospatial interpolation techniques for estimating ambient $PM_{2.5}$ mass concentrations from sugarcane burning in South Florida	*Research Square (preprint)*	Statistical model	Daily $PM_{2.5}$ from a dense low-cost sensor network (46 sites), IDW vs kriging	USA (Florida, Everglades Agricultural Area)
Monsoon aerosol Pb isotopes rainwater offsets	Monsoon-related variability of water-soluble aerosol Pb isotopes and systematic aerosol-rainwater offsets at Peng Chia Yu	*Atmos Environ*	Metadata only (no abstract)	Water-soluble aerosol Pb isotopes vs rainwater	Taiwan (Peng Chia Yu)
Mukhamediyarov et al.	Elemental Composition, Sources, and Health Risks of Ambient $PM_{2.5}$ in Ust-Kamenogorsk (Oskemen) and Pavlodar, Kazakhstan	*Atmosphere*	Source apportionment	$PM_{2.5}$ elemental composition by ICP-OES and ICP-MS with PCA source resolution	Kazakhstan (Ust-Kamenogorsk, Pavlodar)
Mulena dust intrusion regimes north-central Argentina vertical structure optical properties	Dust intrusion regimes over north-central Argentina: Vertical structure, optical properties, and transport pathways	*Atmos Res*	Metadata only (no abstract)	Mineral dust aerosol vertical profiles and optical properties	Argentina (north-central)
Multi-scale $PM_{2.5}$ nonlinear time-series two metros	Deciphering multi-scale $PM_{2.5}$ temporal dynamics and cross-scale interactions in two climatically distinct metropolitan areas: insights from advanced nonlinear time-series analysis	*Atmos Pollut Res*	Metadata only (no abstract)	Monitor $PM_{2.5}$ time series (multi-scale decomposition)	Not stated
Munteanu et al.	Evaluating Graph-Centrality Context for Station-Level Correction of CAMS Air Quality Reanalysis in Lombardy from 2018 to 2021	*Sensors (Basel)*	Machine-learning model	CAMS reanalysis $PM_{2.5}$, PM_{10}, NOx, NO2 corrected to ARPA stations	Italy (Lombardy)
N-containing OA	Unveiling the Precursor-Specific Aqueous Formation Mechanisms of Nitrogen-Containing Organic Aerosols	*Atmos Environ*	Metadata only (no abstract)	Metadata only - precursor-specific aqueous formation of nitrogen-containing organic aerosol (by title)	Not recoverable from metadata
Nassir et al.	A multi-sensor satellite assessment of air quality during the 2026 Middle East Conflict across nine regional countries	*Environ Pollut*	Satellite time-series analysis	AOD (MODIS MAIAC), TROPOMI/OMI NO2, SO2, CO, CH4; CAMS $PM_{2.5}$ (indeterminate)	Middle East (nine countries)
Netto metal distribution $PM_{2.5}$ smelting Rio de Janeiro solubility risk	Metal distribution in atmospheric particulate matter ($PM_{2.5}$) from urban areas with smelting activities in the Rio de Janeiro state, Brazil: Solubility, correlation and risk assessment	*Atmos Pollut Res*	Metadata only (no abstract)	$PM_{2.5}$ metals, soluble fractions	Brazil (Rio de Janeiro state)
Nolles IGP vs NCP $PM_{2.5}$ pollution forming efficiency CAMS	Indo-Gangetic Plain experiences higher aerosol pollution than North China Plain despite lower emissions	*Environ Sci Pollut Res Int*	Reanalysis / inventory analysis	CAMS EAC4 $PM_{2.5}$ vs CAMS-GLOB-ANT emissions	South Asia and China (IGP vs NCP)
North Macedonia biomass PMF	Biomass burning drives urban $PM_{2.5}$ across the Western Balkans: evidence from a six-city source apportionment study in North Macedonia	*Environ Pollut*	Source apportionment	$PM_{2.5}$ from 1,195 daily filters, EPA PMF 5.0	North Macedonia (six cities)
OA molecular composition predicts ROS yield MCR-VK	Linking molecular composition of organic aerosols to reactive oxygen species formation using an MCR-VK framework	*Atmos Chem Phys*	Laboratory + field measurement	Filter $PM_{2.5}$: H2O2 + radical ROS yields; FT-ICR/Orbitrap OA composition	Multi-site (China, Germany, Finland) + chamber SOA
Oliveira et al.	Nanomineralogical and geochemical signatures of coal-derived particles in playground dust	*Atmos Pollut Res*	Metadata only (no abstract)	Metadata only - coal-derived nanoparticles in settled dust (by title)	Not recoverable from metadata
$PM_{2.5}$ and GPP	Climate-dependent nonlinear relationships between $PM_{2.5}$ and ecosystem productivity across China's urban agglomerations	*Environ Pollut*	Machine-learning model	Satellite $PM_{2.5}$ against gross primary productivity across five urban agglomerations	China (five urban agglomerations)

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Pani Southeast Asian biomass burning smoke aerosol light extinction apportionment high-altitude western North Pacific	Impact of peninsular Southeast Asian biomass-burning smoke on chemical apportionment of aerosol light extinction at a high-altitude remote environment in the western North Pacific	*Atmos Res*	Metadata only (no abstract)	Aerosol chemical composition and light extinction (IMPROVE-type apportionment)	Taiwan (inferred from author affiliation; not stated in record)
Park $PM_{2.5}$ composition sources COVID-19 social distancing Seoul	Restructuring of $PM_{2.5}$ Composition and Source Contributions Across COVID-19 Social Distancing Periods: A Case Study at an Urban Monitoring Site in Seoul, South Korea	*Toxics*	Source apportionment	$PM_{2.5}$ mass and speciation, PMF (7 factors), GAM meteorological adjustment	South Korea (Seoul, Gwangjin-gu)
Park Paris UFP number emissions mass-number conserving	Reconciling mass and number emissions through particle size: Implications for urban ultrafine particle modelling	*J Hazard Mater*	Chemical transport modelling	Size-resolved particle number (UFP) from WRF-CHIMERE/MUNICH/SSH-aerosol	France (Greater Paris)
Perez-Vizcaino PM_{10} sources Riotinto mining multi-time-resolution PMF	Characterization of PM(10) sources in the Riotinto copper mining district (SW Spain) using multi-time-resolution receptor modelling.	*J Hazard Mater*	Intensive field campaign, PMF source apportionment	Hourly XRF elements and Aethalometer BC + daily filter PM_{10} chemistry	Spain (Riotinto mining district)
Petrovic Belgrade PMF	Chemical characterization and source apportionment of PM_{10} in Belgrade, Serbia: influence of local and regional anthropogenic and natural sources	*Atmos Chem Phys*	Source apportionment	PM_{10} with 34 species (OC/EC, elements, ions, organic tracers), PMF 5.0	Serbia (Belgrade urban background)
Pourkiaei Santiago CMAQ EDGAR vs INEMA inventories	High-Resolution Air Quality Modeling over Santiago, Chile: A Comparative Assessment of EDGAR and INEMA Anthropogenic Emission Inventories	*Research Square (preprint)*	Chemical transport model evaluation	CMAQ $PM_{2.5}$ (9/3/1 km nests), source apportionment and DDM sensitivities	Chile (Santiago)
Pu Lima hybrid transfer-learning $PM_{2.5}$	A Hybrid Transfer Learning Approach for Ambient $PM_{2.5}$ Exposure Modeling in LMIC Contexts: A Case Study from Lima, Peru	*ACS EST Air*	Machine-learning model	Daily 1 km $PM_{2.5}$ from reference + low-cost monitors, MAIAC AOD, WRF-Chem and covariates	Peru (Lima)
Qu $PM_{2.5}$-bound heavy metals source apportionment Laibin winter	Source Apportionment and Health Risk Assessment of $PM_{2.5}$-Bound Heavy Metals in Wintertime Laibin, Guangxi, China	*Toxics*	Source apportionment	$PM_{2.5}$ mass and nine metals (Zn, Cr, Ni, Pb, V, Cu, Sb, Cd, Co), day/night filters	China (Laibin, Guangxi)
Rocha Volta Redonda	Long-term trends and seasonal variability of particulate matter (PM_{10} and $PM_{2.5}$) in Volta Redonda (RJ), Brazil: a 15-year temporal analysis	*Environ Monit Assess*	Metadata only (no abstract)	Metadata only - 15-year PM_{10} and $PM_{2.5}$ record at a Brazilian steel-industry city (by title)	Brazil (Volta Redonda, RJ)
Rosu multi-instrument Saharan dust event South-Eastern Europe	A multi-instrument analysis of a Saharan dust event in South-Eastern Europe	*Atmos Res*	Field campaign / remote sensing	Saharan dust event (metadata only)	Europe (South-Eastern)
Salem Riyadh $PM_{2.5}$ vegetation	Spatiotemporal variation of $PM_{2.5}$ and its relationship with vegetation cover and meteorological factors during spring (2016-2022) in Riyadh, Saudi Arabia	*Environ Monit Assess*	Monitoring-record analysis	Spring $PM_{2.5}$ from 16 regulatory stations; fractional vegetation cover	Saudi Arabia (Riyadh)
Sao Paulo 2024 fire extremes	Compound fire-meteorology extremes drove megacity air pollution during the 2024 South American fire season	*J Environ Manage*	Chemical transport model	Surface $PM_{2.5}$ and CO, AOT, trajectories; fire-on/fire-off global simulations	Brazil (Sao Paulo State)
Savolainen road traffic emission factors impact factors Helsinki street canyon	From direct emission factors to inverse and indirect impact factors of road traffic: influence of emerging and unregulated pollutants	*Atmos Chem Phys*	Field measurement	Particle number and size distribution, BC, PM_{10}, $PM_{2.5}$, PM chemical components, NOx, CO, VOCs, small ions	Finland (Helsinki street canyon)
Sheidaei et al.	A Multi-Stage Deep Learning Framework for Daily $PM_{2.5}$ Estimation at 100 m Resolution Across the Contiguous United States, 2000-2024	*Environ Sci Technol*	Machine-learning model	Daily $PM_{2.5}$ surface, 100 m, from gap-filled MAIAC-type AOD + meteorology + geospatial predictors	USA (contiguous)

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Shi et al.	Compositional transition and coupled formation pathways of $PM_{2.5}$ pollution in a typical industrial city: An empirical study based on Handan, China	*Environ Pollut*	Metadata only (no abstract)	Metadata only - $PM_{2.5}$ compositional transition and coupled formation pathways (by title)	China (Handan, Hebei)
Shoukat et al.	Suspended Atmospheric Microplastics in the Outskirts of Seoul City: Seasonal Variability, Correlation with Meteorology	*Atmos Pollut Res*	Metadata only (no abstract)	Metadata only - suspended airborne microplastics (by title)	Not recoverable from metadata
Somsunun $PM_{2.5}$-bound metals health risk haze season Chiang Mai	Health Risk Assessment of Ambient $PM_{2.5}$-Bound Metals During Haze Seasons in Urban and Rural Chiang Mai, Thailand	*Toxics*	Filter sampling + composition	24 h $PM_{2.5}$ filters (n=101) and metals, three land-use sites	Thailand (Chiang Mai)
Szalata waste fire toxic emissions Gaussian plume environmental risk	Environmental Risk Assessment of Toxic Emissions into the Air Resulting from Waste Fires	*Toxics*	Physical exposure model	Modelled PM_{10}, $PM_{2.5}$ and gases from a waste-storage fire (Gaussian plume)	Poland
Szczepanik Warsaw wintertime air quality improvement 2016-2025	Improvement in wintertime air quality in urban and suburban areas of Warsaw within 2016-2025	*Aerosol Air Qual Res*	Metadata only (no abstract)	Wintertime ambient air quality (pollutants not stated)	Poland (Warsaw)
Temirzhanova Pavlodar snow aerosol	Geochemical Characteristics of the Pollution of Snow Cover with Aerosols in an Industrial Center	*Atmosphere*	Field measurement	Elemental composition of snow-cover solid fraction (ICP-MS)	Kazakhstan (Pavlodar)
Torii Kalimantan peat fire super-linear scaling	Peat Burning Scales Super-linearly with Fire-Season Magnitude in the West Kalimantan-Sarawak Border Region, 2012-2025	*Research Square* (*preprint*)	Satellite time-series analysis	VIIRS active-fire detections on peat vs non-peat; FRP; back-trajectories to Sarawak receptors	Indonesia / Malaysia (Kalimantan-Sarawak)
Tripathi satellite-reanalysis sulfate aerosol India ground measurements	Improving Satellite-Reanalysis Sulfate Aerosol Estimates over India Using Ground Measurements	*Environ Pollut*	Metadata only (no abstract)	Sulfate aerosol (satellite/reanalysis vs ground)	India
Tsai mineral dust biomass burning aerosol convergence East Asia western North Pacific	Convergence of mineral dust and biomass-burning aerosols over East Asia and the western North Pacific	*Atmos Environ*	Metadata only (no abstract)	Mixed mineral dust and biomass-burning aerosol (transport)	East Asia
UK Air Quality Cell PM_{10} plume back trajectory tyre fire Mexborough vulnerability	Back trajectory modelled PM_{10} plumes: population vulnerability profile from an uncontrolled fire	*Eur J Public Health* (*conference abstract*)	Event-based monitoring	Incident PM_{10} monitoring + ADMS-STAR back-trajectory plume (22 hourly plumes)	United Kingdom (Mexborough, England)
US school wildfire smoke trends	Spatio-Temporal Trends in Wildfire Smoke Exposures at Schools and Childcare Centers in the United States and Implications for Child and Adolescent Health	*Environ Res*	Metadata only (no abstract)	Wildfire smoke $PM_{2.5}$ at school and childcare locations	United States
Upaul Khulna $PM_{2.5}$ meteorology HYSPLIT random forest STL	Meteorological and HYSPLIT-Derived Transport Influences on $PM_{2.5}$ Exposure in a Khulna City, Bangladesh	*Research Square* (*preprint*)	Monitoring-record analysis	Hourly $PM_{2.5}$ from the CAMS-8 station (2024); NASA POWER meteorology; HYSPLIT	Bangladesh (Khulna)
Vikas wavelet background $PM_{2.5}$ India MERRA-2	Estimating background $PM_{2.5}$ concentrations using a wavelet-based convolution framework	*Environ Monit Assess*	Algorithm development	MERRA-2 reanalysis $PM_{2.5}$; CPCB monitors for evaluation	India
Vukic et al.	Integrated assessment of road and maritime traffic impacts on urban air quality and noise in a port-city environment	*Environ Monit Assess*	Field measurement	Short-duration low-cost $PM_{2.5}$/PM_{10} and CO at 10 sites; vessel and vehicle counts	Croatia (Split)
Wang China $PM_{2.5}$ decade	From rapid decline to slower improvement: Decadal nonlinear evolution and regional predictive associations of $PM_{2.5}$ in China	*Atmos Pollut Res*	Metadata only (no abstract)	Ambient $PM_{2.5}$ trends, 2015-2024	China (nationwide)
Wang et al.	Organosulfate-driven amplification of secondary organic aerosol formation from isoprene oxidation intermediates: Coupled effects of SO2 and relative humidity	*J Environ Sci* (*China*)	Chamber / laboratory	SOA from methacrolein/OH photooxidation; organosulfates (GC-MS, ATR-FTIR)	Not stated (laboratory)

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Wang et al.	Physics-inspired inductive spatiotemporal kriging for $PM_{2.5}$ with satellite gradient constraints	*Atmos Environ*	Metadata only (no abstract)	Metadata only - inductive spatiotemporal kriging for $PM_{2.5}$ under satellite gradient constraints (by title)	Not recoverable from metadata
Wang urban morphology street-scale $PM_{2.5}$ PM_{10} change points	Non-linear associations, model-supported change points, and seasonal variations of urban morphology with street-scale $PM_{2.5}$ and PM_{10} concentrations	*Atmos Pollut Res*	Metadata only (no abstract)	Street-scale $PM_{2.5}$ and PM_{10}	Not stated (metadata only)
Wei $PM_{2.5}$ ozone nonlinear coupling Puyang oilfield city GAM 2015-2025	Long-Term Trends and Nonlinear $PM_{2.5}$-O3 Coupling in a Typical Oilfield City of the "2 + 26" Region	*Atmosphere*	Statistical model	Daily $PM_{2.5}$ and MDA8 O3 (fixed monitors), 2015-2025	China (Puyang, Henan)
Wen and Chang	Estimating marine $PM_{2.5}$ concentrations and exploring the drivers over the eastern China seas using multi-source data and machine learning	*Environ Pollut*	Machine-learning model	Marine $PM_{2.5}$ from AOD + meteorology + NO2/CO/O3/SO2, 2019-2024	China (eastern China seas)
Wu NCP brown carbon	Light absorption properties and composition of water-soluble brown carbon in North China Plain: implication for an enhancing role of nitrogenous organic compounds	*Atmos Chem Phys*	Field measurement	Water-soluble BrC absorption (Abs365, MAE) and composition at five sites, winter 2023	China (North China Plain, five sites)
Xiao NCP Sichuan haze reduced-nitrogen NOC oligomerisation $PM_{2.5}$	Aqueous Carbonyl-Amino Aggregation Drives Atmospheric Nitrogen-Containing Oligomer Formation in China Haze	*Environ Sci Technol*	Field measurement	$PM_{2.5}$ nitrogen-containing organic compounds (ESI+ HRMS)	China (North China Plain, Sichuan Basin)
Yadav ENSO India	Air Quality in Flux: The Baffling ENSO Transition in India	*Atmos Pollut Res*	Metadata only (no abstract)	Metadata only - ENSO-phase modulation of Indian air quality (by title)	India (national)
Yingkar $PM_{2.5}$-bound PAHs Central Asia urban trends sources risk	Environment-to-Health Linkages of $PM_{2.5}$-Bound Polycyclic Aromatic Hydrocarbons in Urban Cities of Central Asia: Trends, Sources, and Risk Assessment	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$-bound PAHs	Kazakhstan (Central Asian cities; inferred from title and author group)
Yousaf nanoscale curvature hygroscopic growth respiratory deposition ultrafine	Nanoscale Curvature Controls Hygroscopic Growth and Respiratory Deposition of Atmospheric Nanoparticles	*Environ Pollut*	Laboratory measurement	15-100 nm NaCl particles, HTDMA growth factors fed to MPPD dosimetry	Laboratory (no field site)
Yu Paris urban plume aerosol composition optical properties airborne ground	Urban-to-downwind variations in aerosol composition and optical properties in the Paris urban plume from coordinated airborne and ground-based observations	*Atmos Chem Phys*	Aircraft measurement campaign	PM1 composition (AMS), particle number 80-200 nm, SSA, refractive index, AAE	France (Paris and downwind region)
Zhang Fusion-DeepRSM satellite $PM_{2.5}$	Fusion-DeepRSM for Satellite-Derived $PM_{2.5}$ Estimation and Precursor Sensitivity Prediction	*Environ Sci Technol*	Machine-learning model	Gridded $PM_{2.5}$ from CTM + multisource satellite + ground fusion	China (nationwide)
Zhao $PM_{2.5}$-O3 co-pollution episodes BTH YRD	Episode analysis of the characteristics and causes of $PM_{2.5}$–O3 co-pollution in Beijing-Tianjin-Hebei and Yangtze River Delta, China	*Atmos Pollut Res*	Metadata only (no abstract)	$PM_{2.5}$ and O3 co-pollution episodes (monitoring data)	China (Beijing-Tianjin-Hebei and Yangtze River Delta)
Zhao et al.	Spatial Heterogeneity and Multi-scale Statistical Associations of Urban Air Pollution Revealed by Interpretable Machine Learning in Shenzhen, China	*Atmos Pollut Res*	Metadata only (no abstract)	Metadata only - multi-scale spatial heterogeneity via interpretable ML (by title)	China (Shenzhen)
Zheng et al.	Comparative analysis of multiple health risks from different air pollution events	*J Environ Sci (China)*	Measurement campaign	Size-resolved number 3 nm-10 um, $PM_{2.5}$ composition, OP; lung deposition model	China (Beijing)

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Zhou condensation sink efficiency hygroscopicity	Condensation Sink Efficiency of Atmospheric Particles Regulated by Their Chemical Compositions and Hygroscopicity	*Environ Sci Technol*	Laboratory + field measurement	Pre-existing particle condensation sink toward H2SO4/HIO3; composition and hygroscopicity	Laboratory
de la Cruz automated forced air injection cordwood stove $PM_{2.5}$ particle size	Less Mass of $PM_{2.5}$ Emissions, Smaller Particles: The Trade-Off of Automated Forced Air Injection in a Residential Cordwood Stove	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ mass emission and particle size distribution	Not stated
Burden, policy & mitigation					
Abdella Italy urban-rural	Cleaning the air or shifting the burden? Characterizing urban-rural disparities in air pollution-attributable mortality across Italy	*Atmos Environ*	Metadata only (no abstract)	Metadata only - urban-rural disparity in PM-attributable mortality across Italy (by title)	Italy (national)
Abu Dhabi PM burden 2008-2022	Ambient Air Pollution Health Impacts in High-Income Countries, Part II: A Case Study in the United Arab Emirates	*Risk Anal*	Health impact model	Anthropogenic PM and ozone, 2019-2022 vs 2008	United Arab Emirates (Abu Dhabi)
Air-health web atlases review Italy	PNC project 'Atlante Aria e Salute': exploratory web review and quality assessment of open-access atlases on the health effects of air pollution	*Epidemiol Prev*	Web review	$PM_{2.5}$/PM_{10} (all platforms), NO2 and O3 (fewer)	Global / multi-region
Atlante Aria e Salute Delphi co-creation	Co-creation as a decision-making tool in the PNC 'Atlante Aria e Salute' project to define the contents of an integrated atlas	*Epidemiol Prev*	Delphi consensus	Outdoor $PM_{2.5}$, PM_{10}, NO2, O3 (atlas exposure layers)	Italy
BC outpatient risk	Black carbon as a major contributor to ambient $PM_{2.5}$-related cause-specific outpatient risks: implications for early warning and health benefits evaluation of air quality improvement	*Eco Environ Health*	Time-stratified case-crossover	Daily $PM_{2.5}$ and constituents (black carbon, organic matter, sulfate, nitrate, ammonium), 274 cities	China (274 cities)
BEST-COST all-cause vs disease-specific exposure-response functions burden $PM_{2.5}$ NO2	Effect of all-cause versus disease-specific exposure-response functions on air pollution burden	*Eur J Public Health (conference abstract)*	Health impact model	EEA $PM_{2.5}$ and NO2 exposure; WHO-recommended ERFs	Europe (Belgium, Estonia, France, Norway, Portugal)
Beveridge HeatRisk AQI cardiovascular action algorithm	HeatRisk and air quality risk recommendations: A narrative review and action algorithm for the care of people at risk for and living with cardiovascular disease	*Am J Prev Cardiol*	Narrative review	Air Quality Index ($PM_{2.5}$-driven) alongside NWS HeatRisk	United States
Bhattarai India crop burning inventories	Air Quality Impacts of Post-monsoon Crop Residue Burning in Northern India Based on Regional and Global Emission Inventories	*Atmos Environ*	Metadata only (no abstract)	Modelled $PM_{2.5}$ from crop-residue burning	India (northern)
Bong et al.	Delayed Coal Plant Retirements in the Era of Data Centers Carry a Measurable Mortality Cost	*medRxiv (preprint)*	Exposure model (secondary data)	Coal-plant-attributable $PM_{2.5}$, plant-specific source-receptor modelling	USA (Medicare, 20 coal plants)
Brunori Rome urban vineyard grapevine leaf PM interception	Plant-Atmosphere Interactions in Urban Vineyards of Rome: A Digital Phenotyping Approach for the Functional Selection of Grapevine Cultivars	*Plants (Basel)*	Field measurement	Leaf-surface PM particle counts (SEM, particles/mm2)	Italy (Rome)
Canada wildfire smoke guidance	Public health guidance about wildfire smoke: Four targets for improvement	*Can J Public Health*	Commentary	Wildfire smoke (guidance)	Canada
Cantu air quality across the lifespan nursing response review	Breathing Unequal Air: Global and US Air Quality Across the Lifespan and the Nursing Response.	*Nurs Clin North Am*	Narrative review	$PM_{2.5}$, O3, NO2 (reviewed)	Global / multi-region (with United States focus)
China BC burden	The Hidden Health and Climate Burden of Black Carbon under China's Dual Carbon Strategy	*Nat Commun*	Health impact model	Localized black carbon emission inventory with revised health risk function; CMIP6 comparison	China (national)
East Asia $PM_{2.5}$ O3 warming levels	Uneven Future $PM_{2.5}$ and O3 responses to Emission Reductions under Different Warming Levels in East Asia	*Environ Pollut*	Metadata only (no abstract)	Projected $PM_{2.5}$ and O3 under warming levels	East Asia
GBD 2023 G20 stroke air pollution temperature	Estimates, temporal evolution, and contributors of the stroke burden due to air pollution and non-optimal temperature across G20 nations, 1990-2023	*Thromb Haemost*	GBD secondary analysis	GBD ambient PM and household solid-fuel pollution; non-optimal temperature	Global / multi-region (G20)

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Global atmospheric Pb 2000-2019	Global modelling of atmospheric lead (Pb) concentrations and health risks during 2000-2019	*Environ Pollut*	Metadata only (no abstract)	Modelled airborne (particle-bound) Pb	Global / multi-region
Global heatwave-fire smoke coexposure	Rising global population coexposure to heatwaves and landscape fire-sourced air pollution	*Sci Adv*	Exposure model (secondary data)	Landscape fire-sourced $PM_{2.5}$ (modelled) coincident with heatwave days	Global / multi-region
Grzesczyk-Nojszewska Poland air pollution interventions public attitudes survey	Public attitudes toward personal and population interventions to reduce air pollution exposure: a 2025 nationwide cross-sectional survey in Poland	*Int J Occup Med Environ Health*	Cross-sectional population survey	Air pollution (perceived); no measurement	Poland (nationwide)
Gujral Dominici Memphis datacenter $PM_{2.5}$	Behind-the-Meter Versus Grid Electricity for AI Datacenters: Magnitude, Geography, and Equity of $PM_{2.5}$-Attributable Mortality in Memphis	*Preprints.org (preprint)*	Health impact model	Modelled population-weighted $PM_{2.5}$ from on-site turbines vs grid generation	United States (Memphis)
Guo lung cancer burden	Trends in lung cancer burden, risk factors, and disease projections in China and the United States, 1990-2023	*Chin Med J Pulm Crit Care Med*	Health impact model	GBD 2023 risk-factor attribution including ambient and household air pollution	China and United States
He ozone-dominant air pollution risk ageing inequality China counties	A nationwide effect-based assessment of ozone-dominant air pollution risk patterns and ageing-related inequality in China	*Ecotoxicol Environ Saf*	Ecological / registry analysis	Cold-season $PM_{2.5}$ and warm-season O3 (multi-source county exposures)	China (813 counties)
Idowu medical waste incinerator airborne heavy metals Monte Carlo Nigeria	Monte Carlo-Based Probabilistic Health Risk Assessment of Airborne Heavy Metals from a Medical Waste Incinerator in Southwestern Nigeria	*Research Square (preprint)*	Health impact model	PM-bound heavy metals (passive glass-slide deposition samplers)	Nigeria (Ido-Ekiti)
Imman caregivers $PM_{2.5}$ prevention preschool children northern Thailand qualitative	Caregivers' experiences of $PM_{2.5}$ prevention for preschool children in Thailand: a qualitative study with focus group interviews	*Child Health Nurs Res*	Qualitative / ethnographic study	Seasonal haze $PM_{2.5}$ (perceived); no measurement	Thailand (rural north)
Indo-Gangetic BC	Long-term decline in Black Carbon Aerosol and Associated Health Risk over north-western Indo-Gangetic Plain	*Atmos Pollut Res*	Metadata only (no abstract)	Metadata only - long-term black carbon trend and associated health risk (by title)	India (north-western Indo-Gangetic Plain)
Isfahan synbiotic omega-3 RCT	Effects of synbiotic versus omega-3 supplementation on total antioxidant capacity in adolescents exposed to ambient air pollution: A randomized double-blind controlled trial	*J Res Med Sci*	Randomised controlled trial	Ambient air pollution (not measured at individual level)	Iran (Isfahan)
Izmir maritime	Quantifying Maritime Contributions to Urban Air Quality in Izmir Bay Using Tier 3 Emission Inventory and AERMOD	*Atmos Environ*	Metadata only (no abstract)	Metadata only - Tier 3 ship emission inventory with AERMOD dispersion (by title)	Turkiye (Izmir Bay)
Jiang PLA melt-blown nonwoven filtration durability humidity airflow	Service Durability of PLA Melt-Blown Nonwovens: Roles of Humidity and Airflow Velocity in Degradation and Filtration Performance	*Langmuir*	Chamber / laboratory	Filtration of particulate matter by PLA melt-blown media	Laboratory
K-HEALTHinAIR indoor $PM_{2.5}$ COPD attributable burden Spanish households	Attributable Burden of COPD from Indoor $PM_{2.5}$ Exposure in Spanish Households, 2025	*Eur J Public Health (conference abstract)*	Burden estimation	Measured indoor $PM_{2.5}$ in Spanish homes (K-HEALTHinAIR)	Spain
Landscape fire smoke health action plans review	Health Management Planning for Landscape Fire Smoke Events: A Literature Review and Suggested Human-Centred Framework for Resource Development	*Public Health Chall*	Literature review	Landscape fire smoke (no metric)	Australia, Canada, United States
Lee et al.	Exhaust Emission Characteristics of Gasoline Engines with Secondary Air Systems and Electrically Heated Catalysts in Cold-Start WLTC Mode	*ACS Omega*	Engine dynamometer testing	Particle number (PN) with CO, THC, NOx over the cold phase of the WLTC	South Korea (laboratory)
Lee type 2 diabetes burden Asia GBD 2023 particulate matter leading risk	The evolving burden of type 2 diabetes mellitus in Asia, 1990-2023: A Global Burden of Disease Study 2023	*Diabet Med*	Burden estimation (GBD)	GBD particulate matter pollution risk factor (ambient + household)	Asia (34 countries)

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Liang emergency haze controls causal ML synthetic control Shijiazhuang	Evaluating Emergency Emission Controls During an Unprecedented Regional Haze Event Using Causal Machine Learning	*Environ Sci Technol*	Augmented synthetic control + weather normalisation	Monitor $PM_{2.5}$, PM_{10}, SO2, NO2, CO; $PM_{2.5}$ source apportionment	China (Shijiazhuang)
Lu $PM_{2.5}$ longevity country panel specification analysis	Paradoxical $PM_{2.5}$-longevity estimates in global country panels are sensitive to differential trend control: a specification analysis, 2000-2023	*Front Public Health*	Ecological panel, fixed effects	Population-weighted annual ambient $PM_{2.5}$ (World Bank WDI)	Global / multi-region (186 economies)
Lu kitchen air purifier health-economic outcomes 338 Chinese cities	Heterogeneous health-economic outcomes of kitchen air purifier interventions: A 338-city study in urban China	*Environ Int*	Metadata only (no abstract)	Kitchen (cooking) particulate exposure - not stated in record	China (338 cities)
Ma TRWP China	Spatiotemporal dynamics of tire wear particles reveal escalating emission inequalities across China's road network	*Nat Commun*	Modelling / inventory	Tire and road wear particle (TRWP) $PM_{2.5}$ emissions, road-network resolved, ML-harmonised field data	China (national road network)
Mohamadi Isfahan $PM_{2.5}$ AirQ+ WRF-SMOKE-CMAQ health damage	Assessing the health impacts of $PM_{2.5}$ from anthropogenic and natural sources in Isfahan, Iran	*J Environ Health Sci Eng*	Integrated emission-air quality-health modelling	Monitoring-station annual $PM_{2.5}$; WRF-SMOKE-CMAQ source contributions	Iran (Isfahan)
Mukhamediyarov Kazakhstan code	Air Quality and Emission Trends Following the 2021 Environmental Code Reform in Kazakhstan Across Regions and Industrial Sectors	*Atmosphere*	Monitoring-record analysis	Stationary-source emissions ($PM_{2.5}$, SO2, CO, NO2) and ambient indicators by region and sector	Kazakhstan (national)
NYC congestion pricing	Attributing $PM_{2.5}$ and NO2 Changes One Year after New York City's Congestion Pricing Policy	*Environ Sci Technol Lett*	Difference-in-differences evaluation	Real-time $PM_{2.5}$ from the NYC Community Air Survey plus TROPOMI tropospheric NO2 columns	USA (New York City)
Pannu et al.	Assessment of the significance of urban vegetation in air phytoremediation: Quantitative analysis of particulate matter removal	*Research Square (preprint)*	Field measurement	Leaf-deposited PM in 0.2-2.5, 2.5-11 and 11-100 um fractions; elemental composition	India (Delhi)
Polwiang Bangkok Chiang Mai burning control DLNM GAM $PM_{2.5}$ regimes	Differential sensitivity of urban and biomass-dominated $PM_{2.5}$ regimes to burning control: A distributed-lag generalized additive model analysis of Bangkok and Chiang Mai	*Atmos Pollut Res*	Metadata only (no abstract)	Daily $PM_{2.5}$ (urban vs biomass-dominated regimes)	Thailand (Bangkok, Chiang Mai)
Prieto Perez Huszar future central European air quality emission changes	The role that changes in emissions play in future central European air-quality	*Atmos Environ*	Metadata only (no abstract)	Future PM and O3 under emission scenarios (presumably)	Central Europe
Quadros et al.	Global air quality and human health impacts of growing aircraft emissions	*Commun Earth Environ*	Health impact model	Aviation-attributable $PM_{2.5}$ and O3 from a global CTM, 2019 and 2040 scenarios	Global
Ranzi et al.	Atlante Aria e Salute: methods for estimating the health impacts of air pollution and national application	*Epidemiol Prev*	Health impact model	Modelled annual $PM_{2.5}$ and NO2 at 1x1 km, random forest over monitors, satellite and covariates, 2016-2024	Italy (national, municipality level)
Rodriguez Bogota car-free	Annual trends in ambient particulate matter during car free days in Bogota from 2016 to 2025	*Research Square (preprint)*	Monitoring-record analysis	$PM_{2.5}$ and PM_{10} on Car-Free Day from the city monitoring network, 2016-2025 (2021 excluded)	Colombia (Bogota)
Ruengdit active-flow respirator fit children Chiang Mai	Active-flow systems enhance respirator fit success in children: a comparative study with standard disposable models	*Front Public Health*	Laboratory metrology	Quantitative fit factor (TSI PortaCount, ambient particle counting)	Thailand (Chiang Mai)

First author	Focus	Journal	Design	Metric	Region
Ruwali Lary US infant life expectancy environmental determinants machine learning	Estimating Life Expectancy in the U.S.: Identifying Environmental Determinants and Disparities	*Sensors (Basel)*	Machine-learning model	33 ECMWF reanalysis environmental variables incl. PM, sulphate and dust aerosol, O3, CO, HCHO	USA (contiguous, county level)
Scott equine air quality index Delphi management	Experts recommend management strategies for horses during poor air quality	*J Am Vet Med Assoc*	Delphi consensus	AQI categories ($PM_{2.5}$-driven, wildfire smoke)	USA (expert panel)
Shandong integrated air-pollutant GHG inventory	Strategies for Co-Control of Air Pollutants and Greenhouse Gases: Insights from an Integrated Emission Inventory in Shandong Province	*Atmosphere*	Emission inventory	Primary PM, SO2, NOx, VOCs, NH3, CO plus CO2, CH4, N2O by source and city	China (Shandong)
Sharma France net-zero benefits	Health and economic benefits of air quality improvements in France through diverse net-zero transition scenarios by 2050	*Environ Int*	Metadata only (no abstract)	Scenario-modelled $PM_{2.5}$ and co-pollutants to 2050	France
US vehicle emission burden	Spatially Refined Analysis of Emission Burdens and Equity from Light-, Medium-, and Heavy-Duty Vehicle Traffic in the United States	*Environ Sci Technol*	Modelling / inventory	Census-block NOx and $PM_{2.5}$ emission intensity from MOVES factors and street-level traffic, exhaust vs tyre/brake wear	USA (national, every census block)
Urban tree bark PM retention functional traits	Species-specific PM retention by urban tree bark is associated with functional traits under controlled exposure	*Environ Pollut*	Chamber / laboratory	Gravimetric PM retained on bark, leaves, branches ($PM_{2.5}$, $PM_{2.5}$-10)	China (Hangzhou)
Wang carbon footprint air-pollutant emissions Beijing 2022 Winter Olympics	Linking Carbon Footprints to Air-Pollutant Emissions from Mega Sporting Events: Evidence from the Beijing 2022 Winter Olympics	*Atmosphere*	Emission inventory	Co-emitted $PM_{2.5}$, NOx, SO2, CO, NMVOC from pollutant-to-CO2 ratios	China (Beijing)
Wang climate-adaptive city pilot $PM_{2.5}$ health dividend China	Does climate adaptation policy deliver a health dividend? Exposure, modeled health burden, and defensive capacity in China's climate-adaptive city pilots	*Front Public Health*	Difference-in-differences evaluation	City-level annual $PM_{2.5}$ (gridded product), GEMM burden	China (264 cities)
Wang et al.	What drives long-term variations in biomass burning emissions in China?	*J Environ Sci (China)*	Modelling / inventory	Biomass-burning $PM_{2.5}$, PM_{10}, EC, OC and gas emissions, 2008-2023	China (national, provincial)
Warminski SRC pellets boiler ESP PM CO emissions	Short-Rotation Coppice Pellets as Biofuels in a Modern Boiler with an Electrostatic Precipitator: Particulate Matter and Carbon Monoxide Emissions	*Molecules*	Laboratory measurement	Flue-gas total PM mass (mg/m3) with and without ESP	Not stated in abstract
Yang et al.	Process-resolved characterization and plant-level inventory of multi-pollutant emissions from China's coking industry	*Environ Res*	Modelling / inventory	Size-segregated stack PM, ions, trace elements; plant-level inventory 2012-2024, projections to 2050	China (national, 307 plants)
Yu multicountry $PM_{2.5}$ admissions	Cause-specific hospital admissions associated with short-term $PM_{2.5}$ exposure: a multicountry time-series analysis	*Lancet Planet Health*	Multi-location time-series	Daily ambient $PM_{2.5}$ (community-level), lags to 21 days	Multi-country (Australia, Brazil, Canada, Chile, New Zealand, Taiwan)
Zhang Jiaxing inventory co-control screening $PM_{2.5}$ GHG	Inventory-Based Co-Control Screening of Air Pollutants and Greenhouse Gases in a Typical Yangtze River Delta City	*Atmosphere*	Emission inventory	Inventory $PM_{2.5}$ (with NOx, SO2, VOCs, NH3) vs CO2-eq	China (Jiaxing, Yangtze River Delta)
Zhang US wildfire smoke five-year mortality impulse response	Wildfire smoke leaves a large five-year imprint on mortality in the United States	*medRxiv (preprint)*	Distributed-lag panel deconvolution	Monthly county-level wildfire-attributable $PM_{2.5}$ (smoke $PM_{2.5}$ surface)	United States (all counties)
Zhang early-onset female lung cancer burden GBD 2021	Global Burden and Trends of Early-Onset Tracheal, Bronchus, and Lung Cancer Among Women From 1990 to 2021.	*Asia Pac J Clin Oncol*	Burden estimation (GBD)	Ambient particulate matter pollution (GBD risk factor)	Global / multi-region (GBD 2021)

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Zhang et al.	Impact of strong biomass burning in Northeast China: air pollution, enhancement ratio among pollutants, and health risk	*Atmos Pollut Res*	Metadata only (no abstract)	Metadata only - biomass-burning episode, pollutant enhancement ratios, health risk (by title)	China (Northeast)
Zhang respirable magnetite emissions coal-fired power China	Emissions of Respirable Magnetite from the Coal-Fired Power Sector in China: Spatial Inequalities and Mitigation Pathways	*Environ Sci Technol*	Emission inventory	Respirable magnetite (Fe3O4) particle emissions	China (national, provincial)
Zhao et al.	Pollution Characteristics, Influencing Factors and Health Risk Assessments of Mercury in Atmospheric $PM_{2.5}$ from Xinxiang, China	*Atmosphere*	Source apportionment	$PM_{2.5}$-bound mercury with OC/EC, water-soluble ions and trace elements, full-year 2015 sampling	China (Xinxiang, Henan)
Zhao et al.	Thermal-oxygen evolution and atmospheric pollutant emission characteristics during the vertical propagation of underground coal fires	*Research Square (preprint)*	Chamber / laboratory	$PM_{2.5}$, PM_{10} and TSP with CO2, CO, CH4 and C2H6 through a simulated burn cycle	China (laboratory, physical similarity simulation)
Other clinical endpoints					
Beijing CKD cohort	Association between Long-term Exposure to Air Pollution and Incident Chronic Kidney Disease among Adults in Beijing, China	*Environ Pollut*	Prospective cohort	Annual PM_{10}, $PM_{2.5}$, NO2, CO and O3, with a composite pollution score	China (Beijing)
Bottini antibiotic resistance Ecoli Salmonella $PM_{2.5}$ ecological	Environmental, socioeconomic, and livestock-related factors associated with antibiotic resistance in Escherichia coli, Shigella spp., and Salmonella enterica	*One Health*	Ecological	Country-level $PM_{2.5}$ exposure	Global / multi-region (102 countries)
CLHLS indoor constituents	Association between exposure to indoor $PM_{2.5}$ constituents and mortality risk among Chinese older adults: a nationwide prospective cohort study	*BMC Public Health*	Prospective cohort	Indoor $PM_{2.5}$ and constituents (BC, OM, sulfate, ammonium, nitrate) from ambient fields x infiltration factors	China (nationwide, CLHLS-type cohort)
China fire-sourced $PM_{2.5}$ cancer	Extreme, Accumulated, and Lagged Exposure to Fire-Sourced $PM_{2.5}$ and Site-specific Cancer Risk in China	*Environ Pollut*	Ecological / registry analysis	Fire-sourced vs all-source $PM_{2.5}$; lagged, accumulated and extreme metrics	China (nationwide)
Dahl air pollution bone fractures high-risk population	Air pollution and bone fractures in a high-risk population: differences by age and sex	*Environ Int*	Metadata only (no abstract)	Ambient air pollution (metric not stated in record)	Not stated in record
Deng ultrafine particles kidney disease ED visits New York	A mysterious danger for kidney disease exacerbations: short-term impacts of ultrafine particles on hospital visits	*Environ Res*	Time-stratified case-crossover	Daily UFP number concentration from a validated chemical transport model (street-geocoded residence)	USA (New York State)
Duan landscape fire-sourced $PM_{2.5}$ kidney function eGFR children China	Landscape fire-sourced $PM_{2.5}$ and kidney function in children: evidence from a national cross-sectional study in China	*Environ Res*	Cross-sectional survey	Three-year mean landscape fire-sourced $PM_{2.5}$ at school address (modelled)	China (national surveillance)
Hefei conjunctivitis	Ambient air pollutants and outpatient visits for conjunctivitis in Hefei, China: a time-series study of hospital-based data during 2014-2023	*Front Public Health*	Monitoring-record analysis	Daily $PM_{2.5}$ and PM_{10} with O3, NO2 and SO2 from regulatory monitors	China (Hefei)
Hoang $PM_{2.5}$ minimal residual disease childhood ALL REDIAL	Fine particulate matter and minimal residual disease in childhood acute lymphoblastic leukemia from the REDIAL consortium	*JNCI Cancer Spectr*	Retrospective cohort	CDC daily census-tract $PM_{2.5}$ during induction (residence; centre then residence)	USA (REDIAL consortium, Texas-led)
Hu air pollution built environment lung cancer recurrence after surgery China	Ambient Air Pollution, Residential Built Environment, and Lung Cancer Recurrence After Curative-Intent Surgery: A Retrospective Cohort Study in Three Chinese Cities	*Research Square (preprint)*	Retrospective cohort	Satellite-derived 1 km $PM_{2.5}$, PM_{10}, NO2, CO, SO2, O3 (post-diagnosis first year)	China (three cities)

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Jiang air pollution meteorology hand foot mouth disease Shenzhen DLNM	Ambient air pollution and meteorological factors associated with hand, foot, and mouth disease risk in Shenzhen, China: a time-series study	*Front Public Health*	Time-series	Daily city-wide $PM_{2.5}$ and PM_{10} (fixed monitors)	China (Shenzhen)
Li ambient air pollution gastric cancer incidence 204 countries ecological	Ambient Air Pollution and Gastric Cancer Incidence: A Multinational Ecological Analysis	*Curr Med Sci*	Ecological panel, fixed effects	GBD 2021 national population-weighted $PM_{2.5}$, NO2, O3	Global (204 countries)
Lima et al.	Epigenetic Exposure Signatures Distinguish Early-Onset from Later-Onset Colorectal Cancer in a Racially Diverse US Cohort	*medRxiv (preprint)*	Observational - cross-sectional	Methylation risk score for coarse PM (tumour tissue), not measured exposure	USA (DANCE cohort)
Lin $PM_{2.5}$ frailty trajectories CHARLS	The association of particulate matter 2.5 exposure with frailty trajectories in middle-aged and older Chinese adults and the potential modification by lifestyle and socioeconomic status	*J Occup Environ Med*	Prospective cohort	Satellite-derived $PM_{2.5}$ (Dalhousie/ACAG surface, per 10 $\mu g/m^3$)	China (national, CHARLS)
Lin air pollution cervical cancer recurrence mortality Fudan cohort	Long-Term Exposure to Ambient Air Pollution and Cervical Cancer Outcomes in Chinese Women: A Prospective Study on Recurrence and Mortality Risks	*Environ Health (ACS)*	Prospective cohort	Residential $PM_{2.5}$ and PM_{10} from high-resolution spatiotemporal models (post-surgery)	China (Shanghai, single hospital)
Ma green space pollution aging	Associations of Green Space and Air Pollution Mixture with Heterogeneous Biological Aging Patterns	*Environ Health Perspect*	Prospective cohort	$PM_{2.5}$, PM_{10}, SO2, O3 mixture (WQS); 500 m NDVI	China (single high-pollution city)
Northern Thailand NPC survival $PM_{2.5}$ and coarse PM	The Effect of Fine ($PM_{2.5}$) and Coarse (PM_{10}-2.5) Particulate Matter on Survival Outcomes in Nasopharyngeal Carcinoma Patients: A Retrospective Cohort Study in a Northern Thailand Tertiary Care Center	*J Clin Med*	Metadata only (no abstract)	$PM_{2.5}$ and PM_{10}-2.5 (exposure model not stated)	Thailand (northern)
Ran $PM_{2.5}$ MASLD proteins	Association of protein signature of fine particulate matter with MASLD: the underlying pathways and Mendelian randomization	*J Adv Res*	Prospective cohort	Annual average $PM_{2.5}$ by bilinear interpolation, time-weighted; 700-protein elastic-net signature	Not stated (large prospective cohort, 52,190 participants)
SEER wildfire $PM_{2.5}$ head and neck cancer	Wildfire Air Pollution Exposure and Head and Neck Cancer Incidence	*Laryngoscope*	Ecological / registry analysis	County-level wildfire-specific $PM_{2.5}$ (exceedance days, any exposure, % days > 5 $\mu g/m^3$), 0-10-y lags	United States (SEER counties)
Sandal et al.	Disasters and hazard exposure in end-stage kidney disease: a scoping review of patient outcomes	*Clin Kidney J*	Scoping review	Poor air quality among heat, disaster and conflict hazards (metric varies by study)	Global (30 studies, 10 countries/regions)
Slovakia $PM_{2.5}$ lung cancer mortality gender lag ecological	$PM_{2.5}$ exposure and lung cancer mortality in Slovakia: gender differences and delayed effects	*Eur J Public Health (conference abstract)*	Ecological	National population-weighted $PM_{2.5}$ (Indicator of Population Exposure), 2010-2023	Slovakia
Tian et al.	Annual average exposure to ambient air pollution, polygenic risk, and incidence of major chronic diseases: a large-scale prospective cohort study	*Hum Genet*	Prospective cohort	Annual $PM_{2.5}$, PM_{10}, NO2, NOx at residence from land-use regression	United Kingdom (UK Biobank)
Wang $PM_{2.5}$ lung cancer nonsmokers narrative review IL-1beta EGFR screening	Environmental $PM_{2.5}$ exposure associated with lung cancer in nonsmokers: a narrative review of high-risk groups, mechanisms, and implications for screening	*J Toxicol Environ Health B Crit Rev*	Narrative review	Ambient and microenvironmental $PM_{2.5}$ (incense, cooking, transit, occupational)	Global/multi-region
Wen landscape fire $PM_{2.5}$ renal hospitalisations six-country time-series	Landscape Fire-Sourced $PM_{2.5}$ and Cause-Specific Renal Hospitalizations: A Multicountry Two-Stage Time-Series Analysis of 12.5 Million Admissions	*Environ Sci Technol*	Multi-location time-series	Landscape-fire-sourced vs non-fire $PM_{2.5}$ (per 10 $\mu g/m^3$, lag 0-5)	Multi-country (Australia, Brazil, Canada, Chile, New Zealand, Thailand)

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Xia et al.	Long-term exposure to fine particulate matter constituents and lung cancer risk in Chinese never-smokers	*J Hazard Mater*	Prospective cohort	Annual $PM_{2.5}$ and constituents (NO3-, NH4+, Cl-, SO4 2-, BC, OM), ChinaHighAirPollutants 2004-2018	China (China Kadoorie Biobank, 10 regions)
Yang and Liu	No evidence for large causal effects of six air pollutants on non-suppurative otitis media: a two-sample Mendelian randomization study	*medRxiv* (*preprint*)	Two-sample Mendelian randomisation	Genetic instruments for $PM_{2.5}$, $PM_{2.5}$ absorbance, $PM_{2.5}$-10, PM_{10}, NO2, NOx (UK Biobank GWAS)	Europe (UK Biobank exposure; FinnGen R13 outcome)
Zhang air pollution cataract meta-analysis	Air pollution exposure and cataract outcomes: a systematic review and pollutant-specific meta-analysis	*Front Med* (*Lausanne*)	Systematic review & meta-analysis	Long-term $PM_{2.5}$, PM_{10}, NO2, black carbon (study-specific models)	Global / multi-region (15 studies)
Zhong air pollution years of life lost cancer Pudong Shanghai time series	The short-term impact of ambient air pollution on years of life lost from cancer in Pudong New Area, Shanghai: A time series study	*PLoS One*	Distributed-lag time-series	Daily $PM_{2.5}$, PM_{10}, SO2, NO2, O3 (monitoring network)	China (Shanghai, Pudong New Area, 2013-2021)

Provenance and method

Pure aggregation over `state/corpus/2026-10-*.json` (31 daily issues, entry windows 1–31 October 2026, contiguous), `state/metrics.csv` and `state/trials.json`. No API was re-queried and no DOI re-resolved; every record passed the DOI gate at daily-issue time. Figures are pooled re-renders of the daily set; the forest panels catalogue estimates and do not pool them. Abstracts remain the copyright of their respective publishers.